Physics of the Earth as a Planet

Lectures 13–24: Seismology

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Preface

These notes cover the seismology part of the course, Lectures 13–24, and they will also be available on the Teaching Information System (TiS). The material is close to that of previous years, but the notes have been revised throughout and a number of figures added. Errors will surely remain: if you find one, however small, please email me at da380@cam.ac.uk and the version on the TiS will be updated. The two problem sets for this part of the course are included at the end of these notes, and their solutions are available on the TiS.

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Lecture 13: Finite elasticity

Outline and motivation

In this lecture, we derive the equations of motion for an elastic body. These equations provide a good, though not complete, model for understanding seismic wave propagation and other related topics in geophysics. Most of you will have encountered elasticity within earlier courses, but here we will go back to first principles. In particular, we discuss **finite elasticity**, by which we mean that there is no assumption that the deformation remains small. Within later applications it will actually be sufficient to consider linearised motions about an equilibrium state, but by starting with the exact theory it is easier to appreciate the underlying physics.

Basic definitions

We start by considering the kinematics of a solid body. At an instant in time, t, the body occupies a subset M_t of physical space, $\mathbb{R}^3$. A useful way to describe the motion is through the introduction of a **reference body**, M. Each point $\mathbf{x}$ in the reference body corresponds to a unique particle. At time t, the particle $\mathbf{x}$ sits at a point in physical space that we denote by $\boldsymbol{\varphi}(\mathbf{x}, t)$. In this manner we see that the evolution of the body is completely described by this time-dependent mapping, $\boldsymbol{\varphi}(\cdot, t)$, from the reference body into physical space. We will call $\boldsymbol{\varphi}$ the **motion** of the body, this being the "field" within this classical field theory. We will not consider motions for which the body is torn or fractured. As a result, we suppose that $\boldsymbol{\varphi}$ is continuously differentiable, and that at each fixed time, t, it establishes a one-to-one correspondence between M and M_t. These ideas are summarised graphically in Fig. 1.

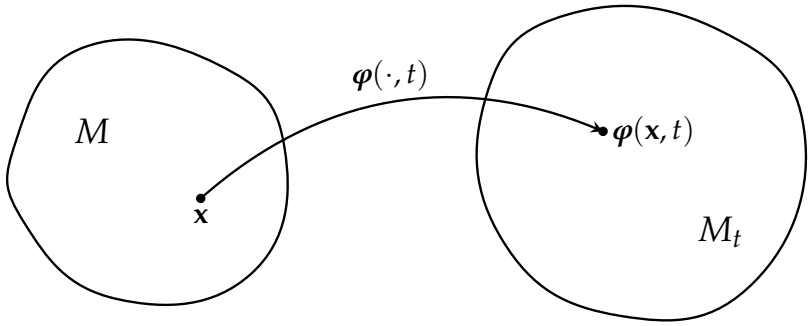


Fig. 1: A schematic illustration of the mapping from the reference body M into the body's instantaneous position in physical space.

In making these definitions we have not specified how the reference body is chosen. Indeed, it turns out that there are infinitely many ways of doing this. Nevertheless, a common approach is to let points in the reference body correspond to the position of the particles at some reference time, t_0. Having done this, we then have

$$\boldsymbol{\varphi}(\mathbf{x}, t_0) = \mathbf{x}, \tag{1}$$

for all $\mathbf{x}$. There are circumstances in which it is useful to select a different reference body, but we will not worry about that within this course. It is, however, worth knowing that this freedom can be put to real use: for example, the equations of motion for an aspherical earth model can be exactly rewritten as those for a spherically symmetric reference body, with the effects of boundary topography and lateral heterogeneity absorbed into modified material parameters.

Velocity, deformation gradient, and Jacobian

If we fix attention on a particle $\mathbf{x}$, then the function $t \mapsto \boldsymbol{\varphi}(\mathbf{x}, t)$ describes its trajectory through space. Differentiating with respect to time, we obtain the **referential velocity**

$$v_i(\mathbf{x}, t) = \frac{\partial \varphi_i}{\partial t}(\mathbf{x}, t). \tag{2}$$

Note that this is the velocity of a fixed particle, and not that at a fixed spatial point. This differs from what is commonly done in fluid mechanics. In what follows, we will generally drop the term "referential" and just say "velocity" for this vector field.[1]

A second quantity that will be important is the **deformation gradient**, $\mathbf{F}$, which is defined by

$$F_{ij} = \frac{\partial \varphi_i}{\partial x_j}. \tag{3}$$

Physically, the deformation gradient quantifies how the motion differs between nearby points in the reference body. For example, if we consider particles $\mathbf{x}$ and $\mathbf{x} + \delta\mathbf{x}$ with $\delta\mathbf{x}$ small, then we can use a Taylor expansion to obtain

$$\varphi_i(\mathbf{x} + \delta\mathbf{x}, t) = \varphi_i(\mathbf{x}, t) + F_{ij}(\mathbf{x}, t)\delta x_j + \cdots, \tag{4}$$

which is accurate to first order in $\delta\mathbf{x}$. The deformation gradient is related to, but not identical with, the concept of **strain**, a correspondence discussed further in the next lecture.

Because we have assumed that the motion establishes a one-to-one correspondence between M and M_t at each time, it can be shown that $\mathbf{F}(\mathbf{x}, t)$ is an invertible matrix at each point.[2] The determinant of the deformation gradient is known as the **Jacobian** of the motion, and is written

$$J(\mathbf{x}, t) = \det \mathbf{F}(\mathbf{x}, t). \tag{5}$$

Invertibility of the deformation gradient implies that J is never equal to zero, while continuity of J, which follows from the continuous differentiability of $\boldsymbol{\varphi}$, implies that it is of one sign throughout the body (assumed connected). We will assume that J is always positive. This implies in particular that a right-handed set of basis vectors in M is mapped to a right-handed set of basis vectors in M_t.

Consider a function f defined in the instantaneous body M_t, and its integral

$$I = \int_{M_t} f(\mathbf{y}) \, \mathrm{d}^3\mathbf{y}. \tag{6}$$

[1] I prefer the descriptive terms "referential" and "spatial" for the two ways in which quantities can be described within a continuum. Some authors instead use "Lagrangian" and "Eulerian" for these concepts, though both perspectives were introduced by Euler.

[2] This follows from a result known as the **inverse function theorem**.

Using the motion at this instant in time, we can change variables to write the integral as one over the reference body

$$I = \int_M f[\boldsymbol{\varphi}(\mathbf{x}, t)] J(\mathbf{x}, t) \, \mathrm{d}^3\mathbf{x}. \tag{7}$$

This ability to easily transform integrals between the instantaneous and reference bodies will be very useful in what follows.

Conservation of mass

Let U be a subset of the reference body and U_t the region of physical space it occupies at time t. The mass of this sub-body can be written

$$\int_{U_t} \varrho(\mathbf{y}, t) \, \mathrm{d}^3\mathbf{y}, \tag{8}$$

where $\varrho(\mathbf{y}, t)$ is the **spatial density** at the point $\mathbf{y}$ in M_t at time t. During the deformation, matter is neither created nor destroyed. This means that the mass of this arbitrary sub-body is constant, and hence

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{U_t} \varrho(\mathbf{y}, t) \, \mathrm{d}^3\mathbf{y} = 0. \tag{9}$$

To simplify this expression we use the motion to transform the integration in eq. (9) to the reference body

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_U J(\mathbf{x}, t) \varrho[\boldsymbol{\varphi}(\mathbf{x}, t), t] \, \mathrm{d}^3\mathbf{x} = 0. \tag{10}$$

This integral is over a fixed domain, and so we can interchange the order of differentiation and integration. Furthermore, as U is an arbitrary subset of M, the resulting equality can hold for every such U if and only if

$$\frac{\partial}{\partial t} \{ J(\mathbf{x}, t) \varrho[\boldsymbol{\varphi}(\mathbf{x}, t), t] \} = 0. \tag{11}$$

It follows that we can introduce a time-independent **referential density** ρ such that

$$\rho(\mathbf{x}) = J(\mathbf{x}, t) \varrho[\boldsymbol{\varphi}(\mathbf{x}, t), t]. \tag{12}$$

This is a statement of mass conservation expressed within the reference body. Note that if the reference body is chosen such that eq. (1) holds at some time $t = t_0$, then at this time eq. (12) simplifies to

$$\rho(\mathbf{x}) = \varrho(\mathbf{x}, t_0). \tag{13}$$

Thus, when using this common choice of reference body, the referential density is precisely the density at the reference time.

Kinetic energy

By analogy with particle mechanics, the **kinetic energy density** of the body, measured per unit volume within the reference body M, is defined as

$$\frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}(\mathbf{x},t)\|^2, \tag{14}$$

and summing up such contributions, we see that the total kinetic energy can be written

$$\mathcal{T}(t) = \frac{1}{2}\int_M \rho(\mathbf{x})\|\mathbf{v}(\mathbf{x},t)\|^2 \,\mathrm{d}^3\mathbf{x}. \tag{15}$$

Potential energy

What should the potential energy of an elastic body look like? The basic idea is that the body has a preferred shape and/or volume, and that energy is required to deform the body away from this equilibrium state. As a starting point we assume that the potential energy can be expressed in the form

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x},t), \mathbf{F}(\mathbf{x},t)] \,\mathrm{d}^3\mathbf{x}, \tag{16}$$

where W is a **potential energy density** measured per unit volume within the reference body. This is not the most general form that the potential energy could take, but defines what is known as a **simple hyperelastic body**. That a body is **hyperelastic** means that it has a well-defined potential energy. A consequence of this assumption, which you will establish in the first problem set, is that energy is conserved during the deformation of the body, and hence within this theory we are discounting dissipative processes.[3] Energy dissipation is relevant to seismology, but it is a relatively minor effect that we will neglect in this course. The term **simple** in the above context means that the potential energy density at a point, $\mathbf{x}$, depends only on the motion and the deformation gradient at the same point. By contrast, we might have allowed the potential energy density to depend on higher-order spatial derivatives of the motion, or on the motion in some non-local manner. But there seems to be no need for such generalisations for geophysical applications.

Simplifying the potential energy using symmetries

Having accepted the form of the potential energy in eq. (16), we are now in a position to obtain the equations of motion using Hamilton's principle. Before doing this, the form of the potential energy density can be simplified by imposing certain physical constraints:

(i) **Time-translation invariance.** Consider a motion $\boldsymbol{\varphi}(\mathbf{x},t)$ and its time-translation $\boldsymbol{\varphi}'(\mathbf{x},t) = \boldsymbol{\varphi}(\mathbf{x},t-\tau)$ for some τ. The second motion takes exactly the same form

[3] The situation is analogous to particle mechanics restricted to the case of conservative forces, i.e. those forces that are derived from a potential energy.

but is just shifted in time, and hence it is natural to require that their respective potential energies satisfy

$$\mathcal{V}'(t) = \mathcal{V}(t-\tau). \tag{17}$$

This condition is automatically met if the potential energy density has no explicit dependence on time, and hence we simplify its form to

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x},t), \mathbf{F}(\mathbf{x},t)]\, \mathrm{d}^3\mathbf{x}. \tag{18}$$

The necessity of this condition can also be established.

(ii) **Invariance under rigid body motions.** As stated above, an elastic body has a preferred shape and/or volume, and the potential energy is associated with deformations of the body away from this. Motions that do not alter its shape or volume should not, therefore, be associated with a change in potential energy. Within Euclidean space, motions that do not change the relative lengths between points arise through a combination of rotations and translations.[4] If $\boldsymbol{\varphi}$ is a motion of the body, we can define a second motion $\boldsymbol{\varphi}'$ by setting

$$\boldsymbol{\varphi}'(\mathbf{x},t) = \mathbf{a}(t) + \mathbf{Q}(t)\boldsymbol{\varphi}(\mathbf{x},t), \tag{19}$$

where $\mathbf{a}(t)$ is a time-dependent translation vector and $\mathbf{Q}(t)$ a time-dependent rotation matrix. Because these motions differ only by a superimposed rigid body motion, they must be associated with the same potential energy. Noting that $\mathbf{F}'(\mathbf{x},t) = \mathbf{Q}(t)\mathbf{F}(\mathbf{x},t)$, it follows that this invariance condition will be met so long as the potential energy density satisfies

$$W[\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x},t), \mathbf{F}(\mathbf{x},t)] = W[\mathbf{x}, \mathbf{a}(t) + \mathbf{Q}(t)\boldsymbol{\varphi}(\mathbf{x},t), \mathbf{Q}(t)\mathbf{F}(\mathbf{x},t)]. \tag{20}$$

Again, the necessity of this condition can also be proven. Setting $\mathbf{Q}(t) = \mathbf{I}$, this identity reduces to

$$W[\mathbf{x}, \boldsymbol{\varphi}(\mathbf{x},t), \mathbf{F}(\mathbf{x},t)] = W[\mathbf{x}, \mathbf{a}(t) + \boldsymbol{\varphi}(\mathbf{x},t), \mathbf{F}(\mathbf{x},t)]. \tag{21}$$

For fixed $(\mathbf{x}, t)$, we can vary $\mathbf{a}(t)$ such that the second argument on the right-hand side takes any possible value. It follows that the potential energy density must be independent of its second argument. Given this result, it is useful once again to redefine the potential energy density to depend only on $\mathbf{x}$ and $\mathbf{F}(\mathbf{x},t)$, and having done this the condition in eq. (20) reduces to

$$W[\mathbf{x}, \mathbf{F}(\mathbf{x},t)] = W[\mathbf{x}, \mathbf{Q}(t)\mathbf{F}(\mathbf{x},t)], \tag{22}$$

for all deformation gradients, and all rotation matrices $\mathbf{Q}(t)$. This condition limits the allowable form of the potential energy density, and we will explore its consequences further within the next lecture.

In summary, we find that the potential energy takes the simpler form

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \mathbf{F}(\mathbf{x},t)]\, \mathrm{d}^3\mathbf{x}, \tag{23}$$

[4] Reflections also preserve lengths, but they are excluded by our assumption that $J > 0$.

where W is subject to eq. (22). Within continuum mechanics this potential energy density is called the **strain energy function**, and this is the term we shall use from now on. The invariance of the potential energy under (i) time-translation, and (ii) superimposed rigid body motions, goes under the collective name of the principle of **material frame indifference**. This idea was developed by Clifford Truesdell and Walter Noll during the 1950s and 1960s (see Truesdell & Noll 1965), and plays a central role within all branches of theoretical continuum mechanics. Here we have only formulated this principle for a simple hyperelastic body, but it can be generalised to other types of material, and to incorporate thermodynamics and electromagnetism.

Hamilton's principle

A Lagrangian density for a hyperelastic body is defined in the usual manner as the difference between the kinetic and potential energy densities

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, \mathbf{F}), \tag{24}$$

while the associated action is

$$\mathcal{S}(\boldsymbol{\varphi}) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t, \tag{25}$$

where $[0, T]$ is the time interval of interest, and it is understood that the referential velocity and deformation gradient on the right-hand side are those associated with the given motion.[5] Note that the Lagrangian density in eq. (24) shows no explicit dependence on either t or $\boldsymbol{\varphi}$, but we shall retain this more general expression in what follows. Hamilton's principle states that the true motion is a stationary point of the action with respect to all smooth variations $\delta\boldsymbol{\varphi}$ which satisfy the fixed end-point condition

$$\delta\boldsymbol{\varphi}(\mathbf{x}, 0) = \delta\boldsymbol{\varphi}(\mathbf{x}, T) = \mathbf{0}. \tag{26}$$

Mathematically, this means that we require the first variation of the action

$$\delta\mathcal{S} = \lim_{h \to 0} \frac{1}{h}[\mathcal{S}(\boldsymbol{\varphi} + h\,\delta\boldsymbol{\varphi}) - \mathcal{S}(\boldsymbol{\varphi})] \tag{27}$$

vanish for all smooth variations $\delta\boldsymbol{\varphi}$ satisfying eq. (26). The first variation depends linearly on $\delta\boldsymbol{\varphi}$, and it is often convenient to write $\delta\mathcal{S} = \langle D\mathcal{S}(\boldsymbol{\varphi}) | \delta\boldsymbol{\varphi} \rangle$, where $D\mathcal{S}(\boldsymbol{\varphi})$ is called the **functional derivative** of the action at $\boldsymbol{\varphi}$, and the angled brackets denote its linear action on a variation; we shall meet this notation again in later lectures. We readily find that this first variation can be written

$$\delta\mathcal{S} = \int_0^T \int_M \left\{ \frac{\partial\mathcal{L}}{\partial\varphi_i}\delta\varphi_i + \frac{\partial\mathcal{L}}{\partial v_i}\delta v_i + \frac{\partial\mathcal{L}}{\partial F_{ij}}\delta F_{ij} \right\} \mathrm{d}^3\mathbf{x} \, \mathrm{d}t, \tag{28}$$

[5] For those with less experience with variational principles, the necessary ideas are developed here from first principles. But please ask in a supervision, or otherwise, if you would like any additional help with this material.

where the variations in the referential velocity and deformation gradient are given by

$$\delta v_i = \frac{\partial \delta \varphi_i}{\partial t}, \tag{29}$$

$$\delta F_{ij} = \frac{\partial \delta \varphi_i}{\partial x_j}, \tag{30}$$

and we have suppressed all arguments for clarity. From the following identities

$$\frac{\partial \mathcal{L}}{\partial v_i}\delta v_i = \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial v_i}\delta \varphi_i\right) - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial v_i}\right)\delta \varphi_i, \tag{31}$$

$$\frac{\partial \mathcal{L}}{\partial F_{ij}}\delta F_{ij} = \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial F_{ij}}\delta \varphi_i\right) - \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial F_{ij}}\right)\delta \varphi_i, \tag{32}$$

along with eq. (26) and the divergence theorem, we obtain

$$\delta \mathcal{S} = \int_0^T \int_M \left\{ \frac{\partial \mathcal{L}}{\partial \varphi_i} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial v_i}\right) - \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial F_{ij}}\right)\right\} \delta \varphi_i \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t + \int_0^T \int_{\partial M} \frac{\partial \mathcal{L}}{\partial F_{ij}} \hat{n}_j \delta \varphi_i \, \mathrm{d}S \, \mathrm{d}t, \tag{33}$$

where $\hat{\mathbf{n}}$ denotes the outward unit normal vector to ∂M, and $\mathrm{d}S$ is the surface element. As $\delta\boldsymbol{\varphi}$ is arbitrary, this first variation can vanish only if the volume and boundary terms are independently equal to zero.[6] Hamilton's principle, therefore, implies that the **Euler–Lagrange equations**

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial v_i}\right) - \frac{\partial}{\partial x_j}\left(\frac{\partial \mathcal{L}}{\partial F_{ij}}\right) = 0, \tag{34}$$

hold within the body, while we also have the **natural boundary conditions**

$$\frac{\partial \mathcal{L}}{\partial F_{ij}} \hat{n}_j = 0, \tag{35}$$

on ∂M. Note that these boundary conditions are called "natural" because they have arisen directly from the variational principle, and not because of some prior restriction on the class of admissible motions.

For the Lagrangian density in eq. (24) we readily find that

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} = 0, \tag{36}$$

$$\frac{\partial \mathcal{L}}{\partial v_i} = \rho v_i, \tag{37}$$

$$\frac{\partial \mathcal{L}}{\partial F_{ij}} = -\frac{\partial W}{\partial F_{ij}}. \tag{38}$$

The **first Piola–Kirchhoff stress tensor** is defined by

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)], \tag{39}$$

[6] If this statement is not obvious, then you might wish to ask in a supervision. But for the purposes of this course it is sufficient to know how to apply this method.

so that the Euler–Lagrange equations can be written

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0, \tag{40}$$

while the boundary condition becomes

$$T_{ij}\hat{n}_j = 0. \tag{41}$$

The vector $\mathbf{t} = \mathbf{T}\hat{\mathbf{n}}$ occurring in the boundary conditions is known as the **traction**. Within the first problem set you will see that it represents a *surface force measured per unit area* within the reference body. Indeed, what the first Piola–Kirchhoff stress does is relate the orientation of a surface within the reference body to the force acting on the instantaneous image of that surface within physical space, this force being measured per unit area within the reference body.

There are other stress tensors that can be defined. For example, within spatial descriptions of continuum mechanics, there is the **Cauchy stress tensor** which relates the orientation of surfaces in the instantaneous body to the force per unit area acting on this instantaneous surface. The Cauchy stress tensor is the one that is most commonly used within fluid mechanics (including in the earlier parts of this course). In the next lecture, we will see that within linearised theories about a stress-free equilibrium state the first Piola–Kirchhoff stress and the Cauchy stress are equal, with this result accounting for the fact that their distinction is not made or emphasised within elementary courses on continuum mechanics.

As a final comment, the equations of motion are defined on the *fixed* reference body, and hence we need not worry about applying boundary conditions on a moving boundary as is sometimes required in fluid mechanics problems that use a spatial formulation of continuum mechanics. This fact considerably simplifies the solution of free-boundary problems in elasticity, and is one of the reasons for working with a referential description of the motion.

What you need to know and be able to do

(i) Understand the terminology needed to describe the kinematics of an elastic body.

(ii) Use the conservation of mass to derive the relation between the spatial and referential densities.

(iii) State expressions for the kinetic and potential energies of a simple hyperelastic body, and explain the underlying physical ideas and assumptions.

(iv) Give a statement of Hamilton's principle in the context of finite elasticity, and use it to derive the equations of motion and associated boundary conditions.

Lecture 14: Constitutive theory

Outline and motivation

In the previous lecture, we obtained the equations of motion for a simple hyperelastic body. The mechanical properties of such bodies are defined by a referential density, ρ, and a strain energy function, W. Using the principle of material frame indifference, we obtained conditions that must be satisfied by the strain energy function. Within the first part of this lecture, we will explore these restrictions in more detail to determine concretely the form the strain energy function must take. We then consider **linearised motions** about a **stress-free equilibrium configuration**. The resulting equations of motion are appropriate for studying most situations within seismology. To conclude we discuss **material symmetries**, which occur in many real materials due to their underlying microscopic structure. By taking account of such symmetries the form of the equations of motion can be significantly simplified.

The polar decomposition theorem

In the previous lecture, we showed that the principle of material frame indifference requires the strain energy function of a simple hyperelastic body to satisfy

$$W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \mathbf{QF}), \tag{1}$$

for any deformation gradient $\mathbf{F}$ and any rotation matrix $\mathbf{Q}$. Physically, this states that the potential energy of the body is invariant under superimposed rigid body rotations. To understand the implications of this condition, we need a result from linear algebra known as the **polar decomposition theorem**. This result states that an invertible matrix $\mathbf{F}$ with $\det \mathbf{F} > 0$ can be written in the form

$$\mathbf{F} = \mathbf{RU} = \mathbf{VR}, \tag{2}$$

where $\mathbf{R}$ is a rotation matrix, while $\mathbf{U}$ and $\mathbf{V}$ are symmetric and positive-definite matrices with the same eigenvalues. The geometric interpretation of this theorem is that the action of a deformation gradient can be viewed as either (i) a stretching about a set of orthogonal axes followed by a rotation, or (ii) the same rotation followed by stretching about co-rotated axes. A graphical representation of this idea can be seen in Fig. 1.

Proof of the polar decomposition theorem (non-examinable)

You do not need to know the proof of this theorem, but it is simple enough to sketch for those interested. First we define the symmetric matrix

$$\mathbf{C} = \mathbf{F}^T\mathbf{F}, \tag{3}$$

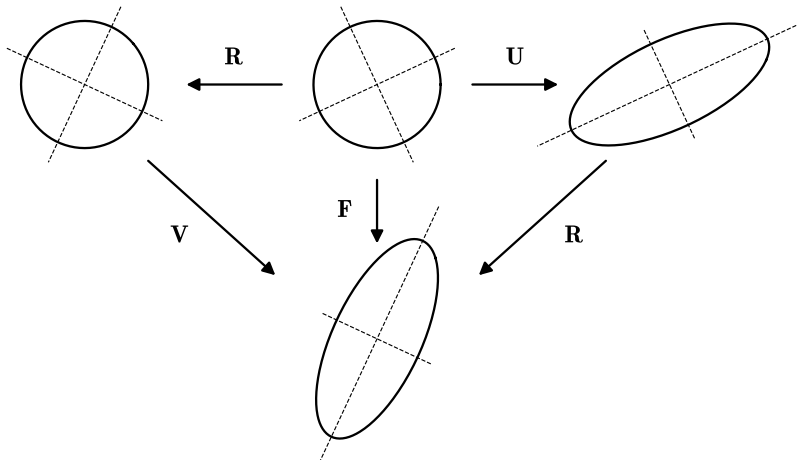


Fig. 1: Under an invertible matrix **F**, a circle (in three dimensions, a sphere) is mapped to an ellipse (ellipsoid). Using this fact, the geometric meaning of the polar decomposition theorem is shown. An undeformed circle is shown in the centre of the upper row, with the dashed lines along the eigenvectors of the symmetric matrix **U**; in the other panels the dashed lines show the images of these axes.

known as the **right Cauchy–Green deformation tensor**, which is positive-definite because **F** is invertible. It follows that the square root of **C** can be defined,[1] and we set $\mathbf{U} = \sqrt{\mathbf{C}}$ which is symmetric and positive-definite by construction. Our claim is that $\mathbf{R} = \mathbf{F}\mathbf{U}^{-1}$ is a rotation matrix. This requires that $\mathbf{R}^T\mathbf{R} = \mathbf{I}$ and also $\det\mathbf{R} = 1$. For the first identity we have

$$\mathbf{R}^T\mathbf{R} = \mathbf{U}^{-1}\mathbf{F}^T\mathbf{F}\mathbf{U}^{-1}, \tag{4}$$

but $\mathbf{F}^T\mathbf{F} = \mathbf{C} = \mathbf{U}^2$, and hence $\mathbf{R}^T\mathbf{R} = \mathbf{I}$ as required. For the determinant condition we note that $\det\mathbf{R} = \det\mathbf{F}/\det\mathbf{U}$, while $\mathbf{U}^2 = \mathbf{F}^T\mathbf{F}$ and $\det\mathbf{U} > 0$ imply $\det\mathbf{U} = |\det\mathbf{F}|$. Since we assume $\det\mathbf{F} > 0$, we have $\det\mathbf{R} = 1$. For the second form of the decomposition, we merely note that

$$\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{R}\mathbf{U}\mathbf{R}^T\mathbf{R}, \tag{5}$$

and define $\mathbf{V} = \mathbf{R}\mathbf{U}\mathbf{R}^T$ which is just **U** rotated and so has the same eigenvalues.

Implications for the strain energy function

Using the polar decomposition theorem, the deformation gradient can be written

$$\mathbf{F} = \mathbf{R}\mathbf{U}, \tag{6}$$

with **R** a rotation matrix, while $\mathbf{U} = \sqrt{\mathbf{C}}$ with **C** the right Cauchy–Green deformation tensor of eq. (3). It follows from eq. (1) that

$$W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \mathbf{R}\mathbf{U}) = W(\mathbf{x}, \mathbf{U}), \tag{7}$$

[1] This is done by first diagonalising $\mathbf{C} = \mathbf{P}\,\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)\mathbf{P}^T$ with **P** orthogonal, and then setting $\mathbf{U} = \mathbf{P}\,\mathrm{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \sqrt{\lambda_3})\mathbf{P}^T$, which is well-defined as the eigenvalues are all positive.

and we conclude that the strain energy depends on the deformation gradient only through the symmetric matrix $\mathbf{U}$, or equivalently through the right Cauchy–Green deformation tensor $\mathbf{C}$ defined in eq. (3).[2] To summarise, we have shown that a strain energy function satisfies the principle of material frame indifference if and only if it takes the form

$$W(\mathbf{x}, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}), \tag{8}$$

where U is some auxiliary strain energy function dependent on the right Cauchy–Green deformation tensor. Note that we use $\mathbf{C}$ instead of $\mathbf{U}$ in this formula because it simplifies later calculations.

Combining eq. (8) with the definition of the first Piola–Kirchhoff stress tensor

$$T_{ij} = \frac{\partial W}{\partial F_{ij}}, \tag{9}$$

we use the chain rule to obtain

$$T_{ij} = \frac{\partial U}{\partial C_{kl}} \frac{\partial C_{kl}}{\partial F_{ij}}. \tag{10}$$

From the definition of the right Cauchy–Green deformation tensor, a routine calculation shows that

$$\frac{\partial C_{kl}}{\partial F_{ij}} = \frac{\partial}{\partial F_{ij}}(F_{mk}F_{ml}) = \delta_{im}\delta_{jk}F_{ml} + F_{mk}\delta_{im}\delta_{jl} = \delta_{jk}F_{il} + \delta_{jl}F_{ik}, \tag{11}$$

and using the symmetry of $\mathbf{C}$ we arrive at the identity

$$T_{ij} = 2F_{ik}\frac{\partial U}{\partial C_{kj}}. \tag{12}$$

If we define the **second Piola–Kirchhoff stress tensor** to be

$$S_{ij} = 2\frac{\partial U}{\partial C_{ij}}, \tag{13}$$

the first Piola–Kirchhoff stress tensor can be decomposed in the form

$$T_{ij} = F_{ik}S_{kj}, \tag{14}$$

or in matrix form as $\mathbf{T} = \mathbf{FS}$. It is clear that $\mathbf{S}$ is a symmetric matrix (with $\partial U/\partial C_{ij}$ understood to be symmetric in i and j), and so we see that the principle of material frame indifference implies that the following combination

$$\mathbf{F}^{-1}\mathbf{T} \tag{15}$$

of the deformation gradient and the first Piola–Kirchhoff stress tensor is symmetric. In the first problem set you will show that this condition is equivalent to the **conservation of angular momentum**. It follows, therefore, that angular momentum is automatically conserved during the motion of a simple hyperelastic body.[3] Within other forms of continuum mechanics, such as for viscous fluids, angular momentum conservation needs to be explicitly imposed within the governing equations.

[2] $\mathbf{FF}^T$ is the **left Cauchy–Green deformation tensor**, which is useful in some circumstances.

[3] For those who know about such things, this also follows from Noether's theorem.

The equilibrium equations

Within the last lecture, we showed that the equations of motion for a simple hyperelastic body can be written

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0, \tag{16}$$

and are subject to the traction-free boundary conditions

$$T_{ij}\hat{n}_j = 0. \tag{17}$$

An **equilibrium configuration** $\boldsymbol{\varphi}^0$ is a time-independent solution of these equations, and therefore satisfies

$$\frac{\partial T^0_{ij}}{\partial x_j} = 0, \tag{18}$$

along with the boundary condition

$$T^0_{ij}\hat{n}_j = 0. \tag{19}$$

Here we have defined the equilibrium first Piola–Kirchhoff stress tensor

$$T^0_{ij} = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, \mathbf{F}^0(\mathbf{x})], \tag{20}$$

with $\mathbf{F}^0$ the deformation gradient of $\boldsymbol{\varphi}^0$. As the choice of reference body M is arbitrary, we are free to require that for all $\mathbf{x} \in M$ we have

$$\boldsymbol{\varphi}^0(\mathbf{x}) = \mathbf{x}. \tag{21}$$

It follows that $\mathbf{F}^0(\mathbf{x}) = \mathbf{I}$, and recalling eq. (14) we see that at equilibrium the first and second Piola–Kirchhoff stresses are equal, and hence both are symmetric. Indeed, within such a configuration these tensors are also equal to the Cauchy stress occurring within spatial formulations of continuum mechanics. For the linearised motions about a stress-free equilibrium considered below, the same is true to first order in the perturbation, since $\mathbf{F}$ then differs from $\mathbf{I}$ only by a term of first order and the stresses are themselves of first order.

We assume for the first part of this course that $\mathbf{T}^0(\mathbf{x}) = \mathbf{0}$ for every $\mathbf{x} \in M$, which is one solution, though by no means the only one, of the equilibrium equations in eq. (18) and the associated boundary conditions in eq. (19). Such an equilibrium configuration is said to be **stress-free**, and corresponds physically to the equilibrium state being one of **minimum elastic potential energy**. Prior to an earthquake, the Earth is certainly not stress-free, and we will return to this issue later. For many applications, however, it turns out that the neglect of equilibrium stress is an acceptable approximation.

Linearised equations of motion

We can now rapidly obtain the equations of motion about a stress-free equilibrium configuration. Suppose that at $t = 0$ the body is displaced from equilibrium, with the initial motion taking the form

$$\varphi_i(\mathbf{x}, 0) = x_i + s u^0_i(\mathbf{x}), \tag{22}$$

for some given vector field $\mathbf{u}^0$, while for some vector field $\mathbf{v}^0$ the initial velocity is given by

$$v_i(\mathbf{x}, 0) = s v_i^0(\mathbf{x}). \tag{23}$$

Here s is a dimensionless number that indicates the order of terms in a perturbation series. We shall assume that the subsequent motion can be expanded in powers of s, and hence seek a solution of the form

$$\varphi_i(\mathbf{x}, t) = x_i + s u_i(\mathbf{x}, t) + O(s^2), \tag{24}$$

where $\mathbf{u}$ will be known as the **displacement vector** of the linearised motion. Working always to first order in s, we find that the referential velocity of the motion is given by

$$v_i = s \frac{\partial u_i}{\partial t} + O(s^2), \tag{25}$$

while the deformation gradient can be written

$$F_{ij} = \delta_{ij} + s \frac{\partial u_i}{\partial x_j} + O(s^2). \tag{26}$$

Similarly, the first Piola–Kirchhoff stress tensor can be expanded as

$$T_{ij}(\mathbf{x}, t) = s \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{x}, \mathbf{I}) \frac{\partial u_k}{\partial x_l} + O(s^2), \tag{27}$$

where we have made use of our assumption that the equilibrium stress vanishes. It will be useful to define the elastic tensor by

$$A_{ijkl}(\mathbf{x}) = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}}(\mathbf{x}, \mathbf{I}), \tag{28}$$

so that we can write the stress tensor more concisely as

$$T_{ij} = s A_{ijkl} \frac{\partial u_k}{\partial x_l} + O(s^2). \tag{29}$$

This linearised relationship is **Hooke's law**.[4] But please note that this is neither a "law" of nature nor merely an empirical result. It arises as an approximation to finite elasticity which is valid for sufficiently small deformations about equilibrium.

Due to the equality of mixed partial derivatives, the elastic tensor A_{ijkl} possesses the so-called **hyperelastic symmetry**

$$A_{ijkl} = A_{klij}. \tag{30}$$

We shall see throughout this course that this innocuous-looking symmetry plays a central role within the theory of elastic wave propagation! In the first problem set you will show that this tensor has additional symmetries

$$A_{ijkl} = A_{jikl} = A_{ijlk} \tag{31}$$

[4] When a seismic source is later added to the strain energy function in Lecture 15, it is the time-independent, purely elastic part of W whose second derivative defines the elastic tensor; the source contributes force terms, but does not alter A_{ijkl}.

that reduce the number of independent components from 45 to 21. In particular, the symmetry of the elastic tensor in its final two indices means that

$$A_{ijkl}\frac{\partial u_k}{\partial x_l} = A_{ijkl}\frac{1}{2}\left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k}\right), \tag{32}$$

where on the right-hand side we identify what is commonly called the strain tensor. More properly, this quantity is the **linearised strain tensor**, which is obtained through linearisation of the **finite strain tensor** $\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I})$, which vanishes in the equilibrium configuration. It is possible to develop the general theory in terms of $\mathbf{E}$ rather than $\mathbf{C}$, but there is little gained by doing this.

Substituting the preceding expansions for $\mathbf{v}$ and $\mathbf{T}$ into the exact equations of motion and boundary conditions, eqs. (16) and (17), and equating first-order terms, we obtain the linearised equations of motion

$$\rho\frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right) = 0, \tag{33}$$

along with the linearised boundary conditions

$$\hat{n}_j A_{ijkl}\frac{\partial u_k}{\partial x_l} = 0. \tag{34}$$

Eqs. (33) and (34) are the basic equations governing elastic wave propagation, and in later lectures we will look at their properties. In these equations there are two physical parameters: the density ρ, and the elastic tensor $\mathbf{A}$. Both are functions that can vary smoothly within the body. There might also be discontinuities in these parameters across internal surfaces such as the core–mantle boundary, across which continuity of displacement and traction must be imposed.

Material symmetries

In general the elastic tensor has 21 independent components. In certain materials, however, this number is reduced due to the presence of **material symmetries**. Let us fix a point $\mathbf{x}$, and suppose there is a rotation matrix $\mathbf{Q}$ such that

$$W(\mathbf{x}, \mathbf{FQ}) = W(\mathbf{x}, \mathbf{F}), \tag{35}$$

for any deformation gradient $\mathbf{F}$. Using the polar decomposition theorem we have $\mathbf{F} = \mathbf{RU}$, and hence $\mathbf{FQ} = (\mathbf{RQ})(\mathbf{Q}^T\mathbf{UQ})$, with $\mathbf{RQ}$ a rotation matrix and $\mathbf{Q}^T\mathbf{UQ}$ symmetric and positive-definite. It follows that $\mathbf{FQ}$ has the same principal stretches as $\mathbf{F}$, but about a rotated set of axes within the reference body. Thus we can say that in eq. (35) there is a symmetry of the strain energy with respect to **orientation of stretching**. The collection of all such rotation matrices forms the **material symmetry group** at the point $\mathbf{x}$.[5]

If the material symmetry group at $\mathbf{x}$ comprises all rotation matrices, we say that the material is **isotropic** at this point. If the material is not isotropic, then it is **anisotropic**. The crystalline materials that make up the Earth's interior are almost always quite strongly anisotropic. However, seismic waves have wavelengths substantially larger than individual crystals, and so they "see" an average over many crystals with many different orientations. The end result is that for most seismological applications the Earth appears to be either isotropic or only slightly anisotropic.

[5] If you care to, check that this really is a group.

Isotropic materials

Given their practical relevance, let us consider isotropic materials in further detail. To simplify notation, we work at a fixed point $\mathbf{x}$, and suppress this dependence within the formulae. By assumption the strain energy of an isotropic material satisfies

$$W(\mathbf{FQ}) = W(\mathbf{F}), \tag{36}$$

for all $\mathbf{F}$ and any rotation matrix $\mathbf{Q}$. From material frame indifference, however, we know that we can always write

$$W(\mathbf{F}) = U(\mathbf{C}), \tag{37}$$

with $\mathbf{C} = \mathbf{F}^T\mathbf{F}$, and hence for an isotropic material we must have

$$U(\mathbf{C}) = U(\mathbf{Q}^T\mathbf{C}\mathbf{Q}). \tag{38}$$

Because $\mathbf{C}$ is symmetric, the matrix $\mathbf{Q}^T\mathbf{C}\mathbf{Q}$ can be made diagonal for an appropriate choice of rotation matrix $\mathbf{Q}$. We conclude that in an isotropic material the strain energy depends on $\mathbf{C}$ only through its eigenvalues. Using this result it is only a matter of tedious algebra (which we will skip) to obtain the general form of the elastic tensor in an isotropic material. The end result is that

$$A_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \tag{39}$$

where λ and μ are collectively known as the **Lamé parameters**, and μ is called the **shear modulus**. If it is also assumed that the Lamé parameters are constant, it can readily be shown that the linearised equations of motion reduce to

$$\rho\frac{\partial^2\mathbf{u}}{\partial t^2} - (\lambda+\mu)\nabla(\nabla\cdot\mathbf{u}) - \mu\nabla^2\mathbf{u} = \mathbf{0}. \tag{40}$$

This is the form of the equations of elasticity covered in the Part IB dynamics course.

For reference, note that the isotropic elastic tensor can also be expressed using the shear modulus along with the **bulk modulus** κ, which is defined in terms of λ and μ through

$$\kappa = \lambda + \frac{2}{3}\mu. \tag{41}$$

In terms of κ and μ, eq. (39) becomes

$$A_{ijkl} = \kappa\delta_{ij}\delta_{kl} + \mu\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \tfrac{2}{3}\delta_{ij}\delta_{kl}\right), \tag{42}$$

which separates the volumetric and shear parts of the response. As a comment, you might ask why I have taken almost two lectures to get to an equation most of you will already know! There are two reasons. First, eq. (40) is not the equation of motion for a general elastic body, but for the linearised motion subject to a whole host of simplifying assumptions (e.g. isotropy, homogeneity, no initial stress, no gravitation). Such assumptions are generally invalid within seismological applications, and hence we do in practice need to consider more general formulations. But beyond this, I think it is always better to have stated clearly and precisely what has and has not been assumed within a physical theory.

Transversely isotropic materials

We describe briefly an important class of anisotropic materials. Let $\hat{\boldsymbol{\nu}}$ be a unit vector, and suppose that eq. (36) holds for all rotation matrices $\mathbf{Q}$ that leave $\hat{\boldsymbol{\nu}}$ fixed. Such a strain energy function is said to be **transversely isotropic** with symmetry axis $\hat{\boldsymbol{\nu}}$. The most general strain energy function consistent with this condition has been determined, and it can be shown that the resulting elastic tensor takes the following form

$$\begin{aligned} A_{ijkl} = {} & (A-2N)\delta_{ij}\delta_{kl} + N(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}) + (F-A+2N)(\delta_{ij}\hat{v}_k\hat{v}_l + \hat{v}_i\hat{v}_j\delta_{kl}) \\ & + (L-N)(\hat{v}_i\hat{v}_k\delta_{jl} + \hat{v}_j\hat{v}_k\delta_{il} + \hat{v}_j\hat{v}_l\delta_{ik} + \hat{v}_i\hat{v}_l\delta_{jk}) + (A+C-2F-4L)\hat{v}_i\hat{v}_j\hat{v}_k\hat{v}_l, \end{aligned} \tag{43}$$

where A, C, F, L and N are five independent elastic moduli, this being the notation introduced by Love. If the symmetry axis is taken along $\hat{\mathbf{x}}_3$, these moduli are the components $A_{1111} = A$, $A_{3333} = C$, $A_{1133} = F$, $A_{1313} = L$ and $A_{1212} = N$. In the isotropic case we have $A = C = \lambda + 2\mu$, $F = \lambda$ and $L = N = \mu$, and eq. (43) reduces to eq. (39). Do not fear: you do not need to remember any such complicated tensor expressions! The point here is just to indicate concretely that there is more to the world than isotropic elasticity.

Elastic fluids

The discussion of material symmetries above actually applies only to solids. There is, however, an extension of these ideas to so-called **elastic fluids**. This means that the fluid is assumed to be compressible, but to have no viscosity. Within seismological applications this is a good approximation when considering wave propagation within the oceans or in the Earth's outer core. All our basic equations already apply to elastic fluids, and the only difference is the form of the strain energy function. We stated in the previous lecture that an elastic body has a preferred shape and/or volume. In an elastic fluid, however, there is only a preferred volume. This means that **deformations that leave the volume fixed have no effect on the strain energy**. Mathematically this is expressed by requiring that

$$W(\mathbf{x}, \mathbf{FQ}) = W(\mathbf{x}, \mathbf{F}), \tag{44}$$

for all $\mathbf{F}$ and all matrices $\mathbf{Q}$ with determinant equal to one.[6] It follows readily that the strain energy depends on $\mathbf{F}$ only through its determinant, and hence we can write

$$W(\mathbf{x}, \mathbf{F}) = V(\mathbf{x}, J), \tag{45}$$

for some function V, where $J = \det(\mathbf{F})$. In terms of the linearised theory, it is then a matter of algebra to show that the elastic tensor takes the form

$$A_{ijkl} = \kappa\delta_{ij}\delta_{kl}, \tag{46}$$

where $\kappa = \frac{\partial^2 V}{\partial J^2}(\mathbf{x}, 1)$ is the bulk modulus of the fluid. This is the elastic tensor of an isotropic solid but with vanishing shear modulus.

[6] If you know some group theory, we can say that an elastic fluid has symmetry group equal to $\mathrm{SL}(3)$, while for a solid the symmetry group is a subgroup of $\mathrm{SO}(3)$. It can be shown that no subgroup of $\mathrm{SL}(3)$ lies strictly between $\mathrm{SO}(3)$ and $\mathrm{SL}(3)$, and hence there is no "intermediate" state between solids and fluids. This approach to classifying elastic materials is due to Walter Noll (1925–2017).

What you need to know and be able to do

(i) The statement of the polar decomposition theorem, and how this can be used to restrict the allowable form of the strain energy function. The proof of the polar decomposition theorem is non-examinable.

(ii) What a stress-free equilibrium configuration is, and how to derive the linearised equations of motion for small disturbances about such a configuration. In this course, you will become very familiar with these linearised equations of motion, and it will be necessary to memorise their form (in the non-gravitating case discussed in this lecture) for the exam.

(iii) The definition of the material symmetry group, and its physical interpretation.

(iv) The meaning of the terms elastic solid, elastic fluid, isotropic, anisotropic, and transversely isotropic. You do not need to remember the form of the elastic tensor in any particular case, but you should memorise the hyperelastic symmetry $A_{ijkl} = A_{klij}$ due to its great physical importance.

Lecture 15: Seismic sources

Outline and motivation

We now turn our attention to the generation of seismic waves during an **earthquake**. In doing so, we see immediately that our equations of motion imply that earthquakes cannot happen. We will not attempt to rectify this by producing a complete **dynamical theory** for earthquakes, as no such theory yet exists. Instead, we simply modify our equations in a **phenomenological** manner to account for a specified displacement across a **fault**. Because this displacement is regarded as a parameter and not as a dynamic variable, this approach is known as the **kinematic source approximation**. The resulting theory has been found to work very well in practice, and it allows us to set up an **inverse problem** to determine the displacement across the fault from recorded seismic waves. In the final part of the lecture we begin to think more about inverse problems, this being a topic we will return to throughout these lectures.

The elastic rebound theory for earthquakes

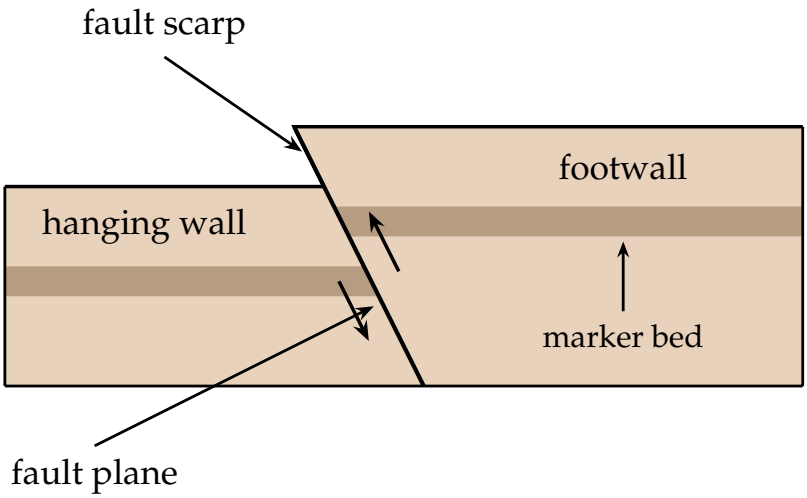


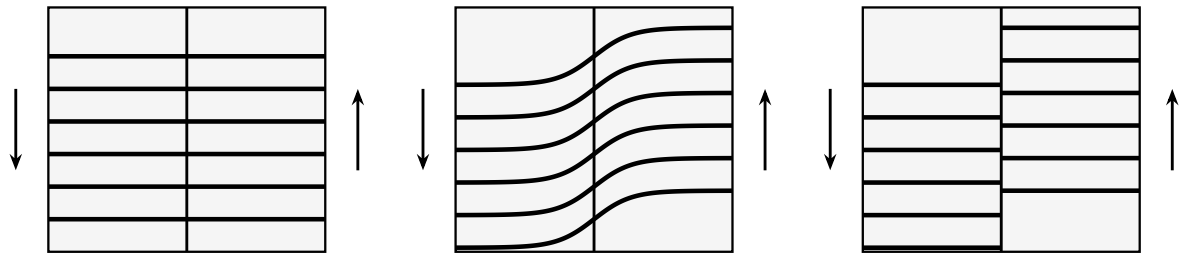
Fig. 1: A schematic cross-section through a thrust fault. You do not need to learn the associated terminology.

The majority of natural earthquakes occur due to motion across **faults** within the Earth (see Fig. 1), with this motion resulting from the slow build-up, and then rapid release, of energy accumulated during long-term tectonic deformation. This long-term deformation is associated with processes such as mantle convection and plate tectonics. Indeed, most earthquakes occur at or near plate boundaries, and the type of motion across the fault is determined broadly by the plate tectonic setting. The exceptions to this picture are (i) earthquakes associated with motion of fluids within volcanic systems and (ii) deep earthquakes occurring within subducted slabs whose cause remains

uncertain.[1] There are also man-made earthquakes, including nuclear weapon tests, explosions generated as part of oil prospecting, and small earthquakes associated with fracking or extraction of natural gas.[2]

The idea that earthquakes are produced by rapid motion across a fault was formulated by Harry Fielding Reid (1859–1944) following a large earthquake in San Francisco in 1906. Based on observations around San Francisco, Reid arrived at his **elastic rebound theory** (Reid 1910) as summarised in Fig. 2. Prior to the earthquake, deformation is gradually built up on either side of the locked fault due to tectonic processes, and this stores energy within the system. At some point, the tractions across the fault become sufficiently large to overcome frictional resistance, rupture occurs rapidly, and seismic waves radiate away some of the stored energy.

While this "theory" is light on details, it is surely correct in a broad sense, and subsequent work has tried to make things more precise. Key questions include: (i) the nature of frictional resistance on the fault, (ii) the conditions under which rupture will occur, and (iii) the factors that determine when rupture stops. Why, for example, are some earthquakes on the same fault small and others big? Such work also informs ideas about earthquake predictability, though a widely held view is that earthquakes are a so-called **critical phenomenon**, with the size of a particular earthquake being so sensitive to initial conditions that meaningful predictions are impossible. The physics of earthquakes is a large and difficult subject, about which much remains to be learned. In this course, however, we will approximate the full physics by regarding the motion on the fault as simply a parameter within the theory, and calculate the seismic waves generated. Within this **kinematic source approximation**, any interaction between the radiated waves and rupture propagation is neglected, but it is thought that this is a good approximation in most circumstances.



(a) Rocks either side of a locked fault are driven in opposite directions

(b) The rocks deform, storing elastic energy

(c) The fault slips, the strain is released and seismic waves radiate

Fig. 2: A graphical representation of Reid's elastic rebound theory.

Introduction of the stress glut

Our aim is to incorporate motion on a fault into the equations of motion for an elastic body. To do this, we will actually take a more general, and yet ultimately simpler,

[1] At these depths it is thought that rocks will be too hot to undergo brittle failure, and so the process must seemingly be different from that for shallow earthquakes. In terms of the seismic waves generated, however, there is little to distinguish deep earthquakes from shallower ones caused by brittle failure.

[2] These are small, but sometimes still quite damaging.

approach. In Lecture 13 we introduced the Lagrangian density

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, \mathbf{F}), \tag{1}$$

for the finite motion of a hyperelastic body, while the associated action is

$$\mathcal{S}(\boldsymbol{\varphi}) = \int_0^T \int_M \mathcal{L}[\mathbf{x}, t, \boldsymbol{\varphi}(\mathbf{x}, t), \mathbf{v}(\mathbf{x}, t), \mathbf{F}(\mathbf{x}, t)] \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t, \tag{2}$$

where $[0, T]$ is the time interval of interest. From the resulting equations of motion it can be shown (and you will do so in the first problem set) that the **total energy** of the elastic body is conserved,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_M E(\mathbf{x}, t) \, \mathrm{d}^3\mathbf{x} = 0, \tag{3}$$

where the **energy density** is defined in the expected manner,

$$E(\mathbf{x}, t) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}(\mathbf{x}, t)\|^2 + W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)]. \tag{4}$$

If the body is initially in a stable equilibrium configuration, then any spontaneous non-trivial deformation will, by definition, raise the potential energy,[3]

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] \, \mathrm{d}^3\mathbf{x}. \tag{5}$$

Because the kinetic energy is non-negative, this must lead to the overall energy of the body being increased. Eq. (3) implies that such an increase in the total energy cannot occur, and so if a body is initially at rest, then it will remain so for all later times.

It is clear that our theory needs to be modified in some way to account for earthquakes. The potential energy is the obvious place to look as this is the part whose form was essentially guessed at. We recall that the principle of material frame indifference requires that the potential energy $\mathcal{V}(t)$ of the body be:

1. invariant under time-translations;
2. invariant under superimposed rigid body translations;
3. invariant under superimposed rigid body rotations.

A kinematic description of a seismic source necessarily breaks the first of these invariance properties, as the onset of motion on the fault gives a **distinguished point in time**. We shall still retain the final two invariance properties of the potential energy. Indeed, they are necessary for the seismic source to represent an internal process that does not alter the net linear or angular momentum of the Earth. Having dropped the time-translation invariance, we can generalise the Lagrangian density to become

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}), \tag{6}$$

[3] By non-trivial we mean a deformation that is not a rigid-body motion, but one that causes a change in shape and/or volume.

where we note that $W(\mathbf{x}, t, \mathbf{F}) = W(\mathbf{x}, t, \mathbf{QF})$ for all rotation matrices $\mathbf{Q}$. It is clear that the resulting Euler–Lagrange equations remain unchanged,

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0, \tag{7}$$

and similarly for the boundary conditions

$$T_{ij}\hat{n}_j = 0 \tag{8}$$

on ∂M. However, the first Piola–Kirchhoff stress tensor now depends on the time-dependent strain energy function through the generalised constitutive relation

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, t, \mathbf{F}(\mathbf{x}, t)]. \tag{9}$$

Within Lecture 14, we saw that the invariance of the potential energy under superimposed rigid body rotations requires that the strain energy function depends on the deformation gradient only through the right Cauchy–Green deformation tensor. Using this result, we shall assume that the time-dependent strain energy function can be decomposed into a sum of two terms,

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - \frac{1}{2}\overline{S}_{ij}(\mathbf{x}, t)C_{ij}, \tag{10}$$

where $\overline{S}_{ij}$ is a given time-dependent symmetric second-order tensor field. Here U describes the assumed constitutive properties of the material, which do not change with time, while $\overline{S}_{ij}$ represents the effects of the seismic source. Though this is not the most general possible modification to the strain energy, more general forms do not change the linearised equations of motion. Using eq. (9), we see that eq. (10) implies

$$T_{ij}(\mathbf{x}, t) = F_{ik}(\mathbf{x}, t)\left\{2\frac{\partial U}{\partial C_{kj}}[\mathbf{x}, \mathbf{C}(\mathbf{x}, t)] - \overline{S}_{kj}(\mathbf{x}, t)\right\}, \tag{11}$$

and we can use this relation to introduce the **stress glut**, $\overline{T}_{ij}$, through the identity

$$\overline{T}_{ij}(\mathbf{x}, t) = F_{ik}(\mathbf{x}, t)\overline{S}_{kj}(\mathbf{x}, t) = 2F_{ik}(\mathbf{x}, t)\frac{\partial U}{\partial C_{kj}}[\mathbf{x}, \mathbf{C}(\mathbf{x}, t)] - T_{ij}(\mathbf{x}, t). \tag{12}$$

Physically, the stress glut represents the difference between the stress predicted by the elastic constitutive relation and its true value. It can, therefore, be interpreted as arising from a **failure of elasticity to correctly determine stresses within the body**. This point of view is consistent with elastic rebound theory, with material near to a fault undoubtedly failing to be elastic.

Linearised equations of motion incorporating a stress glut

We consider a simple hyperelastic body that is initially at rest in a stable equilibrium configuration. At some later time, a non-zero stress glut is initiated, and we wish to

model the subsequent linearised motion of the body. As with Lecture 14, we assume that the motion can be written

$$\varphi_i(\mathbf{x}, t) = x_i + s u_i(\mathbf{x}, t) + O(s^2), \tag{13}$$

where we note that the reference body has been chosen such that the equilibrium configuration is the identity mapping. Without loss of generality, the initial conditions on the displacement vector can be applied at time $t = 0$, and so we have

$$u_i(\mathbf{x}, 0) = 0, \quad \frac{\partial u_i}{\partial t}(\mathbf{x}, 0) = 0. \tag{14}$$

Motivated by eq. (10), we write the strain energy function for the body in the form

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - \frac{s}{2}\overline{S}_{ij}(\mathbf{x}, t) C_{ij}, \tag{15}$$

where we note that the time-dependent term has no zeroth-order part. Following the method in Lecture 14, it is a simple matter to obtain the corresponding linearised equations of motion. In doing so, the elastic tensor A_{ijkl} is obtained as the second derivative with respect to $\mathbf{F}$, evaluated at $\mathbf{F} = \mathbf{I}$, of the time-independent part $U(\mathbf{x}, \mathbf{C})$ of the strain energy function alone; the term involving the stress glut is already of first order in s, and contributes only to the force terms. This is the sense in which the definition of the elastic tensor given in Lecture 14 is to be understood whenever a seismic source is present. We find

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = -\frac{\partial \overline{S}_{ij}}{\partial x_j}, \tag{16}$$

and boundary conditions

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \overline{S}_{ij} \hat{n}_j \tag{17}$$

on ∂M. In this manner, we see that the seismic source enters into the linearised equations of motion as an **effective body force and an effective surface traction**, with each determined from the stress glut. Note that within this linearised theory the tensors $\overline{S}_{ij}$ and $\overline{T}_{ij}$ agree to first order, with **either being called the stress glut**. In particular, within the linearised equations of motion, the **stress glut is necessarily symmetric**.

The stress glut for an idealised fault

We wish to determine the appropriate stress glut for an **idealised fault**. On a real fault, there will be large non-elastic deformation within some **finite fault zone**, as can be seen in Fig. 3. We shall, however, neglect such complexities, and suppose that the elastic constitutive relation fails only on the fault surface across which the displacement is discontinuous. Such an approximation is appropriate so long as we consider waves having wavelength substantially larger than the width of the fault zone, and this is true for most seismological applications (e.g. seismic wavelengths of kilometres and fault zone widths of metres).

We assume that the **fault plane**, Σ, passes through the origin, and lies within the (x_1, x_2)-plane. The discontinuity in the displacement vector on the fault plane is assumed to be only in its tangential component so that there is no separation or interpenetration of material, and hence the displacement vector can be decomposed in the



Fig. 3: A fault in cross section, where the offset of the strata on either side can be clearly seen. It is also apparent that the fault is not a single surface along which the motion is discontinuous, but rather a zone of finite width which has undergone severe localised deformation. Photograph: source unknown; the road at the bottom gives the scale, the exposure being a few metres high.

form

$$u_i(x_1, x_2, x_3, t) = u_i^0(x_1, x_2, x_3, t) + w_i(x_1, x_2, t)H(x_3), \tag{18}$$

where u_i^0 is continuous, w_i is tangential on Σ (i.e. $w_3 = 0$), and $H(\cdot)$ denotes the Heaviside step function. We call **w** the **displacement discontinuity vector**. We wish to determine the linearised first Piola–Kirchhoff stress tensor associated with this displacement vector field using the assumed hyperelastic constitutive relation. To do this, we must first differentiate the displacement vector to obtain

$$\frac{\partial u_i}{\partial x_j} = \frac{\partial u_i^0}{\partial x_j} + \frac{\partial w_i}{\partial x_j}H(x_3) + w_i\delta_{j3}\delta(x_3), \tag{19}$$

where we have used the well-known identity $H'(x) = \delta(x)$. It follows that the linearised stress tensor predicted by the constitutive relation takes the form

$$A_{ijkl}\frac{\partial u_k^0}{\partial x_l} + A_{ijkl}\frac{\partial w_k}{\partial x_l}H(x_3) + A_{ijk3}w_k\delta(x_3). \tag{20}$$

The first term here is everywhere continuous, the second has a jump discontinuity across the fault plane, while the third is non-zero only on the fault plane but is singular there. That the stress might be discontinuous across the fault plane (in its tangential components; continuity of traction is enforced by the equations of motion themselves) is physically acceptable, with this property approximating rapid but continuous variations in stress across the finite width of a fault zone. It is, therefore, only the final singular term within eq. (20) that is non-physical and needs to be removed by inclusion of a stress glut. Specialising eq. (12) to the linearised case, we see that a suitable choice is given by

$$\overline{S}_{ij} = A_{ijk3}w_k\delta(x_3). \tag{21}$$

It is worth emphasising that this result is really **more of a definition than something we have derived**. Indeed, an ideal fault is just a mathematical abstraction, providing a

useful model for something far more complicated. These ideas extend immediately to a general fault whose surface, Σ, need not be planar. The appropriate stress glut is

$$\overline{S}_{ij} = A_{ijkl} w_k \hat{n}_l \delta_\Sigma, \tag{22}$$

where the displacement discontinuity vector is everywhere tangent to Σ, $\hat{\mathbf{n}}$ is the unit normal vector to the surface, and δ_Σ is the Dirac surface distribution for Σ which is defined through the relation

$$\int_M f \, \delta_\Sigma \, \mathrm{d}^3\mathbf{x} = \int_\Sigma f \, \mathrm{d}S, \tag{23}$$

for any smooth f.

Formulating an inverse problem

We have now assembled suitable equations of motion to describe seismic wave generation and propagation within the Earth. For convenience, we recall that the linearised equations of motion are

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = -\frac{\partial \overline{S}_{ij}}{\partial x_j}, \tag{24}$$

along with boundary conditions

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \overline{S}_{ij} \hat{n}_j, \tag{25}$$

while the stress glut for an ideal fault takes the form

$$\overline{S}_{ij} = A_{ijkl} w_k \hat{n}_l \delta_\Sigma. \tag{26}$$

Within these equations we can identify a number of **model parameters**:

1. The referential density, ρ.
2. The elastic tensor, A_{ijkl}.
3. The displacement discontinuity vector, w_i.
4. Geometric parameters describing the shape of the earth model, including any internal discontinuities, and the geometry of the fault surface.

Given all this information it is possible, with sufficient effort, to solve the equations, though this generally requires use of a computer.

Suppose we have made seismic observations following an earthquake. Let $\mathbf{x}^{(k)}$, $k = 1, \ldots, n$, denote the surface locations of the seismometers, with each instrument providing recordings over some time interval $[t_1, t_2]$. As noted above, given suitable model parameters we can simulate seismic wave propagation to produce **synthetic seismograms** at each seismometer location. This is an example of solving a **forward problem**, which is to say performing a mapping from model parameters to observable

quantities. To each forward problem there is an associated **inverse problem** in which we try to work backwards from the observations to the underlying model parameters.

Both the forward and inverse problems are concerned with transforming one kind of information into another. There is, however, a fundamental difference between these two problems, and this is best expressed through the concept of being well- or ill-posed. For a problem to be **well-posed** the following three things must hold:

1. For any given set of inputs, there exists an output.
2. There is a unique output for each given input.
3. The output depends continuously on the input.

Note that these conditions only make sense sequentially; if the first does not hold, then the second is meaningless, and so on. A problem that is not well-posed is said to be **ill-posed**.

These definitions can be illustrated by considering a simple example. Suppose for distinct points x_i, $i = 1, \ldots, n$ on the real line we are given corresponding function values $\{y_i\}_{i=1}^n$. We can then ask if there is a polynomial, $p(x)$, of degree m such that

$$p(x_i) = y_i. \tag{27}$$

Here the inputs to the problem are the values $\{y_i\}_{i=1}^n$, and the desired outputs are the $m+1$ coefficients of the polynomial. Considering first the existence question, it should be clear that if $m+1 \geq n$, then a suitable polynomial can always be found. If $m+1 < n$, then, in general, there is no such polynomial. But a solution is possible for certain inputs (e.g. the y_i all lie on a line, and $m \geq 1$). Assuming existence has been established, we can turn to uniqueness. Here, again, we can readily see that the polynomial is unique if and only if $m+1 \leq n$, while if $m+1 > n$ there are infinitely many possible answers. Finally, assuming that uniqueness has been established, it is possible to show that the coefficients of the polynomial depend continuously on the data, and hence a stable solution can be obtained.

A few additional comments can be made. First, reflecting the fact that all real measurements have uncertainties, we might choose to modify the problem to

$$p(x_i) = y_i + z_i, \tag{28}$$

where z_i are **unknown random errors** drawn from a known distribution. In addressing the existence question we must then broaden what it means for a solution to be found, with the fit to the data no longer required to be exact, but only adequate in a statistical sense. As a result, our statements on existence in terms of n and m might need to be modified. Next, we can see that one potential cause of non-uniqueness is if the problem is **underdetermined**: we have fewer knowns than unknowns. For geophysical inverse problems this is almost always the case. As you might already imagine, uncertainty quantification within underdetermined problems is challenging, and this remains a major research topic.

Any plausible physical theory should be well-posed. If this were not the case, then it would not be possible to test it experimentally. For example, suppose that a physical theory satisfied the first two conditions above, but not the third. We could formulate an experiment to test our theory by comparing its predictions against observations

made from known inputs. But the issue is that we can never know our inputs with complete precision. The lack of continuity within the theory means that tiny errors within these inputs can map onto arbitrarily large differences in the output, and this renders comparison between predictions and observations meaningless.

It can be shown that the seismological forward problem defined above is well-posed so long as the model parameters are subject to reasonable physical constraints (e.g. density should always be positive). By contrast, **the inverse problem is almost always ill-posed**. As noted above, the main reason is a lack of information. Real data sets are necessarily finite-dimensional, but the model parameters are functions, and hence live in infinite-dimensional spaces. Later in this course we will see a number of practical methods for dealing with ill-posed inverse problems. None are perfect, and all must be applied with care. But these techniques do seem to have produced useful and reliable information about the Earth.

What you need to know and be able to do

(i) Know that most natural earthquakes are generated by rapid movement across faults within the Earth.

(ii) Summarise Reid's elastic rebound theory for earthquakes.

(iii) Explain how a seismic source can be introduced into elasticity using a stress glut, and outline the derivation of the stress glut for an idealised fault.

(iv) Understand what is meant by an inverse problem along with the associated terminology and the definition of well- and ill-posed problems.

Lecture 16: Plane wave propagation

Outline and motivation

In this lecture, we step back from realistic earth models to consider the propagation of **plane elastic waves** within a whole space with constant material parameters. We show that under reasonable restrictions on the elastic tensor, up to **three** different types of plane wave can propagate in each direction. The existence of multiple wave types is a fundamental property of elasticity, and accounts for much of the complexity of the observed seismic wavefield. In the case of isotropic materials a simple closed-form solution of the problem can be found, and in this manner we arrive at **P- and S-waves**. We then apply perturbation theory to understand how these solutions are modified by the slight anisotropy present in some regions of the Earth.

Derivation of the Christoffel equation

In Lecture 14 we obtained the linearised equations of motion

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0, \tag{1}$$

where ρ is the referential density, and A_{ijkl} the elastic tensor. We recall that the elastic tensor possesses the symmetry $A_{ijkl} = A_{klij}$, and will start to see the significance of this property as we proceed through this lecture. Throughout this lecture we assume that the reference body is equal to $\mathbb{R}^3$, and that the material parameters are independent of position. We say that a body is **homogeneous** if its material parameters are independent of position, and otherwise it is said to be **heterogeneous**.

Note that though the terms "homogeneous" and "isotropic" are sometimes confused, they mean completely different things. The first refers to the **invariance of properties with spatial position**, while the latter is an **invariance of the properties with respect to orientation at a fixed point**. A homogeneous medium could be anisotropic, while a heterogeneous one can be isotropic.

We begin by looking for plane wave solutions of eq. (1) taking the form

$$u_i(\mathbf{x}, t) = f(t - p_j x_j) a_i, \tag{2}$$

where $\mathbf{p}$ is the **slowness vector**, $\mathbf{a}$ is the **polarisation vector**, and f is an arbitrary function. The fact that the shape of the plane wave does not change during propagation shows that, if such waves exist, they are non-dispersive. The above displacement vector is seen to be constant on the wavefronts

$$t - p_j x_j = \text{constant}, \tag{3}$$

and these move through space in the direction of $\mathbf{p}$ at the phase speed

$$c = \frac{1}{\|\mathbf{p}\|}. \tag{4}$$

Here we see that the slowness vector is used to specify both the **speed and direction of propagation of the plane wavefronts**, while the polarisation vector gives the **orientation of the associated particle motion**.

To see if such plane wave solutions are possible, we simply substitute the displacement vector in eq. (2) into the equations of motion. First, we find readily that

$$\frac{\partial^2 u_i}{\partial t^2} = f''(t - p_j x_j) a_i, \tag{5}$$

where f'' denotes the second derivative of f with respect to its argument. To calculate the remaining terms in the equations of motion, we first note that

$$\frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right) = A_{ijkl}\frac{\partial^2 u_k}{\partial x_j \partial x_l}, \tag{6}$$

because the elastic tensor is constant. From the definition of $\mathbf{u}$ above, we can use the chain rule to show that

$$\frac{\partial u_k}{\partial x_l} = -f'(t - p_m x_m) a_k p_l, \tag{7}$$

and hence that

$$\frac{\partial^2 u_k}{\partial x_j \partial x_l} = f''(t - p_m x_m) a_k p_j p_l. \tag{8}$$

Combining these calculations, we arrive at

$$\frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right) = f''(t - p_m x_m) A_{ijkl} p_j p_l a_k. \tag{9}$$

Cancelling the common factor of f'', it follows that the equations of motion will be satisfied if and only if

$$(\rho \delta_{ik} - A_{ijkl} p_j p_l) a_k = 0. \tag{10}$$

To simplify this result, it will be useful to define the **Christoffel matrix**, $\mathbf{\Gamma}(\mathbf{p})$, for a given slowness vector $\mathbf{p}$ to have components

$$\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l. \tag{11}$$

Using this definition, eq. (10) can be written using matrix-vector notation as

$$[\mathbf{\Gamma}(\mathbf{p}) - \mathbf{I}]\mathbf{a} = \mathbf{0}, \tag{12}$$

which is known as the **Christoffel equation**. This equation acts as a restriction on the possible values of the slowness and polarisation vectors for which plane elastic wave propagation is possible. As a first observation, we note that this is a homogeneous system of linear equations for $\mathbf{a}$. It always has the trivial solution $\mathbf{a} = \mathbf{0}$, while non-trivial solutions will only exist if

$$\det[\mathbf{\Gamma}(\mathbf{p}) - \mathbf{I}] = 0. \tag{13}$$

This is an algebraic condition on the slowness vector alone, and the totality of its solutions defines what is known as the **slowness surface**.

Reduction to a symmetric eigenvalue problem

To obtain solutions of the Christoffel equation it is useful to first note two properties of the Christoffel matrix. The first is that

$$\mathbf{\Gamma}(\gamma\mathbf{p}) = \gamma^2\mathbf{\Gamma}(\mathbf{p}), \tag{14}$$

for any real γ and $\mathbf{p}$.[1] We know that the slowness vector sets both the direction and speed of the plane wave. It is, therefore, useful to write it in the form

$$\mathbf{p} = \frac{1}{c}\hat{\mathbf{p}}, \tag{15}$$

where c is the phase speed, and $\hat{\mathbf{p}}$ a unit vector. Having done this, eq. (14) allows us to rewrite the Christoffel equation as

$$\mathbf{\Gamma}(\hat{\mathbf{p}})\mathbf{a} = c^2\mathbf{a}. \tag{16}$$

If we focus on plane waves in a fixed direction, then we have arrived at an **eigenvalue problem** with c^2 the eigenvalue, and $\mathbf{a}$ the eigenvector. Next, for any $\mathbf{p}$, the Christoffel matrix is symmetric. Indeed, by definition we have

$$\Gamma_{ki}(\mathbf{p}) = \frac{1}{\rho}A_{kjil}p_jp_l = \frac{1}{\rho}A_{ilkj}p_jp_l = \frac{1}{\rho}A_{ijkl}p_jp_l = \Gamma_{ik}(\mathbf{p}), \tag{17}$$

where in the second equality we have used the hyperelastic symmetry, while the third has simply relabelled the dummy indices. This result is very useful, because we can now see that eq. (16) is an eigenvalue problem for a symmetric matrix.

Qualitative solution of the eigenvalue problem

With these facts at hand we can describe in qualitative terms the plane wave solutions possible in a fixed propagation direction. Eq. (16) is a symmetric eigenvalue problem in $\mathbb{R}^3$, and so has three eigenvalues c_1^2, c_2^2, c_3^2, with corresponding eigenvectors $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$. Assuming that the squared phase speeds are positive, this gives six possible phase speeds $\pm c_1, \pm c_2, \pm c_3$, with the pairs corresponding to waves propagating either forwards or backwards along $\hat{\mathbf{p}}$. Moreover, eigenvectors belonging to distinct eigenvalues are necessarily orthogonal. What does this mean physically? In a given propagation direction, $\hat{\mathbf{p}}$, there are up to three different plane wave solutions possible (or six if we include those propagating backwards). The phase speed of each wave is set by the corresponding eigenvalue of the Christoffel matrix, and its polarisation vector is determined by the associated eigenvector. Finally, the polarisation vectors for plane waves with different phase speeds are necessarily orthogonal.

Note that the Christoffel matrix might have **degenerate eigenvalues**, meaning that an eigenvalue has more than one linearly independent eigenvector. If this occurs, then these eigenvectors need not be orthogonal, but we can always pick an orthogonal basis that spans the eigenspace. We will see below that such degeneracies occur in the case of isotropic materials, and hence this situation cannot be discounted.

[1] Mathematically, we can say that the Christoffel matrix is a **homogeneous function** of order two, this being the third different way the word "homogeneous" is used in this lecture!

The above discussion has assumed that the eigenvalues of the Christoffel matrix are all non-negative, and hence lead to well-defined phase speeds. This requirement places a physical constraint on the components of the elastic tensor. We will not look into this issue in any detail, but note that it is met in all real materials. Moreover, in a solid the three phase speeds are always positive, while in a fluid there is one positive phase speed and two that vanish.

Determination of the slowness surface

Having now determined the form of solutions of the Christoffel equation in a fixed propagation direction, all we need do is vary $\hat{\mathbf{p}}$ over the unit sphere to obtain a complete description of the possible plane waves in the body. For each direction we get (allowing for possible degeneracies) three possible phase speeds, and hence three values of $\mathbf{p}$. Let us write $c_1(\hat{\mathbf{p}})$, $c_2(\hat{\mathbf{p}})$, $c_3(\hat{\mathbf{p}})$ for the three non-negative phase speeds along $\hat{\mathbf{p}}$ and order them such that

$$c_1(\hat{\mathbf{p}}) \leq c_2(\hat{\mathbf{p}}) \leq c_3(\hat{\mathbf{p}}), \tag{18}$$

where we have to use $\leq$ to allow for possible degeneracies. The corresponding values of the slowness vector are

$$\mathbf{p} = \frac{1}{c_1(\hat{\mathbf{p}})}\hat{\mathbf{p}}, \quad \mathbf{p} = \frac{1}{c_2(\hat{\mathbf{p}})}\hat{\mathbf{p}}, \quad \mathbf{p} = \frac{1}{c_3(\hat{\mathbf{p}})}\hat{\mathbf{p}}. \tag{19}$$

As $\hat{\mathbf{p}}$ varies over the unit sphere, these slowness vectors build up the slowness surface. This surface is obviously closed, and is symmetric under reflection about the origin. If there are no degeneracies and all phase speeds are positive, then it comprises three distinct **sheets**. If degeneracies occur along certain propagation directions, then two or more sheets will touch. We note also that the defining relation in eq. (13) can now be written

$$[\|\mathbf{p}\|^2 c_1(\hat{\mathbf{p}})^2 - 1][\|\mathbf{p}\|^2 c_2(\hat{\mathbf{p}})^2 - 1][\|\mathbf{p}\|^2 c_3(\hat{\mathbf{p}})^2 - 1] = 0, \tag{20}$$

with each factor defining a sheet of the slowness surface.

For a general elastic tensor the determination of the slowness surface requires solution of the symmetric eigenvalue problem in each possible propagation direction. Such calculations are, of course, very easy to do numerically. In Fig. 1 we illustrate these properties of the slowness surface by plotting slices through two examples that have been calculated numerically. We note, in particular, that the innermost sheet of the slowness surface is always convex in these examples. This is, in fact, a general property, and a proof might be sought if someone is particularly interested.[2]

[2] As a hint, note that eq. (13) is a sextic polynomial in $\mathbf{p}$, and hence along a fixed line in $\mathbb{R}^3$ there can be no more than six roots.

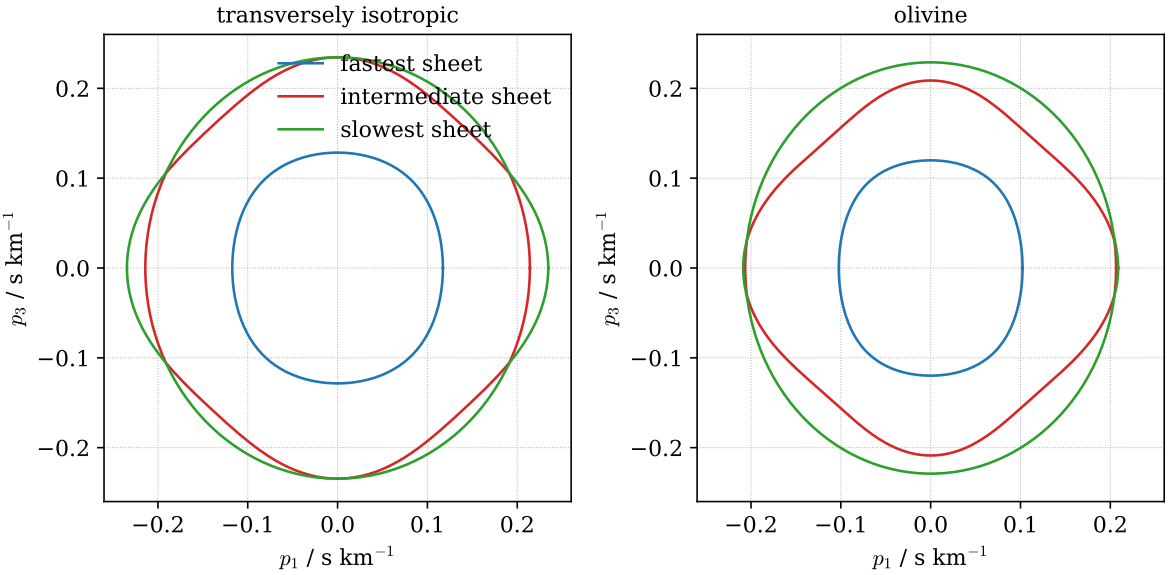


Fig. 1: Plane sections through the slowness surfaces of two anisotropic materials, calculated numerically from the Christoffel equation. On the left is a transversely isotropic medium with Love moduli $A = 240$, $C = 200$, $F = 70$, $L = 60$ and $N = 72$ GPa and density 3300 kg m^{-3}, with the section containing the symmetry axis (which is vertical); on the right is a single crystal of olivine (section in the [100]–[001] plane), an orthorhombic mineral that makes up much of the upper mantle, using the elastic constants of Abramson et al. (1997). The three sheets of the slowness surface are shown in different colours, and there are directions along which two of the sheets touch.

The Christoffel equation in an isotropic material

In the case of an isotropic material a simple closed-form solution of the Christoffel equation can be obtained, and in this manner we arrive at **P-waves** and **S-waves** that you have likely seen before. We recall from the last lecture that in an isotropic material the elastic tensor takes the form

$$A_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \tag{21}$$

with λ and μ the Lamé parameters. A simple calculation then yields

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda+\mu}{\rho}\hat{p}_i\hat{p}_k + \frac{\mu}{\rho}\delta_{ik}. \tag{22}$$

At this stage we could write out $\Gamma_{ik}(\hat{\mathbf{p}})$ explicitly as a matrix and calculate its eigenvalues and eigenvectors in the usual manner. There is, however, a quicker way to proceed which is worth learning. First we rewrite the expression as

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda+2\mu}{\rho}\hat{p}_i\hat{p}_k + \frac{\mu}{\rho}(\delta_{ik} - \hat{p}_i\hat{p}_k). \tag{23}$$

We can now identify two terms with a geometric significance. $\hat{p}_i\hat{p}_k$ are the components of the matrix that projects a vector onto the direction $\hat{\mathbf{p}}$, while $\delta_{ik} - \hat{p}_i\hat{p}_k$ are those of the

matrix that projects orthogonally to $\hat{\mathbf{p}}$. It follows that if we take the polarisation vector equal to the propagation direction we obtain

$$\mathbf{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{p}} = \frac{\lambda + 2\mu}{\rho}\hat{\mathbf{p}}. \tag{24}$$

This shows that $\mathbf{a} \propto \hat{\mathbf{p}}$ is an eigenvector of the Christoffel matrix with phase speed

$$\alpha = \sqrt{\frac{\lambda + 2\mu}{\rho}}. \tag{25}$$

Next, if we let the polarisation vector be orthogonal to $\hat{\mathbf{p}}$ (i.e. $\mathbf{a} \perp \hat{\mathbf{p}}$) we find

$$\mathbf{\Gamma}(\hat{\mathbf{p}})\mathbf{a} = \frac{\mu}{\rho}\mathbf{a}, \tag{26}$$

and hence have found two further linearly independent eigenvectors with a common eigenvalue whose associated phase speed is

$$\beta = \sqrt{\frac{\mu}{\rho}}. \tag{27}$$

Physically, we have now shown that there are two types of plane waves possible in an isotropic whole space. The first are P-waves, which have phase speed α independent of $\hat{\mathbf{p}}$, and whose polarisation vector lies parallel to the propagation direction. The next are S-waves, which have phase speed β, again independent of $\hat{\mathbf{p}}$, and whose polarisation vectors lie in the plane orthogonal to the propagation direction. It can be shown that in any real isotropic solid

$$\mu > 0, \quad \kappa = \lambda + \tfrac{2}{3}\mu > 0, \tag{28}$$

and hence we always have $\alpha > \beta > 0$. The slowness surface, therefore, has two spherical sheets, with the inner one corresponding to P-waves, while the outer one, for S-waves, is two-fold degenerate. Note finally that in the case of an elastic fluid the above solution still applies, but as $\mu = 0$, there are no S-waves that can propagate.

Perturbation theory for slightly anisotropic materials

We noted in the last lecture that seismic waves generally see the Earth as being close to isotropic. It is, therefore, of interest to ask how the plane wave solutions just obtained in the isotropic case are modified by the presence of slight anisotropy. We recall that in considering propagation in direction $\hat{\mathbf{p}}$ we have to solve the symmetric eigenvalue problem

$$\mathbf{\Gamma}(\hat{\mathbf{p}})\mathbf{a} = c^2\mathbf{a}. \tag{29}$$

Let us suppose that the elastic tensor takes the following form

$$A_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + s\,\delta A_{ijkl}. \tag{30}$$

Here s is a perturbation parameter, with $s = 0$ corresponding to the isotropic case, and δA_{ijkl} an anisotropic perturbation. The corresponding Christoffel matrix can then be written

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \Gamma^0_{ik}(\hat{\mathbf{p}}) + s\,\delta\Gamma_{ik}(\hat{\mathbf{p}}), \tag{31}$$

where the zeroth-order part $\mathbf{\Gamma}^0(\hat{\mathbf{p}})$ is that for an isotropic material given in eq. (22), while the first-order term is given by

$$\delta\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{1}{\rho}\delta A_{ijkl}\hat{p}_j\hat{p}_l. \tag{32}$$

The eigenvalue problem to be solved can, therefore, be written

$$[\mathbf{\Gamma}^0(\hat{\mathbf{p}}) + s\,\delta\mathbf{\Gamma}(\hat{\mathbf{p}})]\mathbf{a} = c^2\mathbf{a}. \tag{33}$$

While a numerical solution of the problem is always possible, significant insight can be gained through an application of **first-order eigenvalue perturbation theory**. Some of you will have seen these ideas before (for example, in quantum mechanics), but we will derive the necessary results from first principles. In the isotropic case we know that the possible solutions correspond to P- and S-waves, with the latter being two-fold degenerate. For P-waves we can apply **non-degenerate perturbation theory**. This assumes that the phase speed takes the form

$$c = \alpha + s\,\delta\alpha + \cdots, \tag{34}$$

while the polarisation vector can be written

$$\mathbf{a} = \hat{\mathbf{p}} + s\,\delta\mathbf{a} + \cdots. \tag{35}$$

Putting these expansions into the eigenvalue problem and isolating the first-order terms in s, we find that

$$\mathbf{\Gamma}^0(\hat{\mathbf{p}})\delta\mathbf{a} + \delta\mathbf{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{p}} = \alpha^2\delta\mathbf{a} + 2\alpha\,\delta\alpha\,\hat{\mathbf{p}}. \tag{36}$$

This equation involves both $\delta\alpha$ and $\delta\mathbf{a}$, but we can eliminate the latter by taking the inner product with $\hat{\mathbf{p}}$ and noting that

$$\hat{\mathbf{p}}\cdot[\mathbf{\Gamma}^0(\hat{\mathbf{p}})\delta\mathbf{a}] = [\mathbf{\Gamma}^0(\hat{\mathbf{p}})\hat{\mathbf{p}}]\cdot\delta\mathbf{a} = \alpha^2\hat{\mathbf{p}}\cdot\delta\mathbf{a}, \tag{37}$$

where we have used the symmetry of the Christoffel matrix along with the zeroth-order eigenvalue problem. In this manner we arrive at the simple expression

$$\delta\alpha = \frac{1}{2\alpha}\hat{\mathbf{p}}\cdot\delta\mathbf{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{p}}, \tag{38}$$

for the first-order change to the phase speed. The first-order perturbation to the polarisation vector could then also be determined, but we will not work through the details. What is physically important is that in a slightly anisotropic material we have shown that there will be a plane wave that has phase speed close to that of a P-wave, and whose polarisation vector is almost aligned with the propagation direction. The phase speed, however, now generally depends on the propagation direction as determined by eq. (38). Such a plane wave is known as a **quasi P-wave**.

The S-wave case can be handled similarly, but is complicated by the unperturbed eigenspace being degenerate. Let $\hat{\mathbf{t}}_1$ and $\hat{\mathbf{t}}_2$ be two unit vectors orthogonal to the propagation direction. In the absence of anisotropy we know that the polarisation vector

can be any vector orthogonal to $\hat{\mathbf{p}}$. It therefore seems reasonable to look for solutions of the form

$$c = \beta + s\,\delta\beta + \cdots, \tag{39}$$
$$\mathbf{a} = a_1\hat{\mathbf{t}}_1 + a_2\hat{\mathbf{t}}_2 + s\,\delta\mathbf{a} + \cdots, \tag{40}$$

where a_1 and a_2 are zeroth-order terms to be determined. Putting these expansions into the eigenvalue problem, the first-order term is given by

$$\boldsymbol{\Gamma}^0(\hat{\mathbf{p}})\delta\mathbf{a} + \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})(a_1\hat{\mathbf{t}}_1 + a_2\hat{\mathbf{t}}_2) = \beta^2\delta\mathbf{a} + 2\beta\,\delta\beta\,(a_1\hat{\mathbf{t}}_1 + a_2\hat{\mathbf{t}}_2). \tag{41}$$

In a similar manner we can eliminate $\delta\mathbf{a}$ by taking the inner product with either $\hat{\mathbf{t}}_1$ or $\hat{\mathbf{t}}_2$, and doing this leads to the following pair of equations

$$\hat{\mathbf{t}}_1 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})(a_1\hat{\mathbf{t}}_1 + a_2\hat{\mathbf{t}}_2) = 2\beta\,\delta\beta\,a_1, \tag{42}$$
$$\hat{\mathbf{t}}_2 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})(a_1\hat{\mathbf{t}}_1 + a_2\hat{\mathbf{t}}_2) = 2\beta\,\delta\beta\,a_2. \tag{43}$$

Written in matrix form, we have obtained

$$\begin{pmatrix} \hat{\mathbf{t}}_1 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{t}}_1 & \hat{\mathbf{t}}_1 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{t}}_2 \\ \hat{\mathbf{t}}_2 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{t}}_1 & \hat{\mathbf{t}}_2 \cdot \delta\boldsymbol{\Gamma}(\hat{\mathbf{p}})\hat{\mathbf{t}}_2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 2\beta\,\delta\beta \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}. \tag{44}$$

This is a 2×2 symmetric eigenvalue problem whose solution determines both the zeroth-order polarisation vectors and the first-order changes in phase speed.

It would not be difficult to solve the above eigenvalue problem in closed form, but the qualitative behaviour is more important. In the unperturbed case, S-waves are associated with a two-fold degenerate sheet of the slowness surface. For an anisotropic perturbation, the above eigenvalue problem will almost always have two distinct eigenvalues, and hence the S-wave degeneracy will be split. There will then be two distinct plane waves whose phase speeds depend on the propagation direction. They will, moreover, have orthogonal polarisation vectors lying almost orthogonal to the propagation direction; note that the first-order term $\delta\mathbf{a}$ will in general have a component along $\hat{\mathbf{p}}$. These plane waves are known as **quasi S-waves**, and the process we have described is known as **shear wave splitting**.

This phenomenon occurs for waves in the Earth, and the observation of shear wave splitting provides an important constraint on anisotropy. The reason this is interesting is as follows. Most of the Earth appears to seismic waves to be isotropic. This is because anisotropic grains are generally oriented randomly, and hence the structure seen by seismic waves averages out to be isotropic. In some parts of the mantle, however, the seismic waves do see a slight anisotropy because there is a large-scale statistical alignment of grains due to convective flow. Thus, by making observations of shear wave splitting it has been possible to make inferences about the orientation of flow within parts of the mantle.

What you need to know and be able to do

(i) How to derive the Christoffel equation from the linearised equations of motion.

(ii) Reduction of the problem to a symmetric eigenvalue problem, and a qualitative description of its solution.

(iii) How to solve the Christoffel equation for an isotropic solid, and to deduce the properties of plane P- and S-waves.

(iv) How to use first-order perturbation theory to study the effect of slight anisotropy on plane waves.

Lecture 17: Ray theory

Outline and motivation

In the previous lecture we examined the propagation of plane elastic waves. Here we consider what happens when the material parameters are not constant, developing a ray theory that provides a **good approximation in situations when the length scale of the heterogeneity is large relative to that of the waves**. Essentially, the waves always look locally planar, but as they propagate the wavefronts gradually deform, and the amplitudes can be changed by focusing and defocusing effects. This ray theory is related to elastodynamics in the same manner that geometric optics is related to Maxwell's equations, with both being "high frequency" approximations in a suitable sense. There are also close links with what is known as semi-classical analysis, this being the subject in which the behaviour of quantum mechanical systems is studied in the "classical limit" $\hbar \to 0$.

Relative length scales of heterogeneity

We consider the linearised motion of an elastic body occupying a whole space whose material parameters are constant everywhere except a bounded region in which ρ and A_{ijkl} can vary smoothly. For definiteness, let p_i^0 be the slowness vector of a chosen plane wave in the homogeneous part of the body, and assume that the heterogeneous region is contained within the half-space $p_i^0 x_i \geq 0$. Within the homogeneous half-space $p_i^0 x_i < 0$ we know that plane wave solutions are possible of the form

$$u_i(\mathbf{x}, t) = f(t - p_j^0 x_j) a_i^0, \tag{1}$$

and our aim is to extend this solution suitably into the heterogeneous region. In general there exists no exact closed-form solution to this problem, and so it must be studied either through numerical calculations or using approximate methods. While numerical calculations are of huge practical value, they provide relatively little physical insight, and so we shall focus on physically motivated approximate methods.

In broad terms there are three possible approaches for studying this problem, with the appropriate choice being determined by the relative length scales of the incident wave and the heterogeneous structure. Let λ (not to be confused with the Lamé parameter) denote the **dominant wavelength for the plane wave**, and denote by L the **smallest length scale over which the material parameters vary to an appreciable extent**. If their ratio satisfies

$$\frac{L}{\lambda} \approx 1, \tag{2}$$

then we expect the incident wave to be **scattered by the heterogeneity** (see Fig. 1). There is not sufficient time in this course to discuss scattering of seismic waves further, though it is an important physical process within the Earth, and can be analysed using the **Born approximation** that you may know from quantum mechanics.

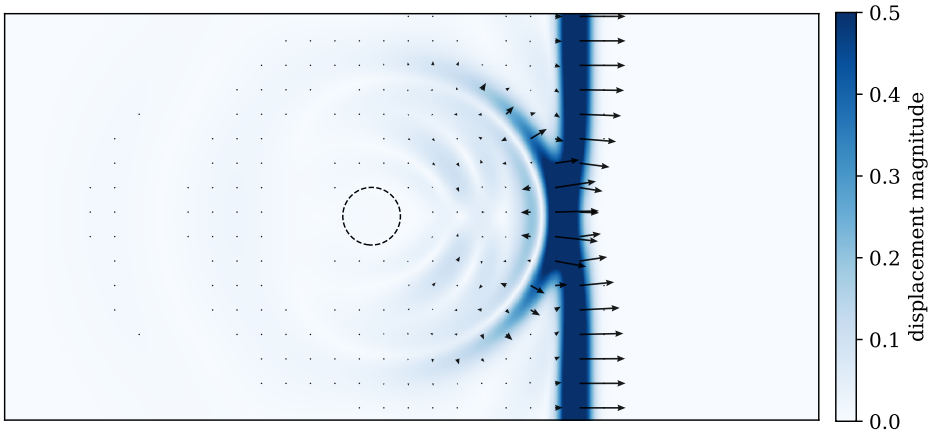


Fig. 1: Snapshot from a numerical simulation of the scattering of a plane P-wave in a homogeneous background model due to its interaction with a heterogeneity having a typical length scale approximately equal to the length scale of the incident wave. The colours show the magnitude of the displacement, the arrows show the particle motion, and the dashed line marks the extent of the heterogeneity; the colour scale is saturated so as to show the weak scattered field. The box is about 13 wavelengths of the incident wave wide and 6 high. We see that the scattered wavefield comprises both P-waves and S-waves.

In this lecture we shall focus on situations when

$$\frac{L}{\lambda} \gg 1, \tag{3}$$

which is to say the length scale of the heterogeneity is much greater than the length scale of the incident wave. Here we do not expect to see significant scattering of the incident wave, but instead a smooth evolution of the wavefront as it propagates through the heterogeneity, along with associated amplitude variations due to focusing effects (see Fig. 2). This is the domain of **geometric ray theory**.

Finally, we might ask what happens in the case that

$$\frac{L}{\lambda} \ll 1. \tag{4}$$

This is actually very applicable to the Earth, as rocks are made up of small grains of different minerals that can have quite distinct physical properties, and yet observed seismic waves have wavelengths on the order of kilometres to thousands of kilometres. We also lack the time to enter into this topic in any detail, but it can be shown that such waves "see" a suitably averaged version of the body that appears smooth on the length scale of the incident waves. The appropriate smoothed medium can be determined using the methods of **homogenisation theory**. Again, there is sadly not time within this course to discuss this topic further.

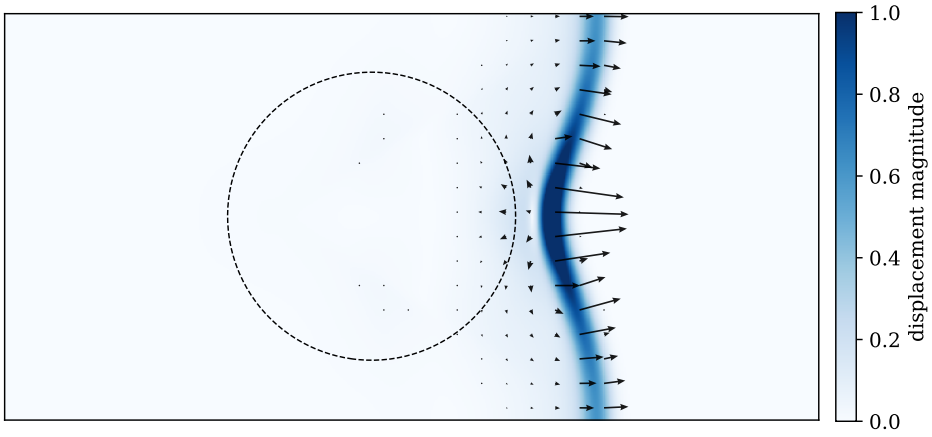


Fig. 2: As for Fig. 1, but for the passage of a plane P-wave through a heterogeneous region having length scale much larger than that of the incident wave. Here we see minimal scattering, along with the smooth distortion of the wavefront and associated amplitude and polarisation variations. This is a situation to which ray theory can be usefully applied.

The ray series ansatz

As noted above, we shall focus on situations where the length scale of the heterogeneity is long relative to the length scale of the incident plane wave. As a first guess, we might expect that the continuation of the plane wave into the heterogeneous region can be approximated in the following manner

$$u_i(\mathbf{x}, t) = f[t - T(\mathbf{x})]a_i(\mathbf{x}), \tag{5}$$

with this expression taking account of the following physical ideas:

1. The linear phase term $t - p_i^0 x_i$ has been replaced with the non-linear term $t - T(\mathbf{x})$. This allows the wavefronts in the heterogeneous region to curve as they propagate through regions with different wave speeds;

2. The amplitude and polarisation term a_i^0 has been replaced with a function of position $a_i(\mathbf{x})$ which allows for (i) focusing and defocusing effects during propagation and (ii) changes in the particle motion associated with both wavefront curvature and variations in anisotropic properties.

We will call the function T the **travel time**, and note that the wavefronts of eq. (5) are, by definition, the level surfaces

$$T(\mathbf{x}) = \text{constant}. \tag{6}$$

It will be useful to define the **local slowness vector** in terms of this travel time by setting

$$p_i = \frac{\partial T}{\partial x_i}, \tag{7}$$

which is everywhere orthogonal to the wavefronts. Note that the local slowness vector is equal to the slowness vector within homogeneous regions where the travel time depends linearly on $\mathbf{x}$.

To proceed, it is useful to write the waveform f as a Fourier integral, so that eq. (5) becomes

$$u_i(\mathbf{x}, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \tilde{f}(\omega) \mathrm{e}^{\mathrm{i}\omega[t - T(\mathbf{x})]} a_i(\mathbf{x}) \, \mathrm{d}\omega, \tag{8}$$

where $\tilde{f}$ denotes the Fourier transform of f. It is clear that eq. (5) is the temporal convolution of the singular plane wave

$$u_i(\mathbf{x}, t) = \delta[t - T(\mathbf{x})] a_i(\mathbf{x}) \tag{9}$$

with the waveform f. In what follows, it will be convenient to focus on such singular waves, and from now on we set $\tilde{f}(\omega) = 1$ or equivalently $f(t) = \delta(t)$.

We could try to use the ansatz in eq. (9) as a starting point for the development of ray theory, but experience has shown that it is necessary to consider instead the following generalisation

$$u_i(\mathbf{x}, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{n=0}^{\infty} \frac{a_i^n(\mathbf{x})}{(\mathrm{i}\omega)^n} \mathrm{e}^{\mathrm{i}\omega[t - T(\mathbf{x})]} \, \mathrm{d}\omega, \tag{10}$$

where we note that the n in a_i^n is a superscript and not an exponent. This expression is known as the **elastodynamic ray series**. The $n = 0$ term within this expression is equivalent to eq. (9). The physical meaning or need for the additional terms within the expansion is less clear, and indeed, these terms are typically ignored within practical applications. But as an initial comment, we note that each successive term within the series is multiplied by a further factor of $1/(\mathrm{i}\omega)$, and within the Fourier domain this is equivalent to a temporal integration. Thus the terms within this series are getting progressively smoother and smoother.

Substitution into the equations of motion

We recall that the linearised equations of motion for an elastic body can be written

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0, \tag{11}$$

where we do not include a body force associated with a seismic source. Within the temporal Fourier domain this equation becomes

$$-\omega^2 \rho u_i - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0, \tag{12}$$

where for simplicity we use the same symbol for the displacement vector and its Fourier transform. From eq. (10) we see that the ray series in the Fourier domain is simply

$$u_i(\mathbf{x}, \omega) = \sum_{n=0}^{\infty} \frac{a_i^n(\mathbf{x})}{(\mathrm{i}\omega)^n} \mathrm{e}^{-\mathrm{i}\omega T(\mathbf{x})}. \tag{13}$$

From eqs. (7) and (13) we find that

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(\mathrm{i}\omega)^n} \left(\frac{\partial a_k^n}{\partial x_l} - \mathrm{i}\omega p_l a_k^n \right) \mathrm{e}^{-\mathrm{i}\omega T}, \tag{14}$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(\mathrm{i}\omega)^n} A_{ijkl} \left(\frac{\partial a_k^n}{\partial x_l} - \mathrm{i}\omega p_l a_k^n \right) \mathrm{e}^{-\mathrm{i}\omega T}. \tag{15}$$

Taking the divergence of the second expression, we obtain

$$\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = \sum_{n=0}^{\infty} \frac{1}{(\mathrm{i}\omega)^n} \left[\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial a_k^n}{\partial x_l} \right) - \mathrm{i}\omega \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^n) \right.$$
$$\left. - \mathrm{i}\omega A_{ijkl} p_j \frac{\partial a_k^n}{\partial x_l} + (\mathrm{i}\omega)^2 A_{ijkl} p_j p_l a_k^n \right] \mathrm{e}^{-\mathrm{i}\omega T}, \tag{16}$$

and substituting into the frequency domain equations of motion in eq. (12) we arrive at

$$\sum_{n=0}^{\infty} \frac{1}{(\mathrm{i}\omega)^n} \left[\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial a_k^n}{\partial x_l} \right) - \mathrm{i}\omega \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^n) \right.$$
$$\left. - \mathrm{i}\omega A_{ijkl} p_j \frac{\partial a_k^n}{\partial x_l} + (\mathrm{i}\omega)^2 A_{ijkl} p_j p_l a_k^n - (\mathrm{i}\omega)^2 \rho a_i^n \right] \mathrm{e}^{-\mathrm{i}\omega T} = 0. \tag{17}$$

The above equation must hold for all frequencies, and this can only be true if the coefficient of each power of $\mathrm{i}\omega$ is separately equal to zero. From the coefficients of the highest-order term $(\mathrm{i}\omega)^2$ we obtain

$$[A_{ijkl} p_j p_l - \rho \delta_{ik}] a_k^0 = 0, \tag{18}$$

while those for the $\mathrm{i}\omega$ term take the form

$$[A_{ijkl} p_j p_l - \rho \delta_{ik}] a_k^1 = \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^0) + A_{ijkl} p_j \frac{\partial a_k^0}{\partial x_l}, \tag{19}$$

and then for a general term with $n \geq 2$

$$[A_{ijkl} p_j p_l - \rho \delta_{ik}] a_k^n = \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^{n-1}) + A_{ijkl} p_j \frac{\partial a_k^{n-1}}{\partial x_l} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial a_k^{n-2}}{\partial x_l} \right). \tag{20}$$

The first of these equations depends on only the local slowness vector and the zeroth-order polarisation vector. We will see shortly that this equation can be solved for the travel time T and also gives the orientation of $\mathbf{a}^0(\mathbf{x})$ at each point. Having done this, solution of eq. (19) gives the amplitude of the zeroth-order term, while eq. (20) can be used to successively determine the higher-order polarisation vectors. We will not, however, discuss the solution of these higher-order equations within this course.

At this stage it is reasonable to ask where exactly we have made an approximation. Assuming that the above set of equations can be solved for T and all the $\mathbf{a}^n$ then it would seem that we have obtained an exact solution to the equations of motion. For this to be a valid conclusion, however, we must then think carefully about the sense in

which the ray series converges. Though we will not enter into this topic in detail, the key point is **the ray series is not convergent in any usual sense**. Indeed, the problem is at low frequencies, with the series being very badly behaved at $\omega = 0$. To force this series to converge, we could zero-out the solution in some neighbourhood of $\omega = 0$, but by doing this we would be left with an approximate solution to the equations of motion. Because, however, the error lies only within a bounded range of frequencies it must be a smooth function. Building on such ideas, it can be shown that the ray theoretic solution **exactly predicts the singular part of the wavefield**, but that there is a **smooth error term whose size need not be small**.[1]

The eikonal equation

Within this section, we show that eq. (18) can be solved for the travel time T using the so-called **method of characteristics**. This is a useful method that can be applied to both linear and non-linear first-order partial differential equations, and reduces their solution to the integration of systems of ordinary differential equations. In fact, it is through solution of these ordinary differential equations that rays will enter into the theory in a natural manner.

Before applying this technique, we must first reduce eq. (18) to a single partial differential equation for the travel time. To do this, it will be useful to define the **local Christoffel matrix** through

$$\Gamma_{ik}(\mathbf{x}, \mathbf{p}) = \frac{1}{\rho(\mathbf{x})} A_{ijkl}(\mathbf{x}) p_j p_l, \tag{21}$$

for all $\mathbf{x} \in \mathbb{R}^3$ and $\mathbf{p}$. We can then write eq. (18) in the form

$$\{\mathbf{\Gamma}[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\}\mathbf{a}^0(\mathbf{x}) = \mathbf{0}, \tag{22}$$

which closely resembles the Christoffel equation we encountered in the study of elastic plane waves. Indeed, the physical intuition underlying ray theory is that locally the solutions behave like plane waves. At each point, $\mathbf{x}$, we know that eq. (22) may be satisfied by a non-trivial polarisation vector $\mathbf{a}^0(\mathbf{x})$ if and only if the local slowness vector $\mathbf{p}(\mathbf{x})$ is a solution of

$$\det\{\mathbf{\Gamma}[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} = 0, \tag{23}$$

which is known in the context of ray theory as an **eikonal equation**. Eq. (23) can be interpreted as requiring that at each $\mathbf{x}$, the local slowness vector $\mathbf{p}(\mathbf{x})$ lies on the **local slowness surface**.

Having determined the appropriate local slowness vector $\mathbf{p}(\mathbf{x})$, eq. (22) then implies that $\mathbf{a}^0(\mathbf{x})$ lies within the eigenspace of $\mathbf{\Gamma}[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]$ with eigenvalue equal to the square of the corresponding local phase speed. For simplicity, we shall only consider materials whose local slowness surfaces are everywhere non-degenerate, and so we can write

$$\mathbf{a}^0(\mathbf{x}) = A_0(\mathbf{x})\hat{\mathbf{a}}^0(\mathbf{x}), \tag{24}$$

where $A_0(\mathbf{x})$ is a scalar amplitude, and $\hat{\mathbf{a}}^0(\mathbf{x})$ is the appropriate normalised eigenvector which is determined uniquely up to its sign. In an isotropic medium the two S-wave

[1] These ideas are formalised within the theory of **Fourier integral operators** developed in the 1970s by Lars Hörmander and others (Hörmander 1971; Duistermaat 1996).

sheets of the slowness surface coincide, and so this assumption fails; the reduction of the eikonal equation described below remains unambiguous in that case, but the polarisation of the S-wave is no longer determined uniquely by eq. (22). Under this non-degeneracy assumption we can therefore always write the eikonal equation unambiguously in the form

$$\{\|\mathbf{p}(\mathbf{x})\|^2 c_1[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1\}\{\|\mathbf{p}(\mathbf{x})\|^2 c_2[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1\}\{\|\mathbf{p}(\mathbf{x})\|^2 c_3[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1\} = 0, \tag{25}$$

where $c_k(\mathbf{x}, \hat{\mathbf{p}})$ denotes the local phase speed at $\mathbf{x}$ corresponding to the kth sheet and with propagation direction $\hat{\mathbf{p}}$. At an initial point only one of these factors will vanish, say the kth one, and by continuity it is the same factor that vanishes elsewhere. As a result, the eikonal equation is reduced to the simpler form

$$\|\mathbf{p}(\mathbf{x})\|^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0, \tag{26}$$

and this is the equation we will aim to solve.

The method of characteristics

To solve the reduced form of the eikonal equation we will proceed in a way that will likely seem rather unmotivated. We introduce the **ray Hamiltonian** for the kth sheet of the local slowness surface

$$H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\|\mathbf{p}\|^2 c_k(\mathbf{x}, \hat{\mathbf{p}})^2, \tag{27}$$

which is defined on a six-dimensional **phase space** comprising points $(\mathbf{x}, \mathbf{p})$, and allows us to write the eikonal equation, eq. (26), in the alternative form

$$H_k[\mathbf{x}, \mathbf{p}(\mathbf{x})] = \frac{1}{2}, \quad \mathbf{p}(\mathbf{x}) = (\nabla T)(\mathbf{x}). \tag{28}$$

Eq. (28) states that the desired solution, T, of the eikonal equation corresponds to a **three-dimensional surface** $S = \{(\mathbf{x}, \nabla T(\mathbf{x}))\}$ lying within the level set $H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2}$ of the ray Hamiltonian. This geometric viewpoint is central to the method of characteristics.[2]

The **initial data** we have for the problem is that on the plane

$$p_i^0 x_i = 0, \tag{29}$$

the travel time is constant, and we are free to set this value equal to zero. We then wish to continue this solution into the heterogeneous region $p_i^0 x_i \geq 0$. Each point $\mathbf{x}^0$ on this initial plane corresponds to a point $(\mathbf{x}^0, \mathbf{p}^0)$ within the phase space, and at such points eq. (28) is satisfied. In this manner, the initial data for the problem specifies a **two-dimensional surface S^0 lying within this level set**, and to solve the eikonal equation we must construct a three-dimensional surface S within the level set that contains S^0.

[2] The treatment of the eikonal equation in this lecture follows Duistermaat (1974), though without the language of symplectic geometry in which it is most naturally expressed; there, caustics appear as the singularities of the projection of the surface S onto the position variables.

Let $(\mathbf{x}^0, \mathbf{p}^0)$ be a point on the initial surface S^0, and consider the solution of **Hamilton's canonical equations** for the ray Hamiltonian H_k as defined by

$$\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{\mathrm{d}p_i}{\mathrm{d}\sigma} = -\frac{\partial H_k}{\partial x_i}, \tag{30}$$

subject to the initial conditions

$$x_i(0) = x_i^0, \quad p_i(0) = p_i^0. \tag{31}$$

Within these equations, σ is a **generating parameter** whose physical interpretation will become clear but initially is just a parameterisation for the resulting integral curves. A key property of Hamilton's equations is that the **Hamiltonian is conserved along their solutions**. Indeed, using the chain rule we obtain

$$\frac{\mathrm{d}}{\mathrm{d}\sigma} H_k[\mathbf{x}(\sigma), \mathbf{p}(\sigma)] = \frac{\partial H_k}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} + \frac{\partial H_k}{\partial p_i}\frac{\mathrm{d}p_i}{\mathrm{d}\sigma} = -\frac{\mathrm{d}p_i}{\mathrm{d}\sigma}\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} + \frac{\mathrm{d}x_i}{\mathrm{d}\sigma}\frac{\mathrm{d}p_i}{\mathrm{d}\sigma} = 0, \tag{32}$$

where on the right-hand side we have suppressed arguments for clarity. As we have $H_k(\mathbf{x}^0, \mathbf{p}^0) = \frac{1}{2}$ at each point on the initial surface, it follows that $H_k[\mathbf{x}(\sigma), \mathbf{p}(\sigma)] = \frac{1}{2}$ along the curve within phase space obtained by integration of eq. (30) subject to the given initial conditions. What we can then do is take in turn each point $(\mathbf{x}^0, \mathbf{p}^0)$ of the initial surface S^0, and integrate Hamilton's equations to produce a curve on which the ray Hamiltonian is equal to one half. The union of all these curves defines a smooth three-dimensional surface S lying within the level set $H_k = \frac{1}{2}$, and this gives us the desired solution of the eikonal equation in geometric form.

In detail, this solution is usually defined only locally near to S^0, and we shall return to this point in a moment. Furthermore, it is not actually clear that such level surfaces need to be formed from the graph of ∇T for some travel time function T. Indeed, through this process, we obtain a value for $\mathbf{p}$ for each $\mathbf{x}$, and hence a vector field $\mathbf{p}(\mathbf{x})$. But this vector field is meant to be the gradient of a scalar function, and hence its first derivatives must possess the non-trivial symmetry

$$\frac{\partial p_i}{\partial x_j} = \frac{\partial p_j}{\partial x_i}. \tag{33}$$

Using the methods of **symplectic geometry** – a beautiful subject that is intrinsically linked to Hamilton's equations – it can be shown that the necessary symmetry of $\mathbf{p}(\mathbf{x})$ does hold, and hence a valid travel time is locally defined.

Having constructed the surface S, how do we recover the travel time T? We first make use of eqs. (7) and (30) to obtain

$$\frac{\mathrm{d}}{\mathrm{d}\sigma} T[\mathbf{x}(\sigma)] = \frac{\partial T}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} = p_i \frac{\partial H_k}{\partial p_i}. \tag{34}$$

To simplify the result, we note that the ray Hamiltonian $H_k(\mathbf{x}, \mathbf{p})$ is a homogeneous function of $\mathbf{p}$ of degree two, meaning

$$H_k(\mathbf{x}, \gamma\mathbf{p}) = \gamma^2 H_k(\mathbf{x}, \mathbf{p}), \tag{35}$$

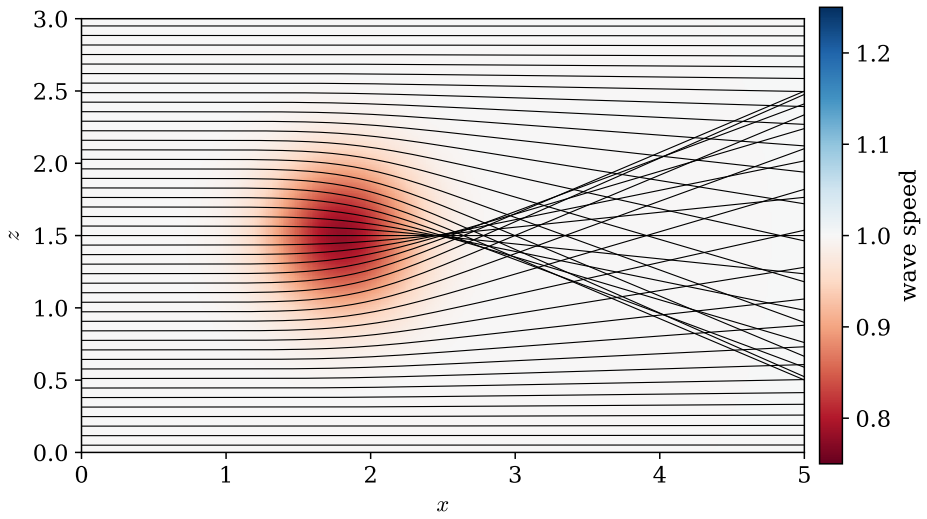


Fig. 3: Rays from an initially plane wavefront passing through a smooth region of low wave speed, obtained by numerical integration of Hamilton's equations. Behind the anomaly the rays cross, and their envelope forms a caustic.

for all $\gamma > 0$, and so Euler's theorem for homogeneous functions can be applied.[3] It implies that

$$p_i \frac{\partial H_k}{\partial p_i} = 2H_k, \tag{36}$$

and from eq. (28) we conclude that

$$\frac{\mathrm{d}}{\mathrm{d}\sigma} T[\mathbf{x}(\sigma)] = 1. \tag{37}$$

On the initial surface S^0 the travel time is everywhere equal to zero, and so eq. (37) has the trivial solution

$$T[\mathbf{x}(\sigma)] = \sigma \tag{38}$$

along the curve starting from each initial point $(\mathbf{x}^0, \mathbf{p}^0)$. In this manner, we can identify the generating parameter with the travel time along an integral curve.

The curves $\sigma \mapsto \mathbf{x}(\sigma)$ formed by projection of solutions of Hamilton's equations in phase space onto the position variable are known as **rays**. In order for the travel time T constructed by the above method to be a **single-valued function** of $\mathbf{x}$, it is necessary that each point $\mathbf{x}$ within the heterogeneous region is joined to a point $\mathbf{x}^0$ on the initial plane by a **unique ray**. Indeed, if there were two such rays, then for each we could calculate a travel time through the above method, and there is no reason why these two times should be equal.

Where the rays cross, points $\mathbf{x}$ within the heterogeneous region are reached by two or more rays starting from different points on the initial plane. Such a region of **multipathing** is bounded by an envelope of the ray family that is known as a **caustic**, and on a caustic ray theory breaks down. Though we do not have time to look quantitatively

[3] A proof of this theorem is given in the appendix to this lecture, but you do not need to know it. In isotropic media the necessary calculation can be performed explicitly, and this is the only case in which you might be asked to reproduce this derivation.

at the behaviour of the scalar amplitude $A_0(\mathbf{x})$ introduced in eq. (24), it can be shown that this ray theoretical amplitude becomes **infinite on a caustic**. In reality, caustics are characterised by large but finite amplitudes (the name comes from the Greek for "burning"), and more sophisticated asymptotic techniques allow them to be handled quantitatively. The formation of a caustic behind a region of low wave speed is shown in Fig. 3, while Fig. 4 shows rays from an initially plane wavefront passing through a smooth random medium. In the latter case there is nothing special about the structure, and yet caustics still form once the rays have propagated far enough. Their occurrence is generic, and not a peculiarity of simple models.

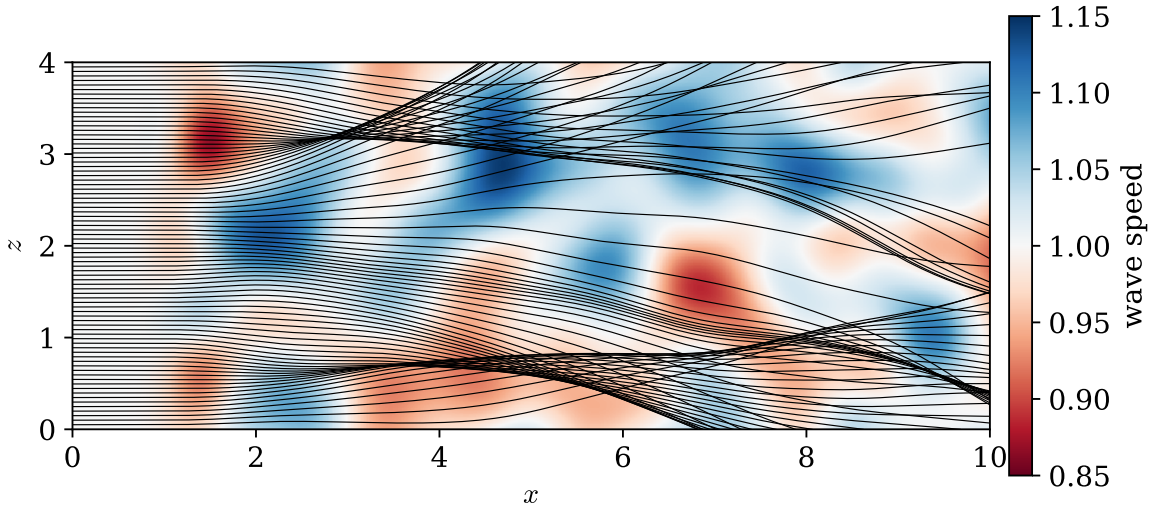


Fig. 4: Rays from an initially plane wavefront passing through a smooth random medium whose wave speed fluctuates by about five per cent (the medium is homogeneous to the left of $x \approx 1$). Caustics form generically once the rays have propagated a few correlation lengths of the heterogeneity.

Though within this lecture we have focused on the problem of an initially plane wave entering into a heterogeneous region, it should be apparent that ray theory can be applied to problems having more general initial conditions. For example, the initial data for the travel time can be specified on a curve or even at a point. Furthermore, it is possible to extend the method to allow for discontinuities in material parameters along internal boundaries so long as they are smooth relative to the wavelength of the waves considered.

What you need to know and be able to do

(i) The form of the ray-series expansion, and understand its underlying physical motivation.

(ii) How to obtain the eikonal equation and transport equations by substituting the ray expansion into the equations of motion. The full derivation given in the lecture is too involved to be set in an exam, but you may be asked to apply these methods within simpler situations.

(iii) Understand how the eikonal equation can be interpreted geometrically within a six-dimensional phase space, and be able to explain how solutions can be obtained through integration of Hamilton's equations.

(iv) Be able to solve the Hamiltonian ray equations in simple cases, and also to deduce some of their properties. Practice in this will be given in the next lecture and within the first problem set.

(v) What a caustic is, and why ray theory breaks down there.

Appendix: Euler's theorem for homogeneous functions (non-examinable)

A function f defined on $\mathbb{R}^n$ is said to be **homogeneous of degree** m if

$$f(\gamma \mathbf{p}) = \gamma^m f(\mathbf{p}), \tag{39}$$

for all $\mathbf{p}$ and all $\gamma > 0$. Euler's theorem states that if such a function is differentiable, then

$$p_i \frac{\partial f}{\partial p_i}(\mathbf{p}) = m f(\mathbf{p}). \tag{40}$$

The proof is almost trivial. Differentiating both sides of eq. (39) with respect to γ, the chain rule applied to the left-hand side gives

$$p_i \frac{\partial f}{\partial p_i}(\gamma \mathbf{p}) = m \gamma^{m-1} f(\mathbf{p}), \tag{41}$$

and setting $\gamma = 1$ yields eq. (40). The ray Hamiltonian $H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\|\mathbf{p}\|^2 c_k(\mathbf{x}, \hat{\mathbf{p}})^2$ is homogeneous of degree two in $\mathbf{p}$ for each fixed $\mathbf{x}$, because the local phase speed c_k depends only on the direction $\hat{\mathbf{p}}$ and not on the length of $\mathbf{p}$, and so eq. (36) follows. In an isotropic medium the result can be checked directly: $H_k = \frac{1}{2} c(\mathbf{x})^2 \|\mathbf{p}\|^2$, and hence $p_i \, \partial H_k / \partial p_i = c(\mathbf{x})^2 \|\mathbf{p}\|^2 = 2 H_k$.

Lecture 18: Spherical Earth structure

Outline and motivation

In this lecture we apply ray theory to explain how quantitative properties of earthquakes and the Earth's internal structure can be determined from observations of seismic waves. Most of the methods and results we describe are more than seventy years old, and are not representative of current research in seismology. They are, nonetheless, still of great value, and later developments in the field have been built on these foundations. Indeed, it is really quite remarkable that most of the knowledge about the Earth and earthquakes contained within this lecture was determined at a time before digital seismometers and electronic computers.

Hypocentre location using high-frequency waves

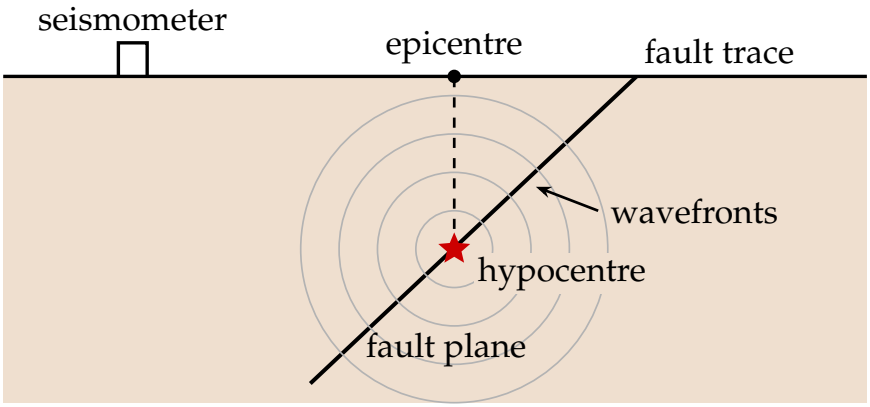


Fig. 1: A diagram showing an idealised planar fault. The hypocentre is the point on the fault where rupture initiates, while the epicentre is simply the projection of this point onto the Earth's surface. It is the hypocentre location that can be estimated by looking at the arrival times of high-frequency seismic waves recorded on seismometers.

Natural earthquakes occur due to rupture on faults within the Earth, and the point on the fault at which this rupture initiates is known as the earthquake's **hypocentre**. Fracture mechanics tells us that the speed of rupture propagation on the fault must be less than the fastest elastic wave speed of the medium (e.g. the P-wave speed in isotropic bodies), and it follows that the **first waves to be recorded at distant seismometers are generated at the hypocentre**. Using the recorded **arrival times** of the first seismic waves at a large number of seismometers, it is possible to estimate the hypocentre location.

We shall focus on the case of an earthquake recorded at a large number of seismometers around the globe, and assume that the Earth's elastic properties are isotropic,

which is a reasonable first approximation. On a global scale, seismic waves can be recorded at frequencies up to about 2 Hz. The reason for this upper frequency limit is two-fold: (i) earthquakes generate energy at a whole range of frequencies, but the spectrum tends to be biased towards lower frequencies, and (ii) attenuation mechanisms for seismic waves (which we have not discussed) act more strongly at higher frequencies, and so higher frequency components of the wavefield are removed more quickly as they propagate through the Earth. At the highest observed frequencies, the first arriving waves on seismograms are usually quite sharp, and so their arrival times can be measured to within a fraction of a second.

Given an accurate model of the Earth's seismic velocity structure, and having guessed the hypocentre location $\mathbf{x}_h$ and hypocentre time t_h for the earthquake, then it is possible to calculate the arrival times we would expect at each of the seismometers using ray theory. The use of ray theory is justified here so long as velocity variations within the Earth are smooth over the length scale of the observed seismic waves. At frequencies of about 2 Hz, this is probably a reasonable approximation, but the real reason ray theory is used for these calculations is that it is computationally efficient. Indeed, calculation of the travel time from the source to receiver only requires the integration of the ordinary differential equations described in Lecture 17. Let us write t_i for the measured arrival time of the first seismic wave at the ith seismometer, which has position $\mathbf{x}_i$. We can then relate this observation to the true source location by

$$t_i = t_h + T(\mathbf{x}_i, \mathbf{x}_h) + e_i, \tag{1}$$

where $T(\mathbf{x}_i, \mathbf{x}_h)$ is the ray theoretic travel time of the fastest ray from $\mathbf{x}_h$ to $\mathbf{x}_i$, and e_i denotes a random measurement error which we assume to have zero mean and known variance ς_i^2 (not to be confused with the ray parameter σ introduced below). To estimate the four unknown parameters $(\mathbf{x}_h, t_h)$, we minimise the following **least-squares misfit**

$$J(\mathbf{x}_h, t_h) = \sum_{i=1}^{N_s} \frac{[t_h + T(\mathbf{x}_i, \mathbf{x}_h) - t_i]^2}{\varsigma_i^2}, \tag{2}$$

where N_s is the total number of seismometers from which we have measurements. The dependence of the calculated travel times on the hypocentre location $\mathbf{x}_h$ is non-linear, but we can apply iterative optimisation methods to find the best-fitting solution given an initial guess. To do this, we need to differentiate the misfit J with respect to the model parameters $(\mathbf{x}_h, t_h)$, but we will not enter into the details.

The accuracy to which earthquake hypocentres can be located depends on a number of things. Clearly errors in the measured arrival times lead to uncertainties in the solution, but this can be assessed using standard error-propagation techniques. Next, the limited frequency content of observed seismic waves sets a fundamental limit on the accuracy of location estimates. If we record waves at a maximum frequency of 2 Hz, then using a typical P-wave velocity of about $5\,\mathrm{km\,s^{-1}}$ for shallow parts of the Earth, we see that the shortest wavelengths present in the data are about 2.5 km. Such waves cannot distinguish sources separated by less than about one half or one quarter of their minimum wavelength.[1] The accuracy is also dependent on the locations of the seismometers relative to the source. Ideally we would like a fairly even spatial distribution in terms of both azimuth and distance from the hypocentre. If, however, there

[1] The precise fraction isn't well-defined, nor is it important.

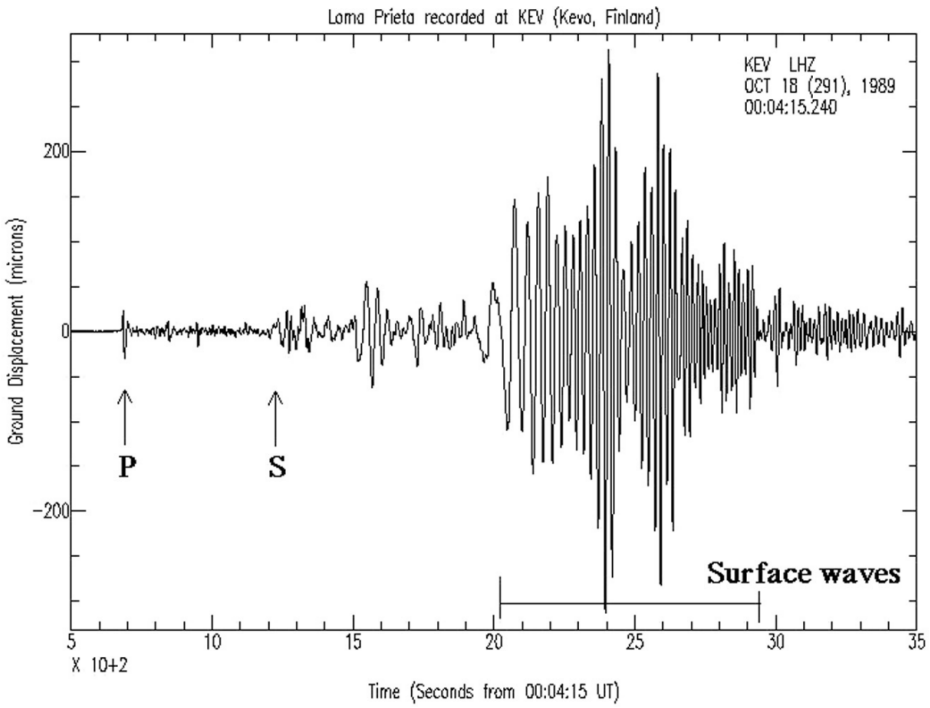


Fig. 2: A vertical component seismogram recorded in Finland following an earthquake in California (the 1989 Loma Prieta earthquake recorded at station KEV). The P-wave arrival can be seen very clearly, while later arriving phases are significantly harder to identify.

are strong biases in the spatial distribution of the seismometers, then large uncertainties in some of the recovered parameters can be introduced. Finally, any uncertainty in the assumed Earth structure will clearly map onto errors in the hypocentre location. In the above discussion we have only considered the arrival times of the first waves seen on the seismograms. It is, however, often possible to identify and measure the arrival times of other seismic phases, and these additional measurements can be used within the above procedure to produce better results.

The spherically averaged velocity structure of the Earth

In the remainder of this lecture we consider how the Earth's spherically averaged velocity structure has been determined. The basic idea is very simple. After an earthquake, seismic waves propagate through the Earth's interior and are recorded at various locations around the globe. The observed seismograms depend on (i) the earthquake source, and (ii) the material that the waves have travelled through. And so, in principle, it is possible to work backwards from such observations to make inferences about both the earthquake source and the structure of the Earth's interior.

Studies of earthquakes and of Earth structure are fundamentally linked, and we cannot learn about one without considering the other. For example, to locate an earthquake accurately we must know the Earth's velocity structure, and yet these earthquake locations are required if we wish to determine the velocity structure! It is necessary either to consider these problems simultaneously, or to proceed in an iterative manner with one part of the problem being considered solved (i.e. that earthquake

locations are known) while the other parameters are refined.

Ray theory in spherically symmetric earth models

To learn about the Earth's velocity structure from travel times, we will use ray theory as discussed in the last lecture. For simplicity we restrict attention to isotropic earth models, in which case we need to consider either P-waves or S-waves, along with a whole range of converted phases that can arise at internal or external boundaries. For definiteness, we will focus on P-waves, but the ideas generalise easily.

The eikonal equation for P-waves in such a model is given by

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2}, \tag{3}$$

where $\mathbf{x}$ is a position vector on a ray, $\mathbf{p}$ is the local slowness vector, and the P-wave Hamiltonian is given by

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2}\alpha(\mathbf{x})^2 \|\mathbf{p}\|^2, \tag{4}$$

with $\alpha(\mathbf{x})$ the P-wave speed at a point $\mathbf{x}$. The geometry of the ray paths is determined by solving Hamilton's canonical equations

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\sigma} = \frac{\partial H}{\partial \mathbf{p}}, \quad \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}\sigma} = -\frac{\partial H}{\partial \mathbf{x}}, \tag{5}$$

and the travel time T along a ray is simply given by $T[\mathbf{x}(\sigma)] = \sigma$. Because the earth model under consideration is spherically symmetric, the P-wave speed, α, only depends on the radial distance $r = \|\mathbf{x}\|$ from the centre of the model. Using this fact, we will show that this problem can be solved using quadratures. This situation is analogous to that for a single particle in a central potential, and to other **integrable systems** which have sufficiently many conserved quantities.

Making use of the P-wave Hamiltonian, we find that the ray equations in eq. (5) in a spherically symmetric earth model take the specific form

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\sigma} = \alpha\hat{\mathbf{p}}, \quad \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}\sigma} = -\frac{1}{\alpha}\frac{\mathrm{d}\alpha}{\mathrm{d}r}\hat{\mathbf{x}}, \tag{6}$$

where we have used the eikonal equation to simplify the form of the right-hand sides, and where "hats" are used to denote unit vectors. If we define $\mathbf{q} = \mathbf{x} \times \mathbf{p}$, then a simple calculation shows that

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}(\mathbf{x} \times \mathbf{p}) = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\sigma} \times \mathbf{p} + \mathbf{x} \times \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}\sigma} = \alpha\hat{\mathbf{p}} \times \mathbf{p} - \frac{1}{\alpha}\frac{\mathrm{d}\alpha}{\mathrm{d}r}\mathbf{x} \times \hat{\mathbf{x}} = \mathbf{0}. \tag{7}$$

It follows that $\mathbf{q}$ is constant along solutions to these equations, and we mention in passing that this is precisely the conserved quantity associated with the spherical symmetry through Noether's theorem. Since $\mathbf{x}$ and $\mathbf{p}$ are both orthogonal to the constant vector $\mathbf{q}$, rays always remain within the plane spanned by the initial values of $\mathbf{x}$ and $\mathbf{p}$. Furthermore, $q = \|\mathbf{q}\|$ is constant along a ray, and this can be written

$$q = \frac{r \sin\theta}{\alpha}, \tag{8}$$

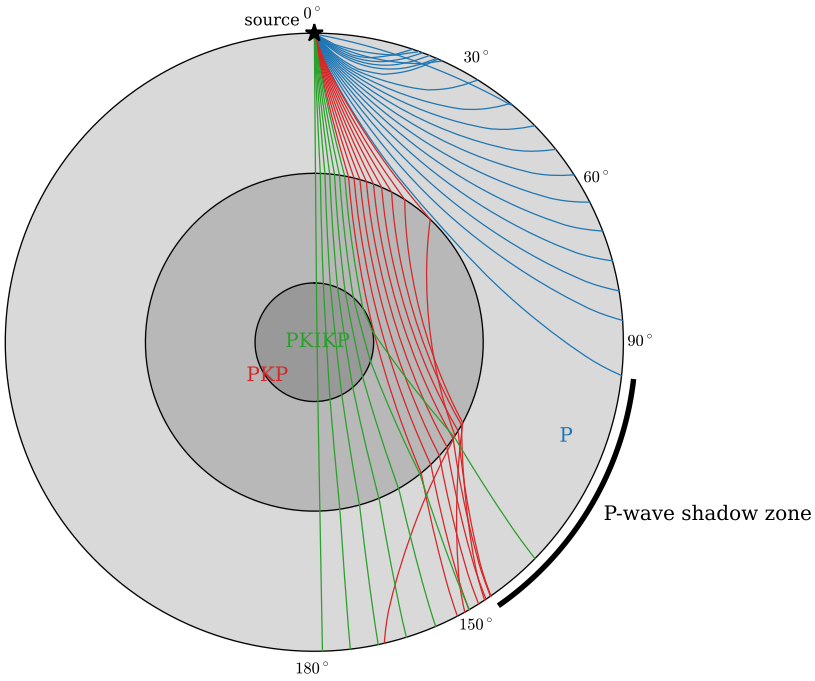


Fig. 3: P-wave ray paths within the earth model PREM for a source at the surface, calculated using the ray equations of this lecture. Rays that turn within the mantle (blue) emerge at epicentral angles up to about 98°. PKP rays (red), which turn within the outer core, emerge beyond about 145°, while PKIKP rays (green), which pass through the inner core, can emerge within the shadow zone. The sudden reduction in P-wave velocity on entering the fluid outer core thus leads to the so-called "P-wave shadow zone", while no S-waves are able to propagate within the outer core at all.

where $\theta \in [0, \pi]$ is the angle between $\mathbf{x}$ and $\mathbf{p}$. Because $\mathbf{x}$ and $\mathbf{p}$ remain within their initial plane, we can write

$$\hat{\mathbf{p}} = \cos[\theta(\mathbf{x})]\hat{\mathbf{x}} + \sin[\theta(\mathbf{x})]\hat{\boldsymbol{\Delta}}(\mathbf{x}), \tag{9}$$

where the epicentral angle Δ is that subtended between $\mathbf{x}$ and its initial value, and $\hat{\boldsymbol{\Delta}}(\mathbf{x})$ the unit vector in the direction of increasing Δ about the point $\mathbf{x}$. Making use of eq. (8), we can then write

$$\sin\theta = \frac{\alpha q}{r}, \quad \cos\theta = \pm\sqrt{1 - \frac{\alpha^2 q^2}{r^2}}, \tag{10}$$

where the sign in the second expression is determined by whether the ray is up-going (+) or down-going (−). For definiteness, we consider an initially down-going ray, and using the first of the equations in eq. (6) we find

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\sigma} = -\alpha\sqrt{1 - \frac{\alpha^2 q^2}{r^2}}\hat{\mathbf{x}} + \frac{\alpha^2 q}{r}\hat{\boldsymbol{\Delta}}. \tag{11}$$

In this manner, we have eliminated the local slowness vector from the problem, and are left with a single equation for the position vector along the ray. To proceed, we

calculate

$$\frac{\mathrm{d}}{\mathrm{d}\sigma} r[\mathbf{x}(\sigma)] = \frac{\partial r}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} = -\alpha\sqrt{1 - \frac{\alpha^2 q^2}{r^2}}, \tag{12}$$

where we have used the familiar relation $\frac{\partial r}{\partial x_i} = \hat{x}_i$. In the same way, we find

$$\frac{\mathrm{d}}{\mathrm{d}\sigma} \Delta[\mathbf{x}(\sigma)] = \frac{\partial \Delta}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}\sigma} = \frac{\alpha^2 q}{r^2}, \tag{13}$$

where we have used the identity $\frac{\partial \Delta}{\partial x_i} = \frac{1}{r}\hat{\Delta}_i$ which is readily established. Combining eqs. (12) and (13), we have now shown that along a ray we can write

$$\frac{\mathrm{d}\Delta}{\mathrm{d}r} = \frac{\mathrm{d}\Delta}{\mathrm{d}\sigma}\left(\frac{\mathrm{d}r}{\mathrm{d}\sigma}\right)^{-1} = \frac{-\alpha q}{r\sqrt{r^2 - \alpha^2 q^2}}. \tag{14}$$

Similarly, using the fact that $\mathrm{d}T/\mathrm{d}\sigma = 1$ along a ray, we find

$$\frac{\mathrm{d}T}{\mathrm{d}r} = \frac{\mathrm{d}T}{\mathrm{d}\sigma}\left(\frac{\mathrm{d}r}{\mathrm{d}\sigma}\right)^{-1} = \frac{-r}{\alpha\sqrt{r^2 - \alpha^2 q^2}}. \tag{15}$$

It is important to note that these equations are only valid in portions of the ray in which the epicentral angle Δ and the travel time T are single-valued functions of the radius r.

Let us now suppose that the P-wave speed, α, increases monotonically with depth into the earth model. Consider a ray starting at the surface $r = b$ that is initially travelling into the earth model in a non-vertical direction (i.e. $\pi/2 < \theta < \pi$ initially). The fact that $q = \frac{r\sin\theta}{\alpha}$ is conserved implies that this ray will travel monotonically downwards into the earth model with Δ monotonically increasing until it reaches a **turning point** r_t such that

$$q = \frac{r_t}{\alpha(r_t)}. \tag{16}$$

Beyond this turning point, the ray moves up towards the surface, with r monotonically increasing, and Δ continuing to monotonically increase. Furthermore, the up- and down-going legs of the ray's path are symmetric about the radial line passing through the turning point. Using eq. (14), which is valid throughout the down-going leg, we obtain

$$\Delta = \int_{r_t}^{b} \frac{2\alpha(r)q}{r\sqrt{r^2 - \alpha(r)^2 q^2}}\,\mathrm{d}r, \tag{17}$$

for the total epicentral angle subtended by the ray as it travels from the surface to the turning point and then up to the surface again. Using eq. (15), we similarly write the travel time along this ray path as

$$T = \int_{r_t}^{b} \frac{2r}{\alpha(r)\sqrt{r^2 - \alpha(r)^2 q^2}}\,\mathrm{d}r. \tag{18}$$

In this manner, we see that the travel time along a ray path in such a spherical model can be obtained through a simple integral over the velocity structure (the integrands are singular at the turning point, but this is an integrable singularity). The extension of these results to models having more complicated velocity structures is straightforward in principle, but takes a little effort.

Travel time inversions for spherical velocity structure

As we have just seen, ray theory can be used to efficiently determine the travel times of various **seismic phases** within a spherically symmetric earth model. At a given epicentral angle, there will generally be many distinct seismic phases that arrive at different times and correspond to different ray paths, as well as non-geometric phases such as surface waves, head waves, and diffracted waves that are not predicted by ray theory. The use of the word "phase" here refers to a distinct waveform on a seismogram, and should not be confused with other uses of "phase" within these lectures or elsewhere.

Assuming that we know an earthquake's location and origin time, it is possible to measure the travel times of various seismic phases to different seismometers. By repeating this process for many events, globally averaged travel time curves have been constructed. Given a radial velocity structure, we know how to calculate travel times for the different phases, and we aim to find an earth model that is consistent with our observations. This is another example of an inverse problem in which we must work backwards from observations to learn about a set of model parameters.

We will have more to say about inverse problems in the next lecture, but historically speaking, the determination of the Earth's spherically symmetric velocity structure largely pre-dates the introduction of inverse theory, and early studies used more or less ad hoc methods. In this field, the most important figures are Richard Oldham (1858–1936), Beno Gutenberg (1889–1960), Harold Jeffreys (1891–1989), and Inge Lehmann (1888–1993). In particular, it was Oldham (1906) who first argued for the existence of the Earth's core, from the anomalously late arrivals of waves at large epicentral angles, while it was Gutenberg (1914) who explained the P-wave shadow zone illustrated in Fig. 3 and used it to determine the depth of the core–mantle boundary to within a few tens of kilometres of its modern value. Jeffreys first convincingly argued that the core of the Earth must be at least partially fluid, though here the relevant observations were of solid-Earth tides, and not seismic waves. Finally, Lehmann later realised that there exists a solid inner core through careful study of waves travelling through the Earth's central region. There is a famous quote from Oldham's (1906) paper on the Earth's core that is worth repeating:

> "Of all regions of the Earth none invites speculation more than that which lies beneath our feet, and in none is speculation more dangerous; yet, apart from speculation, it is little that we can say regarding the constitution of the interior of the earth. Many theories of the earth have been propounded at different times: the central substance of the earth has been supposed to be fiery, fluid, solid, and gaseous in turn, till geologists have turned in despair from the subject, and become inclined to confine their attention to the outermost crust of the Earth, leaving its centre as a playground for mathematicians."

We shall not consider the determination of the spherically symmetric velocity structure further, but will instead turn to discussing its broad features.

An overview of the Earth's spherically averaged velocity structure

Figure 5 shows the variation of P-wave and S-wave velocity with radius in the Earth according to the Preliminary Reference Earth Model (PREM) of Dziewonski & Anderson (1981). This model was obtained by fitting travel times of seismic waves, as well

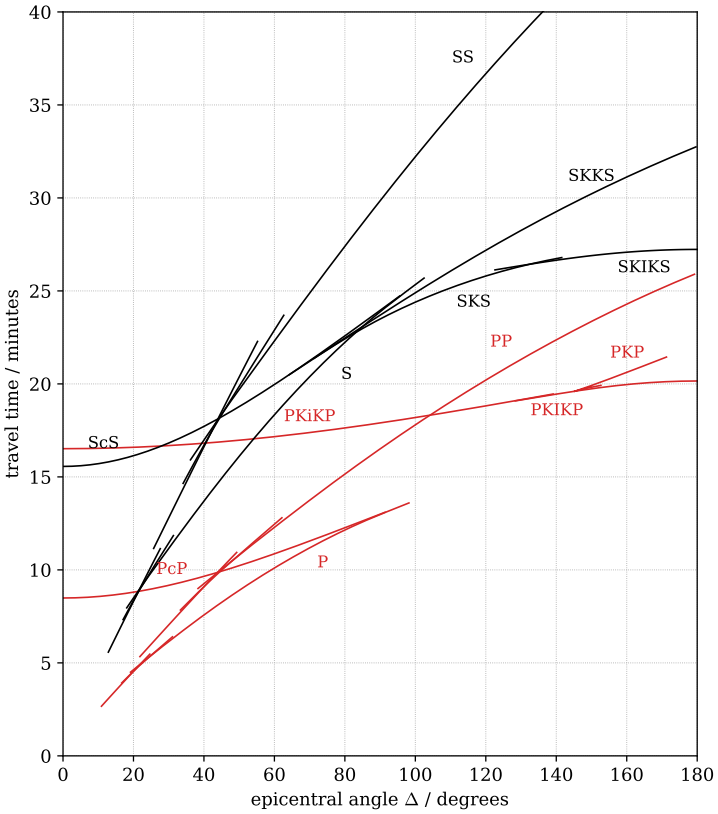


Fig. 4: Travel time curves for the main body-wave phases within the earth model PREM, for a source at the surface, calculated using the ray equations of this lecture (phases travelling as P-waves in the mantle in red, and as S-waves in black). Only rays whose turning or reflection point lies below 220 km depth are shown. The nomenclature for the different phases can be ignored.

as further constraints obtained from observations of the Earth's free oscillations that will be the focus of a later lecture. As a first observation, there is a general tendency for the velocity to increase gradually downwards, though there are also a number of sharp discontinuities within the structure. The P-wave velocity α and S-wave velocity β are given by

$$\alpha = \sqrt{\frac{\kappa + \frac{4}{3}\mu}{\rho}}, \quad \beta = \sqrt{\frac{\mu}{\rho}}, \tag{19}$$

where κ is the bulk modulus, μ the shear modulus, and ρ the density. As we go deeper into the Earth, things are increasingly compressed, and this tends to increase both the elastic moduli and the density; the former effect dominates, so that wave speeds typically increase.

Discontinuities in the seismic velocity are identified principally by the occurrence of reflected seismic waves, and their presence must be due either to sharp changes in composition or to a phase change within the same material. Note here that "sharp" is always defined relative to the wavelength of the seismic waves used to image the Earth, and so these boundaries may, in reality, be somewhat diffuse. The most notable discontinuities within the Earth are at the core–mantle boundary (CMB), which is thought to represent a change from the rocky mantle to a fluid iron core, and at the

inner core boundary (ICB), which is explained by a fluid to solid phase change within the iron core. It is worth emphasising that we do not actually know that the core is made of iron (and likely a few additional minor components). The reasons for thinking this is likely are: (i) iron at the appropriate pressures and temperatures has the right sort of density and wave speeds, (ii) iron is very abundant in the solar system due to its high nuclear binding energy per nucleon, (iii) there exist iron meteorites that are interpreted as fragments from the cores of "proto-planets", and (iv) such a composition is consistent with the generation of the geodynamo in a convecting conductor.

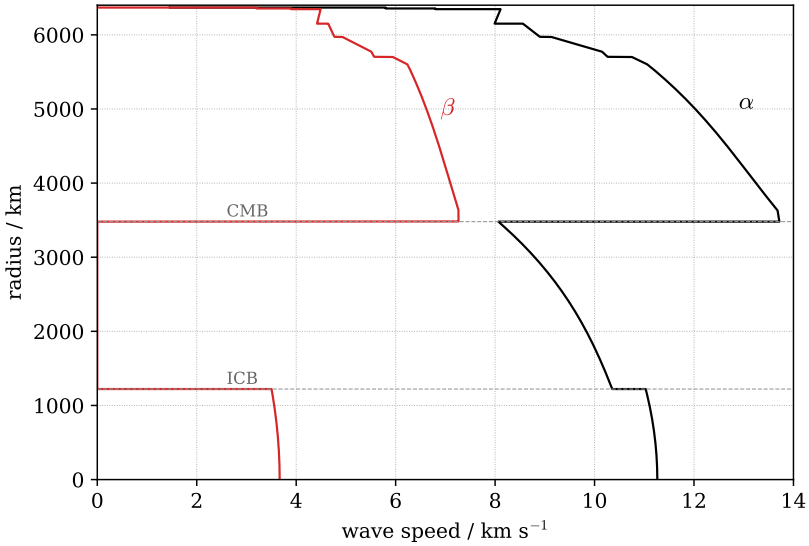


Fig. 5: Variation of P-wave velocity (black curve) and S-wave velocity (red curve) with radius in the model PREM of Dziewonski & Anderson (1981). Here we can see that both wave speeds generally increase with depth, but this increase is interrupted by a number of sharp velocity discontinuities. Such discontinuities are associated with compositional or phase changes within the Earth. The most notable boundaries within the Earth are between the rocky mantle and the fluid iron outer core where the S-wave velocity becomes equal to zero, and at the inner core boundary where the iron of the core undergoes a fluid to solid phase transformation.

That the outer core is a fluid is strongly suggested by (i) observations of solid-Earth tides, (ii) the absence of S-waves propagating through this region ($\mu = 0$ in a fluid), and (iii) the fact that a fluid core is required for the generation of the geodynamo. The likely solidity of the inner core has been deduced by a number of arguments. Firstly, it does not seem plausible that there could be a sharp jump in velocity within a purely fluid region. There is also compelling evidence derived from the observed free oscillation periods (Dziewonski & Gilbert 1971). Lastly, some studies have identified inner core S-waves produced by conversion from a P-wave at the ICB (e.g. Deuss et al. 2000). Such core phases have very small amplitudes, and the robustness of such observations is still debated.

As well as the velocity discontinuities at the CMB and ICB, there are a number of other discontinuities within the shallower Earth. Those within the crust and uppermost mantle are almost certainly due to compositional changes. There are then additional discontinuities at around 410 km and 670 km that are not readily explained in

terms of composition. Instead these discontinuities are thought to represent solid-state phase transformations within the minerals making up the Earth's mantle that result from radial changes in both pressure and temperature. The occurrence of such phase transformations within this part of the mantle is consistent with both laboratory experiments and quantum mechanical calculations. The region between the 410 km and 670 km discontinuities is known as the transition zone, and can be regarded as separating the upper and lower parts of the mantle. Other properties such as viscosity are believed to also be discontinuous across these phase boundaries, with the lower mantle seeming to be significantly more viscous. The viscosity contrast between the upper and lower mantle has important implications for mantle convection, and may lead to some "decoupling" of the two regions.

What you need to know and be able to do

(i) Know what the hypocentre of an earthquake is, and how the hypocentral location and time can be determined using observations of seismic waves.

(ii) Understand how the Hamiltonian ray tracing equations within a spherically symmetric earth model can be reduced to the calculation of appropriate radial integrals. In the second problem set you will see how such travel time integrals can be inverted for the velocity structure.

(iii) Understand in outline the observations and methods that were used to constrain the Earth's spherically symmetric velocity structure. You should also know the basic properties of this spherically symmetric velocity structure, and the physical interpretation of its major features.

Lecture 19: Delay time tomography

Outline and motivation

In the previous lecture we discussed the use of ray theory in determining the Earth's spherically symmetric velocity structure. We now consider an extension of these ideas, known as **delay time tomography**, that can be used to produce quantitative images of lateral velocity variations within the Earth. This method falls into a class of techniques known as **seismic tomography**, and we will say more about some other variants in later lectures. Such work is done to better understand convection in the mantle, with the tomographic models obtained acting as an indirect observational constraint on this process. Delay time tomography is an example of a **linear inverse problem**, and we discuss a practical method for its solution known as **regularised least squares**.

Laterally varying structure from seismology

Though the Earth is close to being spherically symmetric, the small deviations from sphericity are still of great interest. This is because they can tell us about processes such as mantle convection deep within the Earth that are inaccessible to direct observation. The area of seismology concerned with imaging lateral variations in the Earth's interior is known as seismic tomography, and is one of the main areas of current research within solid Earth geophysics. The results of such studies are tomographic models that show variations in seismic wave speed and other parameters within the Earth. These variations are defined **relative to a fixed spherically symmetric reference model**, and so include only the **laterally varying perturbations**. An early example of such a tomographic model is shown in Fig. 1.

The basic idea underlying seismic tomography is that if we knew the Earth's internal structure then we could calculate what would be observed after an earthquake, and so we can try to work backwards to determine this structure from seismic observations. This is a further example of an inverse problem, and in this lecture we shall start to discuss the solution of such problems in detail. In particular, we will focus on travel time measurements that are modelled using ray theory. This was the earliest tomographic method developed, with the first papers written in the late 1970s. Many of the basic ideas, however, carry over to more complicated tomographic techniques including waveform tomography that we will discuss next time.

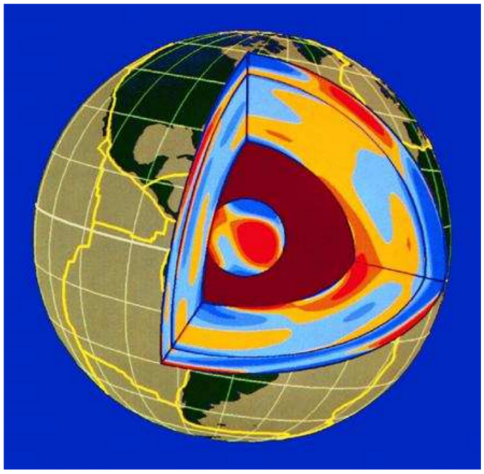


Fig. 1: An image of a tomographic model of S-wave velocity variations within the Earth produced by Woodhouse & Dziewonski (1989). The colours show the perturbation in S-wave velocity relative to a spherical reference model, with blue areas being faster than normal, and red areas slower. Areas of faster velocity are thought to be colder than normal, while those that are slower are hotter, and hence the chosen colour scale. These temperature anomalies have associated changes in density, and so the models tell us something about the buoyancy forces that drive mantle convection.

Delay time measurements

If we know the location and origin time of an earthquake, then we have seen that it is possible to measure the travel times of various seismic phases recorded on seismograms around the globe. If the Earth were spherically symmetric then the observed travel times would (within the limits of observational errors) depend only on the epicentral angle and the source depth. Though this is a reasonable first approximation, there do exist variations in observed travel times due to lateral variations within the Earth. For a given travel time measurement, we define its associated **delay time** by

$$\delta T = T - T_0, \tag{1}$$

where T is the **measured** travel time, and T_0 is the travel time of the seismic phase **predicted** by a spherically symmetric reference model. We note that this definition is such that the delay time will be positive if the observed phase arrives later than expected. Such delay times can be measured for many source–receiver pairs and for many seismic phases, and constitute the data in delay time tomography. To some extent these times can be measured by hand. But more often this is done using signal processing tools such as cross-correlation, or increasingly using forms of machine learning.

Relating delay times to lateral variations

Suppose we have obtained the delay time of a seismic phase for a given source–receiver pair. Neglecting for the moment observational errors, this delay time would vanish if the Earth's velocity structure were equal to that of the spherical reference model. In practice, measured delay times are small (e.g. tens of seconds for a phase whose

travel time is tens of minutes), and so we can conclude that the Earth's true structure cannot be too far from the spherical reference model. We know that the travel time of a seismic phase can be calculated using ray theory. But this travel time is related to the velocity structure in a complicated non-linear manner through solution of the Hamiltonian ray equations, and we cannot in general write an explicit formula for T in terms of the velocity, v. Because, however, lateral velocity variations within the Earth seem to be small we can simplify the problem by seeking an approximate **linearised relation** between the observed delay time and velocity perturbations within the Earth.

Let us write $T(v)$ for the travel time of a certain seismic phase as determined by the velocity model, v. This is a mapping from the velocity, which is a scalar function in the earth model, to the real numbers, and so the mapping $v \mapsto T(v)$ defines a **functional**. To proceed, we need to introduce the **functional derivative** of T with respect to v as defined through a first-order Taylor expansion

$$T(v + \delta v) = T(v) + \langle DT(v)|\delta v\rangle + O(\delta v^2). \tag{2}$$

As in our discussion of functional derivatives in Lecture 13 in the context of Hamilton's principle, we see that this functional derivative, $DT(v)$, is an object that acts on a velocity perturbation, δv, to produce the first-order change in the travel time. We denote the action of the functional derivative on the velocity perturbation using the **inner product** notation $\langle\cdot|\cdot\rangle$.

The desired functional derivative can be obtained very easily using **Fermat's principle** for elastic ray theory, which you established in the first problem set. Having solved the relevant ray tracing equations, we know that the travel time can be expressed as the following line integral

$$T = \int_{\text{ray}} \frac{1}{v}\,\mathrm{d}s, \tag{3}$$

where s denotes the arc length along the ray. Fermat's principle states that this travel time is stationary with respect to all first-order perturbations to the ray geometry that leave the end-points fixed. If we perturb the velocity of the earth model to be $v + \delta v$, then the travel time is affected by both (i) a change in the ray geometry, and (ii) the change in velocity along the new ray path. To first-order accuracy the contributions of these effects can be added, but by Fermat's principle the first-order term associated with the change in ray geometry is zero. It follows that the delay time can be written

$$\delta T = -\int_{\text{ray}} \frac{\delta v}{v^2}\,\mathrm{d}s, \tag{4}$$

where this integral is taken along the unperturbed ray. To identify the functional derivative here, it seems natural to use the conventional inner product

$$\langle u|w\rangle = \int_M uw\,\mathrm{d}^3\mathbf{x}, \tag{5}$$

for any two real-valued scalar fields, where M denotes the region occupied by the earth model. Looking at eq. (4), it follows that the functional derivative can be suitably defined only if we allow it to be a generalised function that is non-zero only along the reference ray path, while along the ray path it is infinite in such a way that

$$\langle DT(v)|\delta v\rangle = -\int_{\text{ray}} \frac{\delta v}{v^2}\,\mathrm{d}s. \tag{6}$$

This definition should be compared with the familiar one for a Dirac delta function. The singular nature of this functional derivative causes difficulties in some approaches to solving the inverse problem, though these issues are avoided (at a cost) within the most common approach discussed below.

Setting up the inverse problem

We now consider how the inverse problem can be formulated. Our data in such a study comprise delay times for a variety of seismic phases measured from many source–receiver pairs. Typically the number of delay times used will be of order 10^5–10^8. For definiteness, we focus on P-wave phases, meaning the data are only sensitive to variations in the P-wave velocity, α. Nearly identical methods can be applied to studies based on S-waves, or those that consider both P- and S-wave measurements simultaneously.

At the outset, we shall assume that (i) velocity variations in the Earth are sufficiently small that they can be related linearly to the delay times, and (ii) the length scale of heterogeneity in the Earth is such that ray theoretic approximations are appropriate. Given these assumptions, each delay time measurement δT_i for $i = 1, \ldots, n$ can then be expressed in terms of the P-wave velocity perturbation $\delta\alpha$ through the equation

$$\delta T_i = -\int_{\text{ray}_i} \frac{\delta\alpha}{\alpha^2}\, \mathrm{d}s + e_i, \tag{7}$$

where the integration is taken along the appropriate unperturbed ray path, α denotes the P-wave velocity in the reference model, and we have written e_i for the associated measurement error. We will assume initially that the e_i represent zero-mean, uncorrelated Gaussian errors with known variances σ_i^2. More complicated statistical models can, of course, be applied, but this is not the place for such a discussion.

From the data set $\{\delta T_i\}_{i=1}^n$ we wish to recover the velocity perturbation $\delta\alpha$, which is a (piecewise) continuous function within the earth model. Clearly a finite amount of data can never be sufficient to recover such a continuous function uniquely, and so we are dealing with a fundamentally **underdetermined problem**. There is a range of methods for dealing with this problem, but we consider one that is both widely used and comparatively simple (the latter point, of course, contributing to the former). First we assume that the velocity perturbation can be expanded using a **finite number** of basis functions

$$\delta\alpha(\mathbf{x}) = \sum_{j=1}^{m} \delta\alpha_j \varphi_j(\mathbf{x}). \tag{8}$$

For example, we might expand the angular dependence of the velocity perturbation in spherical harmonics up to some finite degree, and then use a set of polynomial functions to describe the radial dependence. Equally, some studies split the earth model into a series of small blocks, assuming a constant velocity in each one. Having expanded the velocity perturbations, the problem is reduced to one of linear algebra that can be solved using elementary methods. But clearly we are introducing bias into our solutions because we cannot say a priori what an appropriate basis should be.

Accepting this strong assumption on the form of the velocity model, we can then

write the ith delay time as

$$\delta T_i = \sum_{j=1}^{m} \left(- \int_{\mathrm{ray}_i} \frac{\varphi_j}{\alpha^2} \, \mathrm{d}s \right) \delta\alpha_j + e_i, \tag{9}$$

which gives a system of linear equations relating the velocity expansion coefficients $\{\delta\alpha_j\}_{j=1}^m$ to the delay time data $\{\delta T_i\}_{i=1}^n$. It will be useful to write this equation as

$$\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{e}, \tag{10}$$

where $\mathbf{d}$ is an n-dimensional vector of the measured delay times, $\mathbf{m}$ is an m-dimensional vector of the velocity expansion coefficients, and $\mathbf{e}$ is a vector of the random errors. The elements of the $n \times m$ matrix $\mathbf{A}$ are given by

$$A_{ij} = - \int_{\mathrm{ray}_i} \frac{\varphi_j}{\alpha^2} \, \mathrm{d}s, \tag{11}$$

and can be determined from knowledge of the reference model and the basis functions. For the block parameterisation, in which φ_j is equal to one within the jth block and to zero elsewhere, A_{ij} is simply minus the length of the ith ray path within the jth block divided by α^2 (taking the reference velocity to be uniform within a block). Each ray crosses only a small fraction of the blocks, and so most of the entries of $\mathbf{A}$ are zero: the matrix is **sparse**. Problems with 10^8 data and 10^6 unknowns are solved using iterative methods that require only products of $\mathbf{A}$ and its transpose with vectors, and sparsity makes such products cheap, though it is not essential: what matters most is a good iterative method, and that is a topic beyond this course.

To illustrate the ideas of this lecture we will make use of a toy problem, shown in Fig. 2, in which straight rays cross a square region that has been divided into 24×24 blocks (a "pixel" basis), the data being the delay times along each ray with added Gaussian errors. The model to be recovered is the fractional slowness perturbation in each block, the slowness being $p = 1/\alpha$, so that $\delta\alpha/\alpha = -\delta p/p$, and the input model used to generate the data is a smooth random field.

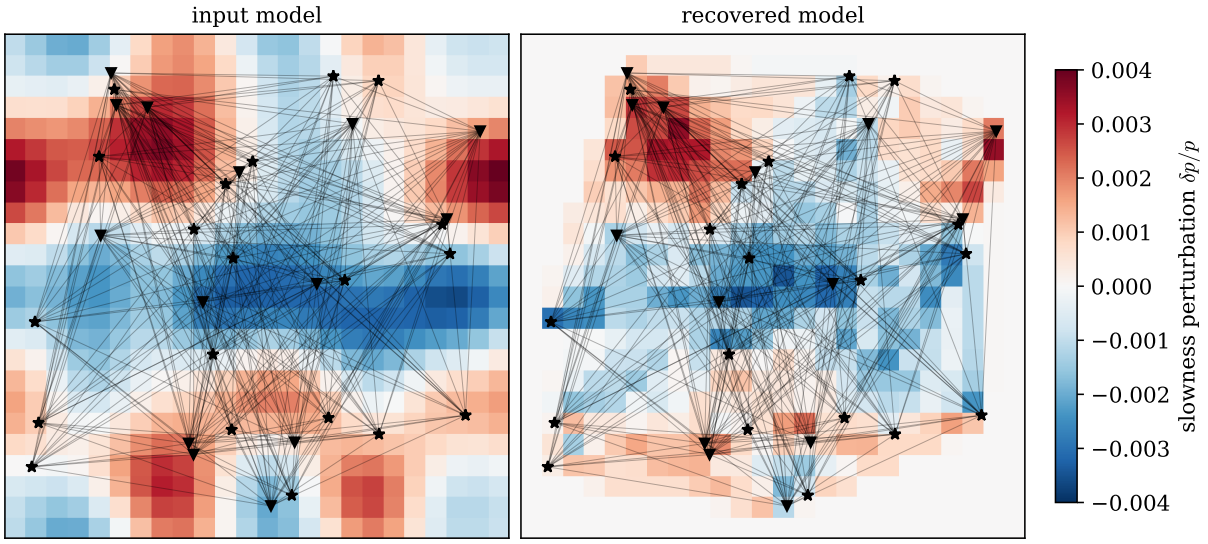


Fig. 2: A toy two-dimensional tomographic problem with a pixel basis of 24×24 blocks. Stars mark "earthquakes", triangles "seismometers", and the straight lines denote the ray paths. The colours show the fractional slowness perturbation. On the left is the input model, and on the right the model recovered by regularised least squares with norm damping, the damping parameter having been chosen so that the data are fit to within their errors. In blocks that no ray crosses the recovered model is zero, while within the covered region the anomalies are smeared out along the ray paths.

Non-uniqueness and the null space

Having reduced delay time tomography to the linear equations of eq. (10), we can start to think about how the model $\mathbf{m}$ might be recovered. If we assume that the observational errors are negligible, then we might be tempted to solve this equation using the inverse of $\mathbf{A}$. However, we recall that $\mathbf{A}$ is an $n \times m$ matrix, where n is the number of delay time measurements, and m is the number of model parameters, and as we will typically have $n \neq m$ the inverse of this matrix cannot be defined.

We could, of course, select the number of model parameters to be equal to the number of data ($n = m$), but there is no physical reason why this should be the case. Indeed, it is better to determine the model parameterisation based on some idea of what the data should reasonably be able to resolve within the Earth. For example, if we are looking at waves of a given dominant frequency, then we can estimate their wavelength in different parts of the Earth using the reference velocity structure. We would not expect to resolve features within the Earth whose size is substantially less than the wavelength of the seismic waves, and can build our model parameterisation accordingly. When this is done in regional studies with limited data, it is often the case that $m \gg n$, which is to say that there are many more model parameters than data.

Considering first the idealised case in which there are no data errors, suppose we have found a solution, $\mathbf{m}$, of the equation

$$\mathbf{d} = \mathbf{A}\mathbf{m}. \tag{12}$$

If there exists a model vector $\mathbf{m}_{\text{null}}$ such that

$$\mathbf{A}\mathbf{m}_{\text{null}} = \mathbf{0}, \tag{13}$$

then it is clear that for any scalar a we have

$$\mathbf{A}(\mathbf{m} + a\mathbf{m}_{\text{null}}) = \mathbf{A}\mathbf{m} + a\mathbf{A}\mathbf{m}_{\text{null}} = \mathbf{A}\mathbf{m} = \mathbf{d}, \tag{14}$$

and so $\mathbf{m} + a\mathbf{m}_{\text{null}}$ is still an exact solution to the problem. The set of all vectors $\mathbf{m}_{\text{null}}$ satisfying eq. (13) forms a linear subspace known as the **null space** of $\mathbf{A}$. If the null space is non-trivial (i.e. contains non-zero vectors) then it is clear that the inverse problem cannot have a unique solution.

When there are more model parameters than data then $\mathbf{A}$ will always have a non-trivial null space, but even if $n \geq m$ there may still be a non-trivial null space due to some of the observations failing to be linearly independent. Almost all inverse problems associated with delay time tomography will have a high-dimensional null space, this largely being due to insufficient ray coverage throughout the Earth. This is illustrated in Fig. 3 for the toy problem of Fig. 2. A model that is non-zero only within cells crossed by no rays obviously lies in the null space, but so do more complicated models within the well-covered region, and adding any such model to the input model produces exactly the same data.

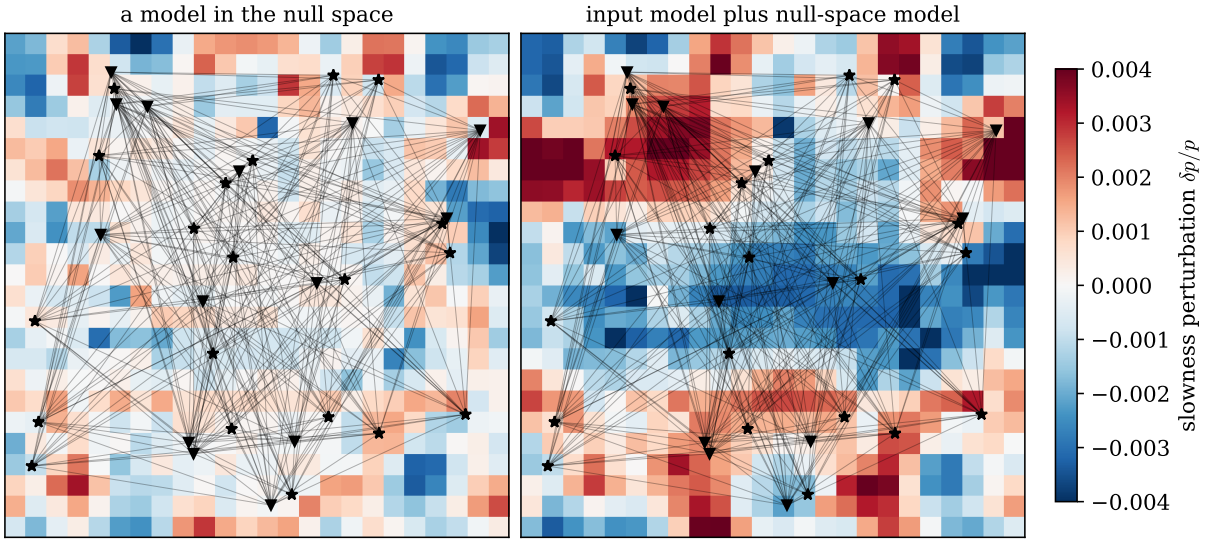


Fig. 3: A model lying in the null space of the toy tomographic problem shown in Fig. 2 (left), and the input model of that figure with this null-space model added (right). The input model on the left of Fig. 2 and the model on the right here produce identical delay times along every ray, and so cannot be distinguished by the data.

The null space can be determined explicitly through an eigenvalue problem. The $m \times m$ matrix $\mathbf{A}^T\mathbf{A}$ is symmetric, and hence possesses real eigenvalues and an orthonormal set of eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_m$. Moreover, if $\mathbf{A}^T\mathbf{A}\mathbf{v} = \mu\mathbf{v}$ with $\mathbf{v}$ normalised, then $\mu = \mathbf{v}^T\mathbf{A}^T\mathbf{A}\mathbf{v} = \|\mathbf{A}\mathbf{v}\|^2 \geq 0$, and so the eigenvalues can be written as s_j^2 with $s_j \geq 0$,

$$\mathbf{A}^T\mathbf{A}\mathbf{v}_j = s_j^2\mathbf{v}_j. \tag{15}$$

The same identity shows that $\mathbf{A}\mathbf{v}_j = \mathbf{0}$ if and only if $s_j = 0$, and hence the null space of $\mathbf{A}$ is precisely the span of those eigenvectors having zero eigenvalue. If we expand a model in the eigenvectors as $\mathbf{m} = \sum_j a_j\mathbf{v}_j$, the data $\mathbf{d} = \mathbf{A}\mathbf{m}$ depend only on those

coefficients with $s_j > 0$, while the remaining coefficients are entirely undetermined. For $s_j > 0$ we can define the unit vectors $\mathbf{u}_j = \mathbf{A}\mathbf{v}_j/s_j$ within the data space, which are orthonormal because $\mathbf{u}_i \cdot \mathbf{u}_j = \mathbf{v}_i^T \mathbf{A}^T \mathbf{A}\mathbf{v}_j/(s_i s_j) = \delta_{ij}$, and then write

$$\mathbf{A} = \sum_{s_j>0} s_j \mathbf{u}_j \mathbf{v}_j^T, \tag{16}$$

which is known as the **singular value decomposition** of $\mathbf{A}$, with the s_j its **singular values**. This decomposition tells us more than just the null space. A coefficient a_j of the model produces a contribution $s_j a_j \mathbf{u}_j$ to the data, and so if s_j is small this contribution is easily lost within the measurement errors. Directions in the model space associated with small singular values are, therefore, effectively unresolved even though they do not lie within the null space, and any attempt to recover them will amplify the noise in the data. Figure 4 shows the singular values of the toy problem along with two of the singular vectors: the first is concentrated where many rays cross, and so is well constrained, while the other, whose singular value is smaller by a factor of more than a thousand, lies almost entirely within a few poorly sampled cells.

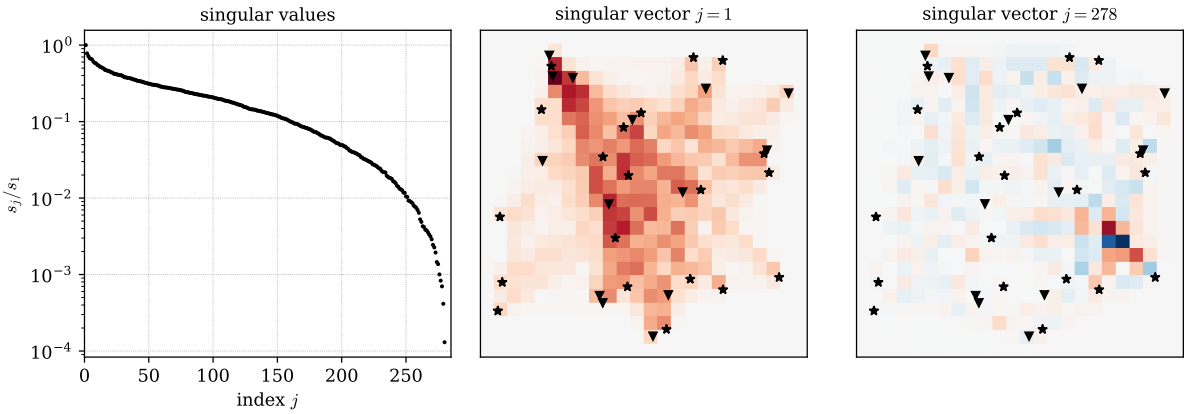


Fig. 4: The non-zero singular values of the toy problem, in decreasing order and relative to the largest (left), together with the singular vectors having the largest singular value (centre) and the 278th (right), colours on an arbitrary scale. There are 280 data and 576 blocks, so at most 280 singular values are non-zero and the null space has dimension at least 296.

Regularised least squares solutions

To solve the inverse problem, we might be tempted to apply a simple least squares method, and hence consider the following weighted **least squares misfit**[1]

$$J(\mathbf{m}) = \frac{1}{2} \sum_{i=1}^{n} \frac{[(\mathbf{A}\mathbf{m})_i - d_i]^2}{\sigma_i^2}, \tag{17}$$

where the factor of one half has been included for later convenience. If we introduce the diagonal **covariance matrix** $\mathbf{C} = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ then the misfit above can be writ-

[1] Some of you might note that this is proportional to the χ^2-statistic.

ten more concisely as

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{Am} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{Am} - \mathbf{d}). \tag{18}$$

This second expression is also valid for a general covariance matrix that accounts for correlations between the different measurement errors. The problem with this approach is that there is not a unique model that minimises this chosen misfit. Indeed, suppose there exists a non-zero vector $\mathbf{m}_{\text{null}}$ lying in the null space of $\mathbf{A}$. It is then clear that $J(\mathbf{m} + a\mathbf{m}_{\text{null}}) = J(\mathbf{m})$ for all scalars a.

A non-ideal but pragmatic way around this problem is to seek a model that not only fits the data well, but possesses a property that we see as desirable. For example, we might think that the true velocity structure must be close to that of the spherical reference model, and hence prefer models that are small in some sense. A common way to measure the "size" of the velocity perturbation is through the square of the L^2 norm

$$\int_M \delta\alpha^2 \, \mathrm{d}^3\mathbf{x}. \tag{19}$$

To proceed we substitute our finite-dimensional expansion of the velocity perturbation into this expression to obtain

$$\int_M \delta\alpha^2 \, \mathrm{d}^3\mathbf{x} = \sum_{j=1}^{m} \sum_{k=1}^{m} \delta\alpha_j \delta\alpha_k \int_M \varphi_j \varphi_k \, \mathrm{d}^3\mathbf{x}, \tag{20}$$

which in matrix-vector notation can be written

$$\int_M \delta\alpha^2 \, \mathrm{d}^3\mathbf{x} = \mathbf{m}^T \mathbf{B} \mathbf{m}, \tag{21}$$

where the matrix $\mathbf{B}$ has components

$$B_{jk} = \int_M \varphi_j \varphi_k \, \mathrm{d}^3\mathbf{x}. \tag{22}$$

Given this result we now consider the modified least squares misfit

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{Am} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{Am} - \mathbf{d}) + \frac{\lambda}{2}\mathbf{m}^T \mathbf{B} \mathbf{m}, \tag{23}$$

where λ is a positive constant. The first term on the right-hand side measures how well the data are fit, while the second is proportional to the quantity we believe will be small for the true model. The **damping parameter**, λ, provides a means of trading off between the two terms. By making λ smaller we favour models that fit the data more closely, while larger values favour models compatible with our preconceptions. The effect of the damping parameter on the toy problem is shown in Fig. 5, along with the corresponding trade-off between data misfit and model norm. How, then, should λ be chosen? A natural criterion is to require that the data are fit to within their errors, but no better, so that $\chi^2 \approx n$, its expected value for the true model. This is known as the **discrepancy principle**, and it was used to fix the damping in Fig. 2; it will be met again in the second problem set. Fitting the data more closely than this serves only to fit the noise.

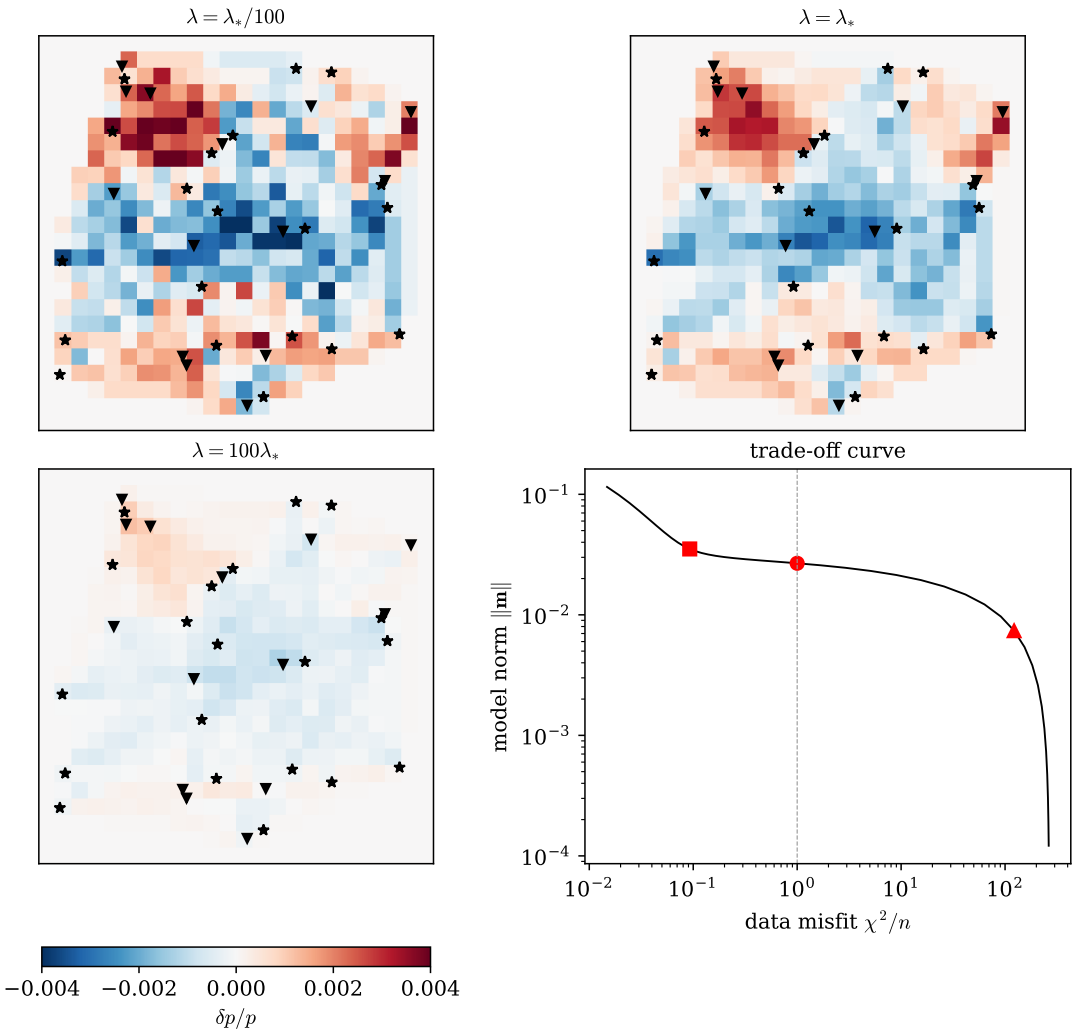


Fig. 5: Regularised least squares solutions of the toy problem for three values of the damping parameter, where λ_* is the value given by the discrepancy principle, along with the trade-off curve of model norm against data misfit as λ is varied (the three models are marked: square $\lambda_*/100$, circle λ_*, triangle $100\lambda_*$). Too small a value of λ fits the noise in the data, while too large a value suppresses the model.

Let us show that minimisation of eq. (23) leads to a unique model for each $\lambda > 0$. To do this we will differentiate J with respect to $\mathbf{m}$, and show that the equation

$$\nabla J(\mathbf{m}) = \mathbf{0} \tag{24}$$

has a unique solution corresponding to a minimum of J. Using index notation we have

$$J(\mathbf{m}) = \frac{1}{2}(A_{ij}m_j - d_i)C^{-1}_{ik}(A_{kl}m_l - d_k) + \frac{\lambda}{2}m_i B_{ij} m_j, \tag{25}$$

and so

$$\begin{aligned}\frac{\partial J}{\partial m_p} =& \frac{1}{2}(A_{ip})C_{ik}^{-1}(A_{kl}m_l - d_k) \\ &+ \frac{1}{2}(A_{ij}m_j - d_i)C_{ik}^{-1}(A_{kp}) \\ &+ \frac{\lambda}{2}\delta_{ip}B_{ij}m_j + \frac{\lambda}{2}m_i B_{ij}\delta_{jp} \\ =& A_{pi}^T C_{ik}^{-1}(A_{kl}m_l - d_k) + \lambda B_{pj}m_j. \end{aligned} \tag{26}$$

Here the symmetry of $\mathbf{C}$ and $\mathbf{B}$ has been used in the final step. Reverting to matrix-vector notation this result becomes

$$\nabla J(\mathbf{m}) = (\mathbf{A}^T\mathbf{C}^{-1}\mathbf{A} + \lambda\mathbf{B})\mathbf{m} - \mathbf{A}^T\mathbf{C}^{-1}\mathbf{d}. \tag{27}$$

Setting this derivative equal to zero we arrive at the linear equations

$$(\mathbf{A}^T\mathbf{C}^{-1}\mathbf{A} + \lambda\mathbf{B})\mathbf{m} = \mathbf{A}^T\mathbf{C}^{-1}\mathbf{d}, \tag{28}$$

which will have a unique solution so long as the $m \times m$ symmetric matrix

$$\mathbf{A}^T\mathbf{C}^{-1}\mathbf{A} + \lambda\mathbf{B} \tag{29}$$

is invertible. That this is the case can be seen by examining the sign of its eigenvalues. To do this we consider the eigenvalue problem

$$(\mathbf{A}^T\mathbf{C}^{-1}\mathbf{A} + \lambda\mathbf{B})\mathbf{m} = \mu\mathbf{m}, \tag{30}$$

with $\mathbf{m}$ the eigenvector and μ the corresponding eigenvalue. Taking the inner product of this equation with $\mathbf{m}$ we obtain

$$(\mathbf{A}\mathbf{m})^T\mathbf{C}^{-1}(\mathbf{A}\mathbf{m}) + \lambda\mathbf{m}^T\mathbf{B}\mathbf{m} = \mu, \tag{31}$$

where we have assumed that the eigenvector is normalised. The first term on the left-hand side is non-negative because $\mathbf{C}$ is a covariance matrix, while the second is positive by eq. (21) since the basis functions are linearly independent. Thus all eigenvalues are positive, and the matrix is invertible. In this manner we have shown that the unique solution of $\nabla J(\mathbf{m}) = \mathbf{0}$ is given by the simple formula

$$\mathbf{m} = (\mathbf{A}^T\mathbf{C}^{-1}\mathbf{A} + \lambda\mathbf{B})^{-1}\mathbf{A}^T\mathbf{C}^{-1}\mathbf{d}. \tag{32}$$

Finally, we note that this model must be a minimum of J. This is because it is a quadratic function that tends to infinity as $\|\mathbf{m}\|$ grows in magnitude, and hence its only stationary point must be a minimum.

What we have obtained in eq. (32) is known as a **regularised least squares solution** of the inverse problem. Such a solution is obtained by choosing a regularisation matrix $\mathbf{B}$ corresponding to a quantity we wish to make small, along with the damping parameter λ which determines how data misfit and model size are balanced. The action of the damping can be seen clearly using the singular value decomposition of eq. (16). Taking $\mathbf{B} = \mathbf{I}$ and $\mathbf{C} = \sigma^2\mathbf{I}$ for simplicity, eq. (32) becomes

$$\mathbf{m} = \sum_{s_j > 0} \frac{s_j}{s_j^2 + \lambda\sigma^2}(\mathbf{u}_j \cdot \mathbf{d})\,\mathbf{v}_j, \tag{33}$$

so that the components of the model along well-resolved directions, for which $s_j^2 \gg \lambda\sigma^2$, are recovered as $(\mathbf{u}_j \cdot \mathbf{d})/s_j$, exactly as they would be by inverting eq. (16) directly, while those along directions with $s_j^2 \ll \lambda\sigma^2$ are suppressed, and the null space is never entered. Damping thus acts by discarding those parts of the model to which the data are least sensitive.

There is a range of different schemes for regularisation including, for example, those that seek as smooth a model as possible. More generally, one may minimise

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{A}\mathbf{m} - \mathbf{d})^T\mathbf{C}^{-1}(\mathbf{A}\mathbf{m} - \mathbf{d}) + \frac{\lambda}{2}(\mathbf{m} - \mathbf{m}_0)^T\mathbf{B}(\mathbf{m} - \mathbf{m}_0), \tag{34}$$

where we can see the regularisation term as asking that the model be close to some preferred value $\mathbf{m}_0$, with "close" being measured by the matrix $\mathbf{B}$. Following the same method as above, the optimal solution in this case can be shown to be

$$\mathbf{m} = (\mathbf{A}^T\mathbf{C}^{-1}\mathbf{A} + \lambda\mathbf{B})^{-1}(\mathbf{A}^T\mathbf{C}^{-1}\mathbf{d} + \lambda\mathbf{B}\mathbf{m}_0). \tag{35}$$

Though such approaches are all rather ad hoc, they are widely used because of their simplicity and applicability to problems in high-dimensional spaces.

Whatever the regularisation used, it is important to assess what a given data set is actually able to resolve. A simple and instructive test is to invert synthetic data calculated from an input model that is zero except within a single small region, a so-called **spike test**, using the same ray paths and regularisation as for the real data. If the data constrained the model perfectly the spike would be recovered exactly; in practice its amplitude is reduced and it is smeared out along the ray paths that cross it, to an extent that depends strongly on the ray coverage, as Fig. 6 shows for the toy problem. Because the inversion is linear, the recovered model for a spike is a column of the matrix that maps input models to recovered models, and this **resolution matrix** will be considered further in the second problem set.

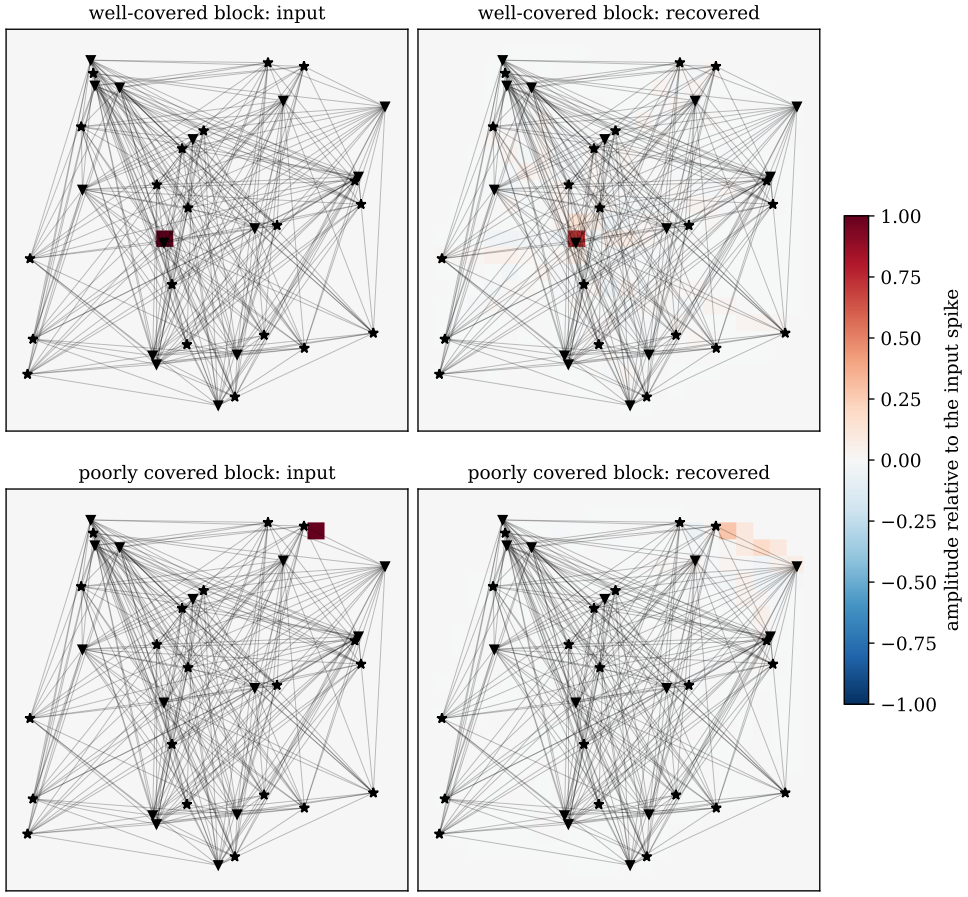


Fig. 6: Spike tests for the toy problem. Noise-free synthetic data are calculated for an input model that is non-zero only within a single block (left), and are inverted with the regularisation used in Fig. 2 (right). For a spike in a well-covered block (upper row, crossed by 38 rays) about 70 per cent of the amplitude is recovered in the right place, with some smearing along the rays that cross it. For a spike in a poorly covered block (lower row, crossed by three rays) only about a quarter of the amplitude is recovered, and the anomaly is spread along the few rays that sample it: such a block is barely constrained by the data, and the damping suppresses it.

A Bayesian perspective

A seemingly very different way of thinking about inverse problems is through **Bayes' theorem**. This states that the **probability density function** (PDF) of the model, $\mathbf{m}$, given that the data, $\mathbf{d}$, have been observed can be written

$$p(\mathbf{m}|\mathbf{d}) = \frac{p(\mathbf{d}|\mathbf{m})p(\mathbf{m})}{\int p(\mathbf{d}|\mathbf{m})p(\mathbf{m})\,\mathrm{d}\mathbf{m}}, \tag{36}$$

where $p(\mathbf{d}|\mathbf{m})$, which is known as the **likelihood**, is the PDF for observing $\mathbf{d}$ from the model $\mathbf{m}$, and $p(\mathbf{m})$ is the PDF of $\mathbf{m}$ prior to observing any data. In words, this theorem tells us how to update uncertain knowledge of the model in light of our observations. This result is fully general, and can be applied to non-linear inverse problems with complicated priors. To do so in practice, however, requires random sampling tech-

niques such as **Markov chain Monte Carlo** (MCMC) methods, and this is generally possible only for relatively low-dimensional problems for which forward calculations have a low computational cost.

As a special case, suppose we consider a linear inverse problem with Gaussian data errors, and assume a Gaussian prior on the model with mean $\mathbf{m}_0$ and covariance $\mathbf{C}_m$. The prior PDF then takes the form

$$p(\mathbf{m}) = \frac{1}{\sqrt{(2\pi)^m \det \mathbf{C}_m}} \exp\left[-\frac{1}{2}(\mathbf{m}-\mathbf{m}_0)^T \mathbf{C}_m^{-1}(\mathbf{m}-\mathbf{m}_0)\right], \tag{37}$$

where we recall that m is the number of basis functions for the model. Similarly with Gaussian errors, the likelihood can be written

$$p(\mathbf{d}|\mathbf{m}) = \frac{1}{\sqrt{(2\pi)^n \det \mathbf{C}}} \exp\left[-\frac{1}{2}(\mathbf{A}\mathbf{m}-\mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m}-\mathbf{d})\right]. \tag{38}$$

From Bayes' theorem we then find

$$p(\mathbf{m}|\mathbf{d}) \propto \exp\left[-\frac{1}{2}(\mathbf{m}-\mathbf{m}_0)^T \mathbf{C}_m^{-1}(\mathbf{m}-\mathbf{m}_0) - \frac{1}{2}(\mathbf{A}\mathbf{m}-\mathbf{d})^T \mathbf{C}^{-1}(\mathbf{A}\mathbf{m}-\mathbf{d})\right], \tag{39}$$

and a little effort shows that the exponentiated term can be rearranged, by completing the square, into the form

$$p(\mathbf{m}|\mathbf{d}) \propto \exp\left[-\frac{1}{2}(\mathbf{m}-\mathbf{m}_p)^T \mathbf{C}_p^{-1}(\mathbf{m}-\mathbf{m}_p)\right], \tag{40}$$

where we have defined the **posterior mean**

$$\mathbf{m}_p = (\mathbf{A}^T \mathbf{C}^{-1}\mathbf{A} + \mathbf{C}_m^{-1})^{-1}(\mathbf{A}^T \mathbf{C}^{-1}\mathbf{d} + \mathbf{C}_m^{-1}\mathbf{m}_0), \tag{41}$$

and the **posterior model covariance**

$$\mathbf{C}_p = (\mathbf{A}^T \mathbf{C}^{-1}\mathbf{A} + \mathbf{C}_m^{-1})^{-1}. \tag{42}$$

We conclude that in this special case the posterior distribution is Gaussian, and can be determined in closed form. Moreover, the posterior mean is precisely that obtained via the regularised least squares method subject to the choice $\lambda\mathbf{B} = \mathbf{C}_m^{-1}$. The posterior covariance provides, in addition, a measure of the uncertainty in the recovered model. Figure 7 shows the posterior mean and standard deviation for the toy problem using a Gaussian prior whose covariance between two cells decays with their separation over a correlation length of 0.15 times the box side, this encoding the expectation that the slowness perturbation is smooth. The posterior mean recovers the input model closely, including within regions crossed by few rays, where the prior correlation carries information from neighbouring cells, and the posterior standard deviation shows where this recovery can be trusted.

In this manner we see that solving a regularised least squares problem returns the mean model corresponding to a certain prior distribution. It is sometimes argued that this result justifies the use of regularised least squares, with the prior distribution determining the appropriate choice of regularisation. It is better to read the result the other way round. Any choice of regularisation amounts to the choice of a prior, but one that

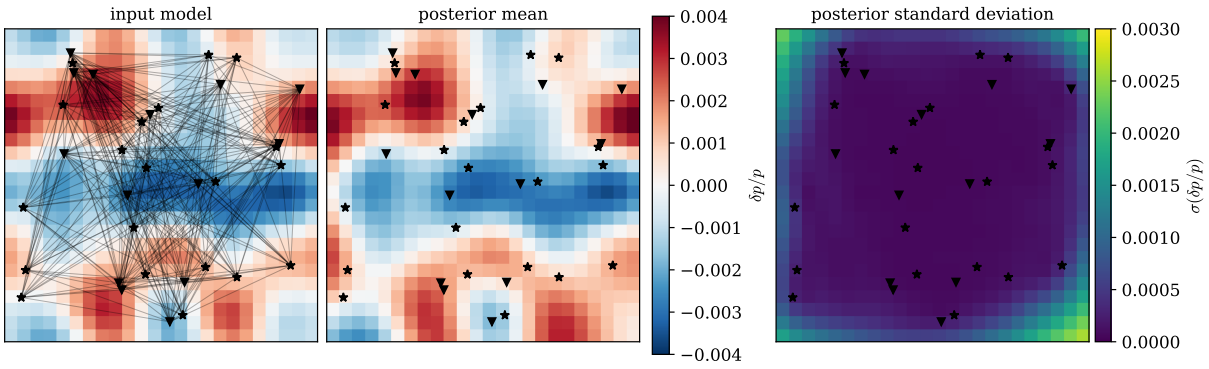


Fig. 7: Bayesian solution of the toy problem using a Gaussian prior with a correlation length of 0.15 times the box side and a standard deviation of 0.003: the input model (left), the posterior mean (centre), and the posterior standard deviation (right), which approaches that of the prior only in the corners far from any ray.

is made implicitly, and then adjusted through a damping parameter in a rather ad hoc manner. Within the Bayesian approach the assumptions made about the model are stated explicitly through the prior, their consequences follow from the rules of probability, and the result is a distribution rather than a single model, so that the uncertainty in what has been recovered is quantified. That the prior remains a matter of judgement is not a weakness of the method: some such choice is unavoidable in any underdetermined problem, and the Bayesian formulation simply makes it honest and visible.

Inversion in function spaces

Throughout this lecture the model has been reduced at the outset to a finite set of coefficients through the expansion of eq. (8), and the choice of basis made there is arbitrary. An alternative, developed over roughly the past fifteen years, is to pose and solve the inverse problem for the function $\delta\alpha$ itself. The model is taken to lie within a suitable space of functions, one in which smoothness can be measured, the prior becomes a Gaussian probability measure on this space, and the data remain finite in number. Bayes' theorem can be formulated in this setting, and for a linear forward problem with Gaussian errors the posterior is again a Gaussian measure, with mean and covariance given by expressions that mirror eqs. (41) and (42). Only once the problem has been solved in this form is it discretised for the purposes of computation, and the essential point is that as the discretisation is refined the discrete answers converge to the function-space solution. The result is a method whose answers do not depend on an arbitrary parameterisation, and in which the prior states directly what is being assumed about the function. You are not expected to understand these ideas in any detail; what you should take away is that the problem need not be discretised arbitrarily at the outset, and that when a discretisation is introduced it approximates a well-defined answer. Figure 8 shows a tomographic problem of the kind considered in this lecture solved in this way, using the Python package `pygeoinf`: the true model is a random draw from the prior, and the posterior expectation and standard deviation are shown.[2]

[2] The theory is set out in the review by Stuart (2010).

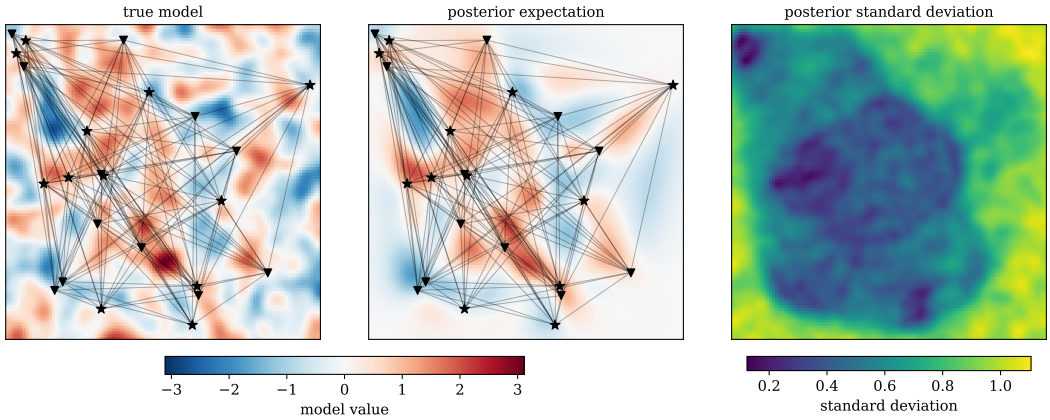


Fig. 8: A straight-ray delay time problem of the kind considered in this lecture, posed and solved in a function space: the true model, which is a random draw from the prior (left), the posterior expectation (centre), and the posterior standard deviation (right). The prior has unit standard deviation. Stars mark sources and triangles receivers. The calculations have, of course, still been discretised, as they must be on a computer, but the discretisation is of a well-defined problem and has been taken far enough for the results to have converged in practice. The results are therefore independent of the particular discretisation used, in contrast to those of the pixel basis in the earlier figures. Calculated with `pygeoinf`.

What you need to know and be able to do

(i) Why the construction of tomographic models is of interest: it gives the main quantitative constraint on mantle convection and hence on the dynamics of the Earth's deep interior. Here you should be familiar with the idea that seismic slow regions are likely to be hotter than average, while fast regions will be cooler.

(ii) How Fermat's principle is used to obtain a simple linearised relation between delay times and velocity perturbations. From this, you should be able to outline how a tomographic problem is reduced to the standard form $\mathbf{d} = \mathbf{A}\mathbf{m} + \mathbf{e}$.

(iii) What the null space of a linear(ised) inverse problem is and why it matters.

(iv) How regularised least squares is used to obtain solutions of linearised tomographic problems. Here you should be able to derive the least squares equations from first principles.

(v) How Bayes' theorem can be applied in the solution of inverse problems, and that the results obtained coincide with regularised least squares for linear(ised) problems with Gaussian errors and priors.

Lecture 20: Waveform tomography

Outline and motivation

In the previous lecture we discussed delay time tomography, this being the first and simplest tomographic method developed. While this approach is still widely used, it is increasingly being superseded by **waveform tomography**. Delay time tomography is based on the measurement of a small number of delay times from each seismogram, with these data modelled using linearised ray theory. By contrast, waveform tomography uses more of the data contained in the seismograms, fully accounts for the complexities of elastic wave propagation in the Earth, and does not linearise the inverse problem. This lecture will outline the ideas behind waveform tomography, focusing on the solution of the inverse problem using **gradient-based optimisation**.

Numerical wavefield simulations

Waveform tomography depends on the ability to solve the full elastic wave equation numerically within realistic earth models. This has been possible routinely on a global scale for around 20 years, with this process facilitated by both algorithmic and hardware advances. These lectures are not about numerical methods. But it will be seen that within this subject computational costs must be carefully considered. As a number to bear in mind, the simulation of elastic wave propagation on a global scale within a realistic 3D earth model takes of order ten minutes running on a modest GPU cluster. Within a tomographic study, it would be typical to use data from around 10^3 earthquakes recorded at thousands of stations (see Fig. 1), and a separate numerical simulation is required for each earthquake. This means that the time for solving the necessary forward problems is of order hours or more.

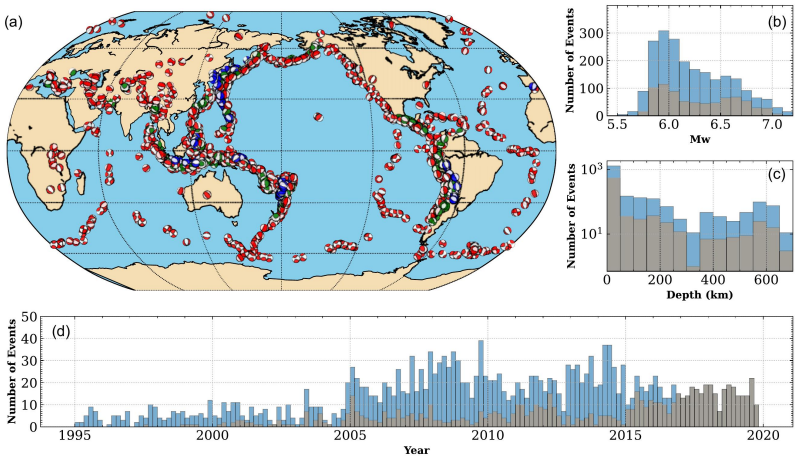


Fig. 1: The 2160 earthquakes used to construct the global waveform tomographic model GLAD-M35 (see Fig. 6): (a) locations, each event being drawn as a visual representation of its moment tensor, coloured by depth; (b) the distribution of magnitudes; (c) the distribution of hypocentre depths; (d) the number of events by year. From Cui et al. (2024).

Waveform measurements

Let $\mathbf{u}^{\text{obs}}(t)$ denote the displacement vector recorded by a seismograph at the surface location, $\mathbf{x}_r$, following an earthquake. In reality, seismographs do not record the displacement vector directly, with the signal instead being modified due to the mechanical and electronic properties of the instrument along with additional processing done by the seismologist. For simplicity, however, we neglect such effects, but they can be incorporated at the cost only of more complicated notation.

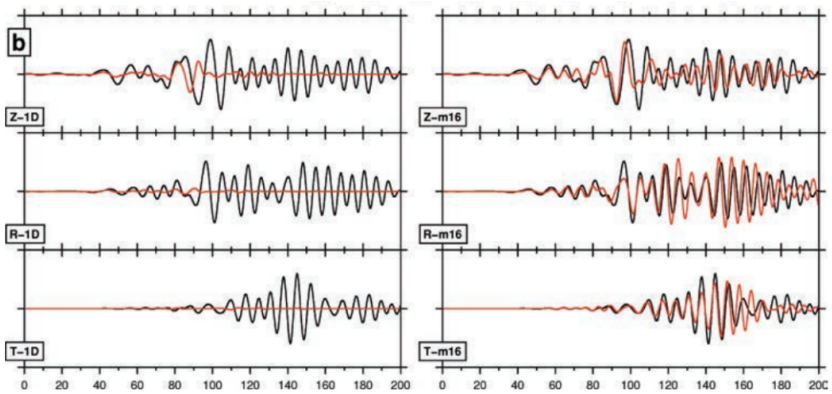


Fig. 2: An example of observed and synthetic three-component seismograms from a waveform tomographic study of Southern California. Data are shown in black, and synthetics in red; the rows show the vertical (Z), radial (R) and transverse (T) components, and time is in seconds. On the left (1D) the comparison is made prior to inversion, with the synthetics calculated in the initial, laterally homogeneous, model. The right-hand side (m16) then shows the data fit following inversion involving 16 iterative model updates. Taken from Tape et al. (2010).

Suppose that we have models for both the Earth and earthquake, and so can nu-

merically solve the elastic wave equation to obtain a synthetic seismogram $\mathbf{u}(\mathbf{x}_r, t)$ at this location. To quantify the misfit between the observed and synthetic seismograms we can define the least-squares misfit

$$J = \frac{1}{2} \int_0^T \|\mathbf{u}(\mathbf{x}_r, t) - \mathbf{u}^{\text{obs}}(t)\|^2 \, \mathrm{d}t, \tag{1}$$

with $[0, T]$ being the time interval over which measurements have been made. Recorded seismograms are, of course, discretely sampled, and so really we should write a sum and not an integral over time. But in practice sampling rates are sufficiently high that this distinction can be ignored.

The value of J depends on the synthetic seismogram, $\mathbf{u}$, and hence ultimately on the underlying parameters for the earth model and the earthquake. We could, therefore, try to minimise J in the hope that the model parameters are then closer to their true values. In practice, of course, we would not actually use J as defined above, but sum such terms over many different locations and earthquakes, and regularisation terms might also be added. Fig. 2 shows the sort of improvement in the fit between observed and synthetic seismograms that can be achieved by such an inversion. In fact, the least-squares misfit between seismograms is rarely used in practice, with other more complicated measures of misfit being preferred. Nevertheless, this simple functional is sufficient to motivate and illustrate waveform inversion.

Functional derivatives and sensitivity kernels

The waveform misfit, J, defined above depends on the synthetic wavefield, $\mathbf{u}$, and hence we can write it as $J(\mathbf{u})$. But the value of $\mathbf{u}$ is determined through solution of the equations of motion from the underlying model parameters. To simplify notation, let us write $\mathbf{m}$ for these model parameters. Here $\mathbf{m}$ could be all of ρ, A_{ijkl} and $\overline{S}_{ij}$, but in practice it is often some subset of these parameters. For example, within a tomographic problem it might be assumed that the source parameters are known.

For a given value of $\mathbf{m}$ we can compute the corresponding synthetic wavefield, $\mathbf{u}$, and from this the value of the misfit, $J(\mathbf{u})$. To indicate that the misfit can be viewed as an implicit function of $\mathbf{m}$, we will write it as $J(\mathbf{m})$. This is a slight abuse of notation, but it should always be clear from context which variables the misfit depends on.

Within waveform tomography, the goal is to find model parameters that explain the observed data, and this is done by minimising $\mathbf{m} \mapsto J(\mathbf{m})$. For a small model perturbation, $\delta\mathbf{m}$, we can write to first-order accuracy

$$J(\mathbf{m} + \delta\mathbf{m}) = J(\mathbf{m}) + \langle DJ(\mathbf{m}) | \delta\mathbf{m} \rangle + O(\|\delta\mathbf{m}\|^2), \tag{2}$$

where $DJ(\mathbf{m})$ denotes the functional derivative of J with respect to $\mathbf{m}$ and $\langle \cdot | \cdot \rangle$ is an appropriate inner product. As an example, supposing that the shear wave speed, β, is the only model parameter, we can concretely write

$$\langle DJ(\mathbf{m}) | \delta\mathbf{m} \rangle = \int_M K_\beta \, \delta\beta \, \mathrm{d}^3\mathbf{x}, \tag{3}$$

where the function K_β is known as a **sensitivity kernel** for J with respect to β; note that in this context the terms functional derivative, gradient, and sensitivity kernel are used interchangeably. We discuss later how these functional derivatives can be obtained practically, but for the moment we just assume that this can be done.

Gradient-based optimisation

If $\overline{\mathbf{m}}$ is a minimum of J, then for any sufficiently small $\delta\mathbf{m}$ we should have

$$J(\overline{\mathbf{m}} + \delta\mathbf{m}) \geq J(\overline{\mathbf{m}}), \tag{4}$$

which requires that

$$DJ(\overline{\mathbf{m}}) = \mathbf{0}. \tag{5}$$

This condition is, of course, familiar from calculus in finite-dimensional spaces.

There are a number of approaches to solve this optimisation problem. One is a systematic "grid search" of the model space. Doing this would require the model space to be discretised using a finite-dimensional basis as in Lecture 19. The issue with this method is the computational cost. Suppose, for example, that the discretised model space had m dimensions, and that along each dimension experience had shown that N points needed to be tested. The number of forward calculations required would then be N^m. In realistic problems, m is of order thousands or millions, while we have already noted the high cost of solving the forward problem. The result is that grid search methods are not viable.

Random sampling methods such as Markov chain Monte Carlo (MCMC) simulations perform better within high-dimensional spaces (the number of forward calculations scales with something like m^2), but the costs are still far too high to be used practically in waveform tomography.

As a result, tomographers must rely on local optimisation methods that seek a point such that eq. (5) holds. An obvious downside to this approach is that the model obtained may only be a **local and not a global minimum of the misfit**. But this is typically all that is currently possible, and there do exist approximate but useful methods available for uncertainty quantification.[1]

Suppose we start from an initial model $\mathbf{m}_0$. At this point we can calculate the value of the misfit, $J(\mathbf{m}_0)$, along with its gradient, $DJ(\mathbf{m}_0)$. The gradient points in the direction along which the misfit most rapidly increases, and the negative gradient gives the direction of most rapid decrease. This suggests that we can look for a new model of the form

$$\mathbf{m}_1 = \mathbf{m}_0 - \lambda\, DJ(\mathbf{m}_0), \tag{6}$$

where $\lambda > 0$ is some step length to be determined. To find a suitable value for λ, we need only consider a one-dimensional minimisation problem for the function

$$f(\lambda) = J[\mathbf{m}_0 - \lambda\, DJ(\mathbf{m}_0)], \tag{7}$$

which can be solved quite easily. Having now arrived at a new model, $\mathbf{m}_1$, with a lower misfit, we can simply repeat the whole process until the gradient vanishes and a local minimum has been found.[2] The above method is the simplest example of gradient-based optimisation, and is known as **the method of steepest descent**. More sophisticated algorithms for choosing the descent direction, which have superior convergence properties, tend to be used in practice. In all cases, however, this direction is built from linear combinations of the current and past gradients of the misfit.

[1] There is not sufficient time to discuss these here.

[2] In practice, the process might be stopped when the misfit has decreased sufficiently from a statistical perspective.

How not to calculate the gradient of J

Gradient-based optimisation depends on being able both to solve the forward problem and to calculate the gradient of the misfit. As noted, the first task can be done using numerical wavefield simulations at a high but manageable cost. Schematically, such a code provides a means for performing the mapping $\mathbf{m} \mapsto J(\mathbf{m})$. With this at hand, one way to get the required gradient is to make use of the **finite-difference approximation**

$$\langle DJ(\mathbf{m}) | \delta\mathbf{m} \rangle \approx J(\mathbf{m} + \delta\mathbf{m}) - J(\mathbf{m}). \tag{8}$$

Using this formula, we can get the derivative of the misfit along any direction at the cost of two forward calculations. Suppose we introduce an m-dimensional parameterisation for the model space. Using $m + 1$ forward calculations we could then approximate the gradient relative to this basis. The advantage of this approach is that it requires no additional theory or coding once a numerical wave propagation code has been developed. The problem is, as noted before, that the number of model parameters, m, is typically large, and so the overall cost is too high to compute a single gradient in this manner, let alone the many different gradients needed in the iterative solution of the optimisation problem.

How you should calculate the gradient of J

Our aim is to calculate the gradient of $J(\mathbf{m})$. This is equivalent to finding the gradient of $J(\mathbf{u})$ subject to the constraint that $\mathbf{u}$ is related to $\mathbf{m}$ through solution of the forward problem. To proceed, we make use of the method of **Lagrange multipliers**, which many of you will have seen previously, though not necessarily applied to fields. Within this course it is necessary to know how to apply this method, but not to explain why it works. For those interested, a non-examinable outline is provided in the appendix to this lecture. We begin by defining a **Lagrangian** for the problem, which takes the following form

$$\begin{aligned} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = J(\mathbf{u}) &- \int_0^T \int_M \left[\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t \\ &- \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t. \end{aligned} \tag{9}$$

Here we see first on the right-hand side the unconstrained misfit, $J(\mathbf{u})$. The second term is associated with the constraint that $\mathbf{u}$ should satisfy the equations of motion and includes a Lagrange multiplier field $\mathbf{u}'$. The final term does the same for the boundary conditions, and features a second Lagrange multiplier field $\mathbf{w}'$.

The **Lagrange multiplier theorem** tells us that the equality

$$DJ(\mathbf{m}) = D_{\mathbf{m}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) \tag{10}$$

holds so long as the following conditions are met

$$D_{\mathbf{u}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = 0, \tag{11}$$

$$D_{\mathbf{u}'} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = 0, \tag{12}$$

$$D_{\mathbf{w}'} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) = 0. \tag{13}$$

Here subscripts are used for partial functional derivatives in an obvious manner. To understand this result, we begin with eq. (12). The dependence of the Lagrangian on the field $\mathbf{u}'$ is very simple, and so we readily find that

$$\langle D_{\mathbf{u}'} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) | \delta \mathbf{u}' \rangle = - \int_0^T \int_M \left[\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \right] \delta u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t. \quad (14)$$

We need this expression to vanish for all possible choices of $\delta \mathbf{u}'$, and hence we conclude that $\mathbf{u}$ must satisfy the expected equations of motion

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) = 0. \quad (15)$$

Similarly, the final condition in eq. (13) gives the boundary conditions for $\mathbf{u}$.

Turning now to the first condition in eq. (11), we find that for any $\delta \mathbf{u}$

$$\begin{aligned} \langle D_{\mathbf{u}} L(\mathbf{u}, \mathbf{u}', \mathbf{w}', \mathbf{m}) | \delta \mathbf{u} \rangle = \langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle &- \int_0^T \int_M \left[\rho \frac{\partial^2 \delta u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t \\ &- \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t, \end{aligned} \quad (16)$$

must vanish. This is a more complicated expression, and understanding its consequences will take some effort. But we know that $\mathbf{u}$ is fixed uniquely by the previous constraints, and hence the above expression must somehow determine the Lagrange multiplier fields. As a first step, we can use eq. (1) to explicitly write the first term on the right-hand side as

$$\langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle = \int_0^T [u_i(\mathbf{x}_r, t) - u_i^{\mathrm{obs}}(t)] \delta u_i(\mathbf{x}_r, t) \, \mathrm{d}t. \quad (17)$$

Making use of a Dirac delta function on the surface ∂M, this can be conveniently written

$$\langle DJ(\mathbf{u}) | \delta \mathbf{u} \rangle = \int_0^T \int_{\partial M} h_i' \delta u_i \, \mathrm{d}S \, \mathrm{d}t, \quad (18)$$

where we have set

$$h_i'(\mathbf{x}, t) = [u_i(\mathbf{x}_r, t) - u_i^{\mathrm{obs}}(t)] \delta(\mathbf{x} - \mathbf{x}_r). \quad (19)$$

With this done, we can write the required condition as

$$\begin{aligned} \int_0^T \int_{\partial M} h_i' \delta u_i \, \mathrm{d}S \, \mathrm{d}t - \int_0^T \int_M &\left[\rho \frac{\partial^2 \delta u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \right] u_i' \, \mathrm{d}^3\mathbf{x} \, \mathrm{d}t \\ &- \int_0^T \int_{\partial M} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) \hat{n}_j w_i' \, \mathrm{d}S \, \mathrm{d}t = 0, \end{aligned} \quad (20)$$

for all $\delta \mathbf{u}$.

To proceed, we need to successively integrate by parts to shift derivatives from the perturbed displacement field to the Lagrange multipliers. Starting with the time derivatives, we have

$$\int_0^T \frac{\partial^2 \delta u_i}{\partial t^2} u_i' \, \mathrm{d}t = - \int_0^T \frac{\partial \delta u_i}{\partial t} \frac{\partial u_i'}{\partial t} \, \mathrm{d}t + \left[\frac{\partial \delta u_i}{\partial t} u_i' \right]_{t=T}, \quad (21)$$

where we have used the vanishing initial conditions on u_i (and hence δu_i) to eliminate one of the boundary terms. Integrating by parts once more gives

$$\int_0^T \frac{\partial^2 \delta u_i}{\partial t^2} u_i' \,\mathrm{d}t = \int_0^T \delta u_i \frac{\partial^2 u_i'}{\partial t^2} \,\mathrm{d}t + \left[\frac{\partial \delta u_i}{\partial t} u_i'\right]_{t=T} - \left[\delta u_i \frac{\partial u_i'}{\partial t}\right]_{t=T}. \tag{22}$$

Looking now at the spatial derivatives, we first obtain

$$\begin{aligned}\int_M \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial \delta u_k}{\partial x_l}\right) u_i' \,\mathrm{d}^3\mathbf{x} - \int_{\partial M}\left(A_{ijkl}\frac{\partial \delta u_k}{\partial x_l}\right)\hat{n}_j w_i' \,\mathrm{d}S \\ = -\int_M A_{ijkl}\frac{\partial \delta u_k}{\partial x_l}\frac{\partial u_i'}{\partial x_j}\,\mathrm{d}^3\mathbf{x} - \int_{\partial M}\left(A_{ijkl}\frac{\partial \delta u_k}{\partial x_l}\right)\hat{n}_j(w_i' - u_i')\,\mathrm{d}S.\end{aligned} \tag{23}$$

Making use of the hyperelastic symmetry, $A_{ijkl} = A_{klij}$, we can then integrate by parts once more to arrive at

$$\begin{aligned}\int_M \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial \delta u_k}{\partial x_l}\right) u_i' \,\mathrm{d}^3\mathbf{x} - \int_{\partial M}\left(A_{ijkl}\frac{\partial \delta u_k}{\partial x_l}\right)\hat{n}_j w_i' \,\mathrm{d}S \\ = \int_M \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k'}{\partial x_l}\right)\delta u_i \,\mathrm{d}^3\mathbf{x} - \int_{\partial M}\left(A_{ijkl}\frac{\partial \delta u_k}{\partial x_l}\right)\hat{n}_j(w_i' - u_i')\,\mathrm{d}S \\ - \int_{\partial M}\left(A_{ijkl}\frac{\partial u_k'}{\partial x_l}\right)\hat{n}_j \delta u_i \,\mathrm{d}S.\end{aligned} \tag{24}$$

Putting these results together, we have shown that eq. (20) is equivalent to

$$\begin{aligned}&-\int_0^T\int_M \left[\rho\frac{\partial^2 u_i'}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k'}{\partial x_l}\right)\right]\delta u_i \,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t \\ &-\int_0^T\int_{\partial M}\left(A_{ijkl}\frac{\partial \delta u_k}{\partial x_l}\right)\hat{n}_j(w_i' - u_i')\,\mathrm{d}S\,\mathrm{d}t \\ &+\int_0^T\int_{\partial M}\left[h_i' - \left(A_{ijkl}\frac{\partial u_k'}{\partial x_l}\right)\hat{n}_j\right]\delta u_i \,\mathrm{d}S\,\mathrm{d}t \\ &-\int_M \rho\left\{\left[\frac{\partial \delta u_i}{\partial t}u_i'\right]_{t=T} - \left[\delta u_i\frac{\partial u_i'}{\partial t}\right]_{t=T}\right\}\mathrm{d}^3\mathbf{x} = 0,\end{aligned} \tag{25}$$

for all $\delta\mathbf{u}$. From the first term in this equation, we conclude that the Lagrange multiplier field $\mathbf{u}'$ must satisfy the elastic wave equation

$$\rho\frac{\partial^2 u_i'}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k'}{\partial x_l}\right) = 0. \tag{26}$$

In dealing with the boundary terms, we need the fact that, for an arbitrary function, its boundary value and that of its normal derivative can be chosen separately. As a result, the second term implies that $\mathbf{w}'$ is the restriction of $\mathbf{u}'$ to ∂M, the third term implies

$$\left(A_{ijkl}\frac{\partial u_k'}{\partial x_l}\right)\hat{n}_j = h_i', \tag{27}$$

and finally we have the terminal conditions

$$u_i'(\mathbf{x}, T) = \frac{\partial u_i'}{\partial t}(\mathbf{x}, T) = 0. \tag{28}$$

It has taken some effort, but we have now shown that eq. (11) requires the Lagrange multiplier field, $\mathbf{u}'$, to satisfy an elastic wave equation, with this problem driven by a traction applied at the observation point that is equal to the difference between the predicted and observed seismograms. We have also shown that the second Lagrange multiplier field, $\mathbf{w}'$, is just the restriction of $\mathbf{u}'$ to the boundary, and hence no additional work is necessary for its calculation.

We can now use eq. (10) to obtain sensitivity kernels with respect to the different model parameters. For arbitrary variations in these parameters we have

$$\begin{aligned}\langle DJ(\mathbf{m})|\delta\mathbf{m}\rangle = &-\int_0^T\int_M\left[\delta\rho\frac{\partial^2 u_i}{\partial t^2}-\frac{\partial}{\partial x_j}\left(\delta A_{ijkl}\frac{\partial u_k}{\partial x_l}-\delta\overline{S}_{ij}\right)\right]u_i'\,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t\\ &-\int_0^T\int_{\partial M}\left(\delta A_{ijkl}\frac{\partial u_k}{\partial x_l}-\delta\overline{S}_{ij}\right)\hat{n}_j u_i'\,\mathrm{d}S\,\mathrm{d}t.\end{aligned}\tag{29}$$

Integrating by parts once, this expression can be simplified to

$$\langle DJ(\mathbf{m})|\delta\mathbf{m}\rangle = \int_0^T\int_M\left[\delta\rho\frac{\partial u_i}{\partial t}\frac{\partial u_i'}{\partial t}-\delta A_{ijkl}\frac{\partial u_i'}{\partial x_j}\frac{\partial u_k}{\partial x_l}+\delta\overline{S}_{ij}\frac{\partial u_i'}{\partial x_j}\right]\mathrm{d}^3\mathbf{x}\,\mathrm{d}t,\tag{30}$$

from which we can read off the various sensitivity kernels

$$K_\rho(\mathbf{x}) = \int_0^T\frac{\partial u_i}{\partial t}(\mathbf{x},t)\frac{\partial u_i'}{\partial t}(\mathbf{x},t)\,\mathrm{d}t,\tag{31}$$

$$K_{A_{ijkl}}(\mathbf{x}) = -\int_0^T\frac{\partial u_i'}{\partial x_j}(\mathbf{x},t)\frac{\partial u_k}{\partial x_l}(\mathbf{x},t)\,\mathrm{d}t,\tag{32}$$

$$K_{\overline{S}_{ij}}(\mathbf{x},t) = \frac{\partial u_i'}{\partial x_j}(\mathbf{x},t).\tag{33}$$

Note that the kernel for the stress glut depends on time as well as position, the stress glut itself being a function of both.

The problem for $\mathbf{u}'$ differs from the forward problem in being subject to terminal rather than initial conditions, so that it must be solved backwards in time from $t = T$. This is easily dealt with. Let us define the time-reversed field

$$u_i^\dagger(\mathbf{x},t) = u_i'(\mathbf{x},T-t).\tag{34}$$

Because eq. (26) involves only second time derivatives, $\mathbf{u}^\dagger$ satisfies exactly the same wave equation, along with the boundary condition

$$\left(A_{ijkl}\frac{\partial u_k^\dagger}{\partial x_l}\right)\hat{n}_j = h_i^\dagger,\quad h_i^\dagger(\mathbf{x},t) = h_i'(\mathbf{x},T-t),\tag{35}$$

and the terminal conditions in eq. (28) become the initial conditions

$$u_i^\dagger(\mathbf{x},0) = \frac{\partial u_i^\dagger}{\partial t}(\mathbf{x},0) = 0.$$

The problem for $\mathbf{u}^\dagger$ is therefore an initial value problem of precisely the same form as the forward problem, driven at the receiver by the time-reversed residual $h_i^\dagger$, and it can be solved with the same numerical code. In terms of $\mathbf{u}^\dagger$ the sensitivity kernels read

$$K_\rho(\mathbf{x}) = -\int_0^T \frac{\partial u_i}{\partial t}(\mathbf{x},t)\frac{\partial u_i^\dagger}{\partial t}(\mathbf{x},T-t)\,\mathrm{d}t, \tag{36}$$

$$K_{A_{ijkl}}(\mathbf{x}) = -\int_0^T \frac{\partial u_i^\dagger}{\partial x_j}(\mathbf{x},T-t)\frac{\partial u_k}{\partial x_l}(\mathbf{x},t)\,\mathrm{d}t, \tag{37}$$

$$K_{\overline{S}_{ij}}(\mathbf{x},t) = \frac{\partial u_i^\dagger}{\partial x_j}(\mathbf{x},T-t), \tag{38}$$

where the sign change in K_ρ comes from the time derivative of the reversed field. These expressions make the structure of the kernels clear: the forward field at time t is paired with the adjoint field after it has evolved for a time $T-t$, which is just the time remaining for anything happening at $(\mathbf{x},t)$ to be felt within the observation window. The time reversal is nothing more than a convenient way of converting a terminal value problem into an initial value one, and plays no fundamental role in the theory.

The approach we have described is known as the **adjoint method**, and in this context $\mathbf{u}^\dagger$ is known as the **adjoint wavefield** and $h_i^\dagger$ the **adjoint traction** or **adjoint source**.[3] The steps needed to calculate the gradient of the misfit can be summarised as follows:

1. Solve the forward equations for $\mathbf{u}$;
2. From this solution determine J and the associated adjoint traction $h_i^\dagger$;
3. Solve the adjoint equations for $\mathbf{u}^\dagger$ as an initial value problem;
4. Construct the sensitivity kernels by combining $\mathbf{u}$ and $\mathbf{u}^\dagger$ appropriately.

The adjoint problem is identical in form to the original, and hence the cost of this method is just **two numerical solutions of the elastic wave equation**. In fact, within practical applications it is usual to perform three solutions, this being done so that the forward field $\mathbf{u}$ need not be stored at all times when assembling the sensitivity kernels: because the kernels pair $\mathbf{u}$ at time t with $\mathbf{u}^\dagger$ at time $T-t$, the forward field is instead reconstructed backwards in time alongside the adjoint calculation. Even so, the saving relative to the finite-difference approach is by a factor of $\frac{3}{m+1}$ where m is the number of model parameters. Given that m in practice is of order 10^4–10^6, this is an enormous improvement and justifies a few pages of algebra! It is the adjoint method that allows waveform inversion to be a practical tool within seismology and also in a range of other fields. Indeed, it is this method that underlies data assimilation in numerical weather prediction.

It is worth commenting that though we have obtained the adjoint equations for a specific choice of misfit, the form of J only enters into the problem through the adjoint traction h_i' (or, equivalently, its time reverse $h_i^\dagger$). This means that the theory can be readily applied to other measures of misfit without having to go through the lengthy derivation each time.

[3] Some authors reserve these names for the un-reversed field $\mathbf{u}'$ and traction h_i'. Nothing depends on this choice so long as the time arguments in the kernels are kept track of.

Banana doughnut kernels

Nothing in the adjoint method is specific to the least-squares misfit, or indeed to misfits at all. It provides the derivative with respect to the model of any differentiable functional of the wavefield at the cost of one additional wavefield simulation, and so it is as useful for asking how a given measurement depends on Earth structure, which is to say for **sensitivity analysis**, as it is for the estimation of the model or the state of a system from data. As a first application, before returning to inversions, we use it to find the sensitivity of the most basic seismological measurement of all, a travel time, and compare the result with the ray-theoretic prediction of Lecture 19.

Let $\mathbf{u}^{\mathrm{obs}}(t)$ again denote the displacement vector recorded by a seismograph at the surface location, $\mathbf{x}_r$, following an earthquake. Suppose we define

$$s^{\mathrm{obs}}(t) = w(t)\hat{\nu}_i u_i^{\mathrm{obs}}(t), \tag{39}$$

with $\hat{\boldsymbol{\nu}}$ a unit vector that selects a component of the displacement of interest, and $w(t)$ a windowing function that zeroes everything out except for a particular seismic phase of interest. Exactly the same thing can be done with a synthetic seismogram

$$s(\mathbf{x}_r, t) = w(t)\hat{\nu}_i u_i(\mathbf{x}_r, t), \tag{40}$$

with $\mathbf{u}$ calculated in some reference model. Consider the **cross-correlation** of these two time series as defined by

$$C(\tau) = \int_{-\infty}^{\infty} s^{\mathrm{obs}}(t-\tau) s(\mathbf{x}_r, t)\,\mathrm{d}t. \tag{41}$$

This quantity will be largest when s^{obs} and s are maximally aligned, and hence the associated value of τ gives a measurement of the delay time between the observed and synthetic waveforms. Indeed, this is typically what is done in practice to automate delay time measurements. Note that with the definition in eq. (41) an observed phase that arrives later than the synthetic gives a negative value of $\overline{\tau}$, the opposite sign to the delay time of Lecture 19; nothing below depends on this choice, since it is only the perturbation $\delta\overline{\tau}$ that enters the sensitivity kernels. The condition for $\overline{\tau}$ to maximise the cross-correlation of the waveforms is given by

$$\frac{\mathrm{d}C}{\mathrm{d}\tau}(\overline{\tau}) = 0, \tag{42}$$

and this acts to define $\overline{\tau}$ implicitly as a function of the synthetic wavefield $\mathbf{u}$, and hence of the underlying model parameters.

The adjoint method can be applied to determine the functional derivatives of $\overline{\tau}$, and as remarked above we need only determine the appropriate adjoint source. To do this we perturb the synthetic waveform, $s \to s + \delta s$, within eq. (42), noting that $\overline{\tau}$ changes in response. Differentiating eq. (41) gives

$$\frac{\mathrm{d}C}{\mathrm{d}\tau}(\tau) = -\int_{-\infty}^{\infty} \dot{s}^{\mathrm{obs}}(t-\tau) s(\mathbf{x}_r, t)\,\mathrm{d}t, \tag{43}$$

and so, to first order, the perturbed form of eq. (42) reads

$$-\int_{-\infty}^{\infty} \dot{s}^{\mathrm{obs}}(t-\overline{\tau})\,\delta s(\mathbf{x}_r, t)\,\mathrm{d}t + \delta\overline{\tau}\int_{-\infty}^{\infty} \ddot{s}^{\mathrm{obs}}(t-\overline{\tau}) s(\mathbf{x}_r, t)\,\mathrm{d}t = 0. \tag{44}$$

If the observed and synthetic waveforms differ only by the time shift, so that $s^{\mathrm{obs}}(t - \overline{\tau}) = s(\mathbf{x}_r, t)$, which is a reasonable approximation when the reference model is close to the truth, then $\dot{s}^{\mathrm{obs}}(t - \overline{\tau}) = \dot{s}(\mathbf{x}_r, t)$ in the first term, while an integration by parts (the window removing any boundary terms) gives $\int \ddot{s} s \, \mathrm{d}t = -\int \dot{s}^2 \, \mathrm{d}t$ in the second, and hence

$$\delta\overline{\tau} = -\frac{1}{N} \int_{-\infty}^{\infty} \dot{s}(\mathbf{x}_r, t) \, \delta s(\mathbf{x}_r, t) \, \mathrm{d}t, \quad N = \int_{-\infty}^{\infty} \dot{s}(\mathbf{x}_r, t)^2 \, \mathrm{d}t. \tag{45}$$

Recalling that $\delta s = w(t) \hat{v}_i \delta u_i(\mathbf{x}_r, t)$, and comparing with eqs. (18) and (19), we can read off the adjoint traction

$$h_i'(\mathbf{x}, t) = -\frac{1}{N} w(t) \dot{s}(\mathbf{x}_r, t) \hat{v}_i \delta(\mathbf{x} - \mathbf{x}_r), \tag{46}$$

which is the windowed velocity of the synthetic seismogram applied at the receiver, and the sensitivity kernels then follow from eqs. (31)–(33) exactly as before. Note that the observed seismogram has dropped out entirely: the kernel depends only on the synthetic wavefield, and hence only on the reference model. In this manner we can write

$$\delta\overline{\tau} = \int_M (K_\alpha \, \delta\alpha + K_\beta \, \delta\beta) \, \mathrm{d}^3\mathbf{x}, \tag{47}$$

for the first-order perturbation in this cross-correlation delay time with respect to the P- and S-wave speeds, holding other model parameters fixed. This expression can be compared to the ray-theoretic result

$$\delta T = -\int_{\mathrm{ray}} \frac{\delta c}{c^2} \, \mathrm{d}s, \tag{48}$$

obtained using Fermat's principle, with c the appropriate phase speed.

Figs. 3 and 4 show how such a kernel is built up in a two-dimensional numerical experiment using the adjoint method exactly as derived above. An explosive point source generates a P-wave that is recorded at a receiver, the delay time of the arrival is defined through cross-correlation, and the corresponding adjoint source is the time-reversed windowed velocity of the synthetic seismogram applied at the receiver. The adjoint field $\mathbf{u}^\dagger$ spreads out from the receiver as an ordinary initial value problem, and the kernel for the P-wave speed, which follows from eq. (37) since $\delta A_{ijkl} = 2\rho\alpha \, \delta\alpha \, \delta_{ij}\delta_{kl}$ when ρ and β are held fixed,

$$K_\alpha(\mathbf{x}) = -2\rho\alpha \int_0^T \nabla \cdot \mathbf{u}(\mathbf{x}, t) \, \nabla \cdot \mathbf{u}^\dagger(\mathbf{x}, T - t) \, \mathrm{d}t,$$

accumulates wherever the forward field at time t overlaps the adjoint field at time $T - t$. At any instant this overlap of the two wavefronts is confined to a pair of points, and as these points sweep from source to receiver they trace out the region of sensitivity, which is an ellipse with the source and receiver at its foci.[4]

The form of these sensitivity kernels deserves some comment, and Fig. 5 shows examples calculated for the Earth. First, within ray theory all sensitivity lies along

[4] The kernel was checked by perturbing the wave speed within a small region and measuring the resulting change in the cross-correlation delay time directly; for a perturbation of two per cent within the first Fresnel zone the kernel predicts this change to within about two per cent, the remaining difference being of second order in the size of the perturbation.

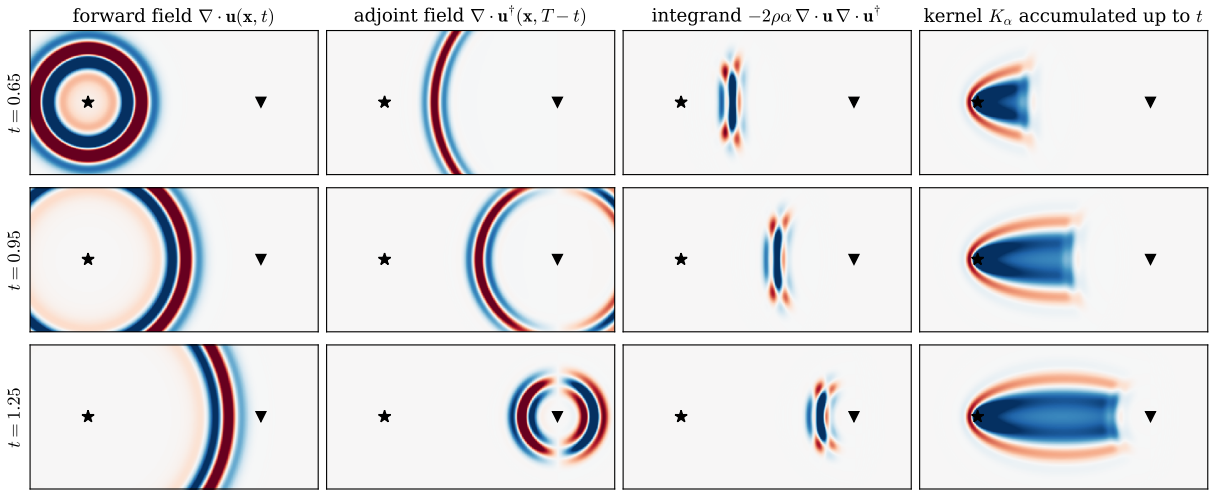


Fig. 3: Construction of a cross-correlation delay-time kernel by the adjoint method in a two-dimensional homogeneous body. The star marks an explosive source and the triangle a receiver. The kernel is $K_\alpha(\mathbf{x}) = -2\rho\alpha \int_0^T \nabla \cdot \mathbf{u}(\mathbf{x}, t)\, \nabla \cdot \mathbf{u}^\dagger(\mathbf{x}, T-t)\, \mathrm{d}t$, and each row shows one instant t (times are in nondimensional units, with $T \approx 2$): the divergence of the forward field $\mathbf{u}(\mathbf{x}, t)$; the divergence of the adjoint field $\mathbf{u}^\dagger(\mathbf{x}, T-t)$ after it has evolved for a time $T - t$ from the receiver; the integrand, that is the instantaneous contribution to the kernel; and the kernel accumulated over times up to t. The integrand, and hence the kernel, is non-zero only where the two fields overlap.

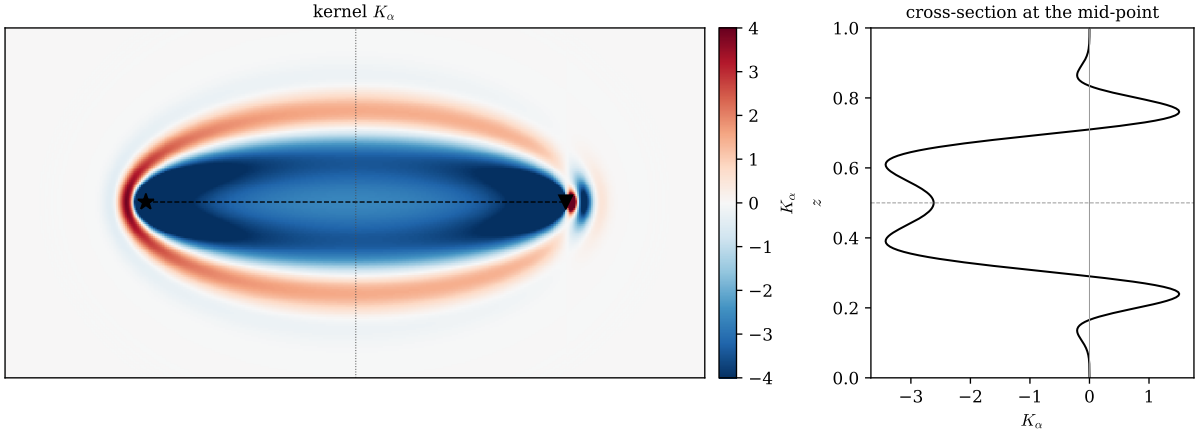


Fig. 4: The finished two-dimensional kernel of Fig. 3 (left; the colour scale is clipped near the source and receiver, where the kernel is singular) and a cross-section through it at the mid-point of the ray (right), where z is the coordinate across the ray, in units of the box height. The kernel is negative throughout the first Fresnel zone, since an increase in wave speed there advances the arrival, and changes sign in the surrounding zones. Unlike the three-dimensional kernels of Fig. 5, it does not vanish on the ray itself, though it is somewhat smaller there than on either side.

the reference ray path. Within the adjoint theory, however, there is a broad region of sensitivity around the ray path. This region can be shown to be the first Fresnel zone for the wave under consideration, i.e. the region in which a scattered wave arrives in phase with the main arrival. This is not too surprising, as ray theory neglects diffraction effects. What is perhaps more curious is that the true kernels vanish along the reference ray path. When this was first observed from numerical calculations, this result seemed counter-intuitive to some people: ray theory says that all the sensitivity is along the ray path, but the exact theory says none of it is!

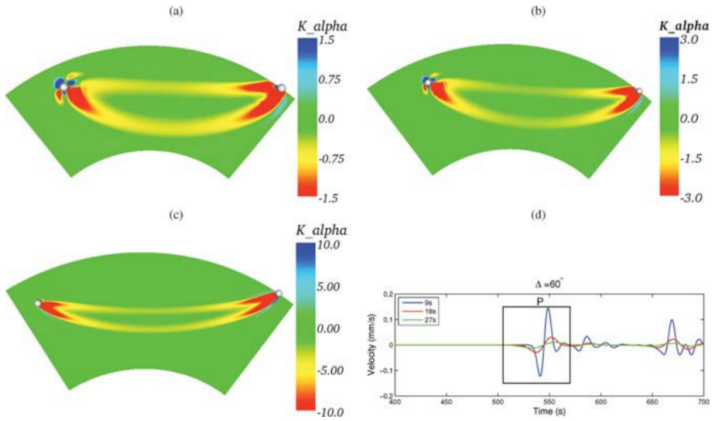


Fig. 5: Cross-sections through the functional derivative of a cross-correlation delay time for a P-wave phase with respect to α as calculated by the adjoint method. The delay time of the same phase can be measured at different frequencies by filtering the data and synthetic waveforms: panels (a)–(c) show the kernel for three different period bands, each on its own colour scale, and (d) the filtered synthetic seismograms at an epicentral distance of 60°, with the box marking the window around the P arrival used for the measurement. The dominant sensitivity lies within the first Fresnel zone, but it curiously vanishes along the ray path. Figure taken from Liu & Tromp (2008).

Subsequent work showed that there is no contradiction. The exact kernels vanish along the ray path because a scatterer on the ray path is associated with no phase delay relative to the reference wave. This argument is specific to three dimensions, and the two-dimensional kernel of Fig. 4 has no such hole. The reason lies in the different character of wave propagation in even and odd numbers of dimensions, but this is a subtle matter that we will not go into. Moreover, it can be shown that as the dominant frequency of the wave increases, the width of the Fresnel zone shrinks, and in the limit as the frequency goes to infinity the two kernels agree in the sense of generalised functions, this meaning that their action on a smooth model perturbation produces the same delay time.

The final point to mention is that you might think that the kernels in Fig. 5 look a little like a banana cut lengthwise, while in cross section they are hollow and can be said to resemble a doughnut. The first person to work on this theory, Tony Dahlen (1942–2007) from Princeton, certainly thought so, and as a result he jokingly named them **banana doughnut kernels**, and this remains the usual term used.

Global waveform tomographic models

An example of a global tomographic model obtained by the methods of this lecture is shown in Fig. 6.

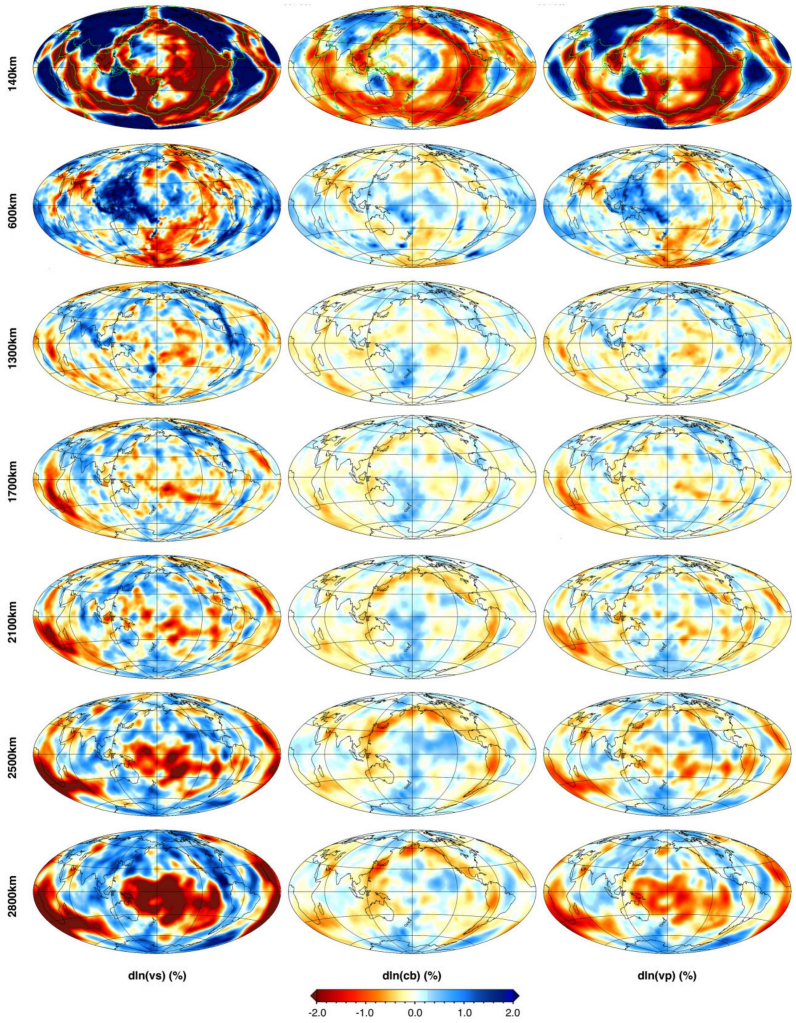


Fig. 6: The global waveform tomographic model GLAD-M35 of Cui et al. (2024), the third generation of models produced at Princeton using the adjoint method (the first, GLAD-M15, is described by Bozdağ et al. 2016), at depths of (from the top) 140, 600, 1300, 1700, 2100, 2500 and 2800 km. The columns show the perturbations in shear wave speed, bulk sound speed and compressional wave speed relative to a spherically symmetric reference model. The inversion used waveforms from 2160 earthquakes recorded at 11 575 stations and 35 iterations of gradient-based optimisation in total. From Cui et al. (2024).

Such a model is the result of gradient-based optimisation, and as remarked earlier the question of its uncertainty is a difficult one. It can, however, be addressed approximately: the second derivative of the misfit at the minimum (the Hessian) plays the role of the inverse posterior covariance in the Bayesian picture of Lecture 19, and although it is far too large to form, its action on a vector can be computed with a few further forward and adjoint simulations, which is enough to estimate its largest eigenvalues and hence the directions in model space that the data constrain best. Fig. 7 shows the

resulting uncertainty estimates for the model of Fig. 6.

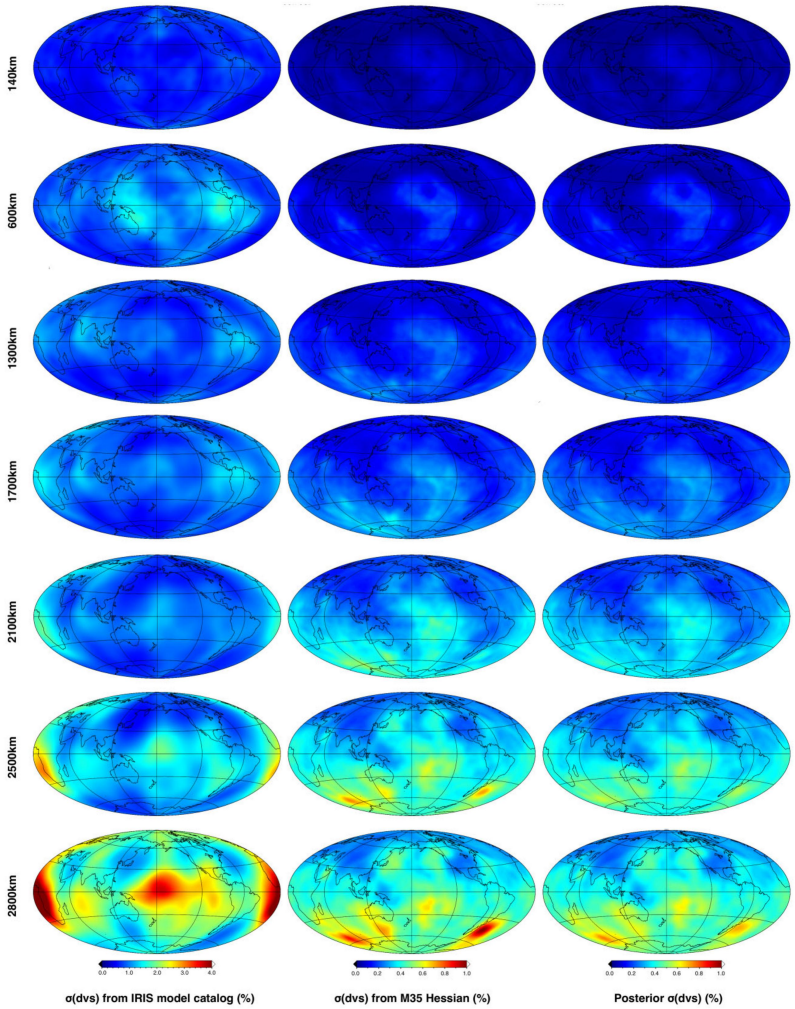


Fig. 7: Uncertainty in the shear wave speed of GLAD-M35 at the depths of Fig. 6: the prior standard deviation assumed before inversion (left), the reduction in uncertainty implied by the data through a low-rank approximation of the Hessian (centre), and the resulting posterior standard deviation (right). The uncertainty is reduced most where the data coverage is best. From Cui et al. (2024).

Another way to judge a tomographic model is to compare it with models obtained by other groups using different data and methods. Fig. 8 makes such a comparison at the base of the mantle for six models that provide both P- and S-wave speeds. Three of the models are the product of waveform tomography as described in this lecture, with the full non-linear inverse problem solved iteratively: GLAD-M35 itself, together with LOWE and its successor REVEAL, which come from a different group and were built with different choices of misfit, data and optimisation scheme, and with very different numbers of iterations (35 for GLAD-M35, 72 for LOWE and 305 for REVEAL). The other three are linearised inversions of the kind described in Lecture 19, in which the forward problem is linearised about a spherically symmetric reference model and solved by regularised least squares. GyPSuM and TX2019 are based on ray-theoretic body-wave travel times, the former combined with geodynamic constraints such as the gravity field and plate motions, and the latter with prior information on the po-

sitions of subducted slabs. SP12RTS combines travel times with surface-wave phase speeds and normal-mode observations of the kind discussed in Lectures 23 and 24. The models agree on the largest features, such as the two regions of low shear wave speed beneath Africa and the Pacific, but differ considerably in their detail and, notably, in the size and sign of the perturbations to the bulk sound speed, on which more will be said in Lecture 24. Such disagreements between models built from comparable data are a reminder that the uncertainties in tomographic models are larger than any single study would suggest. The same point is made by the vertical cross-sections of Fig. 9, which pass beneath a number of volcanic hot spots: the models differ markedly in whether they show continuous low-velocity conduits, or plumes, rising through the mantle beneath them.

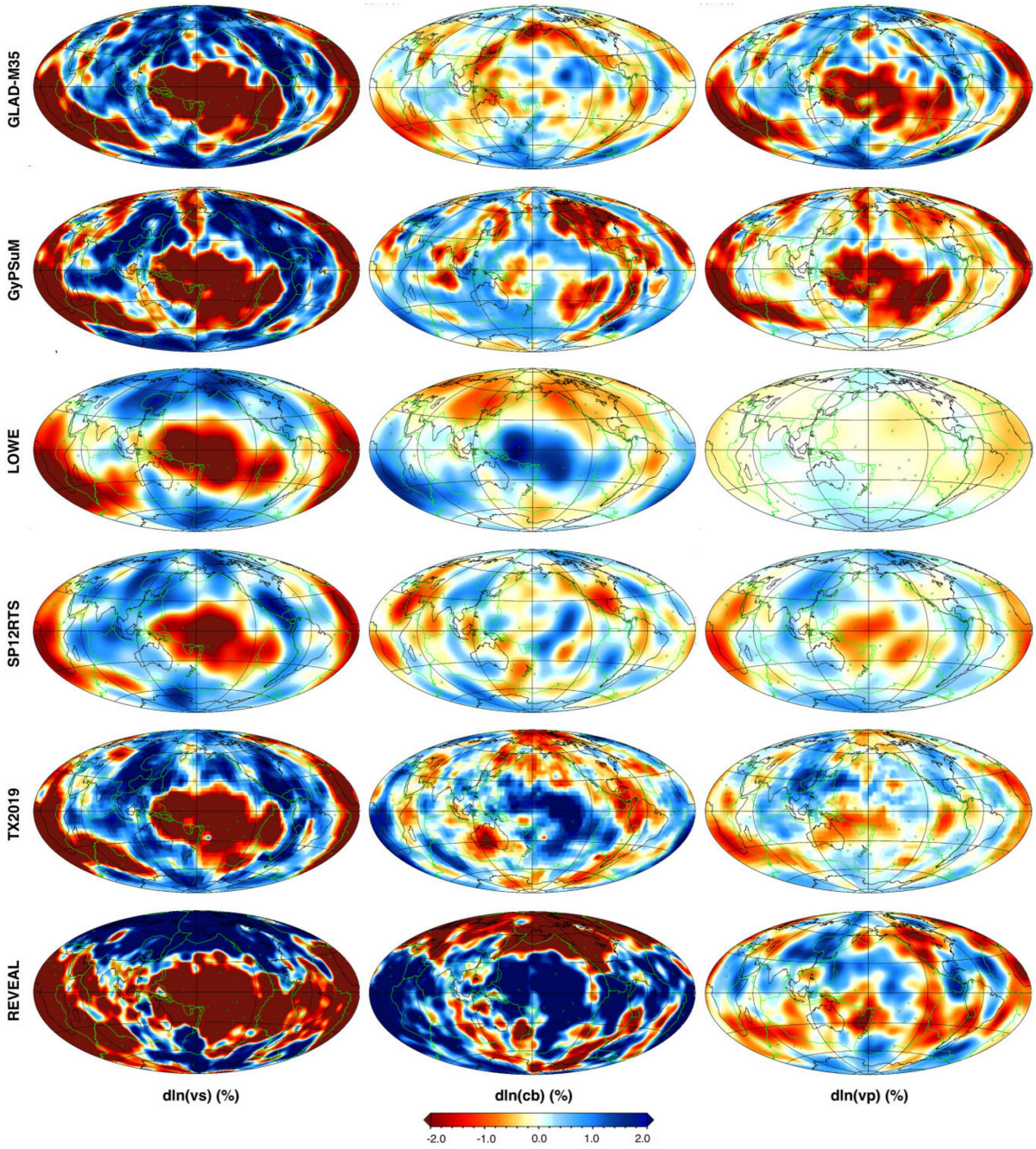


Fig. 8: Six global models that provide both P- and S-wave speeds, at a depth of 2800 km. From the top: GLAD-M35 (Cui et al. 2024), GyPSuM (Simmons et al. 2010), LOWE (van Herwaarden et al. 2020), SP12RTS (Koelemeijer et al. 2016), TX2019 (Lu et al. 2019) and REVEAL (Thrastarson et al. 2024). The columns show the perturbations in shear wave speed, bulk sound speed and compressional wave speed. From Cui et al. (2024).

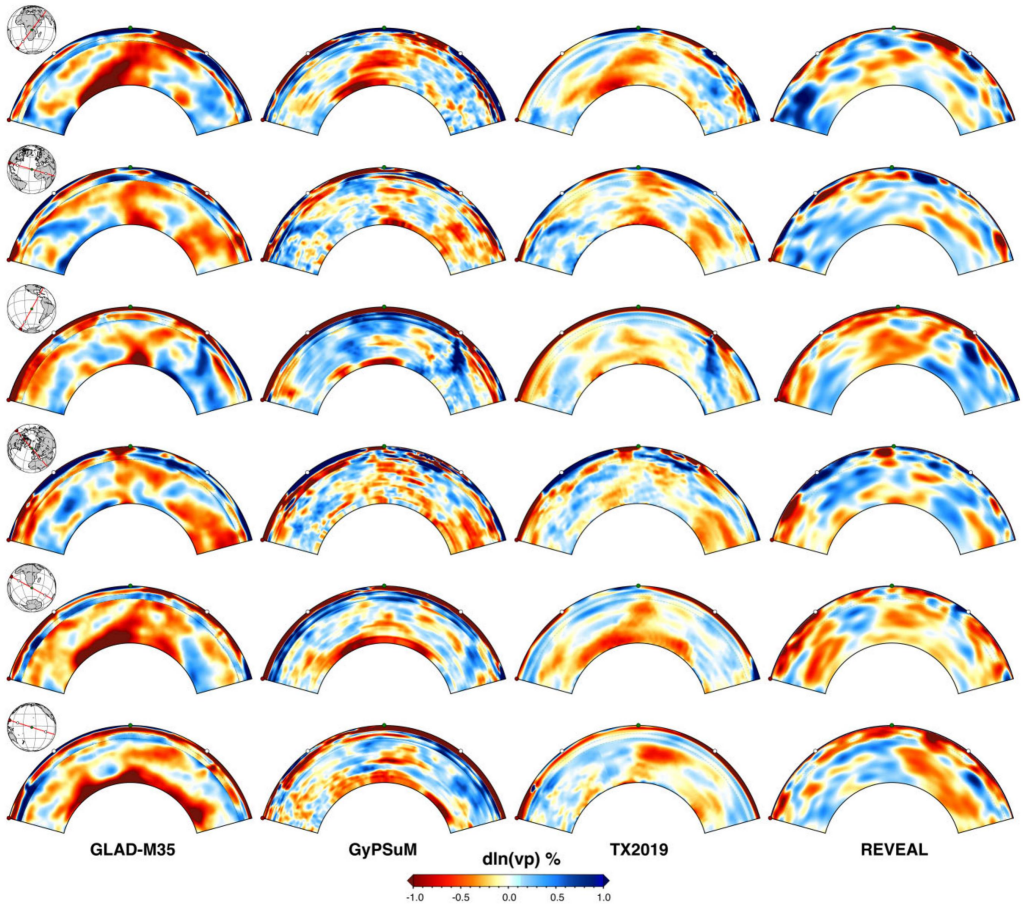


Fig. 9: Vertical cross-sections through the mantle showing the perturbation to the P-wave speed in four of the models of Fig. 8: GLAD-M35, GyPSuM, TX2019 and REVEAL. The small maps show the location of each section, which in every case passes beneath one or more hot spots. From Cui et al. (2024).

What you need to know and be able to do

(i) That waveform tomography depends on the ability to efficiently solve the elastic wave equation in realistic earth models.

(ii) How least-squares waveform misfits can be defined, and what is meant by their gradients.

(iii) How gradient-based optimisation works, why it is useful, and what its limitations are.

(iv) How the adjoint method can be used to calculate gradients, and what its advantages are. The derivation in this lecture is too lengthy for you to reproduce in full. But you may be asked to perform parts of the calculations with suitable guidance or to apply the method to simpler systems.

The Lagrange multiplier theorem (non-examinable)

A sketch proof of the Lagrange multiplier theorem is provided for those interested. Let $\mathbf{u} \mapsto J(\mathbf{u})$ be a functional of the state vector, $\mathbf{u}$. Suppose that $\mathbf{u}$ is determined through solution of the following equation

$$\mathbf{a}(\mathbf{u}, \mathbf{m}) = \mathbf{0}, \tag{49}$$

with $\mathbf{m}$ the model vector. Here $\mathbf{a}$ is some potentially non-linear vector-valued function representing abstractly the partial differential equation satisfied by $\mathbf{u}$ and in which $\mathbf{m}$ acts as a parameter. We assume that for each $\mathbf{m}$ there is a unique solution to this equation, and hence it serves to define a mapping $\mathbf{m} \mapsto \hat{\mathbf{u}}(\mathbf{m})$ such that

$$\mathbf{a}[\hat{\mathbf{u}}(\mathbf{m}), \mathbf{m}] = \mathbf{0}. \tag{50}$$

Using this mapping, we can define a new functional

$$\hat{J}(\mathbf{m}) = J[\hat{\mathbf{u}}(\mathbf{m})]. \tag{51}$$

Within the main notes we do not distinguish notationally between $J(\mathbf{u})$ and $\hat{J}(\mathbf{m})$, but here it is useful to be more explicit. Our aim is to calculate the functional derivative, $D\hat{J}(\mathbf{m})$, this being the linear term within the first-order Taylor expansion

$$\hat{J}(\mathbf{m} + \delta\mathbf{m}) = \hat{J}(\mathbf{m}) + \langle D\hat{J}(\mathbf{m}) | \delta\mathbf{m} \rangle + O(\|\delta\mathbf{m}\|^2). \tag{52}$$

We will obtain this functional derivative first using a "brute-force" method via the chain rule, and then show that the Lagrange multiplier theorem leads to the same result. The advantage of the latter is the relative ease of its application in practice.

Using eq. (51), we can perturb $\mathbf{m}$ and expand to first order to obtain

$$\hat{J}(\mathbf{m} + \delta\mathbf{m}) = J[\hat{\mathbf{u}}(\mathbf{m} + \delta\mathbf{m})] = J[\hat{\mathbf{u}}(\mathbf{m}) + \delta\mathbf{u} + \cdots] = \hat{J}(\mathbf{m}) + \langle DJ(\mathbf{u}) | \delta\mathbf{u} \rangle + \cdots, \tag{53}$$

where $\delta\mathbf{u}$ denotes the first-order perturbation to $\mathbf{u}$ due to the given change in $\mathbf{m}$. To determine the concrete form of $\delta\mathbf{u}$, we perturb eq. (50) to obtain

$$\mathbf{a}(\mathbf{u} + \delta\mathbf{u} + \cdots, \mathbf{m} + \delta\mathbf{m} + \cdots) = \mathbf{0}, \tag{54}$$

where it is understood that $\mathbf{u} = \hat{\mathbf{u}}(\mathbf{m})$. Expanding this out to first order we find

$$D_{\mathbf{u}}\mathbf{a}(\mathbf{u}, \mathbf{m})\delta\mathbf{u} + D_{\mathbf{m}}\mathbf{a}(\mathbf{u}, \mathbf{m})\delta\mathbf{m} = \mathbf{0}. \tag{55}$$

Here we see the partial derivatives of $\mathbf{a}$ with respect to $\mathbf{u}$ and $\mathbf{m}$, and note that these are linear operators that act on the perturbations to $\mathbf{u}$ and $\mathbf{m}$ to return an appropriate vector. If this notation is confusing, think of $\mathbf{u}$ as an n-dimensional vector and $\mathbf{m}$ as an m-dimensional one. Then the values of $\mathbf{a}(\mathbf{u}, \mathbf{m})$ are also n-dimensional vectors, and the linear operators $D_{\mathbf{u}}\mathbf{a}(\mathbf{u}, \mathbf{m})$ and $D_{\mathbf{m}}\mathbf{a}(\mathbf{u}, \mathbf{m})$ are matrices having dimensions $n \times n$ and $n \times m$, respectively. Assuming that the linear operator, $D_{\mathbf{u}}\mathbf{a}(\mathbf{u}, \mathbf{m})$, is invertible, we can solve the above equation to obtain

$$\delta\mathbf{u} = -[D_{\mathbf{u}}\mathbf{a}(\mathbf{u}, \mathbf{m})]^{-1} D_{\mathbf{m}}\mathbf{a}(\mathbf{u}, \mathbf{m})\delta\mathbf{m}. \tag{56}$$

The assumption that this matrix is invertible is not actually an assumption at all. It can be shown via the implicit function theorem that this is a necessary (and locally

sufficient) condition for eq. (50) to admit unique solutions. Substituting this expression for $\delta\mathbf{u}$ into eq. (53) we find

$$\hat{J}(\mathbf{m}+\delta\mathbf{m}) = \hat{J}(\mathbf{m}) - \langle DJ(\mathbf{u})|[D_{\mathbf{u}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{-1}D_{\mathbf{m}}\mathbf{a}(\mathbf{u},\mathbf{m})\delta\mathbf{m}\rangle + \cdots . \tag{57}$$

Recalling the definition of the adjoint of a linear operator, we can transform the inner product term to obtain

$$\hat{J}(\mathbf{m}+\delta\mathbf{m}) = \hat{J}(\mathbf{m}) - \langle [D_{\mathbf{m}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{\dagger}[D_{\mathbf{u}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{-\dagger}DJ(\mathbf{u})|\delta\mathbf{m}\rangle + \cdots , \tag{58}$$

where $(\cdot)^{-\dagger}$ is a shorthand for the inverse of the adjoint. By inspection, we have arrived at the desired result

$$D\hat{J}(\mathbf{m}) = -[D_{\mathbf{m}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{\dagger}[D_{\mathbf{u}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{-\dagger}DJ(\mathbf{u}). \tag{59}$$

We now apply the Lagrange multiplier theorem and show that it leads to an identical result. The starting point is defining the appropriate Lagrangian

$$L(\mathbf{u},\mathbf{u}',\mathbf{m}) = J(\mathbf{u}) + \langle \mathbf{a}(\mathbf{u},\mathbf{m})|\mathbf{u}'\rangle, \tag{60}$$

with $\mathbf{u}'$ the Lagrange multiplier field used to impose eq. (50). Our claim is that

$$D\hat{J}(\mathbf{m}) = D_{\mathbf{m}}L(\mathbf{u},\mathbf{u}',\mathbf{m}), \tag{61}$$

so long as the following conditions hold

$$D_{\mathbf{u}}L(\mathbf{u},\mathbf{u}',\mathbf{m}) = 0, \quad D_{\mathbf{u}'}L(\mathbf{u},\mathbf{u}',\mathbf{m}) = 0. \tag{62}$$

As in the main lecture, we readily see that the second condition implies eq. (50), and hence we have $\mathbf{u} = \hat{\mathbf{u}}(\mathbf{m})$. To examine the first condition, we note that

$$\langle D_{\mathbf{u}}L(\mathbf{u},\mathbf{u}',\mathbf{m})|\delta\mathbf{u}\rangle = \langle DJ(\mathbf{u})|\delta\mathbf{u}\rangle + \langle D_{\mathbf{u}}\mathbf{a}(\mathbf{u},\mathbf{m})\delta\mathbf{u}|\mathbf{u}'\rangle = 0, \tag{63}$$

for any $\delta\mathbf{u}$. Again using the adjoint of a linear operator, this is equivalent to

$$\langle DJ(\mathbf{u}) + [D_{\mathbf{u}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{\dagger}\mathbf{u}'|\delta\mathbf{u}\rangle = 0, \tag{64}$$

and as $\delta\mathbf{u}$ is arbitrary, we see that the first condition in eq. (62) implies

$$DJ(\mathbf{u}) + [D_{\mathbf{u}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{\dagger}\mathbf{u}' = \mathbf{0}. \tag{65}$$

This is a linear equation for the Lagrange multiplier $\mathbf{u}'$, and it has solution

$$\mathbf{u}' = -[D_{\mathbf{u}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{-\dagger}DJ(\mathbf{u}). \tag{66}$$

This result can be substituted into eq. (61), but first we note that

$$\langle D_{\mathbf{m}}L(\mathbf{u},\mathbf{u}',\mathbf{m})|\delta\mathbf{m}\rangle = \langle D_{\mathbf{m}}\mathbf{a}(\mathbf{u},\mathbf{m})\delta\mathbf{m}|\mathbf{u}'\rangle = \langle \delta\mathbf{m}|[D_{\mathbf{m}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{\dagger}\mathbf{u}'\rangle, \tag{67}$$

for any $\delta\mathbf{m}$, and hence

$$D_{\mathbf{m}}L(\mathbf{u},\mathbf{u}',\mathbf{m}) = [D_{\mathbf{m}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{\dagger}\mathbf{u}' = -[D_{\mathbf{m}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{\dagger}[D_{\mathbf{u}}\mathbf{a}(\mathbf{u},\mathbf{m})]^{-\dagger}DJ(\mathbf{u}). \tag{68}$$

The final result is precisely what was obtained using the direct method, and hence we have established the validity of the Lagrange multiplier theorem. It is worth noting that eq. (65) for the Lagrange multiplier, $\mathbf{u}'$, involves the adjoint of the linearised forward operator. It is this fact that gives the adjoint method its name.

Lecture 21: Self-gravitation and rotation

Outline and motivation

In previous lectures we have discussed how travel time and waveform observations can be used to determine variations in elastic wave speeds within the Earth. Nothing has, however, been said about **density**, with this parameter being key to understanding both the Earth's interior composition and dynamics. To learn about density, we need to study the **long-period free oscillations of the Earth**. This is because **self-gravitation**, and hence density, plays a vital role in their physics. Within this lecture we will return to basic elastodynamics, and show how gravitational forces can be included into the equations of motion. We also explain how the **Earth's rotation** enters into the problem. Having obtained the equations of motion, we perform a **scaling analysis** to determine the relative importance of the different terms, and conclude that at short enough periods the effects of self-gravitation and rotation can be safely neglected, and hence the earlier parts of this course remain valid.

A review of Newtonian gravitation

Consider the motion $\boldsymbol{\varphi}$ of an elastic body relative to a reference body M, and write M_t for the instantaneous volume of space occupied at time t. At this instant in time, the gravitational potential of the body satisfies Poisson's equation

$$\nabla^2 \phi = 4\pi G \varrho, \tag{1}$$

in $\mathbb{R}^3$, where G is the universal gravitational constant, and ϱ is the spatial density. The gravitational acceleration associated with this potential is given by

$$g_i = -\frac{\partial \phi}{\partial x_i}. \tag{2}$$

Poisson's equation is subject to the conditions that ϕ tends to zero at infinity, while across any jumps in density we have the continuity conditions

$$[\phi]^+_- = 0, \quad [g_i \hat{n}_i]^+_- = 0, \tag{3}$$

where $[\cdot]^+_-$ denotes the jump in a quantity in the direction of the outward normal. The closed form solution to this problem is given by

$$\phi(\mathbf{y}, t) = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)}{[(y_i - y_i')(y_i - y_i')]^{\frac{1}{2}}} \, \mathrm{d}^3\mathbf{y}', \tag{4}$$

and differentiating, we similarly find

$$g_i(\mathbf{y}, t) = -G \int_{M_t} \frac{\varrho(\mathbf{y}', t)(y_i - y_i')}{[(y_j - y_j')(y_j - y_j')]^{\frac{3}{2}}} \, \mathrm{d}^3\mathbf{y}'. \tag{5}$$

For the Earth, the equilibrium gravitational acceleration and the pressure that it produces are shown in Fig. 1. These were calculated from the radial density profile of the model PREM; where that density comes from is a question we return to in Lecture 24, since it is the free oscillations discussed there that provide the main constraint on it. Note that g is close to $10\,\mathrm{m\,s^{-2}}$ throughout the mantle, reaching its largest value at the core–mantle boundary, and decreases roughly linearly through the core.

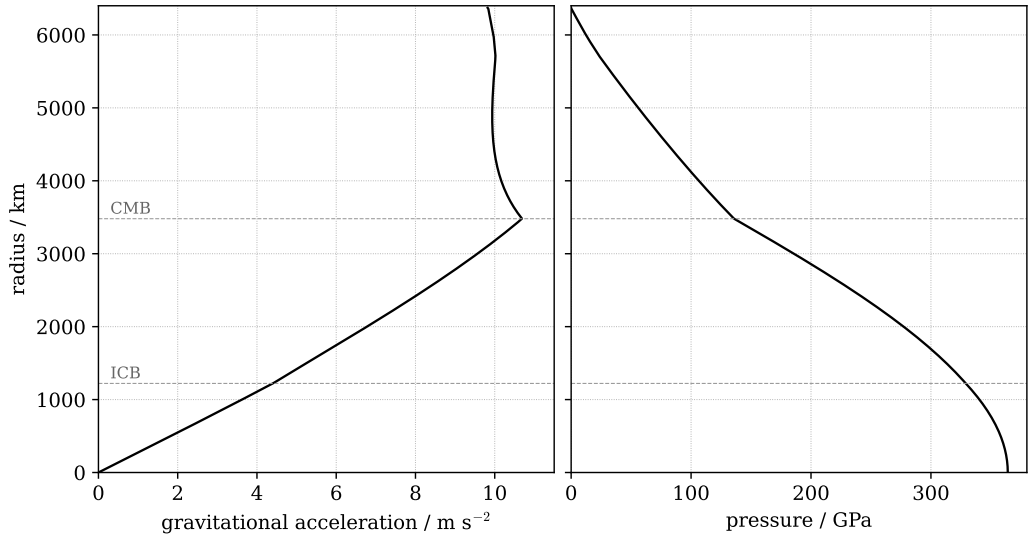


Fig. 1: Gravitational acceleration (left) and pressure (right) as functions of radius in the earth model PREM, calculated from its density profile. The dashed lines mark the core–mantle boundary (CMB) and inner core boundary (ICB).

It will be convenient to define the **referential gravitational potential** by

$$\zeta(\mathbf{x},t) = \phi[\boldsymbol{\varphi}(\mathbf{x},t),t], \tag{6}$$

which is a time-dependent field on the reference body, M. We recall from Lecture 13 that the volume elements in the reference body and physical space are related by $\mathrm{d}^3\mathbf{y} = J(\mathbf{x},t)\,\mathrm{d}^3\mathbf{x}$ with $J = \det(\mathbf{F})$ the Jacobian of the motion. We can then change variables within eq. (4) and use the definition of ζ in eq. (6) to obtain

$$\zeta(\mathbf{x},t) = -G\int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x},t)-\varphi_i(\mathbf{x}',t)][\varphi_i(\mathbf{x},t)-\varphi_i(\mathbf{x}',t)]\}^{\frac{1}{2}}}\,\mathrm{d}^3\mathbf{x}', \tag{7}$$

where from Lecture 13 we know that conservation of mass requires

$$\rho(\mathbf{x}) = J(\mathbf{x},t)\varrho[\boldsymbol{\varphi}(\mathbf{x},t),t], \tag{8}$$

with ρ the **referential density**. In a similar manner, we define the **referential gravitational acceleration** by

$$\gamma_i(\mathbf{x},t) = g_i[\boldsymbol{\varphi}(\mathbf{x},t),t], \tag{9}$$

which from eq. (5) can be written

$$\gamma_i(\mathbf{x},t) = -G\int_M \frac{\rho(\mathbf{x}')[\varphi_i(\mathbf{x},t)-\varphi_i(\mathbf{x}',t)]}{\{[\varphi_j(\mathbf{x},t)-\varphi_j(\mathbf{x}',t)][\varphi_j(\mathbf{x},t)-\varphi_j(\mathbf{x}',t)]\}^{\frac{3}{2}}}\,\mathrm{d}^3\mathbf{x}'. \tag{10}$$

Gravitational binding energy

At each instant of time we can associate a **gravitational binding energy** with the elastic body. This is the energy that would be required to assemble the body from matter dispersed at infinity, assuming that its constituent parts are acted on only by their mutual gravitational attraction. Because gravity is a conservative force, the binding energy is independent of the details of the assembly.

Suppose that the mass of the body is already in place, and that its spatial density and gravitational potential are denoted by ϱ and ϕ. To determine an expression for the binding energy we will consider how this quantity changes due to an infinitesimal addition of mass to the body. By definition of the gravitational potential, the work required to bring in an additional mass element $\delta\varrho\,\mathrm{d}^3\mathbf{y}$ from infinity to a point $\mathbf{y}$ within the body is $\delta\varrho\,\phi\,\mathrm{d}^3\mathbf{y}$. The change in the binding energy associated with a perturbation $\delta\varrho$ to the body's density is, therefore, given by

$$\delta\mathcal{V}_g = \int_{M_t} \delta\varrho\,\phi\,\mathrm{d}^3\mathbf{y}, \tag{11}$$

to first-order accuracy. Using the perturbed Poisson's equation $\nabla^2\delta\phi = 4\pi G\delta\varrho$ for the resulting change in gravitational potential, we can write this increment to the binding energy as

$$\delta\mathcal{V}_g = \frac{1}{4\pi G}\int_{M_t}\left(\frac{\partial^2\delta\phi}{\partial y_i\partial y_i}\right)\phi\,\mathrm{d}^3\mathbf{y} = \frac{1}{4\pi G}\int_{\mathbb{R}^3}\left(\frac{\partial^2\delta\phi}{\partial y_i\partial y_i}\right)\phi\,\mathrm{d}^3\mathbf{y}, \tag{12}$$

where the final equality follows from the fact that $\nabla^2\delta\phi = 0$ outside of the instantaneous body M_t. Using the identity

$$\left(\frac{\partial^2\delta\phi}{\partial y_i\partial y_i}\right)\phi = \frac{\partial}{\partial y_i}\left(\phi\frac{\partial\delta\phi}{\partial y_i}\right) - \frac{\partial\phi}{\partial y_i}\frac{\partial\delta\phi}{\partial y_i}, \tag{13}$$

and applying the divergence theorem we obtain

$$\delta\mathcal{V}_g = -\frac{1}{4\pi G}\int_{\mathbb{R}^3}\frac{\partial\phi}{\partial y_i}\frac{\partial\delta\phi}{\partial y_i}\,\mathrm{d}^3\mathbf{y}, \tag{14}$$

where we have used the continuity of the potentials and their normal derivatives across ∂M_t, while the surface integral at infinity vanishes since, at large distances r, ϕ decays like r^{-1} and $\|\nabla\delta\phi\|$ like r^{-2}. It follows that the increment to the gravitational binding energy can be written

$$\delta\mathcal{V}_g = \delta\left(-\frac{1}{8\pi G}\int_{\mathbb{R}^3}\frac{\partial\phi}{\partial y_i}\frac{\partial\phi}{\partial y_i}\,\mathrm{d}^3\mathbf{y}\right), \tag{15}$$

and so we arrive at the result

$$\mathcal{V}_g = -\frac{1}{8\pi G}\int_{\mathbb{R}^3}\frac{\partial\phi}{\partial y_i}\frac{\partial\phi}{\partial y_i}\,\mathrm{d}^3\mathbf{y}. \tag{16}$$

Using Poisson's equation for ϕ, we can write the integrand of the above expression in the form

$$\frac{\partial\phi}{\partial y_i}\frac{\partial\phi}{\partial y_i} = \frac{\partial}{\partial y_i}\left(\phi\frac{\partial\phi}{\partial y_i}\right) - \phi\left(\frac{\partial^2\phi}{\partial y_i\partial y_i}\right) = \frac{\partial}{\partial y_i}\left(\phi\frac{\partial\phi}{\partial y_i}\right) - 4\pi G\varrho\phi, \tag{17}$$

and applying the divergence theorem we obtain

$$\mathcal{V}_g(t) = \frac{1}{2}\int_{M_t} \varrho(\mathbf{y},t)\phi(\mathbf{y},t)\,\mathrm{d}^3\mathbf{y}, \tag{18}$$

where, as above, we have used the continuity of ϕ and its normal derivative across ∂M_t, and the surface integral at infinity again vanishes. To include this binding energy into Hamilton's principle, it will be necessary to transform eq. (18) to the following integral over the reference body

$$\mathcal{V}_g(t) = \frac{1}{2}\int_{M} \rho(\mathbf{x})\zeta(\mathbf{x},t)\,\mathrm{d}^3\mathbf{x}, \tag{19}$$

where we have used the definition of the referential gravitational potential in eq. (6) and the statement of conservation of mass during the deformation in eq. (8).

The equations of motion relative to inertial space

Having obtained an appropriate expression for the gravitational binding energy, the equations of motion can be readily obtained using Hamilton's principle. The Lagrangian for the problem is

$$\mathcal{L}(\mathbf{x},t,\boldsymbol{\varphi},\mathbf{v},\mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x},t,\mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x},t), \tag{20}$$

where we note that the elastic strain energy is allowed to be time-dependent to accommodate a seismic source. An important point about this Lagrangian is that it is **non-local** due to the gravitational binding energy. This means that the value of the Lagrangian at a point and time cannot be expressed only in terms of the values of the various fields and their derivatives at this point. The physical reason for this non-locality is, of course, the **action-at-a-distance property of Newtonian gravitation**.

The action for the body takes the usual form

$$\mathcal{S}(\boldsymbol{\varphi}) = \int_0^T \int_M \mathcal{L}[\mathbf{x},t,\boldsymbol{\varphi}(\mathbf{x},t),\mathbf{v}(\mathbf{x},t),\mathbf{F}(\mathbf{x},t)]\,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t, \tag{21}$$

and Hamilton's principle requires that this quantity be stationary about the true motion for all perturbations with fixed end points. In obtaining the Euler–Lagrange equations, however, we have to account for the non-local nature of the Lagrangian. As ever, the mathematical statement of Hamilton's principle is the vanishing of the functional derivative of the action

$$\langle D\mathcal{S}(\boldsymbol{\varphi})|\delta\boldsymbol{\varphi}\rangle = 0, \tag{22}$$

for all variations, $\delta\boldsymbol{\varphi}$, subject to the usual fixed-endpoint conditions. Here the left-hand side is, by definition, just the linear term in the expansion of $\mathcal{S}(\boldsymbol{\varphi}+\delta\boldsymbol{\varphi})$ about $\boldsymbol{\varphi}$. Looking at the form of the action, it is clear that this linear term takes the form

$$\langle D\mathcal{S}(\boldsymbol{\varphi})|\delta\boldsymbol{\varphi}\rangle = \int_0^T \int_M \left(\frac{\delta\mathcal{L}}{\delta\varphi_i}\delta\varphi_i + \frac{\delta\mathcal{L}}{\delta v_i}\delta v_i + \frac{\delta\mathcal{L}}{\delta F_{ij}}\delta F_{ij}\right)\,\mathrm{d}^3\mathbf{x}\,\mathrm{d}t, \tag{23}$$

where we have introduced the convenient notation $\frac{\delta \mathcal{L}}{\delta \varphi_i}$, $\frac{\delta \mathcal{L}}{\delta v_i}$ and $\frac{\delta \mathcal{L}}{\delta F_{ij}}$ for the factors multiplying the various perturbations. In the case of a local Lagrangian these factors reduce to normal partial derivatives, and hence it is only the term $\frac{\delta \mathcal{L}}{\delta \varphi_i}$ that requires additional thought. Applying the usual integration by parts steps, we arrive at the Euler–Lagrange equation

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} - \frac{\partial}{\partial t}\left(\frac{\delta \mathcal{L}}{\delta v_i}\right) - \frac{\partial}{\partial x_j}\left(\frac{\delta \mathcal{L}}{\delta F_{ij}}\right) = 0, \tag{24}$$

along with the natural boundary condition

$$\frac{\delta \mathcal{L}}{\delta F_{ij}} \hat{n}_j = 0 \tag{25}$$

on ∂M.

As in the non-gravitating case we have

$$\frac{\delta \mathcal{L}}{\delta v_i} = \rho v_i, \quad \frac{\delta \mathcal{L}}{\delta F_{ij}} = -\frac{\partial W}{\partial F_{ij}}, \tag{26}$$

while in the second problem set you will show that

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho \gamma_i, \tag{27}$$

with γ_i the referential gravitational acceleration as defined above. The equations of motion, therefore, take the expected form

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} - \rho \gamma_i = 0, \tag{28}$$

with $T_{ij} = \frac{\partial W}{\partial F_{ij}}$ the first Piola–Kirchhoff stress tensor. The form of these equations is not, of course, unexpected. To the balance of linear momentum obtained previously, we have simply added the gravitational force acting on the body due to its own mass.

Use of a co-rotating reference frame

We have obtained the equations of motion for a self-gravitating body with respect to an **inertial reference** frame defined by the "fixed stars", and these results could be applied directly to the Earth. As, however, the Earth undergoes a nearly steady rotation about its polar axis, it is useful to reformulate the equations of motion with respect to a **non-inertial frame that rotates with the Earth**.

To do this we express the motion as

$$\varphi_i(\mathbf{x}, t) = \Phi_i(t) + R_{ij}(t)\overline{\varphi}_j(\mathbf{x}, t), \tag{29}$$

where $\mathbf{R}(t)$ is a rotation matrix. Here $\mathbf{\Phi}(t)$ denotes the translation of the reference frame relative to inertial space, and $\mathbf{R}(t)$ changes of orientation. The term $\overline{\boldsymbol{\varphi}}(\mathbf{x}, t)$ describes the motion relative to the non-inertial frame, and will be called the **relative motion**.

Differentiating eq. (29) with respect to time we see that the velocity relative to inertial space can be written

$$v_i = \frac{\mathrm{d}\Phi_i}{\mathrm{d}t} + R_{ij}\left(\frac{\partial\overline{\varphi}_j}{\partial t} + R_{kj}\frac{\mathrm{d}R_{kl}}{\mathrm{d}t}\overline{\varphi}_l\right), \tag{30}$$

where we have used the identity $R_{ij}R_{kj} = \delta_{ik}$ for rotation matrices. If we differentiate the companion identity $R_{kj}R_{kl} = \delta_{jl}$ we find

$$\frac{\mathrm{d}R_{kj}}{\mathrm{d}t}R_{kl} + R_{kj}\frac{\mathrm{d}R_{kl}}{\mathrm{d}t} = 0, \tag{31}$$

and hence conclude that $R_{kj}\frac{\mathrm{d}R_{kl}}{\mathrm{d}t}$ is an **anti-symmetric matrix**. It follows that we can use the alternating tensor to write

$$R_{kj}\frac{\mathrm{d}R_{kl}}{\mathrm{d}t} = \epsilon_{jml}\Omega_m, \tag{32}$$

for a uniquely determined vector $\boldsymbol{\Omega}$. Using this result, the velocity becomes

$$v_i = \frac{\mathrm{d}\Phi_i}{\mathrm{d}t} + R_{ij}(\overline{v}_j + \epsilon_{jkl}\Omega_k\overline{\varphi}_l). \tag{33}$$

Here $\overline{\mathbf{v}}$ is the velocity as seen from the non-inertial frame, while $\boldsymbol{\Omega}$ is the **angular velocity of the frame**. Differentiating again and applying the same ideas, the acceleration relative to inertial space can be written

$$\frac{\partial v_i}{\partial t} = \frac{\mathrm{d}^2\Phi_i}{\mathrm{d}t^2} + R_{ij}\left(\frac{\partial\overline{v}_j}{\partial t} + 2\epsilon_{jkl}\Omega_k\overline{v}_l + \epsilon_{jkl}\frac{\mathrm{d}\Omega_k}{\mathrm{d}t}\overline{\varphi}_l + \epsilon_{jkl}\epsilon_{lmn}\Omega_k\Omega_m\overline{\varphi}_n\right). \tag{34}$$

On the right-hand side we can identify various terms. The first is the acceleration of the frame relative to inertial space. Within the brackets we then see, relative to the non-inertial frame, accelerations associated with the motion along with the expected **Coriolis**, **Euler**, and **centrifugal** contributions.

We have not yet specified *how* the non-inertial frame is to be chosen. Within seismology the most common approach is to take

$$\frac{\mathrm{d}\boldsymbol{\Phi}}{\mathrm{d}t} = \mathbf{0}, \quad \frac{\mathrm{d}\boldsymbol{\Omega}}{\mathrm{d}t} = \mathbf{0}, \tag{35}$$

which corresponds to a frame whose **origin is static** relative to the "fixed stars" but **rotates steadily**, with the angular velocity chosen so that the frame **approximately co-rotates with the Earth**. The advantage of this choice is its simplicity. But it does mean that orbital, rotational, and relative motion of the Earth are mixed up within the term $\overline{\varphi}$. A preferable approach is to use the so-called **Tisserand frame** which is defined such that the relative motion, $\overline{\varphi}$, carries no net linear or angular momentum.[1] But we will stick to the simpler and more common choice in this course.

To complete the derivation of the equations of motion in the non-inertial frame, we note from eq. (29) that

$$F_{ij} = R_{ik}\overline{F}_{kj}, \tag{36}$$

[1] See Maitra & Al-Attar (2024) for details.

with $\overline{\mathbf{F}}$ the deformation gradient associated with relative motion. Due to material frame indifference, we then have $W(\mathbf{x}, t, \mathbf{F}) = W(\mathbf{x}, t, \overline{\mathbf{F}})$ as should be physically expected. Using the chain rule, we find

$$\overline{T}_{ij} = \frac{\partial W}{\partial \overline{F}_{ij}}(\overline{\mathbf{F}}) = \frac{\partial}{\partial \overline{F}_{ij}} W(\mathbf{R}\overline{\mathbf{F}}) = \frac{\partial W}{\partial F_{kl}}(\mathbf{R}\overline{\mathbf{F}}) \frac{\partial}{\partial \overline{F}_{ij}}(R_{km}\overline{F}_{ml}) = R_{ki}T_{kj}, \tag{37}$$

with the left-hand side the stress associated with $\overline{\boldsymbol{\varphi}}$. Passing the rotation matrix to the other side, we have then obtained the useful identity

$$T_{ij} = R_{ik}\overline{T}_{kj}, \tag{38}$$

between the first Piola–Kirchhoff stress relative to the inertial and non-inertial reference frames. Finally, we put eq. (29) into eq. (10) to find the relationship

$$\gamma_i = R_{ij}\overline{\gamma}_j, \tag{39}$$

between the referential gravitational acceleration in the two frames, with $\overline{\gamma}$ defined exactly as in eq. (10) but with $\overline{\boldsymbol{\varphi}}$ replacing $\boldsymbol{\varphi}$. Using these various results we have shown that the equations of motion are equivalent to

$$\rho\left(\frac{\partial \overline{v}_i}{\partial t} + 2\epsilon_{ijk}\Omega_j\overline{v}_k + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l\overline{\varphi}_m\right) - \frac{\partial \overline{T}_{ij}}{\partial x_j} - \rho\overline{\gamma}_i = 0. \tag{40}$$

These equations are identical in form to those in an inertial frame except for the inclusion of the Coriolis and centrifugal accelerations – the Euler acceleration has been removed because we have restricted attention to a steadily rotating frame. From this point on we will consistently work in the approximately co-rotating frame, and hence will simplify notation by dropping overlines on terms linked to the relative motion.

Linearised equations of motion

As ever, our main concern is with the linearised equations of motion about an equilibrium state. Strictly, we should use the term **relative equilibrium**, as the body is now undergoing a steady rotation relative to inertial space. We consider a disturbance to the body caused by a stress glut, and hence take the strain energy to be

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - s\frac{1}{2}\overline{S}_{ij}(\mathbf{x}, t)C_{ij}, \tag{41}$$

with s a perturbation parameter, and $\overline{\mathbf{S}}$ a given stress glut. As in Lecture 15, the elastic tensor that appears below is the second derivative of the time-independent part $U(\mathbf{x}, \mathbf{C})$ of this strain energy function. In a now standard manner we can obtain linearised equations of motion for the displacement vector defined as the first-order term in the expansion

$$\varphi_i(\mathbf{x}, t) = x_i + s\, u_i(\mathbf{x}, t) + O(s^2). \tag{42}$$

At zeroth order, the equations of motion take the form

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m - \frac{\partial T^0_{ij}}{\partial x_j} - \rho\gamma^0_i = 0, \tag{43}$$

where T^0_{ij} and γ^0_i denote the equilibrium values of the stress tensor and gravitational acceleration. We will have more to say about these equations in the following lecture, but for the moment note that there must be a balance between the centrifugal and gravitational forces on the body and the resulting equilibrium stress; the simplifying assumption of a stress-free equilibrium is no longer an option.

From the first-order term in the expansion we arrive at the linearised equations of motion for the displacement field

$$\rho\left(\frac{\partial^2 u_i}{\partial t^2}+2\epsilon_{ijk}\Omega_j\frac{\partial u_k}{\partial t}+\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m\right)-\frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right)-\rho\gamma^1_i=-\frac{\partial \overline{S}_{ij}}{\partial x_j}, \tag{44}$$

where γ^1_i denotes the first-order perturbation to the referential gravitational acceleration. Using a binomial expansion, this term can be written explicitly as

$$\gamma^1_i=-G\int_M \rho'\left(\frac{\delta_{ij}}{[(x_k-x'_k)(x_k-x'_k)]^{3/2}}-\frac{3(x_i-x'_i)(x_j-x'_j)}{[(x_k-x'_k)(x_k-x'_k)]^{5/2}}\right)(u_j-u'_j)\,\mathrm{d}^3\mathbf{x}', \tag{45}$$

where it is understood that primed terms are functions of the integration variable. The boundary conditions for the problem take the expected form

$$A_{ijkl}\frac{\partial u_k}{\partial x_l}\hat{n}_j=\overline{S}_{ij}\hat{n}_j. \tag{46}$$

The key point is that the inclusion of gravitation and rotation into the problem only requires some additional terms in the equations of motion considered so far. It is notable, however, that these are no longer partial differential equations, but **integro-differential equations**, with both derivatives and integrals of the unknown functions occurring within the equations of motion. Again, this feature derives from the action-at-a-distance property of Newtonian gravitation.

The neglect of self-gravitation and rotation

We have obtained the linearised equations of motion within a steadily rotating and self-gravitating body. It is natural to ask if and why it has been reasonable to neglect these physical phenomena up till this point. To do so, we will perform a scaling analysis of the equations.

We assume that the displacement vector field is characterised by a typical amplitude $\|\mathbf{u}\|\sim U$, a length scale L and a time scale T. The following order-of-magnitude scaling estimates are then immediate

$$\rho\frac{\partial^2 u_i}{\partial t^2}\sim\frac{\overline{\rho}U}{T^2}, \tag{47}$$

$$\rho\epsilon_{ijk}\Omega_j\frac{\partial u_k}{\partial t}\sim\frac{\overline{\rho}\Omega U}{T}, \tag{48}$$

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m\sim\overline{\rho}\Omega^2 U, \tag{49}$$

$$\frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right)\sim\frac{\overline{\mu}U}{L^2}, \tag{50}$$

where $\overline{\rho}$ and $\overline{\mu}$ denote typical values of density and elastic moduli within the body, while Ω is the magnitude of the rotation vector. The scaling estimate for the gravitational term within the equations of motion requires more thought. The referential gravitational potential at equilibrium is given by

$$\zeta_0 = -G \int_M \frac{\rho'}{[(x_k - x'_k)(x_k - x'_k)]^{1/2}} \, \mathrm{d}^3\mathbf{x}'. \tag{51}$$

Taking the gradient of this expression twice, we obtain

$$\frac{\partial^2 \zeta_0}{\partial x_i \partial x_j} = G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) \mathrm{d}^3\mathbf{x}', \tag{52}$$

Comparing eqs. (45) and (52), the contribution of u_j to γ_i^1 is $-u_j \, \partial^2 \zeta_0 / \partial x_i \partial x_j$, whose magnitude is set by the trace $\nabla^2 \zeta_0 = 4\pi G \rho$ given by Poisson's equation, and hence we arrive at the desired scaling estimate

$$\rho \gamma_i^1 \sim 4\pi G \overline{\rho}^2 U. \tag{53}$$

Using these results, we can quantify the relative importance of the terms within the force balance. First we have

$$\frac{\text{gravitational forces}}{\text{elastic forces}} = \frac{4\pi G \overline{\rho}^2 U}{\frac{\overline{\mu} U}{L^2}} = \frac{4\pi G \overline{\rho} L^2}{\overline{v}^2}, \tag{54}$$

where $\overline{v} = \sqrt{\overline{\mu}/\overline{\rho}}$ represents a typical elastic wave speed in the model. This ratio scales like L^2 where L is the characteristic length scale of the deformation. We conclude that for deformations occurring over sufficiently small length scales, gravitational forces will be negligible relative to elastic ones. We should then ask whether gravitational forces will ever be of importance within the Earth. To answer this, we note that the largest-scale deformation within the Earth can have L equal to its average radius 6371 km. Recalling that $G = 6.67 \times 10^{-11} \, \mathrm{m}^3 \, \mathrm{kg}^{-1} \, \mathrm{s}^{-2}$, and taking as typical values $\overline{\rho} = 5000 \, \mathrm{kg \, m}^{-3}$ and $\overline{v} = 8000 \, \mathrm{m \, s}^{-1}$, we find

$$\frac{\text{gravitational forces}}{\text{elastic forces}} \sim 3. \tag{55}$$

For such planetary-scale deformations, it follows that gravitational forces are of the same order of magnitude as elastic forces, and so cannot be neglected within the dynamics.

Next, we consider the ratio of the Coriolis and inertial terms, and find

$$\frac{\text{Coriolis term}}{\text{inertial term}} = \frac{\frac{\overline{\rho} \Omega U}{T}}{\frac{\overline{\rho} U}{T^2}} = \Omega T. \tag{56}$$

It follows that the Coriolis terms will be negligible for motions whose time scale is short relative to the rotation period of the Earth. A similar conclusion is readily shown to hold for the centrifugal term. It is known that seismic deformation is limited to periods shorter than about one hour, and so rotational effects play a relatively small (but not

negligible) role in the problem. There are, however, processes at longer periods, e.g. those associated with solid-Earth tides, for which rotation becomes very important.

The conclusions of this scaling analysis can be checked directly by computing synthetic seismograms in a spherically symmetric earth model with gravitation and with all gravitational effects switched off, i.e. with neither the equilibrium gravitational field nor its perturbation included in the equations of motion, as shown in Fig. 2, where attenuation has been neglected so that the long-period waves persist over the six hours shown.[2] Over the full frequency band of the calculation, which extends to 50 mHz, the two seismograms are indistinguishable for the first few hours, and they remain so when band-pass filtered to periods shorter than a few minutes, in keeping with the estimate that gravitational forces are negligible for deformations on length scales much smaller than the radius of the Earth. As the pass band is moved to longer periods, however, differences appear and steadily grow, and in the lowest band, at periods between eight and seventeen minutes, they are substantial: the oscillations are slightly but systematically slower without gravitation, so that the two waveforms drift apart as the long-period surface waves circle the Earth. Even at the shortest periods small differences eventually become visible in the surface waves that have circled the Earth several times, since a fractional change in wave speed of only 10^{-4} accumulates into a noticeable phase shift over the hundreds of cycles they have then completed. These are precisely the deformations whose length scale is comparable to that of the planet, and we shall see in Lecture 23 that the periods of the corresponding free oscillations of the Earth are altered by self-gravitation.

[2] The calculations use the program `yspec`, which solves the equations of motion of this lecture in a spherically symmetric earth model by direct numerical integration in the frequency domain (Al-Attar & Woodhouse 2008).

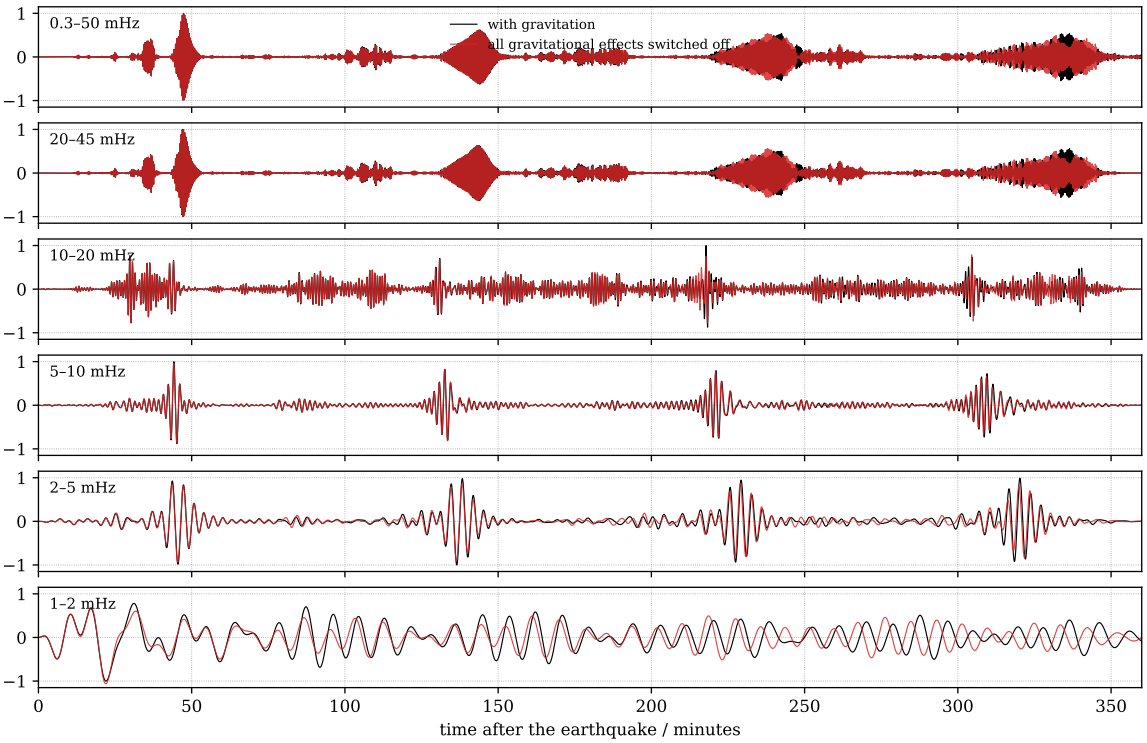


Fig. 2: Synthetic vertical-component acceleration seismograms in the earth model PREM for an earthquake of moment magnitude 8 at 20 km depth, recorded at an epicentral angle of 90°, computed with gravitation (black) and with all gravitational effects switched off (red), i.e. with neither the equilibrium gravitational field nor its perturbation. The top trace shows the full band of the calculation, from 0.3 to 50 mHz, and those below it the same seismogram band-pass filtered in successively lower frequency bands. In each panel both traces are normalised by the maximum of the gravitating one. The calculations neglect attenuation, so that the surface waves continue to circle the Earth undiminished over the six hours shown. The two calculations differ appreciably only in the lowest bands, and there the differences grow with time as the long-period surface waves circle the Earth.

What you need to know and be able to do

(i) Understand the relevant parts of Newtonian gravity, including the manner in which the gravitational potential and acceleration can be transformed into quantities defined on the reference body.

(ii) Know physically what the gravitational binding energy of a planet represents, and how it can be determined.

(iii) Know how the gravitational binding energy is incorporated into the Lagrangian, and how the equations of motion can be obtained. The full derivation is too long to be examined, but you may be asked to reproduce parts of it given appropriate information and guidance.

(iv) Perform scaling arguments to determine the relative size of terms within an equation, and to determine whether some of them might be negligible. This is a tech-

nique required throughout this course, and you should ensure you are happy with it.

Lecture 22: Equilibrium figures

Outline and motivation

In the previous lecture we obtained the equations of motion for a self-gravitating and rotating earth model, and showed that these effects are relevant at sufficiently long periods. A particular conclusion was that the equilibrium state of such an earth model **cannot be stress-free** as has been assumed in earlier parts of the course. In this lecture we investigate the constraints imposed by the equilibrium equations in further detail, and show that they are key to understanding the **observed shapes of planets**.

The equilibrium equations

Recall the equilibrium equations obtained there,

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m - \frac{\partial T^0_{ij}}{\partial x_j} - \rho\gamma^0_i = 0, \tag{1}$$

for a self-gravitating and steadily rotating planet.[1] Here ρ is the density, Ω_i the angular velocity, T^0_{ij} the stress, and γ^0_i the gravitational acceleration. To simplify notation, let us write σ_{ij} for the equilibrium stress, and g_i for the gravitational acceleration. The equilibrium equations then become

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i = 0, \tag{2}$$

which are subject to the boundary conditions $\sigma_{ij}\hat{n}_j = 0$ on ∂M. Though these equations were derived from elasticity, the force balance expressed is valid in more general materials. This is important because over long time scales the Earth or Earth-like planets are not elastic.

If we knew the shape, density distribution, and angular velocity of the planet, then all terms in the equilibrium equations would be fixed except for the stress tensor. Recalling from Lecture 14 that equilibrium stress tensors are necessarily symmetric, we see that we have only three equations for six unknown functions. The problem is, therefore, underdetermined, and we cannot expect to uniquely determine the stress tensor.[2]

[1] Though our focus in this course is on the Earth, the material of this lecture applies so naturally to other planets that it seems sensible to generalise the discussion somewhat.

[2] This counting argument is suggestive but not entirely correct. Rather, the relevant point is that there exist non-zero symmetric tensor fields having vanishing divergence. This can be compared to divergence-free vector fields with which you will be very familiar (i.e. the divergence of a curl is zero). In fact, an extension of the curl operator exists that generates divergence-free symmetric tensor fields.

Hydrostatic equilibrium

You have seen in the first part of this course that over thousands of years the interior of the Earth flows like a fluid, this being due to solid-state creep. Considering for simplicity a Newtonian fluid, we know that the stress tensor takes the form

$$\sigma_{ij} = -p\,\delta_{ij} + \eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\frac{\partial v_k}{\partial x_k}\delta_{ij}\right), \tag{3}$$

where p is the pressure, η the viscosity, and v_i the velocity.[3] Note that the given expression is valid in a **compressible fluid**, but it reduces to that for an incompressible one by requiring the velocity to be divergence-free. If the fluid is not flowing, then the stress tensor takes the simple hydrostatic form

$$\sigma_{ij} = -p\,\delta_{ij}, \tag{4}$$

and the same is true for non-Newtonian fluids. Such an expression must also be approximately valid if the fluid is flowing sufficiently slowly. To get a sense of this approximation for the Earth, we note that the size of the non-hydrostatic stress associated with mantle convection scales like $\eta V/L$ with $\eta \sim 10^{21}\,\mathrm{Pa\,s}$ a typical viscosity, $V \sim 10^{-9}\,\mathrm{m\,s^{-1}}$ a typical speed of a few centimetres per year, and $L \sim 10^{6}\,\mathrm{m}$ the length scale. From this we find that deviatoric stresses are around $\eta V/L \sim 10^{6}\,\mathrm{Pa}$. Pressures in the mantle are, however, known to be of order $10^{10}\,\mathrm{Pa}$,[4] some four orders of magnitude larger, and hence the stress is quite close to being hydrostatic.

This motivates us to look for **hydrostatic** solutions of the equilibrium equations. Putting eq. (4) into eq. (2) we obtain

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m + \frac{\partial p}{\partial x_i} - \rho g_i = 0, \tag{5}$$

along with the boundary condition $p = 0$ on ∂M. In assuming the stress is hydrostatic we have moved from an underdetermined problem to an overdetermined one, with three equations for a single unknown, p. The form of this hydrostatic equilibrium equation can be usefully simplified by noting first that

$$g_i = -\frac{\partial \phi}{\partial x_i}, \tag{6}$$

with ϕ the equilibrium gravitational potential. Similarly, the centrifugal term can also be written as the gradient of a potential

$$\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m = \frac{\partial \psi}{\partial x_i}, \tag{7}$$

where we have defined

$$\psi = \frac{1}{2}(\Omega_i\Omega_j - \Omega_k\Omega_k\delta_{ij})x_i x_j. \tag{8}$$

[3] The stress here is the Cauchy stress that occurs within spatial formulations of continuum mechanics. But note that at equilibrium the referential and spatial pictures coincide due to our choice of reference body.

[4] Values for the pressure can be calculated once the density profile is known.

Moving to vector notation, the hydrostatic equilibrium equation takes the simple form

$$\nabla p + \rho \nabla \gamma = \mathbf{0}, \tag{9}$$

where we have introduced the **gravity potential**, $\gamma = \phi + \psi$. Note that "gravity" is used here for the **combined effect of gravitational and centrifugal forces**.

An immediate consequence of eq. (9) is that ∇p and $\nabla \gamma$ are parallel, and hence the level surfaces of these fields are coincident. In particular, pressure vanishes on the surface of the planet, and so this must also be an equipotential of gravity. If we take the curl of eq. (9), recalling $\nabla \times \nabla f = \mathbf{0}$ for any scalar field, we find

$$\nabla \rho \times \nabla \gamma = \mathbf{0}, \tag{10}$$

and so ρ and γ also have coincident level surfaces. In conclusion we have seen that for a planet to be in hydrostatic equilibrium the level surfaces of p, ρ, and γ must all be the same, as illustrated in Fig. 1.

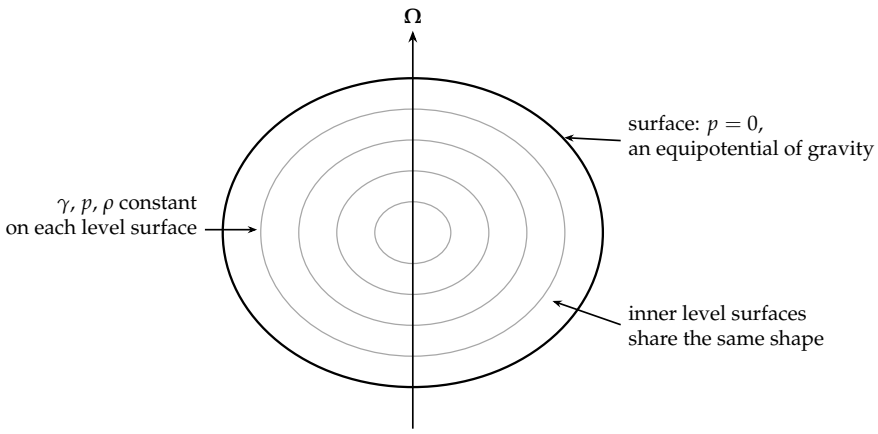


Fig. 1: In a planet in hydrostatic equilibrium the level surfaces of the pressure, the density and the gravity potential all coincide. The outer surface, on which the pressure vanishes, is itself an equipotential of gravity, and every interior level surface has the same character.

Suppose we arbitrarily specified the shape, density variations, and angular velocity of a planet. We could then, in principle, determine its gravitational potential by solving Poisson's equation, and hence find the gravity potential. There would be no reason, however, that this field would have level surfaces that agree with those for the chosen density. This suggests that in requiring a planet to be in hydrostatic equilibrium we are placing rather strong restrictions on, and interrelations between, its shape, density distribution, and angular velocity.

Non-rotating hydrostatic planets

As a starting point, suppose that the planet does not rotate, and hence $\psi = 0$. Assume also that the planet is spherical in shape, and that density varies only with radius. We then know that ϕ is also spherically symmetric, and hence eq. (9) reduces to

$$\nabla p + \rho \frac{\mathrm{d}\phi}{\mathrm{d}r} \hat{\mathbf{r}} = \mathbf{0}, \tag{11}$$

with $\hat{\mathbf{r}}$ a radial unit vector. To solve this equation we can clearly take p to be a function of r alone, and hence need only integrate

$$\frac{\mathrm{d}p}{\mathrm{d}r} + \rho \frac{\mathrm{d}\phi}{\mathrm{d}r} = 0, \tag{12}$$

using the boundary condition $p = 0$ on the surface $r = b$. This argument shows that for a non-rotating planet it is possible to find at least one class of hydrostatic equilibrium states, namely those that are spherically symmetric. From a physical perspective, it seems reasonable that these are the only hydrostatic equilibria possible. This is indeed true, but a mathematical proof of this result is not easy.[5]

Slowly rotating hydrostatic planets

Given that a non-rotating and hydrostatic planet must be spherically symmetric, we would expect that a slowly rotating planet should have an equilibrium figure that is close to spherical. In this section, we show that this is indeed the case. First, however, we should clarify what is meant by **slowly rotating**.

The essential idea is that the ratio of the centrifugal to gravitational potentials should be small. Looking at these quantities on the planet's surface we have approximately

$$\psi \sim \frac{1}{2}\Omega^2 b^2, \tag{13}$$

where Ω is the rotation rate and b the mean radius, while also

$$\phi \sim \frac{4}{3}\pi G \overline{\rho} b^2, \tag{14}$$

with $\overline{\rho}$ the mean density. The ratio of these terms is given by

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim \frac{3\Omega^2}{8\pi G\overline{\rho}}. \tag{15}$$

For the Earth, we have $\overline{\rho} \sim 5000\,\mathrm{kg\,m^{-3}}$ and $\Omega \sim 7 \times 10^{-5}\,\mathrm{s^{-1}}$, and hence we find

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim 2 \times 10^{-3}, \tag{16}$$

which is indeed a fairly small quantity.

We model the rotating planet through a first-order perturbation of a spherical reference model. The density in the reference model will be written ρ_0, and similarly for other quantities. Within the rotating planet we then have

$$\rho = \rho_0 + s\rho_1 + \cdots, \tag{17}$$
$$p = p_0 + s\,p_1 + \cdots, \tag{18}$$
$$\phi = \phi_0 + s\,\phi_1 + \cdots, \tag{19}$$
$$\psi = s\,\psi_1, \tag{20}$$

[5] The case of a homogeneous planet is summarised in the book by Chandrasekhar (1969). The general result, that a non-rotating fluid body in hydrostatic equilibrium under its own gravitation must be spherically symmetric, is due to Lichtenstein (1918, 1933). It can also be seen as an instance of a more general symmetry property of solutions of quasi-linear elliptic equations, which is what the hydrostatic equations amount to once the pressure has been eliminated in favour of the potential.

with s a perturbation parameter introduced for book-keeping purposes. We must also consider the aspherical shape of the planet. For simplicity we neglect internal boundaries,[6] and so need only parameterise the outer surface in the form

$$r(\theta, \varphi) = b + s\, h_1(\theta, \varphi) + \cdots , \tag{21}$$

with (θ, φ) spherical polar co-ordinates.

Putting the above expansions into the equilibrium equations we find at first order

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0}, \tag{22}$$

where $\gamma_1 = \phi_1 + \psi_1$, and we have written $g_0 = \mathrm{d}\phi_0/\mathrm{d}r$ for convenience (so that $g_0 > 0$ is the magnitude of the reference gravity, $\mathbf{g}_0 = -g_0\hat{\mathbf{r}}$). The boundary condition on pressure is applied on the perturbed surface, and hence

$$p[b + s\, h_1(\theta, \varphi) + \cdots , \theta, \varphi] = 0. \tag{23}$$

Expanding this out to first order in s, we arrive at

$$p_1 + \frac{\mathrm{d}p_0}{\mathrm{d}r} h_1 = 0 \tag{24}$$

on the reference surface $r = b$. The zeroth-order equilibrium equation tells us that

$$\frac{\mathrm{d}p_0}{\mathrm{d}r} + \rho_0 g_0 = 0, \tag{25}$$

and hence the boundary condition simplifies to

$$p_1 - \rho_0 g_0 h_1 = 0. \tag{26}$$

We will show that eqs. (22) and (26) allow the density, pressure, and boundary perturbations to be expressed in terms of that for the gravity potential. First, we take the cross product of eq. (22) with the unit vector $\hat{\mathbf{r}}$ to obtain

$$\hat{\mathbf{r}} \times \nabla p_1 + \hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 = \mathbf{0}. \tag{27}$$

Noting that

$$\hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 = \hat{\mathbf{r}} \times \left[\nabla(\rho_0 \gamma_1) - \gamma_1 \frac{\mathrm{d}\rho_0}{\mathrm{d}r} \hat{\mathbf{r}} \right] = \hat{\mathbf{r}} \times \nabla(\rho_0 \gamma_1), \tag{28}$$

eq. (27) can be written

$$\hat{\mathbf{r}} \times \nabla(p_1 + \rho_0 \gamma_1) = \mathbf{0}. \tag{29}$$

This relation tells us that the gradient of $p_1 + \rho_0\gamma_1$ is parallel to $\hat{\mathbf{r}}$, and hence this function depends on radius alone. We are free to assume that the perturbations all average to zero over spherical surfaces,[7] and hence we arrive at

$$p_1 = -\rho_0 \gamma_1, \tag{30}$$

[6] But these can be easily included in the theory.

[7] If not, the spherically symmetric parts of the perturbations could be absorbed into the spherical reference model. In particular, the degree-zero part of ψ_1 is absorbed in this way, and henceforth ψ_1 denotes only its aspherical part, which is proportional to Y_{20} as discussed below.

which expresses the pressure perturbation in terms of that to the gravity potential. From the perturbed boundary condition we then immediately obtain

$$h_1 = -\frac{\gamma_1}{g_0}. \tag{31}$$

For the second stage of the argument, we take the curl of eq. (22) to obtain

$$\frac{\mathrm{d}\rho_0}{\mathrm{d}r}\hat{\mathbf{r}} \times \nabla\gamma_1 + g_0 \nabla\rho_1 \times \hat{\mathbf{r}} = \mathbf{0}. \tag{32}$$

Using the same trick as before, this is equivalent to

$$\hat{\mathbf{r}} \times \nabla\left(\frac{\mathrm{d}\rho_0}{\mathrm{d}r}\gamma_1 - g_0\rho_1\right) = \mathbf{0}, \tag{33}$$

from which we see that the density perturbation is given by

$$\rho_1 = \frac{1}{g_0}\frac{\mathrm{d}\rho_0}{\mathrm{d}r}\gamma_1. \tag{34}$$

To summarise, we started from the hydrostatic equilibrium equations in a slowly rotating planet, and have obtained the following identities

$$p_1 = -\rho_0\gamma_1, \quad h_1 = -\frac{\gamma_1}{g_0}, \quad \rho_1 = \frac{1}{g_0}\frac{\mathrm{d}\rho_0}{\mathrm{d}r}\gamma_1. \tag{35}$$

To complete the discussion, we recall that ϕ_1 is also related to ρ_1 and h_1 through solution of Poisson's equation. Indeed, we have

$$\nabla^2\phi_1 = \begin{cases} 4\pi G\rho_1 & r < b \\ 0 & r > b \end{cases} \tag{36}$$

while the appropriate continuity conditions at $r = b$ are

$$[\phi_1]^+_- = 0, \quad [\hat{\mathbf{r}} \cdot \nabla\phi_1 + 4\pi G\rho_0 h_1]^+_- = 0, \tag{37}$$

whose derivation is set out in the appendix to this lecture. Using eq. (35), we see that Poisson's equation is transformed into

$$\nabla^2\phi_1 = \begin{cases} 4\pi G\frac{1}{g_0}\frac{\mathrm{d}\rho_0}{\mathrm{d}r}(\phi_1 + \psi_1) & r < b \\ 0 & r > b \end{cases} \tag{38}$$

along with the continuity conditions

$$[\phi_1]^+_- = 0, \quad \left[\hat{\mathbf{r}} \cdot \nabla\phi_1 - 4\pi G\rho_0\frac{(\phi_1 + \psi_1)}{g_0}\right]^+_- = 0. \tag{39}$$

Eqs. (38) and (39) constitute a boundary value problem that can be solved for ϕ_1 in terms of ψ_1. Having done this we can then use eq. (35) to determine the other parameters for the perturbed planet. These equations, albeit in a less general form, were first obtained by Clairaut (1743), building on Newton's original discussion of equilibrium

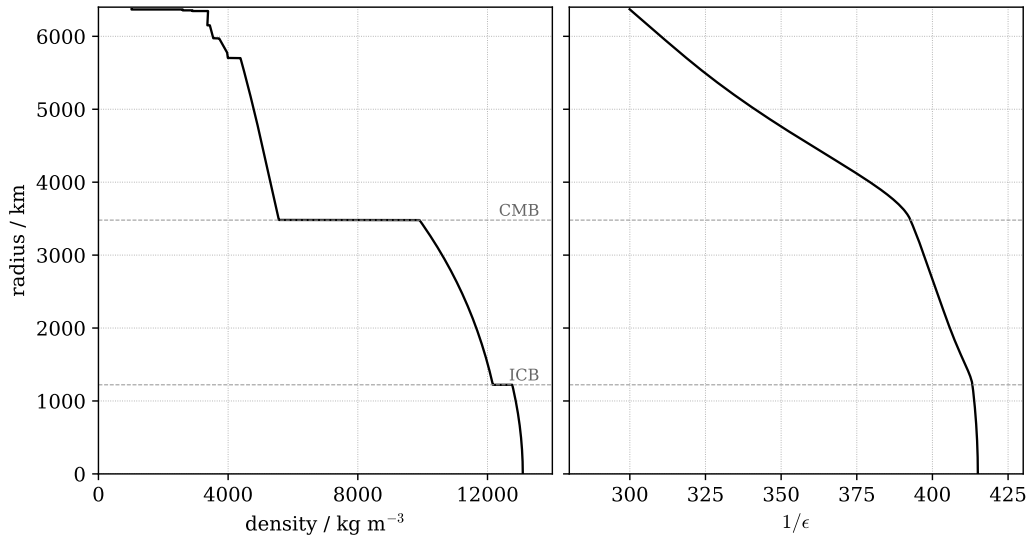


Fig. 2: Left: the Earth's spherically averaged density according to the model PREM, which was determined from free oscillation data along with the mass and moment of inertia. Right: the reciprocal of the ellipticity obtained by numerical solution of Clairaut's equation for this density profile. At the surface $1/\epsilon \approx 300$, while at the centre it is about 415.

figures in the *Principia* (Newton 1687). To implement these equations in practice we must know the variation of spherically averaged density with radius, and in the next two lectures we will see that free oscillation data provide the principal constraint on this profile.

It is worth noting these arguments apply to more general potentials, ψ_1, and not only that linked to the centrifugal force. The theory can, therefore, also be applied to model things like equilibrium tides within gaseous planets. Another point to make is that the equations are linear in ϕ_1 and ψ_1, and hence symmetries of ψ_1 are directly mirrored in ϕ_1 and the other perturbed quantities. In terms of spherical harmonics, the aspherical part of the centrifugal potential, ψ_1, is proportional to Y_{20} when the polar axis is taken along $\mathbf{\Omega}$. It is for this reason that equilibrium figures of slowly rotating planets are found to be oblate spheroids, and characterised in terms of a single radial function, ϵ, known as the ellipticity and defined such that

$$\phi_1 + \psi_1 = \sqrt{\frac{4\pi}{5}}\frac{2}{3} r g_0 \epsilon \, Y_{20}. \tag{40}$$

At the mean surface $r = b$ it can be shown that the ellipticity is such that

$$\epsilon(b) = \frac{\text{equatorial radius} - \text{polar radius}}{\text{mean radius}}, \tag{41}$$

and for the Earth this value is around $1/300$. Fig. 2 shows the ellipticity obtained by solving Clairaut's equation for the Earth's spherically averaged density. The hydrostatic value at the surface is $1/299.9$, slightly smaller than the observed flattening of $1/298.3$, and this small discrepancy is a measure of the non-hydrostatic stresses supported by the mantle.

Using properties of spherical harmonics, it can be shown that eq. (38) reduces to

$$\frac{\mathrm{d}^2\epsilon}{\mathrm{d}r^2} + 8\pi G\rho_0 g_0^{-1}\left(\frac{\mathrm{d}\epsilon}{\mathrm{d}r} + \frac{\epsilon}{r}\right) - \frac{6\epsilon}{r^2} = 0, \tag{42}$$

which is known as Clairaut's equation and is subject to the boundary conditions

$$\left.\frac{\mathrm{d}\epsilon}{\mathrm{d}r}\right|_{r=0} = 0, \quad \left.\frac{\mathrm{d}\epsilon}{\mathrm{d}r}\right|_{r=b} = \frac{1}{b}\left[\frac{5\Omega^2 b^3}{2GM} - 2\epsilon(b)\right], \tag{43}$$

with M the planet's mass. This problem can be solved numerically using a shooting method, whereby the equation is integrated outwards from $r = 0$ and the result rescaled so that the condition at $r = b$ is met, the equation being linear.

Rapidly rotating bodies (non-examinable)

The theory above is restricted to slow rotation, but it is worth describing, without derivations, what is known about the equilibrium figures of a homogeneous fluid body rotating at an arbitrary rate; you will not be asked to reproduce this material. The problem has a long history, beginning with Newton in the seventeenth century and Maclaurin in the eighteenth, and the results are summarised in the book by Chandrasekhar (1969). For a homogeneous body the natural measure of the rotation rate is the dimensionless ratio $\Omega^2/\pi G\rho$, which is the quantity estimated in eq. (15) up to a numerical factor, and it is helpful to describe the figures in terms of the angular momentum of the body rather than its rotation rate, since the angular momentum is what is conserved as a body contracts or spins down.

Maclaurin spheroids. Maclaurin showed in 1742 that an oblate spheroid of any eccentricity is an exact equilibrium figure of a homogeneous fluid for a suitable rotation rate, with $\Omega^2/\pi G\rho$ a function of the eccentricity alone. This function is shown in Fig. 3. It rises from zero for a sphere, in agreement with the slow-rotation theory, but it reaches a maximum of $\Omega^2/\pi G\rho \approx 0.449$ for an eccentricity of about 0.93 and then falls again, so that beyond a certain rotation rate no spheroidal figure exists at all, while below it there are two spheroids for each rotation rate. Along the sequence the angular momentum increases without bound, the highly flattened spheroids rotating slowly but having enormous moments of inertia.

Jacobi ellipsoids. It came as a surprise when Jacobi showed in 1834 that a rotating homogeneous fluid need not be axially symmetric: there is a second family of exact equilibrium figures, the triaxial ellipsoids that now bear his name, which rotate about their shortest axis. As shown in Fig. 3, the Jacobi sequence **bifurcates** from the Maclaurin sequence at an eccentricity of 0.8127, where $\Omega^2/\pi G\rho \approx 0.374$ and the polar axis is 0.58 of the equatorial one. Beyond this point there are therefore two distinct equilibrium figures with the same angular momentum, and the Jacobi ellipsoid has the lower energy. The Maclaurin spheroids become unstable beyond the bifurcation, in the sense that a small triaxial perturbation grows, so long as there is some dissipation to allow the energy to fall, and a slowly contracting fluid body would be expected to pass from the spheroidal to the ellipsoidal sequence as its angular momentum grows. Some members of the two sequences are drawn in Fig. 3.

Pear-shaped figures and fission. Poincaré (1885) showed that the Jacobi sequence has in turn a point of bifurcation, at an axis ratio $a_2/a_1 \approx 0.43$, where $a_1 \geq a_2 \geq$

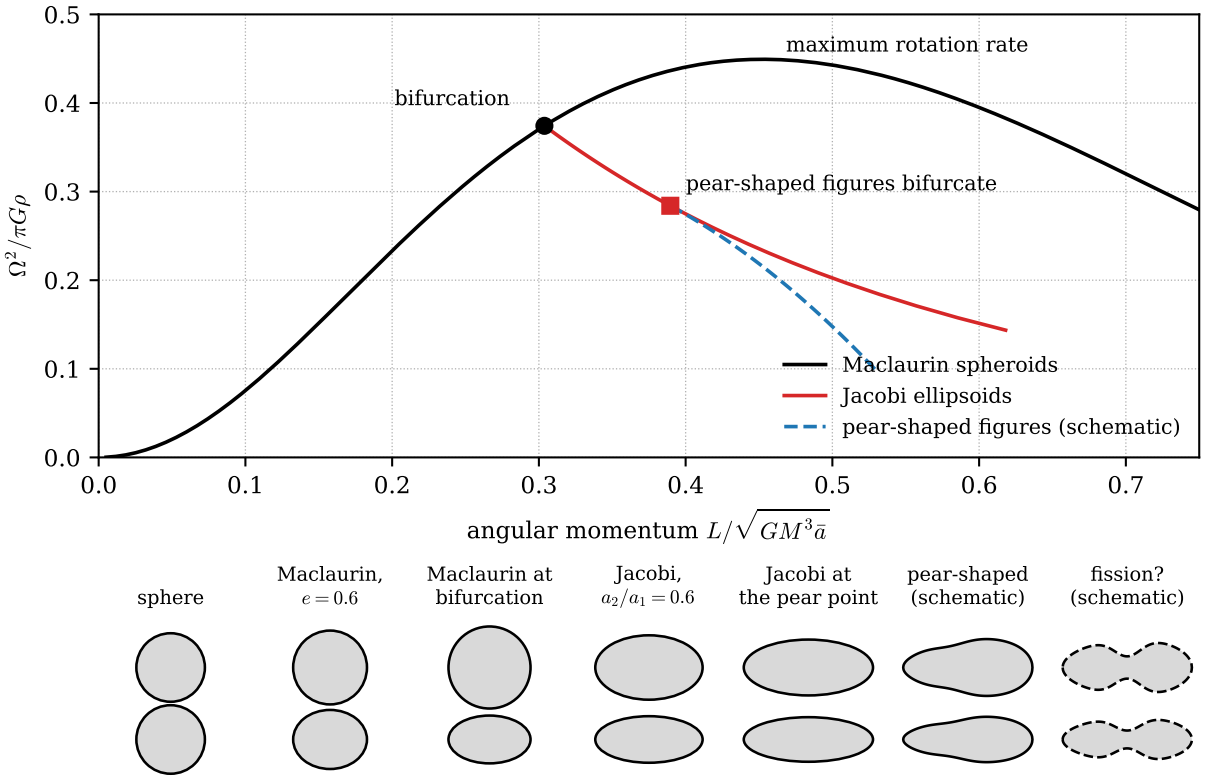


Fig. 3: The Maclaurin and Jacobi sequences of equilibrium figures of a homogeneous rotating fluid, shown as the dimensionless rotation rate $\Omega^2/\pi G\rho$ against the dimensionless angular momentum $L/\sqrt{GM^3\bar{a}}$, where $\bar{a}$ is the mean radius. The Jacobi sequence branches from the Maclaurin sequence at the circle, and the pear-shaped figures of Poincaré branch from the Jacobi sequence at the square. The pear-shaped sequence is known only approximately and is drawn schematically; its figures are unstable. Below are the equatorial (upper) and meridional (lower) sections of some members of the sequences, all with the same volume, followed by sketches of a pear-shaped figure and of the fissioning figure that was once imagined to lie beyond it; no closed form exists for these, and the sketches are only schematic.

a_3 are the semi-axes, from which a sequence of **pear-shaped figures** branches off. It was widely supposed at the time that this sequence might continue to figures with a narrow waist, and finally to two separate bodies, and on this basis George Darwin developed the **fission hypothesis**, in which the Moon was born from a rapidly rotating early Earth and close binary stars from the splitting of a single star. The pear-shaped figures were shown by Liapounov and later by Cartan to be unstable, however, and the fission hypothesis fell from favour; the standard view today is that the Moon formed from the debris of a giant impact on the early Earth. That hypothesis has its own difficulties, notably that the Moon and the Earth are almost indistinguishable in their isotopic composition, which requires the impactor to have been remarkably similar to the Earth. The main objection to fission, that the early Earth would have needed far more angular momentum than the Earth–Moon system now possesses, requires in turn some mechanism to have removed the excess. Neither account is free of ad hoc

elements, and fission has recently attracted renewed interest. Whatever the origin of the Moon, the bifurcation diagram remains the classic example of how a symmetry can be broken as a single parameter, here the angular momentum, is increased.

Haumea. Are there any real bodies that resemble Jacobi ellipsoids? For a planet like the Earth $\Omega^2/\pi G\rho$ is of order 10^{-3}, far below the bifurcation, but in the outer Solar System there is a striking example. The dwarf planet Haumea, which orbits beyond Neptune, rotates once every 3.9 hours, faster than any other large body in the Solar System, and a stellar occultation observed in 2017 showed it to be a triaxial ellipsoid with axes of about 2322, 1704 and 1138 km (Ortiz et al. 2017). As Fig. 4 shows, these axes are almost exactly those of the Jacobi ellipsoid with the same ratio of equatorial axes. Its measured mean density of about 1885 kg m^{-3} gives $\Omega^2/\pi G\rho \approx 0.50$, however, which exceeds the largest rotation rate that any homogeneous equilibrium figure can sustain, and so Haumea cannot be a homogeneous fluid body in equilibrium: it is thought instead to have a denser rocky core beneath an icy mantle, for which the equilibrium figures differ from the classical ones. Its shape is thus a reminder both of the reality of the Jacobi figures and of the importance of internal structure in interpreting the shapes of planets.

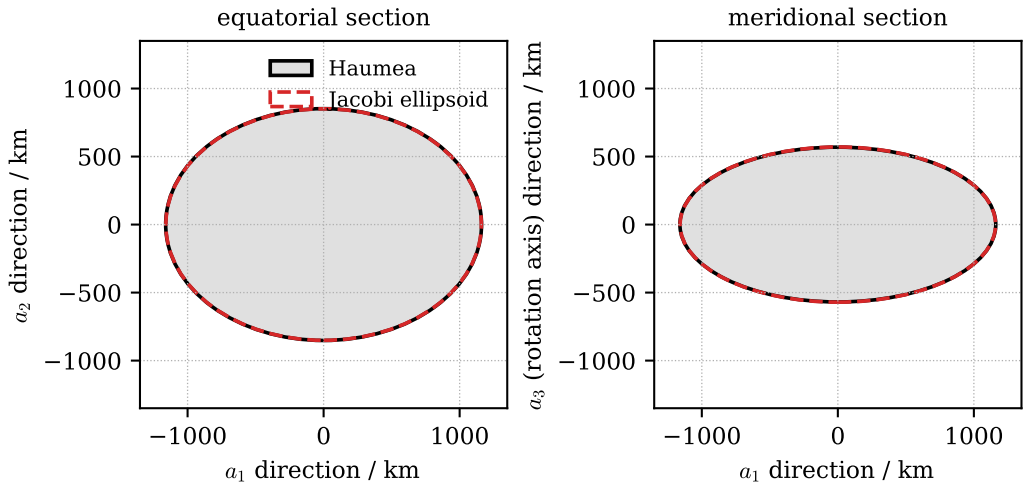


Fig. 4: Equatorial (left) and meridional (right) sections of the dwarf planet Haumea, using the axes measured by Ortiz et al. (2017), compared with those of the Jacobi ellipsoid having the same volume and the same ratio of equatorial axes. The measured axes are $2322 \times 1704 \times 1138$ km, and those of the Jacobi ellipsoid $2319 \times 1702 \times 1141$ km; the two agree to better than one per cent in every axis.

What you need to know and be able to do

(i) Know the form of the equilibrium equations, and that they hold in a general planet and not just an elastic one.

(ii) That the equilibrium equations are underdetermined, and why.

(iii) What a hydrostatic equilibrium state is, and be able to derive the constraints imposed on the level surfaces of p, ρ, and γ.

(iv) Derive the hydrostatic equilibrium state in a non-rotating planet.

(v) Know in outline how solutions can be obtained within a slowly rotating planet. Here you will not need to reproduce the full derivation, but may be asked to perform related calculations with appropriate guidance.

Appendix: perturbation of a boundary condition (non-examinable)

Within the discussion of slowly rotating planets we needed the conditions satisfied by the perturbation to the gravitational potential on a slightly deformed boundary. The calculation is short, but it is easy to get wrong, and so it is set out here. Consider a planet at equilibrium occupying the volume M and with density ρ. Its gravitational potential, ϕ, is a solution of Poisson's equation

$$\nabla^2\phi = 4\pi G\rho,$$

where it is understood that ρ vanishes outside of M. Across the surface ∂M we then have continuity conditions on ϕ and its normal derivative, and finally require that ϕ tend to zero at infinity. Suppose that the planet is only slightly aspherical, with boundary

$$r = b + sh_1(\theta, \varphi) + \cdots,$$

in spherical polar co-ordinates, and its density decomposed as

$$\rho = \rho_0 + s\rho_1 + \cdots,$$

with ρ_0 spherically symmetric. Here s is a perturbation parameter, and the dots denote higher-order terms that are to be neglected. Similarly writing the potential in the form $\phi = \phi_0 + s\phi_1 + \cdots$, we wish to show that across the reference surface $r = b$ we require continuity of both ϕ_1 and

$$\hat{\mathbf{r}} \cdot \nabla\phi_1 + 4\pi G\rho_0 h_1.$$

To see this, consider first the continuity of the potential across the boundary. We require

$$[\phi]^+_- = 0$$

across $r = b + sh_1(\theta, \varphi) + \cdots$, where $[\cdot]^+_-$ denotes the jump in a quantity across the relevant boundary in the direction of the outward unit normal. Expanding this to first-order accuracy we obtain

$$\left[\phi_0 + s\frac{\mathrm{d}\phi_0}{\mathrm{d}r}h_1 + s\phi_1 + \cdots\right]^+_- = 0,$$

across the reference boundary $r = b$. Using the continuity of ϕ_0 and its normal derivative across the reference boundary, this reduces to

$$[\phi_1]^+_- = 0,$$

for $r = b$. To deal with the continuity of the normal derivative, we first write

$$\hat{\mathbf{n}} = \hat{\mathbf{r}} + s\hat{\mathbf{n}}_1 + \cdots,$$

for the outward unit normal to the boundary. The term $\hat{\mathbf{n}}_1$ can be readily expressed in terms of h_1, but we will see that this is not required. Similar to above, the continuity of $\hat{\mathbf{n}} \cdot \nabla\phi$ across the aspherical boundary can be expanded to

$$[\hat{\mathbf{r}} \cdot \nabla\phi_0 + s(\hat{\mathbf{r}} \cdot \nabla)(\hat{\mathbf{r}} \cdot \nabla\phi_0)h_1 + s\hat{\mathbf{n}}_1 \cdot \nabla\phi_0 + s\hat{\mathbf{r}} \cdot \nabla\phi_1 + \cdots]_-^+ = 0,$$

across $r = b$. The zeroth-order term vanishes immediately. We also note that $\hat{\mathbf{n}}_1 \cdot \nabla\phi_0 = (\hat{\mathbf{n}}_1 \cdot \hat{\mathbf{r}})\, \mathrm{d}\phi_0/\mathrm{d}r$ vanishes, because $\hat{\mathbf{n}}_1$ must be orthogonal to $\hat{\mathbf{r}}$ for the perturbed normal to remain a unit vector. The first-order condition therefore becomes

$$[(\hat{\mathbf{r}} \cdot \nabla)(\hat{\mathbf{r}} \cdot \nabla\phi_0)h_1 + \hat{\mathbf{r}} \cdot \nabla\phi_1]_-^+ = 0,$$

at $r = b$. To simplify the first term, we recall that ϕ_0 satisfies the spherically symmetric Poisson equation

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}\phi_0}{\mathrm{d}r}\right) = 4\pi G\rho_0.$$

Using this result, we see that

$$(\hat{\mathbf{r}} \cdot \nabla)(\hat{\mathbf{r}} \cdot \nabla\phi_0) = \frac{\mathrm{d}^2\phi_0}{\mathrm{d}r^2} = 4\pi G\rho_0 - \frac{2}{r}\frac{\mathrm{d}\phi_0}{\mathrm{d}r},$$

which can be substituted into the jump condition to obtain

$$[\hat{\mathbf{r}} \cdot \nabla\phi_1 + 4\pi G\rho_0 h_1]_-^+ = 0,$$

where again the continuity of the normal derivative of ϕ_0 has been applied.

Lecture 23: Free oscillations

Outline and motivation

Having obtained the equations of motion for a self-gravitating earth model, we begin our study of the Earth's **free oscillations** or **normal modes** (the terms are synonymous). The starting point is the development of a concise operator notation that closely resembles what you will know from quantum mechanics. We analyse the associated **eigenvalue problem** in a non-rotating earth model. In particular, the forced response of such an earth model is obtained as an **eigenfunction expansion**, with the frequency-domain response having **resonances** at each of the **eigenfrequencies**. It is due to this fact that observations of the eigenfrequencies can be made from sufficiently long time series, with these data providing important constraints on Earth structure. Indeed, such frequencies form the principal constraint on the Earth's density.

Linearised equations of motion

Within Lecture 21 we obtained the following linearised equations of motion

$$\rho\left(\frac{\partial^2 u_i}{\partial t^2}+2\epsilon_{ijk}\Omega_j\frac{\partial u_k}{\partial t}+\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m\right)-\frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}\right)-\rho\gamma_i^1=-\frac{\partial\overline{S}_{ij}}{\partial x_j}, \quad (1)$$

for a self-gravitating and rotating elastic earth model relative to a co-rotating reference frame. Here ρ is the equilibrium density, u_i the displacement vector, Ω_i the angular velocity, A_{ijkl} the elastic tensor, and γ_i^1 the perturbation to the gravitational acceleration, which is given concretely by

$$\gamma_i^1=-G\int_M\rho'\left(\frac{\delta_{ij}}{[(x_k-x_k')(x_k-x_k')]^{3/2}}-\frac{3(x_i-x_i')(x_j-x_j')}{[(x_k-x_k')(x_k-x_k')]^{5/2}}\right)(u_j-u_j')\,\mathrm{d}^3\mathbf{x}', \quad (2)$$

where it is understood that primed terms are functions of the integration variable, and finally $\overline{S}_{ij}$ is the stress glut associated with a seismic source. The boundary conditions for the problem are

$$A_{ijkl}\frac{\partial u_k}{\partial x_l}\hat{n}_j=\overline{S}_{ij}\hat{n}_j \quad (3)$$

on ∂M. While these equations express quite simple physics, they do have a complicated form, and hence it will help things greatly to find a simpler notation.

As a first step to doing this, we will express the equations in what is known as the **weak form**. Letting w_i be an arbitrary vector field we find from eq. (1) that

$$\int_M\left[\rho\left(\frac{\partial^2 u_i}{\partial t^2}+2\epsilon_{ijk}\Omega_j\frac{\partial u_k}{\partial t}+\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m\right)-\frac{\partial}{\partial x_j}\left(A_{ijkl}\frac{\partial u_k}{\partial x_l}-\overline{S}_{ij}\right)-\rho\gamma_i^1\right]w_i^*\,\mathrm{d}^3\mathbf{x}=0, \quad (4)$$

where we note that $\mathbf{w}$ has been allowed to be complex-valued and * denotes a complex conjugate.[1] Performing an integration by parts and using the boundary conditions this becomes

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m - \gamma_i^1 \right) w_i^* + \left(A_{ijkl}\frac{\partial u_k}{\partial x_l} - \overline{S}_{ij} \right) \frac{\partial w_i^*}{\partial x_j} \right] \mathrm{d}^3\mathbf{x} = 0. \tag{5}$$

We have obtained this result for arbitrary $\mathbf{w}$ whenever $\mathbf{u}$ satisfies eqs. (1) and (3). Conversely, the argument can be reversed to show that if eq. (5) holds for all $\mathbf{w}$, then the field $\mathbf{u}$ must satisfy eqs. (1) and (3), this being the weak form of the problem.

Eq. (5) involves several terms that depend linearly on $\mathbf{u}$ and conjugate-linearly on $\mathbf{w}$, with such objects known as **sesquilinear forms**.[2] For example, if we define

$$\langle \mathbf{w}|P|\mathbf{u}\rangle = \int_M \rho w_i^* u_i \, \mathrm{d}^3\mathbf{x}, \tag{6}$$

then we can write the acceleration term in the weak form as

$$\langle \mathbf{w}|P|\ddot{\mathbf{u}}\rangle, \tag{7}$$

where time derivatives are denoted using dots. The notation is intended to recall bra-ket notation. We will call $\langle \mathbf{w}|P|\mathbf{u}\rangle$ the **kinetic energy form**, and note that it may be associated with an operator, P, that acts on displacement vectors by $(P\mathbf{u})_i = \rho u_i$. In a similar manner we define

$$\langle \mathbf{w}|W|\mathbf{u}\rangle = \int_M 2\rho\epsilon_{ijk} w_i^* \Omega_j u_k \, \mathrm{d}^3\mathbf{x}, \tag{8}$$

which we call the **Coriolis form**, this being associated with an operator, W, defined by $(W\mathbf{u})_i = 2\rho\epsilon_{ijk}\Omega_j u_k$. We finally have the **potential energy form**

$$\langle \mathbf{w}|H|\mathbf{u}\rangle = \int_M \left(\rho\epsilon_{ijk}\epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl}\frac{\partial w_i^*}{\partial x_j}\frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) \mathrm{d}^3\mathbf{x}, \tag{9}$$

and write the force term as

$$\langle \mathbf{w}|\mathbf{f}\rangle = \int_M \overline{S}_{ij}\frac{\partial w_i^*}{\partial x_j} \, \mathrm{d}^3\mathbf{x}. \tag{10}$$

In this manner, we can concisely express the weak form of the equations of motion as

$$\langle \mathbf{w}|P|\ddot{\mathbf{u}}\rangle + \langle \mathbf{w}|W|\dot{\mathbf{u}}\rangle + \langle \mathbf{w}|H|\mathbf{u}\rangle = \langle \mathbf{w}|\mathbf{f}\rangle, \tag{11}$$

which is to hold for all **test functions**, $\mathbf{w}$. From this equation it would be very tempting to write the problem in operator form as

$$P\ddot{\mathbf{u}} + W\dot{\mathbf{u}} + H\mathbf{u} = \mathbf{f}. \tag{12}$$

Here it is, however, important to remember that the problem's boundary conditions are incorporated within the potential energy form and force term. As a result, we should see eq. (12) as only a formal notation; the weak form incorporates the boundary conditions directly, and is generally to be preferred.

[1] The reason for this generality is that later we will consider an associated eigenvalue problem, and in this context complex-valued fields naturally occur.

[2] The prefix "sesqui" (one and a half) is used instead of "bi" to indicate that the form is linear in one argument and conjugate-linear in the other.

Properties of the sesquilinear forms

If we look at the kinetic energy form, then it is clear that

$$\langle \mathbf{w}|P|\mathbf{u}\rangle = \langle \mathbf{u}|P|\mathbf{w}\rangle^*, \tag{13}$$

and we say that the form is **self-adjoint**. Equivalently we can say that the associated operator, P, is self-adjoint. We also note that $\langle \mathbf{u}|P|\mathbf{u}\rangle$ is positive for any non-zero vector field $\mathbf{u}$, and hence the kinetic energy form defines an **inner product**. In contrast, for the Coriolis form, we find that

$$\begin{aligned}\langle \mathbf{w}|W|\mathbf{u}\rangle &= \int_M 2\rho\epsilon_{ijk} w_i^* \Omega_j u_k \,\mathrm{d}^3\mathbf{x} \\ &= -\left(\int_M 2\rho\epsilon_{ijk} u_i^* \Omega_j w_k \,\mathrm{d}^3\mathbf{x}\right)^* = -\langle \mathbf{u}|W|\mathbf{w}\rangle^*, \end{aligned} \tag{14}$$

this meaning that it is **anti-self-adjoint**. This implies, in particular, that $\langle \mathbf{u}|W|\mathbf{u}\rangle$ is purely imaginary for any $\mathbf{u}$.

For the potential energy form it is readily checked that the term associated with elasticity is self-adjoint due to the hyperelastic symmetry $A_{ijkl} = A_{klij}$. Similarly for the centrifugal term we have

$$\begin{aligned}\int_M \rho\epsilon_{ijk}\epsilon_{klm} w_i^* \Omega_j \Omega_l u_m \,\mathrm{d}^3\mathbf{x} &= \left(\int_M \rho\epsilon_{ijk}\epsilon_{klm} w_i \Omega_j \Omega_l u_m^* \,\mathrm{d}^3\mathbf{x}\right)^* \\ &= \left(\int_M \rho\epsilon_{ijk}\epsilon_{klm} u_i^* \Omega_j \Omega_l w_m \,\mathrm{d}^3\mathbf{x}\right)^*, \end{aligned} \tag{15}$$

where to obtain the final equality we have just relabelled and permuted the indices appropriately. What takes more work is to show that

$$\int_M \rho w_i^* \gamma_i^1(\mathbf{u}) \,\mathrm{d}^3\mathbf{x} = \int_M \rho \gamma_i^1(\mathbf{w})^* u_i \,\mathrm{d}^3\mathbf{x}, \tag{16}$$

where we have included arguments to show which displacement vector the gravitational acceleration term corresponds to. The proof of this identity is not difficult, but nor is it enlightening. The end result is that the potential energy form is also self-adjoint.

Eigenfunctions and eigenvalues in a non-rotating earth model

Recall from the scaling arguments in Lecture 21 that while self-gravitation is important for low-frequency seismology, rotation remains only a relatively small effect. For the moment let us, therefore, suppose that the earth model is non-rotating. The equations of motion then read in weak form

$$\langle \mathbf{w}|P|\ddot{\mathbf{u}}\rangle + \langle \mathbf{w}|H|\mathbf{u}\rangle = \langle \mathbf{w}|\mathbf{f}\rangle, \tag{17}$$

which is to hold for any $\mathbf{w}$, and where it is understood that the centrifugal contribution to the potential energy form has been removed.

Suppose first that we set the stress glut equal to zero, and consider a time-harmonic displacement field

$$u_i(\mathbf{x}, t) = s_i(\mathbf{x})\mathrm{e}^{\mathrm{i}\omega t}, \tag{18}$$

with $\mathbf{s}$ the spatial part of the solution, and ω an angular frequency. Putting this into eq. (17) we arrive at

$$-\omega^2\langle \mathbf{w}|P|\mathbf{s}\rangle + \langle \mathbf{w}|H|\mathbf{s}\rangle = 0. \tag{19}$$

This equation has the trivial solution, $\mathbf{s} = 0$, but it is interesting to ask whether it has any non-trivial solutions. As you no doubt recognise, this is an eigenvalue problem with $\mathbf{s}$ the eigenfunction and ω the eigenfrequency. Note, in particular, that the boundary conditions for the problem are built into the weak form of the equations of motion, and hence these eigenfunctions satisfy traction-free boundary conditions on ∂M.

Due to the self-adjointness of the kinetic and potential energy forms, we can quickly establish some expected properties of the eigenfunctions and eigenfrequencies. Let $\mathbf{s}_1$ be an eigenfunction associated with eigenfrequency ω_1, and $\mathbf{s}_2$ an eigenfunction associated with eigenfrequency ω_2. By definition we then have

$$-\omega_1^2\langle \mathbf{w}|P|\mathbf{s}_1\rangle + \langle \mathbf{w}|H|\mathbf{s}_1\rangle = 0, \tag{20}$$

for any $\mathbf{w}$. Taking this test function equal to $\mathbf{s}_2$, and using the self-adjointness of the forms, it follows that

$$0 = -\omega_1^2\langle \mathbf{s}_2|P|\mathbf{s}_1\rangle + \langle \mathbf{s}_2|H|\mathbf{s}_1\rangle = -\omega_1^2\langle \mathbf{s}_1|P|\mathbf{s}_2\rangle^* + \langle \mathbf{s}_1|H|\mathbf{s}_2\rangle^*. \tag{21}$$

Using the eigenvalue equation for $\mathbf{s}_2$ we know that

$$\langle \mathbf{s}_1|H|\mathbf{s}_2\rangle = \omega_2^2\langle \mathbf{s}_1|P|\mathbf{s}_2\rangle, \tag{22}$$

and hence from eq. (21) we have shown that

$$(\omega_1^2 - \omega_2^{2*})\langle \mathbf{s}_1|P|\mathbf{s}_2\rangle^* = 0. \tag{23}$$

Taking $\mathbf{s}_1 = \mathbf{s}_2$, and recalling that $\langle \mathbf{u}|P|\mathbf{u}\rangle$ is positive for any non-zero displacement field, we conclude that $\omega_1^{2*} = \omega_1^2$, which is to say that the **squared eigenfrequencies are real**. Next, if we suppose that $\omega_1^2 \neq \omega_2^2$, then we must have

$$\langle \mathbf{s}_1|P|\mathbf{s}_2\rangle = 0, \tag{24}$$

this meaning that eigenfunctions with different squared eigenfrequencies are **orthogonal relative to the inner product defined by the kinetic energy form**. Orthogonality relative to such a "mass-weighted" inner product should be familiar from your prior study of linear oscillations of finite-dimensional mechanical systems.

The following additional properties of these eigenfunctions and eigenfrequencies will be stated without proof:[3]

1. Each squared eigenfrequency is associated with a **finite-dimensional eigenspace**. Eigenfunctions within the same eigenspace need not be orthogonal, but we can always select a basis such that they are.

[3] The validity of these results in solid earth models can be established rigorously based on the theory of elliptic partial differential operators. Their proof, and indeed general validity, within models containing inviscid fluid regions like the outer core remains an open problem.

2. There is a **countable infinity** of squared eigenfrequencies. Their values are **bounded from below**, and **accumulate only at infinity**. The latter point means that in a finite interval there can only be a finite number of squared eigenfrequencies.

3. The eigenfunctions form a **complete basis** for vector fields in the earth model. This means that an arbitrary vector field can be expressed as a convergent sum of the eigenfunctions, with the necessary expansion coefficients readily determined due to their orthogonality.

To proceed, we let $k = 0, 1, \ldots$ label the different squared eigenfrequencies, ω_k^2. For each k, there is then a finite-dimensional eigenspace, and we use a second index m to label the elements of its basis. Note that the range of m can vary with k, but we leave this implicit within the notation. The mth eigenfunction belonging to the kth eigenspace will be written $|km\rangle$ in a manner familiar from quantum mechanics. These eigenfunctions can be orthonormalised relative to the kinetic energy form, this meaning that

$$\langle km|P|k'm'\rangle = \delta_{kk'}\delta_{mm'}, \tag{25}$$

for all possible indices. Using this notation, we can then express the completeness of the eigenfunctions by writing an arbitrary vector field $\mathbf{u}$ in the form

$$\mathbf{u} = \sum_{km} \langle km|P|\mathbf{u}\rangle |km\rangle, \tag{26}$$

where $\langle km|P|\mathbf{u}\rangle$ is the expansion coefficient of the (k, m)th term.

Eigenfunction expansions

Recall that the equations of motion in weak form are given by

$$\langle \mathbf{w}|P|\ddot{\mathbf{u}}\rangle + \langle \mathbf{w}|H|\mathbf{u}\rangle = \langle \mathbf{w}|\mathbf{f}\rangle, \tag{27}$$

for any test function, $\mathbf{w}$. These equations can be usefully studied in the frequency domain, where we use the convention

$$\tilde{\mathbf{u}}(\mathbf{x}, \omega) = \int_0^\infty \mathbf{u}(\mathbf{x}, t) \mathrm{e}^{-\mathrm{i}\omega t}\, \mathrm{d}t, \quad \mathbf{u}(\mathbf{x}, t) = \frac{1}{2\pi} \int_{-\infty}^\infty \tilde{\mathbf{u}}(\mathbf{x}, \omega) \mathrm{e}^{\mathrm{i}\omega t}\, \mathrm{d}\omega, \tag{28}$$

noting that the initial conditions that $\mathbf{u}$ and $\dot{\mathbf{u}}$ vanish for $t \leq 0$ have been built into the forward transformation. It is readily seen that the Fourier transformed equations of motion take the form

$$-\omega^2 \langle \mathbf{w}|P|\tilde{\mathbf{u}}\rangle + \langle \mathbf{w}|H|\tilde{\mathbf{u}}\rangle = \langle \mathbf{w}|\tilde{\mathbf{f}}\rangle. \tag{29}$$

Using the completeness of the eigenfunctions, we know that we can write

$$\tilde{\mathbf{u}} = \sum_{k'm'} \langle k'm'|P|\tilde{\mathbf{u}}\rangle |k'm'\rangle, \tag{30}$$

for some frequency-dependent coefficients, $\langle k'm'|P|\tilde{\mathbf{u}}\rangle$, that are to be determined. Putting this eigenfunction expansion into eq. (29) we find

$$\sum_{k'm'} [-\omega^2 \langle \mathbf{w}|P|k'm'\rangle + \langle \mathbf{w}|H|k'm'\rangle] \langle k'm'|P|\tilde{\mathbf{u}}\rangle = \langle \mathbf{w}|\tilde{\mathbf{f}}\rangle, \tag{31}$$

which can be simplified using the eigenvalue problem satisfied by $|k'm'\rangle$ to read

$$\sum_{k'm'}(-\omega^2+\omega_{k'}^2)\langle \mathbf{w}|P|k'm'\rangle\langle k'm'|P|\tilde{\mathbf{u}}\rangle = \langle \mathbf{w}|\tilde{\mathbf{f}}\rangle. \tag{32}$$

Finally, we take the test function equal to $|km\rangle$, and use the orthonormality condition in eq. (25) to find that the (k,m)th coefficient of $\tilde{\mathbf{u}}$ is given by

$$\langle km|P|\tilde{\mathbf{u}}\rangle = \frac{\langle km|\tilde{\mathbf{f}}\rangle}{\omega_k^2-\omega^2}. \tag{33}$$

In this manner, we have obtained the **eigenfunction expansion solution**

$$\tilde{\mathbf{u}} = \sum_{km}\frac{\langle km|\tilde{\mathbf{f}}\rangle}{\omega_k^2-\omega^2}|km\rangle, \tag{34}$$

for the frequency-domain problem. Inversion of this solution into the time domain can be readily done using contour integration. Such techniques are not necessary for this course, but in the second problem set you will be given and make use of the resulting time-domain solution.

Looking at eq. (34), it is notable that there are singularities at $\pm\omega_k$ for each k. Supposing for the moment that the squared eigenfrequencies are all positive, these singularities lie along the real line. Physically, we are seeing **resonances** in the frequency-domain response when the system is driven at one of its natural frequencies. From an actual seismic record we could take its Fourier transform, look for associated resonance peaks (see Fig. 2), and hence measure some of the eigenfrequencies of the Earth. These frequencies are determined solely by Earth structure, and hence their measurement provides a possible means to avoid the **source-structure trade-off** that is ever present in seismology.

To consider such measurements further, it is important to note that recorded seismograms are not of an infinite length, and so their Fourier transform will not yield eq. (34) exactly. To understand what happens with a finite-length time series, we can introduce a windowing function, $h(t)$, that is non-zero only in the range $[0,T]$, and so write the observed time series as

$$\mathbf{u}_{\text{obs}}(t) = h(t)\mathbf{u}(t), \tag{35}$$

where we neglect spatial arguments for simplicity. Taking the Fourier transform of such a time series and applying the convolution theorem we find

$$\tilde{\mathbf{u}}_{\text{obs}}(\omega) \propto \int_{-\infty}^{\infty}\tilde{\mathbf{u}}(\omega')\tilde{h}(\omega-\omega')\,\mathrm{d}\omega', \tag{36}$$

with the right-hand side being proportional to the convolution of the two Fourier transforms. The end result is that the singularities of the Fourier-domain solution in eq. (34) are smoothed out within real observations. The extent of the smoothing is controlled primarily by the length of the time series, with the support of $\tilde{h}$ being largely confined to an interval whose width scales like T^{-1}. This is, of course, a familiar result; longer time series provide better resolution in the frequency domain. Within a more complete treatment there is also slight attenuation that shifts the singularities into the upper half-plane, and this too smooths the resonance peaks.

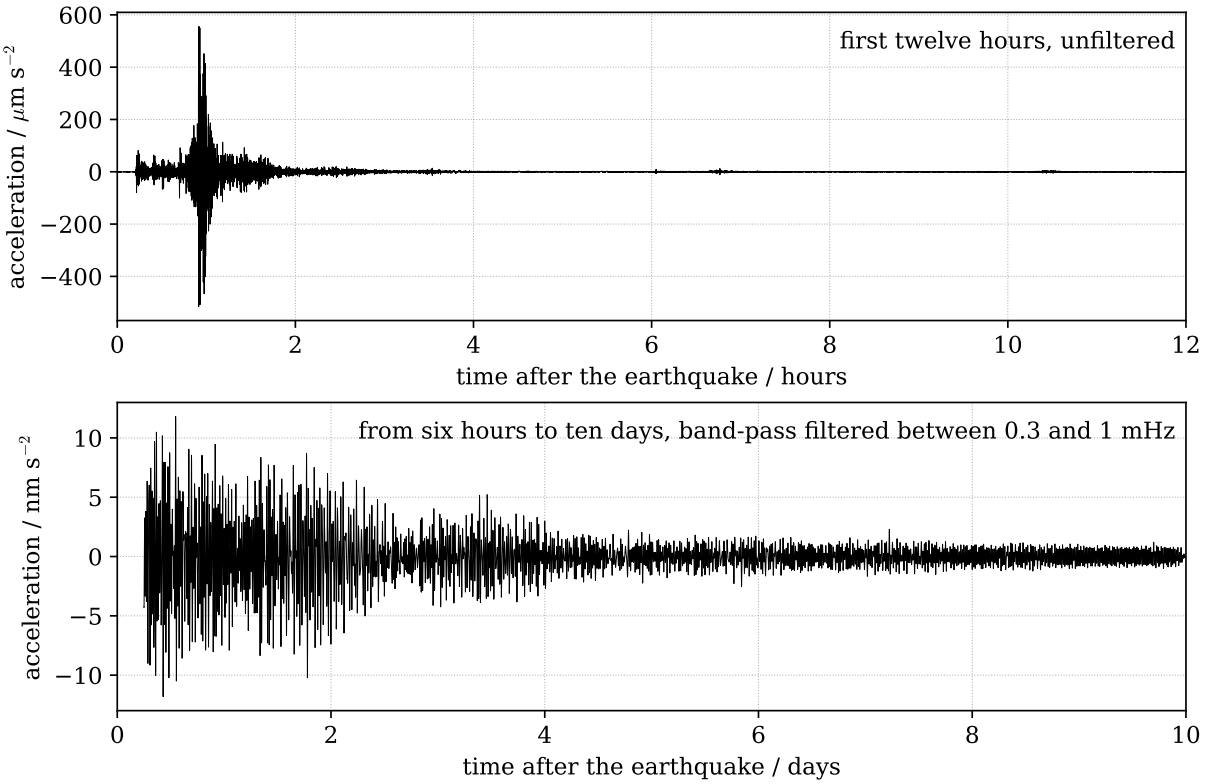


Fig. 1: Vertical-component acceleration recorded at the Black Forest Observatory in Germany following the 2011 Tohoku earthquake in Japan (moment magnitude 9.1). The upper panel shows the first twelve hours of the record, dominated by the surface waves, which circle the Earth several times. The lower panel shows the record from six hours to ten days after the earthquake, band-pass filtered between 0.3 and 1 mHz to isolate the lowest-frequency free oscillations. Their amplitude decays in a roughly exponential fashion over several days. This decay is due to attenuation within the Earth, but we ignore this effect within our current discussion.

It is for this reason that observational work on the Earth's free oscillations is concerned only with the largest earthquakes, the response to which can remain above background noise for many days, as illustrated in Fig. 1. Even so, within observed spectra we cannot always identify individual mode peaks, and this is especially true in cases of near degeneracy that we will examine in more detail in the next lecture.

Gravitational stability

Within the previous section we assumed that the squared eigenfrequencies were all positive, with the result that each term in the mode sum is associated with two resonances on the real axis, and hence oscillatory contributions to the time-domain solution. If, instead, a squared eigenfrequency, ω_k^2, were negative then there would be two resonances on the imaginary axis, resulting in exponentially growing and decaying terms in the time-domain solution. If a squared eigenfrequency vanished, then there would be a double root of the denominator at the origin, and this can be shown to result in a term in the time-domain solution that grows linearly with time. The under-

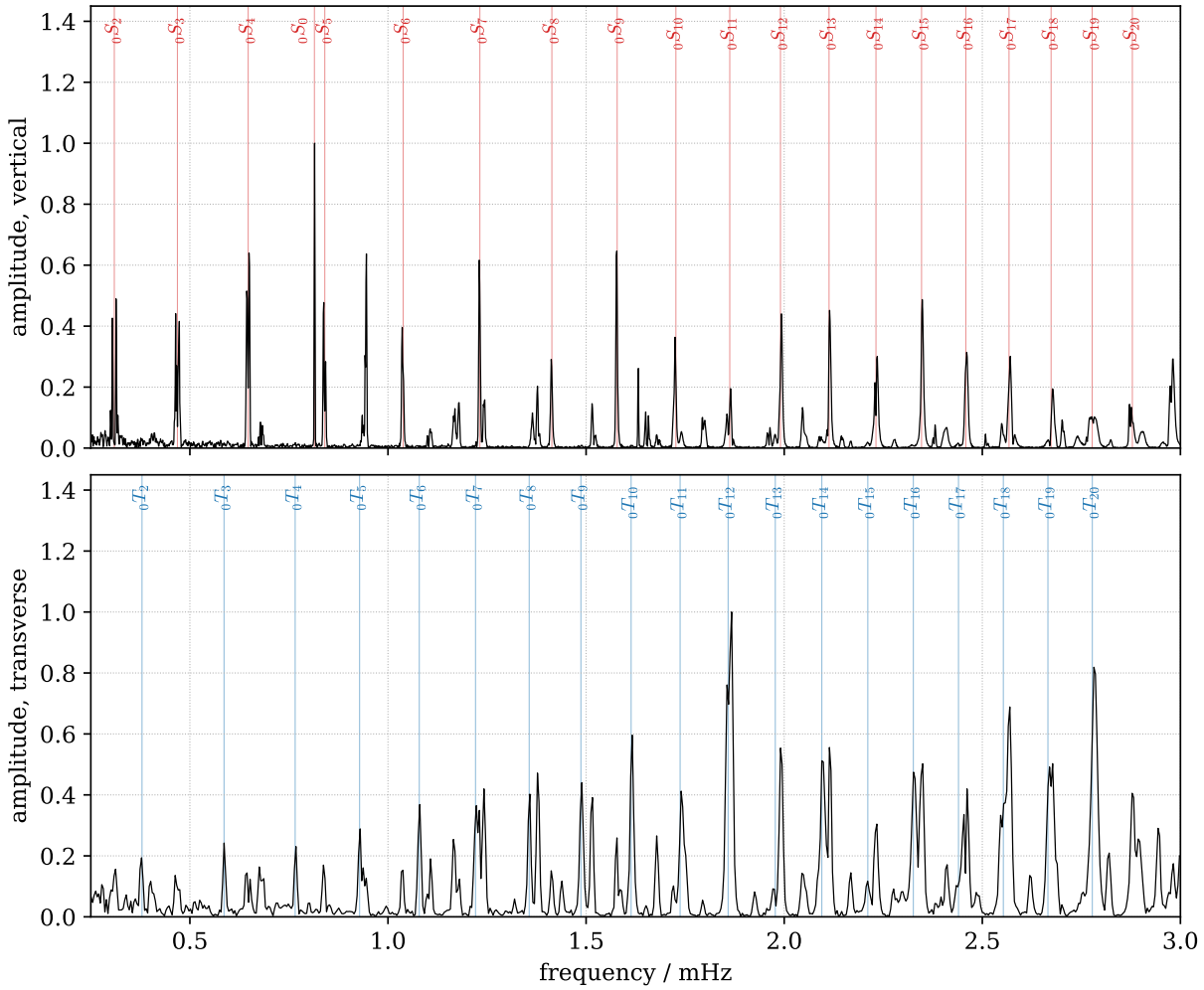


Fig. 2: Amplitude spectra of the record in Fig. 1. The upper panel is for the full ten days of the vertical component, and the lower panel for the first three days of the transverse horizontal component (perpendicular to the great circle from the earthquake), a shorter window being used because horizontal records are noisier at low frequencies. The vertical lines mark the frequencies of the fundamental spheroidal modes ${}_0S_l$ (upper panel) and toroidal modes ${}_0T_l$ (lower panel) in the earth model PREM; the largest peak in the upper panel is the radial mode ${}_0S_0$, and most of the unlabelled peaks are overtones. In some cases individual mode peaks can be identified, but at higher frequencies there is substantial overlap of the peaks due both to attenuation and an insufficiently long time series. The lowest-frequency peaks are also visibly split into several closely spaced peaks, an effect of the Earth's rotation and asphericity that we return to in the next lecture.

lying issue here is whether the earth model is **stable to small perturbations**, with its behaviour being **encoded within the sign of its squared eigenfrequencies**.

Recalling the form of the eigenvalue problem

$$-\omega_k^2 \langle \mathbf{w}|P|km\rangle + \langle \mathbf{w}|H|km\rangle = 0, \tag{37}$$

we can set the test function equal to $|km\rangle$ to obtain

$$\omega_k^2 = \langle km|H|km\rangle, \tag{38}$$

where we have made use of the eigenfunction normalisation condition. It follows that the sign of the squared eigenfrequency is set by the potential energy form.

In the non-gravitating case, this term is given by

$$\langle \mathbf{u}|H|\mathbf{u}\rangle = \int_M A_{ijkl}\frac{\partial u_i^*}{\partial x_j}\frac{\partial u_k}{\partial x_l}\,\mathrm{d}^3\mathbf{x}, \tag{39}$$

which is real due to the hyperelastic symmetry. We know from the first problem set that in the absence of pre-stress the elastic tensor has the additional symmetries $A_{ijkl} = A_{jikl} = A_{ijlk}$, and hence the above expression becomes

$$\langle \mathbf{u}|H|\mathbf{u}\rangle = \frac{1}{4}\int_M A_{ijkl}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right)^*\left(\frac{\partial u_k}{\partial x_l}+\frac{\partial u_l}{\partial x_k}\right)\mathrm{d}^3\mathbf{x}, \tag{40}$$

where we have explicitly symmetrised the spatial derivatives. We will say that the elastic tensor is pointwise stable at $\mathbf{x}$ if

$$A_{ijkl}(\mathbf{x})e_{ij}^*e_{kl} \geq c_0\, e_{ij}^*e_{ij}, \tag{41}$$

for some constant $c_0 > 0$, and all symmetric tensors e_{ij}. If such a condition holds at all points in the earth model, then we say that the elastic tensor is **uniformly pointwise stable**. Assuming this is true, we then have

$$\langle \mathbf{u}|H|\mathbf{u}\rangle = \frac{1}{4}\int_M A_{ijkl}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right)^*\left(\frac{\partial u_k}{\partial x_l}+\frac{\partial u_l}{\partial x_k}\right)\mathrm{d}^3\mathbf{x} > 0, \tag{42}$$

so long as the linearised strain $e_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right)$ is not everywhere equal to zero. It is not difficult to show that e_{ij} vanishes identically for a non-trivial displacement vector if and only if it takes the form

$$u_i(\mathbf{x}) = a_i + B_{ij}x_j, \tag{43}$$

with a_i a constant vector, and B_{ij} an anti-symmetric matrix. Such a displacement vector corresponds to an **infinitesimal rigid body motion**, and hence this result makes physical sense if we recall the principle of material frame indifference. It can be shown that the pointwise stability condition holds in an isotropic medium if and only if the bulk and shear moduli are both positive, and this in turn implies that P- and S-waves have positive phase speeds with $\alpha > \beta$.

To summarise, we have seen that in the case of a non-gravitating earth model the squared eigenfrequencies are all non-negative so long as the elastic tensor is uniformly pointwise stable. There is a six-dimensional eigenspace at zero frequency associated with rigid body motions. Recalling, however, that the force term in the problem is

$$\langle \mathbf{w}|\mathbf{f}\rangle = \int_M \overline{S}_{ij}\frac{\partial w_i^*}{\partial x_j}\,\mathrm{d}^3\mathbf{x}, \tag{44}$$

with $\overline{S}_{ij}$ symmetric, we see that these **trivial modes** cannot be excited by a seismic source. Thus under a reasonable restriction on the elastic tensor, we see that a non-gravitating earth model is always stable to internal perturbations.

With gravitation added to the problem there remains a six-dimensional eigenspace at zero frequency associated with rigid body motions. The issue of stability of the remaining modes is, however, complicated by the fact that the potential energy has a gravitational contribution. While it is possible to examine this problem quantitatively, it requires some fairly advanced techniques. The physically important point, however, is that **self-gravitation is an inherently destabilising force**. Consider, for example, a spherically symmetric and self-gravitating planet at equilibrium. If we could compress this planet uniformly, then there would be an increase in elastic potential energy. At the same time, the mass of the planet has been brought into a smaller volume, and hence the gravitational binding energy must decrease.[4] The net effect on the potential energy depends on the relative magnitude of these terms. If the elastic constants are sufficiently large, then the elastic term will dominate and the planet will be stable with respect to such a perturbation. But if gravity is sufficiently strong then the planet will keep pulling itself inwards until non-linear effects take over. A simple version of this argument for a homogeneous planet is worked through in the second problem set. In practice the question is settled numerically: for realistic earth models such as PREM, calculations of the kind shown in the next lecture return no negative squared eigenfrequencies.

What you need to know and be able to do

(i) The weak formulation of the equations of motion using bra-ket-like notation, and the properties of the different sesquilinear forms.

(ii) How to derive the reality of the squared eigenfrequencies and orthogonality of eigenfunctions in distinct eigenspaces. State without proof the remaining important properties of the eigenvalue problem.

(iii) How to obtain a solution of the forced equations of motion in a non-rotating earth model as an eigenfunction expansion in the frequency domain. You do not need to know how to recover the time-domain solution, but may be asked to verify that such a solution works.

(iv) That eigenfrequencies are associated with resonance peaks in the amplitude spectra obtained from seismograms, and why it is necessary to look at data from large earthquakes to obtain adequate resolution.

(v) What it means for a non-rotating earth model to be stable in terms of the eigenfrequencies. You should understand the quantitative argument for the non-gravitating case, and the qualitative one when gravity is included.

[4] Recall that this is a negative quantity, and hence it has increased in magnitude.

Lecture 24: Mode splitting and coupling

Outline and motivation

In this final lecture on seismology we consider the free oscillations of the Earth in further detail. Last time we established the general theory, and saw how eigenfrequencies can be observed from the Fourier transforms of sufficiently long seismograms. Today we first consider the eigenvalue problem in a **spherically symmetric earth model** for which the eigenfunctions can be calculated with relative ease. An important point is that the eigenspaces in a spherically symmetric model are **degenerate**, each having dimension $2l+1$ for some integer degree $l \geq 0$. This material depends on the properties of **vector spherical harmonics**, but you are not expected to know the details. We then show using **first-order perturbation theory** how in the real Earth this degeneracy is **split by rotation and lateral heterogeneity**, and how the resulting first-order theory is generalised by mode coupling. To conclude, an overview is given of radial and lateral variations of density within the Earth, with some current research questions highlighted.

Vector spherical harmonics

To discuss the eigenfunctions of a spherically symmetric earth model we need to review **spherical harmonic expansions** of scalar and vector functions. The emphasis here is only on establishing notations that will be useful later. You will **not be expected to carry out any detailed calculations with either scalar or vector spherical harmonics**.

Using spherical polar co-ordinates (r, θ, φ) the gradient operator is given by

$$\nabla = \hat{\mathbf{r}}\frac{\partial}{\partial r} + \frac{1}{r}\hat{\boldsymbol{\theta}}\frac{\partial}{\partial \theta} + \frac{1}{r\sin\theta}\hat{\boldsymbol{\varphi}}\frac{\partial}{\partial \varphi}, \tag{1}$$

where, for example, $\hat{\mathbf{r}}$ is a unit vector in the radial direction. This expression can be conveniently written

$$\nabla = \hat{\mathbf{r}}\frac{\partial}{\partial r} + \frac{1}{r}\nabla_1, \tag{2}$$

which acts to define the **tangential gradient operator**, ∇_1, for functions defined on the unit sphere, $\mathbb{S}^2$. In terms of ∇_1 we can define related operators $\hat{\mathbf{r}} \times \nabla_1$ and ∇_1^2. Up to suitable scale factors, the first is the quantum mechanical **angular momentum operator**, $\hat{\mathbf{L}}$, and the second the **total squared angular momentum**, $\hat{L}^2$.

Spherical harmonic functions are simultaneous eigenfunctions for the **commuting** and **self-adjoint operators** ∇_1^2 and $\frac{1}{\mathrm{i}}\frac{\partial}{\partial \varphi}$, the latter being proportional to the z-**component** of $\hat{\mathbf{r}} \times \nabla_1$. The defining relations are

$$\nabla_1^2 Y_{lm} = -l(l+1)Y_{lm}, \quad \frac{1}{\mathrm{i}}\frac{\partial Y_{lm}}{\partial \varphi} = mY_{lm}, \tag{3}$$

where l is the **degree** of the spherical harmonic which takes non-negative integer values, while for fixed l the **order**, m, is an integer such that $-l \leq m \leq l$. Note that the eigenspace of ∇_1^2 with degree l is $(2l+1)$-fold degenerate, this being characteristic of the eigenspaces of **rotationally invariant operators**. The spherical harmonics satisfy the orthonormality relation

$$\int_{\mathbb{S}^2} Y_{lm}^* Y_{l'm'} \, \mathrm{d}S = \delta_{ll'}\delta_{mm'}. \tag{4}$$

They also form a **complete basis** for suitably regular functions f on $\mathbb{S}^2$, and hence we can form the spherical harmonic expansion

$$f = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} f_{lm} Y_{lm}, \tag{5}$$

where the expansion coefficients are

$$f_{lm} = \int_{\mathbb{S}^2} Y_{lm}^* f \, \mathrm{d}S. \tag{6}$$

Usually we will write such spherical harmonic expansions simply as $\sum_{lm} f_{lm} Y_{lm}$ leaving implicit the range of the degree and order.

To work with vector fields it is convenient to introduce the **vector spherical harmonics** which are defined by

$$\mathbf{A}_{lm} = Y_{lm}\hat{\mathbf{r}}, \quad \mathbf{B}_{lm} = \nabla_1 Y_{lm}, \quad \mathbf{C}_{lm} = -\hat{\mathbf{r}} \times \nabla_1 Y_{lm}. \tag{7}$$

The following orthogonality relations can be established:

$$\int_{\mathbb{S}^2} \mathbf{A}_{l'm'}^* \cdot \mathbf{A}_{lm} \, \mathrm{d}S = \delta_{ll'}\delta_{mm'}, \tag{8}$$

$$\int_{\mathbb{S}^2} \mathbf{B}_{l'm'}^* \cdot \mathbf{B}_{lm} \, \mathrm{d}S = l(l+1)\delta_{ll'}\delta_{mm'}, \tag{9}$$

$$\int_{\mathbb{S}^2} \mathbf{C}_{l'm'}^* \cdot \mathbf{C}_{lm} \, \mathrm{d}S = l(l+1)\delta_{ll'}\delta_{mm'}, \tag{10}$$

where $\cdot$ denotes the usual real inner product $\mathbf{a} \cdot \mathbf{b} = a_i b_i$ for vectors, while all mixed integrals, such as $\int_{\mathbb{S}^2} \mathbf{A}_{l'm'}^* \cdot \mathbf{C}_{lm} \, \mathrm{d}S$, vanish. Note that $\mathbf{B}_{lm}$ and $\mathbf{C}_{lm}$ could be scaled such that these functions are orthonormal, but this is not necessary. It can be shown that the vector spherical harmonics are complete for vector fields defined on $\mathbb{S}^2$, this meaning that such a $\mathbf{u}$ can be expanded as

$$\mathbf{u} = \sum_{lm} (U_{lm}\mathbf{A}_{lm} + V_{lm}\mathbf{B}_{lm} + W_{lm}\mathbf{C}_{lm}), \tag{11}$$

with coefficients U_{lm}, V_{lm} and W_{lm} that can be expressed in terms of $\mathbf{u}$ using the above orthogonality relations.

Spheroidal and toroidal modes

In a non-rotating earth model, we recall that the eigenvalue problem takes the form

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0, \tag{12}$$

where ω is the eigenfrequency, $\mathbf{s}$ the eigenfunction, and $\mathbf{w}$ is an arbitrary test function. When the material parameters ρ and A_{ijkl} are spherically symmetric,[1] it can be shown that the eigenfunctions are of two possible types known as **toroidal** and **spheroidal modes**. This can be established either through a direct but lengthy calculation using spherical harmonics, or using representation theory for the rotation group. For reasons of both time and difficulty, we here just summarise the eventual results.

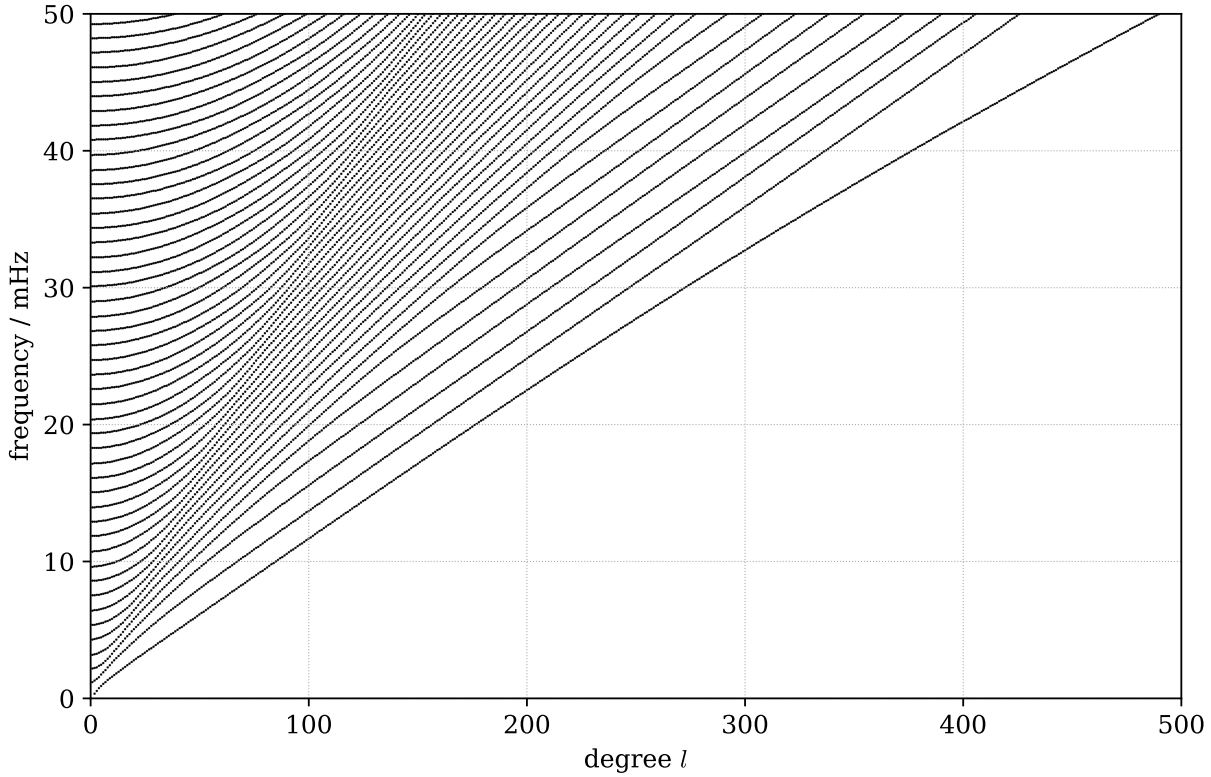


Fig. 1: Eigenfrequencies of toroidal modes within the earth model PREM of Dziewonski & Anderson (1981), plotted against degree l for frequencies up to 50 mHz. For each degree we see a set of different overtones that can be labelled using the overtone number n, the lowest curve being the fundamental branch $n = 0$. Note also that PREM comprises a solid inner core, fluid outer core, and solid mantle and crust. The modes seen here are mantle toroidal modes, but one could also calculate toroidal modes confined to the inner core.

First there are toroidal modes which have eigenfunctions of the form

$$\mathbf{s}(r, \theta, \varphi) = W(r)\mathbf{C}_{lm}(\theta, \varphi), \tag{13}$$

with W a function of radius that depends on l but not m. The corresponding eigenfrequency is also independent of m, and hence such modes lie in $(2l + 1)$-fold degenerate eigenspaces. It can be shown that toroidal modes are associated with no perturbation to the gravitational potential, and also that their eigenfunctions are non-zero only in solid parts of the earth model.

[1] The most general elastic tensor that is consistent with spherical symmetry is **transversely isotropic** with a radial symmetry axis. There are then **five elastic moduli**, which can be taken to be Love's A, C, F, L and N of Lecture 14, each of which can vary with radius. In an isotropic model these reduce to the bulk and shear moduli κ and μ, and for simplicity we assume isotropy in what follows.

The second type of eigenfunction is the spheroidal modes that take the form

$$\mathbf{s}(r,\theta,\varphi) = U(r)\mathbf{A}_{lm}(\theta,\varphi) + V(r)\mathbf{B}_{lm}(\theta,\varphi), \tag{14}$$

where U and V are functions of radius that depend on l but not m. Again, such an eigenfunction lies in a $(2l+1)$-fold degenerate eigenspace. Spheroidal modes are associated with gravitational perturbations, and their eigenfunctions can be non-zero in both solid and fluid regions of the earth model.

Let us consider toroidal modes further. If we put eq. (13) into the kinetic energy form, then due to the orthogonality of vector spherical harmonics, we can only get a non-zero contribution from the part of the test function proportional to $\mathbf{C}_{lm}$. The same turns out to be true for the potential energy form, and hence it is sufficient to consider test functions

$$\mathbf{w}(r,\theta,\varphi) = w(r)\mathbf{C}_{lm}(\theta,\varphi). \tag{15}$$

Using eq. (10) it is then easy to obtain

$$\langle \mathbf{w}|P|\mathbf{s}\rangle = \zeta^2 \int_0^b \rho w^* W r^2 \,\mathrm{d}r, \tag{16}$$

where b is the radius of the earth model, and we have set $\zeta = \sqrt{l(l+1)}$ and assumed the earth model is entirely solid. In a similar manner, it can be shown that

$$\langle \mathbf{w}|H|\mathbf{s}\rangle = \zeta^2 \int_0^b \mu \left(r\frac{\mathrm{d}w^*}{\mathrm{d}r} - w^*\right)\left(r\frac{\mathrm{d}W}{\mathrm{d}r} - W\right)\,\mathrm{d}r + \zeta^2(\zeta^2-2)\int_0^b \mu w^* W \,\mathrm{d}r. \tag{17}$$

Note that derivation of this result is not simple, as you need to know how to differentiate vector spherical harmonics. Nonetheless, the end result is that the eigenvalue problem for a toroidal mode of degree l reduces to the form

$$\begin{aligned} &-\omega^2 \int_0^b \rho w^* W r^2 \,\mathrm{d}r + \int_0^b \mu \left(r\frac{\mathrm{d}w^*}{\mathrm{d}r} - w^*\right)\left(r\frac{\mathrm{d}W}{\mathrm{d}r} - W\right)\,\mathrm{d}r \\ &+ (\zeta^2-2)\int_0^b \mu w^* W \,\mathrm{d}r = 0, \end{aligned} \tag{18}$$

where W is the radial eigenfunction, and w the radial part of the test function. Through a relatively simple calculation, it can be shown that this is an ODE eigenvalue problem of **Sturm–Liouville type** that many of you will have seen previously. In the case of a homogeneous planet closed-form solutions can be obtained using spherical Bessel functions, though such a solution is of little practical use. More generally, the resulting ODE eigenvalue problem can be solved numerically, or the weak form directly converted into an algebraic eigenvalue problem using finite element methods. For our purposes, the key result is that for each degree $l \neq 0$ there exists a countable infinity of toroidal modes, with these different solutions being labelled by an **overtone number**, $n = 0, 1, 2, \ldots$, with $n = 0$ being the **fundamental mode** and eigenfrequency then increasing with n.

For spheroidal modes a similar approach can be applied, with the eigenvalue problem reducing for each degree, l, to one involving the radial eigenfunctions U and V. Solution of this latter problem is considerably more involved than that for toroidal

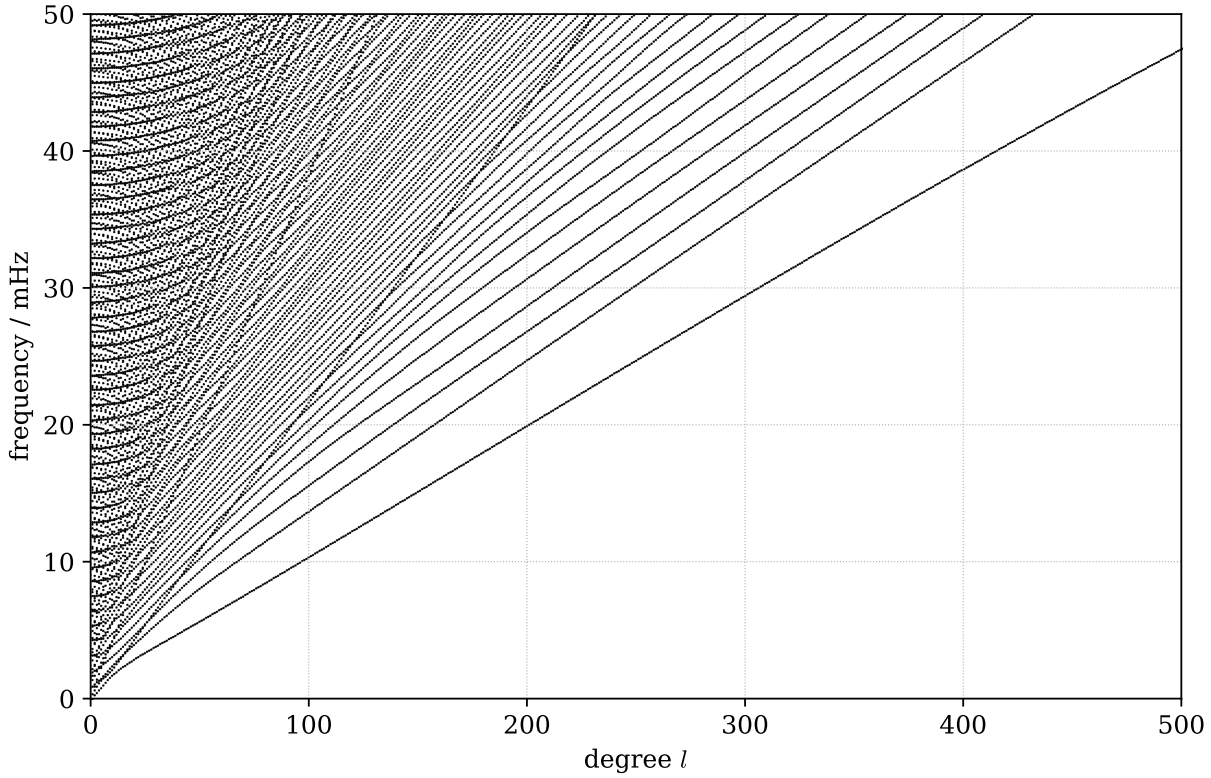


Fig. 2: As for Fig. 1, but here showing spheroidal mode eigenfrequencies within PREM, including the radial modes with $l = 0$. The branches are less regular than in the toroidal case, with modes of different character (for example those whose energy is concentrated in the mantle, in the fluid outer core, or in the inner core) crossing and interacting with one another.

modes, in part due to the need to model the associated gravitational potential perturbations. Nonetheless, numerical solutions of this problem have been routinely determined for decades.

Fig. 3 shows the radial eigenfunctions of a few representative modes. Several general features are worth noting. For toroidal modes the overtone number counts the nodes of the radial eigenfunction, so that the fundamental modes have none; for spheroidal modes there is no such simple rule.[2] As the degree increases at fixed overtone number the eigenfunctions become confined ever closer to the surface; the fundamental modes of high degree are the standing-wave equivalents of surface waves, a topic that is not covered within this course. Toroidal modes of the mantle have vanishing shear traction at the core–mantle boundary, so that $r\ \mathrm{d}W/\mathrm{d}r - W$ vanishes there, while spheroidal modes penetrate the fluid outer core and the inner core, and hence carry information about the deep Earth. The spheroidal eigenfunctions have two radial components, U for the radial displacement and V for the tangential one; for the radial modes with $l = 0$ only U is present. Not all modes are so regular, and Fig. 4 shows some of the more curious kinds: a mode dominated by a rigid translation of the inner core within the fluid outer core, known as the **Slichter mode**, whose period of several hours is

[2] A generalisation of the node-counting property to spheroidal modes does exist, formulated in terms of the so-called minor vectors of the radial equations (Woodhouse 1988), but it lies far beyond this course.

set by gravity rather than elasticity; **Stoneley modes**, whose energy is trapped along a fluid–solid interface, here either the core–mantle boundary or the inner-core boundary, and which are the standing-wave equivalents of waves guided along such an interface; and a mode whose energy is concentrated within the inner core. Much of the art of free oscillation seismology lies in identifying which modes are sensitive to which parts of the Earth.

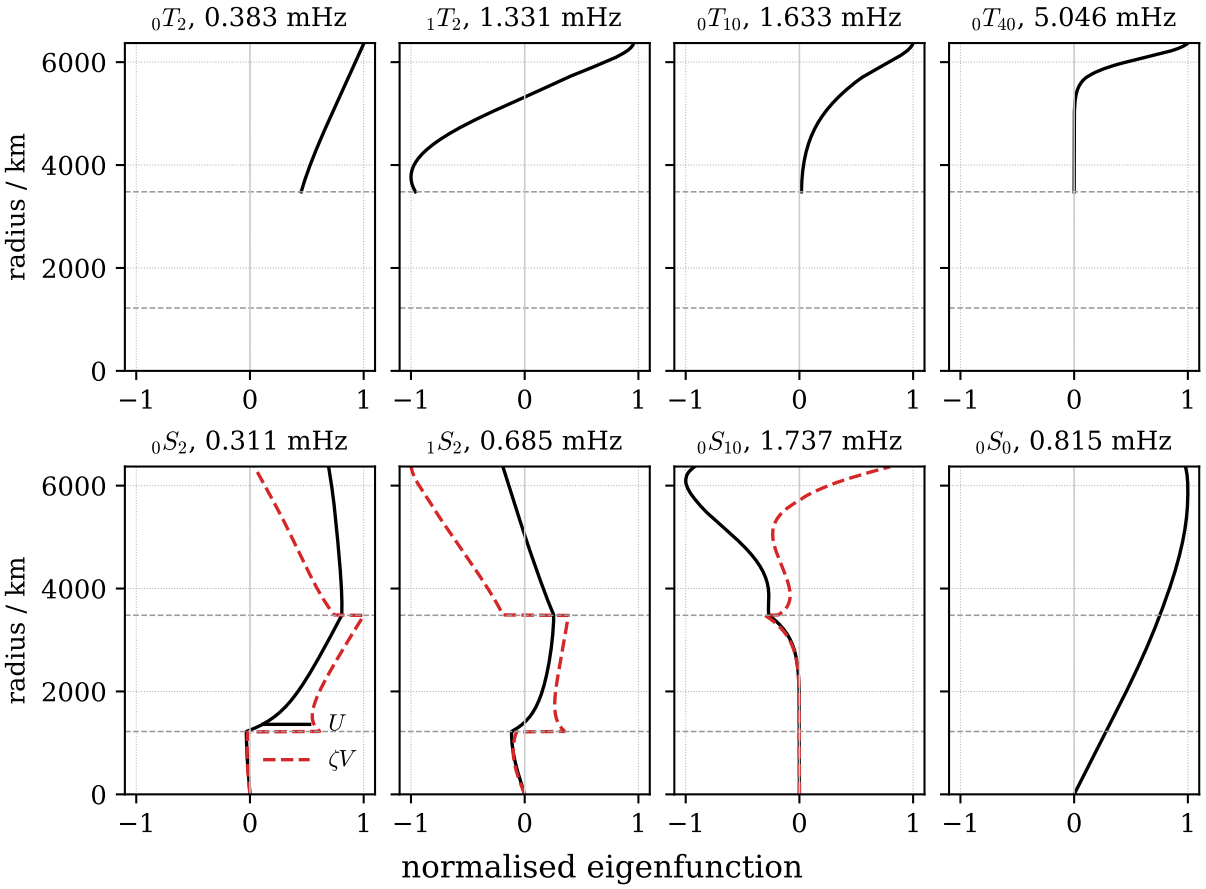


Fig. 3: Radial eigenfunctions of some toroidal (upper row) and spheroidal (lower row) modes of PREM, each normalised to unit maximum amplitude. For the spheroidal modes both the radial component U and the tangential component ζV, where $\zeta = \sqrt{l(l+1)}$, are shown. The dashed lines mark the core–mantle and inner-core boundaries. Calculated using the spectral-element normal-mode code of A. Myhill.

We now summarise the results of this section. Within a spherically symmetric earth model there are two families of normal modes: toroidal and spheroidal. We can introduce an index, q, to distinguish these with, say, $q = 1$ meaning spheroidal, and $q = 2$ toroidal.[3] Having fixed the mode type, q, we can then consider different degrees, l. For each such degree there is an infinite set of eigenfrequencies and radial eigenfunctions labelled by their overtone number, n. Finally, we then have the order, m, of a mode, this labelling an orthogonal basis for the $(2l+1)$-fold degenerate eigenspace.

[3] Along with further values if we need to distinguish toroidals in different solid parts of the model such as the mantle and inner core.

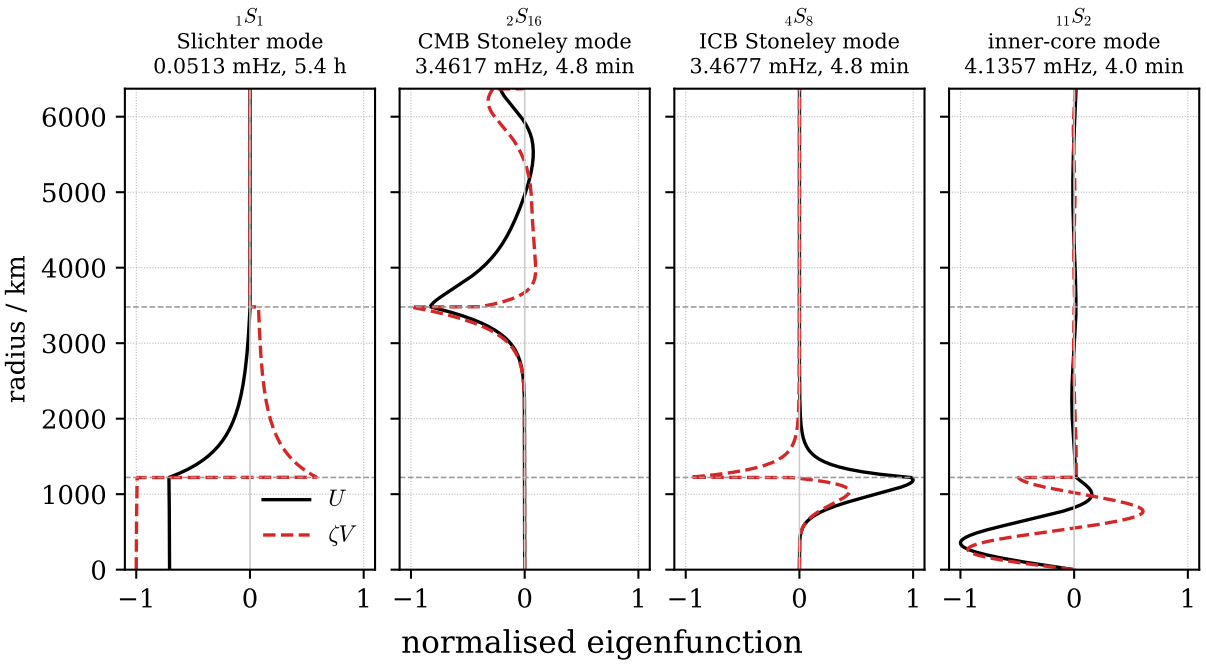


Fig. 4: Radial eigenfunctions of some less typical spheroidal modes of PREM: the Slichter mode ${}_1S_1$, in which the inner core translates within the fluid outer core; Stoneley modes trapped on the core–mantle boundary and on the inner-core boundary; and a mode confined almost entirely to the inner core.

In this manner, we see that **four integers** (q, l, n, m) are sufficient to label all the eigenfunctions of a spherically symmetric earth model, while the corresponding eigenfrequencies depend only on (q, l, n). As a useful shorthand, we can collapse the triplet (q, l, n) into a single index, k, and hence write ω_k for the eigenfrequencies and $|km\rangle$ for the eigenfunctions. Indeed, this is precisely the notation used in the last lecture. When needed, however, we can also write $|qlnm\rangle$ to explicitly characterise an eigenfunction. As a final bit of terminology, for each (q, l, n) we call the eigenspace a **multiplet**, while individual elements of its basis, as labelled by m, are known as **singlets**. We might then, for example, talk about a degree two toroidal multiplet which comprises five singlets. A common labelling scheme for toroidal multiplets is to write ${}_nT_l$ for the multiplet of degree l and overtone number n. Similarly, spheroidal multiplets are written ${}_nS_l$.

Splitting of a multiplet due to rotation and lateral heterogeneity

Within a spherically symmetric earth model, the normal modes come in $(2l + 1)$-fold degenerate multiplets, with l the degree which is a non-negative integer. Such degeneracy is a direct consequence of the spherical symmetry of the problem, and analogous results are seen in other parts of physics. In particular, you know that the energy eigenspaces for the hydrogen atom are similarly degenerate, though note that there the degeneracy is larger still, owing to the "hidden symmetry" of the Coulomb problem associated with the Runge–Lenz vector.

If such a spherically symmetric system undergoes an aspherical perturbation we expect that this degeneracy will be broken, with the singlets that comprise the multiplet

splitting apart. This is precisely what happens in the **Zeeman effect for the hydrogen atom**, and we will apply very similar mathematics to examine what rotation and lateral heterogeneities do to the free oscillations of the Earth.

We consider a spherically symmetric earth model along with the non-rotating eigenvalue problem

$$-\omega^2\langle \mathbf{w}|P_0|\mathbf{s}\rangle + \langle \mathbf{w}|H_0|\mathbf{s}\rangle = 0, \tag{19}$$

where we note that the subscript 0 has been added to the kinetic and potential energy forms. A multiplet of this earth model is identified by the triplet (q, l, n), and we write ω_k and $|km\rangle$ for its eigenfrequency and eigenfunctions using the simplified notations discussed above. We then have, in particular, that

$$-\omega_k^2\langle \mathbf{w}|P_0|km\rangle + \langle \mathbf{w}|H_0|km\rangle = 0, \tag{20}$$

for all test functions $\mathbf{w}$.

Suppose now that we add a laterally heterogeneous perturbation to the model such that the kinetic energy form can be written

$$\langle \mathbf{w}|P|\mathbf{s}\rangle = \langle \mathbf{w}|P_0|\mathbf{s}\rangle + \epsilon\langle \mathbf{w}|P_1|\mathbf{s}\rangle + \cdots, \tag{21}$$

where ϵ is a perturbation parameter. Using a similar notation for the potential energy form, we wish to consider the eigenvalue problem

$$-\omega^2[\langle \mathbf{w}|P_0|\mathbf{s}\rangle + \epsilon\langle \mathbf{w}|P_1|\mathbf{s}\rangle + \cdots] + [\langle \mathbf{w}|H_0|\mathbf{s}\rangle + \epsilon\langle \mathbf{w}|H_1|\mathbf{s}\rangle + \cdots] = 0. \tag{22}$$

We assume that the perturbed eigenfrequency can be written

$$\omega = \omega_k + \epsilon\,\delta\omega + \cdots, \tag{23}$$

while for the eigenfunctions we assume

$$\mathbf{s} = \sum_{m=-l}^{l} a_m|km\rangle + \epsilon\,\delta\mathbf{s} + \cdots, \tag{24}$$

where the a_m are coefficients to be determined. Note here that we are allowing for the zeroth-order eigenspace to be degenerate, this method having been applied earlier in the course when considering shear wave splitting.

Putting these expansions into the eigenvalue problem, we find at order one that

$$\begin{aligned}&\sum_{m=-l}^{l}[-2\omega_k\delta\omega\langle \mathbf{w}|P_0|km\rangle - \omega_k^2\langle \mathbf{w}|P_1|km\rangle + \langle \mathbf{w}|H_1|km\rangle]a_m\\ &- \omega_k^2\langle \mathbf{w}|P_0|\delta\mathbf{s}\rangle + \langle \mathbf{w}|H_0|\delta\mathbf{s}\rangle = 0.\end{aligned} \tag{25}$$

If we take the test function, $\mathbf{w}$, to equal $|km'\rangle$ then the terms involving $\delta\mathbf{s}$ cancel due to the self-adjointness and the zeroth-order eigenvalue problem, and hence we are left with

$$\frac{1}{2\omega_k}\sum_{m=-l}^{l}[-\omega_k^2\langle km'|P_1|km\rangle + \langle km'|H_1|km\rangle]a_m = \delta\omega a_{m'}, \tag{26}$$

where we have used the orthonormality of the singlets. Such an equation is to hold for all $-l \le m' \le l$ and hence we have arrived at a $(2l+1)$-dimensional Hermitian eigenvalue problem for $\delta\omega$ along with the coefficients, a_m, of the zeroth-order

eigenfunction. Assuming that the matrix elements $\langle km'|P_1|km\rangle$ and $\langle km'|H_1|km\rangle$ can be calculated, such an eigenvalue problem can be readily solved either in closed form for small l, or otherwise on a computer. The important qualitative conclusion of this analysis is that there will generically be $2l+1$ distinct eigenfrequency perturbations $\delta\omega$, this showing that the **multiplet is split into the corresponding number of non-degenerate eigenspaces**.

The above argument can be readily extended to allow for rotation of the earth model as a first-order perturbation. Indeed, we generalise eq. (22) to

$$-\omega^2[\langle \mathbf{w}|P_0|\mathbf{s}\rangle + \epsilon\langle \mathbf{w}|P_1|\mathbf{s}\rangle + \cdots] + \mathrm{i}\omega\epsilon\langle \mathbf{w}|W|\mathbf{s}\rangle + [\langle \mathbf{w}|H_0|\mathbf{s}\rangle + \epsilon\langle \mathbf{w}|H_1|\mathbf{s}\rangle + \cdots] = 0. \tag{27}$$

A similar expansion to first order then leads to the Hermitian eigenvalue problem

$$\frac{1}{2\omega_k}\sum_{m=-l}^{l}[-\omega_k^2\langle km'|P_1|km\rangle + \mathrm{i}\omega_k\langle km'|W|km\rangle + \langle km'|H_1|km\rangle]a_m = \delta\omega a_{m'}, \tag{28}$$

where we see on the left-hand side an additional set of matrix elements associated with the Coriolis force. Note also that terms associated with the centrifugal force need to be incorporated into the perturbation to the potential energy form.

The diagonal sum rule and spherically symmetric density variations

Within the first-order splitting calculations earlier, we considered the Hermitian eigenvalue problem, eq. (28). The perturbations to the singlet eigenfrequencies are equal to the eigenvalues of the matrix on the left-hand side. Based on so-called **selection rules** for the matrix elements,[4] it can be shown that this matrix has zero trace. Because the trace of a matrix is invariant under similarity transforms, it follows that the singlet eigenfrequency perturbations sum to zero. This result is known as the **diagonal sum rule** and was obtained by Gilbert (1971) and played a key role in early studies of Earth structure using observations of free oscillations.

In the previous lecture we saw that eigenfrequencies can be measured from the amplitude spectra of sufficiently long seismograms. The first such observations were made following the 1960 Great Chilean earthquake,[5] which remains the largest event that has occurred during the instrumental period. By this time there existed good models of the Earth's spherically symmetric velocity structure obtained from travel time observations. There also existed plausible models of radial density variations that were built using knowledge of the Earth's mass, moments of inertia, and simple thermodynamic models of density variations in a homogeneous solid compressed under its own weight. As a consequence, the first spectra observed could be immediately compared to eigenfrequencies calculated in the different earth models available.

An example of this can be seen in Fig. 5. Here two reasonably clear resonance peaks can be identified, with their centre frequencies measured. Also shown on the plot are the predicted eigenfrequencies from two different spherical earth models for the same multiplet. It is notable, firstly, that these predictions are in reasonable agreement with an observed eigenfrequency, but also that while in a spherical earth model we would

[4] These are nearly identical to those arising in the quantum theory of angular momentum.

[5] "Great" is a technical term used for especially large earthquakes; it does not suggest that the earthquake was a good thing.

expect only one eigenfrequency in this range, there seem to be at least two in the data. The latter fact is due to splitting associated with rotation and lateral heterogeneity. This fact was immediately recognised, and motivated George Backus and Freeman Gilbert to publish the first theoretical treatment (Backus & Gilbert 1961) of the splitting of free oscillations due to rotation. Indeed, as we have seen, such calculations mirror almost exactly those known from quantum mechanics for decades at that point.

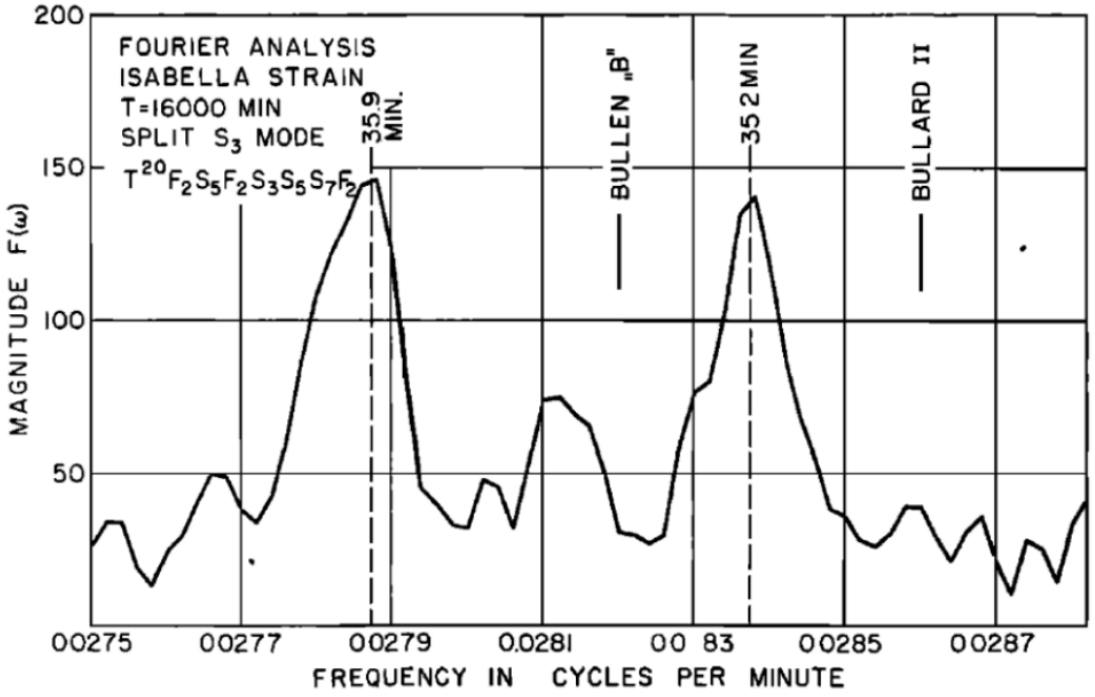


Fig. 5: A part of the first free oscillation spectrum observed following the 1960 Chile earthquake. Taken from Benioff et al. (1961).

Fig. 6 shows the modern equivalent, using twenty-day records of the 2011 Tohoku earthquake at three stations. Here ${}_0S_2$ is resolved into four of its five singlets, the central one being weak at all three stations, and ${}_0S_3$ into most of its seven, with the relative amplitudes of the singlets depending on the position of the station and the orientation of the source. The singlets are spread almost symmetrically about the degenerate frequency of PREM, and their nearly uniform spacing is the signature of the Earth's rotation, which splits a multiplet in much the same manner as a magnetic field splits atomic energy levels in the Zeeman effect.

During these early days of free oscillation seismology, the main emphasis was on better constraining the Earth's spherically averaged structure. With travel time observations this could be easily done. For many source–receiver pairs the travel time of different seismic phases could be measured. This information could then be reduced to plots of travel time against epicentral angle. Due to lateral heterogeneity, along with observational error, there was some scatter in these travel time curves, but this could be averaged out. Note that the latter step is not rigorously justified, but it was at least easy to do. With free oscillations, however, the situation was quite different. What we observe are split eigenfrequencies of the singlets, but what we really want is the eigenfrequency of the associated multiplet prior to this splitting. Supposing that all the singlet frequencies for a split multiplet have been measured, Gilbert's diagonal

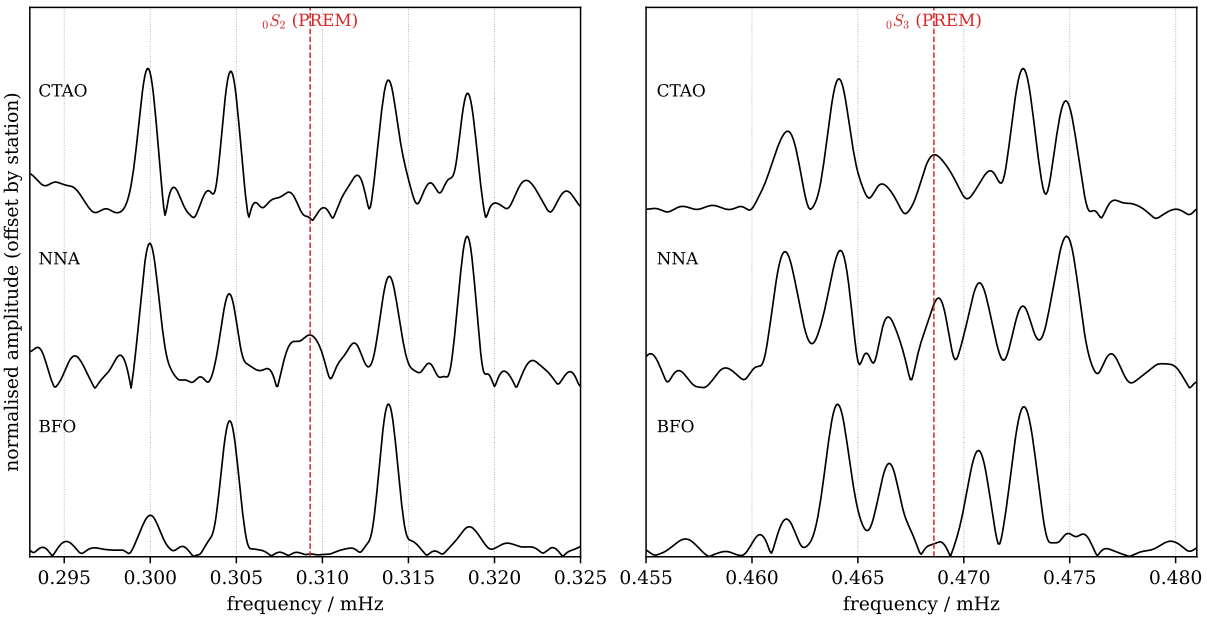


Fig. 6: Splitting of ${}_0S_2$ (left) and ${}_0S_3$ (right) seen in the amplitude spectra of twenty-day vertical-component records of the 2011 Tohoku earthquake at the stations CTAO (Australia), NNA (Peru) and BFO (Germany). The spectra are offset vertically, and the dashed lines mark the degenerate eigenfrequencies of the multiplets in PREM.

sum rule provides an immediate answer. We need only take the arithmetic average of the split eigenfrequencies to obtain the degenerate eigenfrequency corresponding to the Earth's spherically averaged structure. For some low frequency and low degree multiplets it is indeed possible to identify all the singlet eigenfrequencies from a single spectrum. In other cases the singlet frequencies cannot be seen clearly in any one spectrum, but by combining observations from many different earthquakes and stations the singlet frequencies can be recovered using various signal processing methods.

How do the degenerate eigenfrequencies constrain the Earth's structure? The answer is through **sensitivity kernels** of exactly the kind met in Lecture 20. Suppose that the density and elastic moduli of a spherically symmetric earth model are perturbed by small amounts $\delta\rho$, $\delta\kappa$ and $\delta\mu$. Because the eigenfrequencies are stationary with respect to variations of the eigenfunctions, an argument identical to that used for the first-order splitting above (with the perturbation now spherically symmetric, so that the multiplet is not split) shows that the change in a degenerate eigenfrequency is

$$\delta\omega_k = \int_0^b \left(K_\kappa\,\delta\kappa + K_\mu\,\delta\mu + K_\rho\,\delta\rho\right)\,\mathrm{d}r, \tag{29}$$

where the kernels are quadratic in the unperturbed eigenfunctions of the mode.[6] For toroidal modes the kernels can be read off from eq. (18), and with the normalisation

[6] This is Rayleigh's principle. Taking the test function equal to the eigenfunction in the perturbed eigenvalue problem, and using the unperturbed problem to eliminate the change in the eigenfunction, gives $2\omega_k\delta\omega_k = \langle k|H_1|k\rangle - \omega_k^2\langle k|P_1|k\rangle$ with $|k\rangle$ normalised.

$\int_0^b \rho W^2 r^2 \, \mathrm{d}r = 1$ they are

$$K_\mu = \frac{1}{2\omega_k}\left[\left(r\frac{\mathrm{d}W}{\mathrm{d}r} - W\right)^2 + (\zeta^2 - 2)W^2\right], \quad K_\rho = -\frac{\omega_k}{2}W^2 r^2. \tag{30}$$

The spheroidal kernels are more complicated, involving both U and V together with the perturbation to the gravitational potential, and we do not write them out (see Dahlen & Tromp 1998). Fig. 7 shows kernels for a few modes. Several features are important. The density kernel can be of either sign: added mass lowers the frequency through the kinetic energy but also increases the gravitational forces, and for ${}_0S_2$ the two effects compete, the kernel changing sign within the outer core. The kernels for the moduli are positive, as they must be, stiffer material raising the frequency. And a given mode is sensitive only where its eigenfunction is large, so that the high-degree fundamental modes constrain only the upper mantle, while modes such as ${}_0S_2$ and the radial modes sample the whole planet. Because the kernels for κ, μ and ρ have different shapes, the frequencies of a sufficiently large and varied set of modes can, in principle, be inverted for all three parameters, and this is how the radial density structure of the Earth is constrained by seismology.

Combining such observations with the diagonal sum rule, a large database of eigenfrequencies corresponding to the Earth's spherically averaged structure had been determined by the late 1970s. Taken along with travel time information, this allowed existing earth models to be substantially refined, and, in particular, for radial density variations to be found. The resulting density structure is shown in Fig. 8. These observed densities can be compared with predictions made by mineral physicists subject to assumptions about the Earth's interior composition. For example, it has been shown that the density of the outer core is significantly lower than expected if it were composed of an iron–nickel alloy as is suggested by observations of meteorites. This discrepancy is telling us something about how planets form and differentiate. As yet the identity of this "light element" in the core is uncertain, with oxygen or silicon being perhaps the most likely candidates. But what is worth emphasising is that such discussions are only possible due to the highly accurate constraints on the interior structure provided by seismology.

Lateral density variations and mode coupling

As noted in Lectures 19 and 20, recent work in seismology is focused on lateral heterogeneity in the mantle, this being done to better understand mantle convection. In Fig. 9, images from a range of S-wave velocity models are shown at different depths within the Earth, and a broad agreement can be seen. In particular, all models show at the base of the mantle two large regions with low shear wave velocity, one below the Pacific, and the other below Africa.

On the assumption that these velocity variations reflect temperature variations, we conclude that these so-called **Large Low Shear Velocity Provinces (LLSVPs)** are hotter than the surrounding mantle. If this is true, then we would also expect that these regions are lighter than the ambient mantle, and hence will experience an upwards buoyancy force. When LLSVPs were first identified in the early 1990s such an interpretation was taken for granted, and they were known as "super plumes". A range

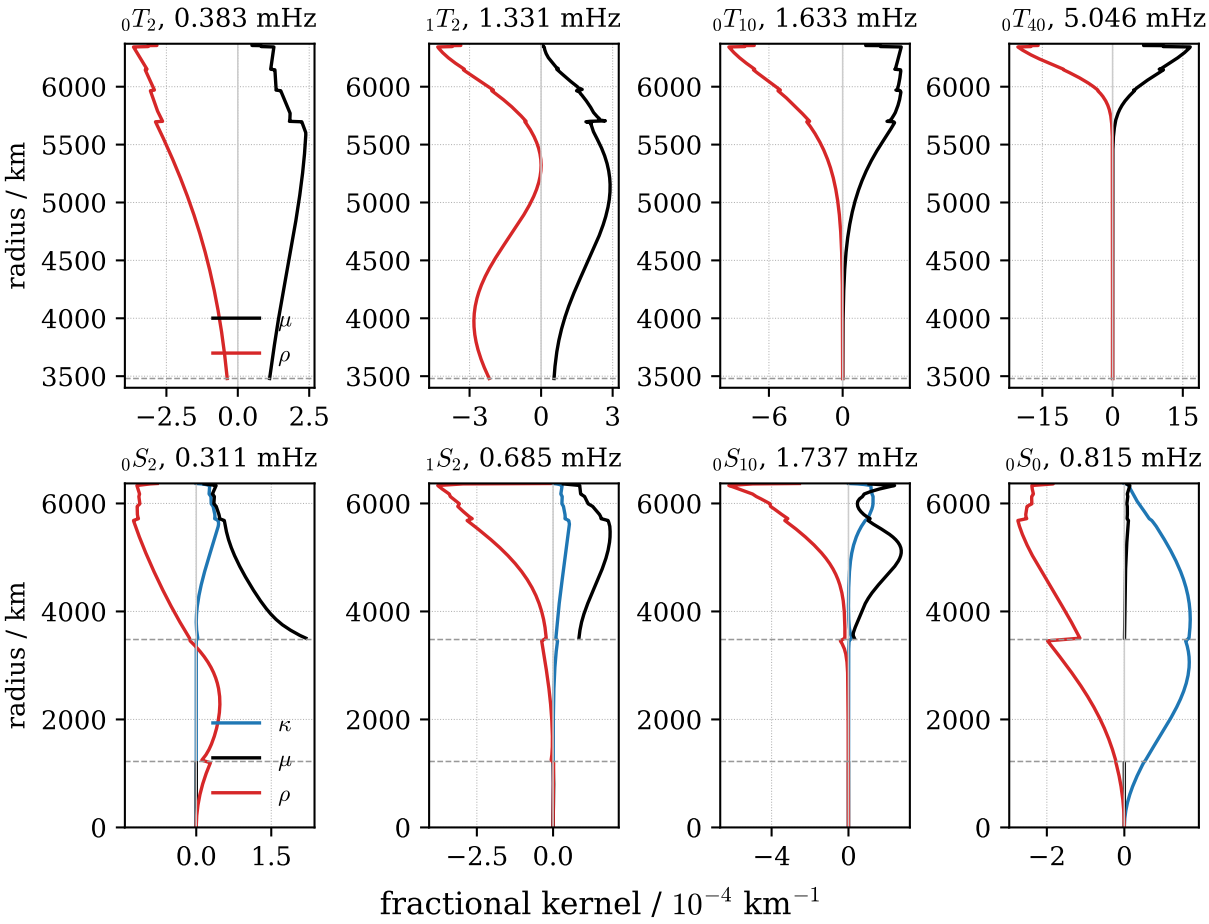


Fig. 7: Sensitivity kernels of the toroidal (upper row) and spheroidal (lower row) modes of Fig. 3. The kernels are shown in fractional form, so that $\delta\omega_k/\omega_k = \int_0^b (\tilde{K}_\kappa\,\delta\kappa/\kappa + \tilde{K}_\mu\,\delta\mu/\mu + \tilde{K}_\rho\,\delta\rho/\rho)\,\mathrm{d}r$. The toroidal kernels follow from eq. (30); the spheroidal ones, which include the effects of self-gravitation, were obtained numerically. The density kernel of ${}_0S_2$ changes sign in the outer core, gravity and inertia competing for this gravest mode.

of subsequent work, however, has brought into question this conclusion. For example, joint S- and P-wave tomographic models have shown that the LLSVPs seem to be associated with a positive perturbation to the bulk sound speed

$$v_\phi = \sqrt{\frac{\kappa}{\rho}}. \tag{31}$$

Note that this is not a speed of any elastic wave, but rather a parameter that is useful because it depends on the bulk modulus but not the shear modulus.

If the lateral variations are caused by temperature, then we would expect that the shear and bulk sound velocities would **positively correlate**. To see this, we assume that temperature variations are sufficiently small to allow us to write

$$\delta\beta = \frac{\partial\beta}{\partial T}\delta T, \quad \delta v_\phi = \frac{\partial v_\phi}{\partial T}\delta T, \tag{32}$$

where δT is the temperature perturbation. The relevant partial derivatives can be esti-

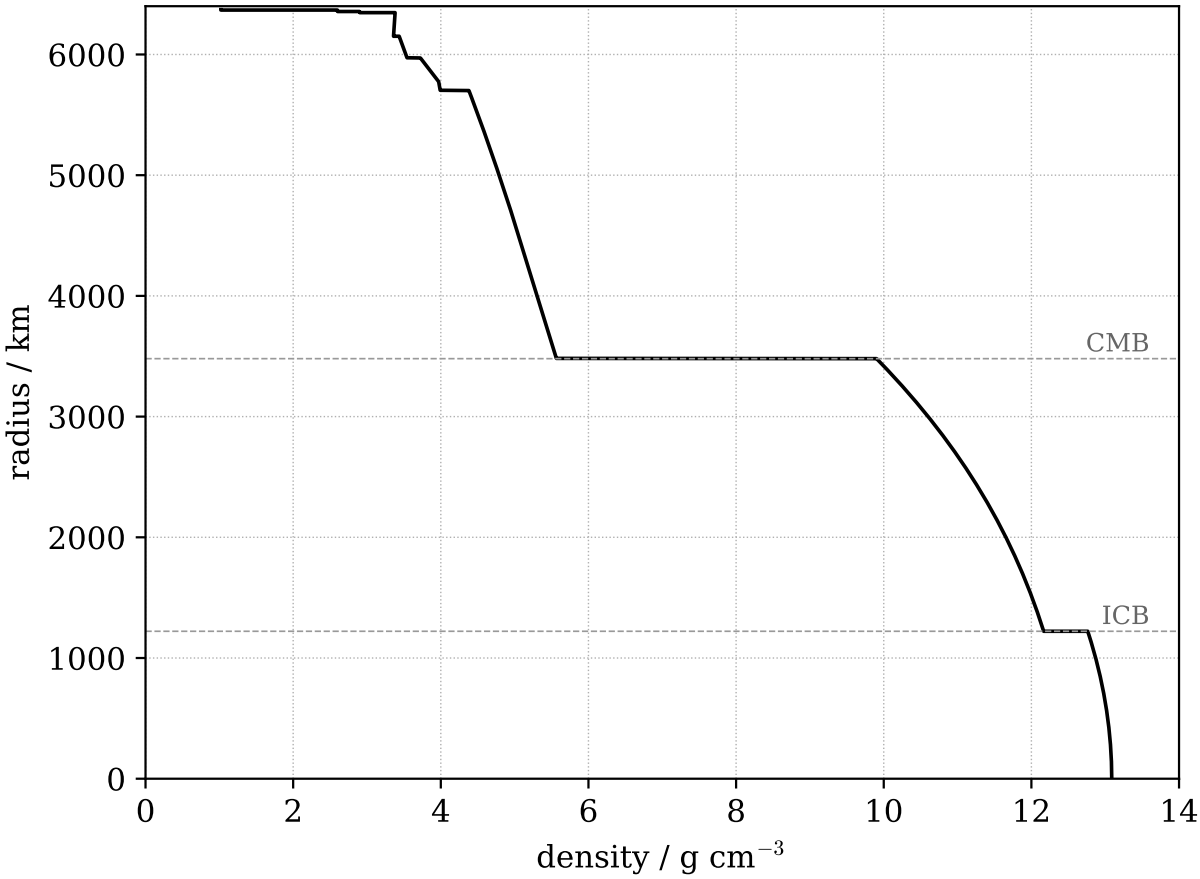


Fig. 8: Variation of density with radius in the model PREM of Dziewonski & Anderson (1981) as determined from the Earth's mass, moment of inertia, and over 1000 measured eigenfrequencies.

mated from mineral physics, and are negative for all crystalline materials, and hence

$$\frac{\delta v_\phi}{\delta \beta} = \frac{\partial v_\phi / \partial T}{\partial \beta / \partial T} > 0, \tag{33}$$

as required. The fact that shear and bulk sound speeds seem to anti-correlate in the lower mantle suggests that temperature is not the only cause for these lateral variations. The other possible explanation is that **compositional variations** in the mantle play an important role in its dynamics. To see how this might reconcile the observations, we write X for a compositional variable. The velocity perturbations can then be expressed in terms of temperature and composition in the form

$$\delta\beta = \frac{\partial \beta}{\partial T}\delta T + \frac{\partial \beta}{\partial X}\delta X, \quad \delta v_\phi = \frac{\partial v_\phi}{\partial T}\delta T + \frac{\partial v_\phi}{\partial X}\delta X, \tag{34}$$

from which it is clear that the ratio $\delta v_\phi / \delta \beta$ need not always be positive. If it is accepted that significant compositional variations exist in the lower mantle, then we cannot determine from the tomographic studies discussed whether LLSVPs are light or dense, and hence the role they play in mantle dynamics. To address this latter point what is needed are direct estimates of their density, and it is free oscillations that have the potential to provide this information.

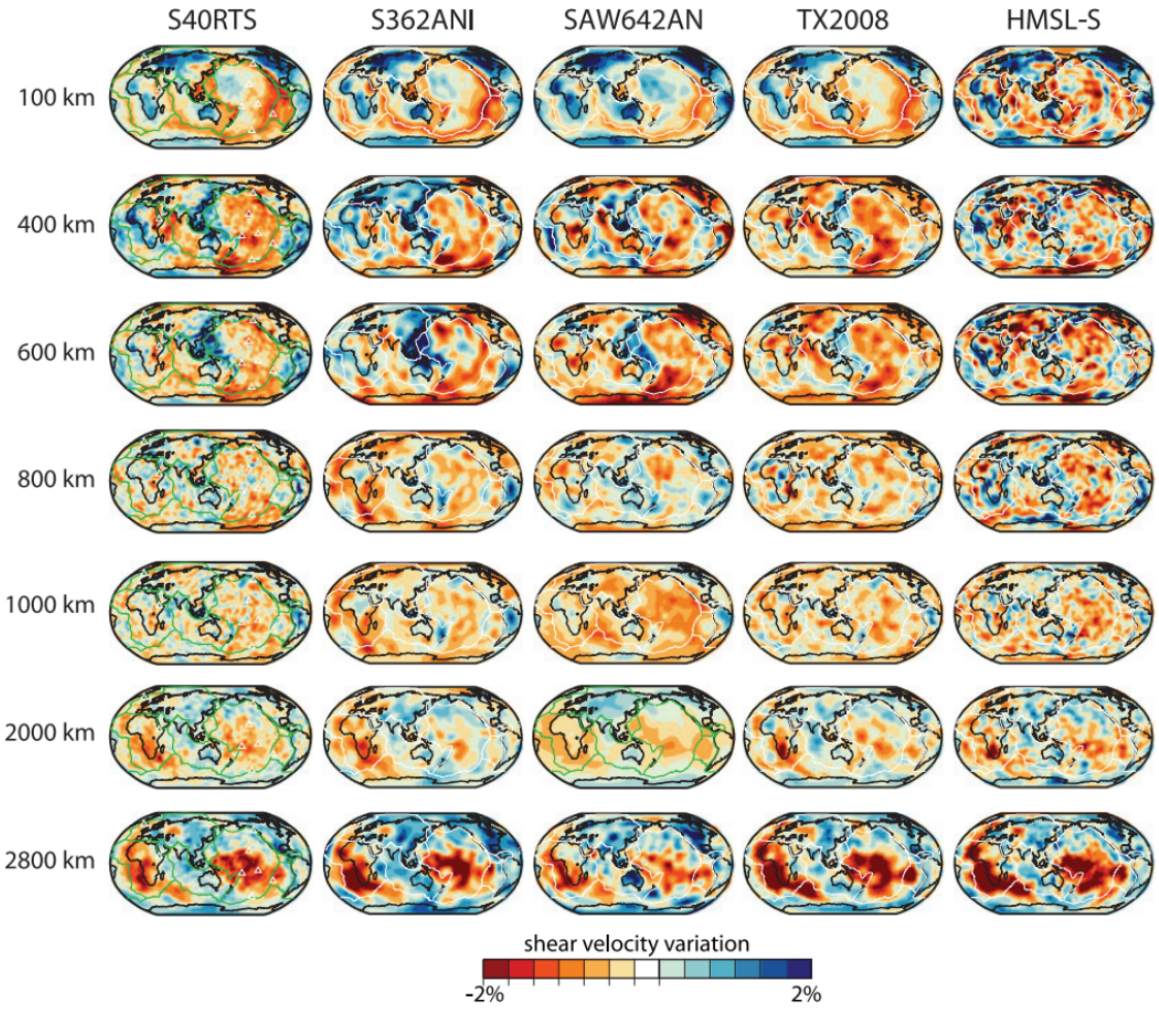


Fig. 9: Depth slices from a range of different S-wave tomographic models, taken from Ritsema et al. (2011). In all cases the Pacific and African LLSVPs can be seen at the base of the mantle. The dynamical significance of these features remains unclear.

The first study of lateral density variations in the lower mantle using free oscillation data was undertaken in the late 1990s by Miaki Ishii and Jeroen Tromp, then both at Harvard (Ishii being a PhD student at the time; Ishii & Tromp 1999). This study was based on a slight elaboration of the first-order splitting theory discussed earlier in the lecture, and the resulting models showed that LLSVPs are denser than the surrounding mantle. This is consistent with the presence of compositional variations, and hence leads to interesting questions about what these variations are, and how they were produced. Do, for example, LLSVPs represent a "slab graveyard" composed of compositionally distinct material accumulated at the base of the mantle through subduction? Or are they instead ancient iron-enriched features produced as the Earth differentiated into a mantle and core? Such questions continue to engage a wide range of Earth scientists.

But the density of LLSVPs is actually not a settled issue. This is because later work has shown that the simple first-order degenerate perturbation theory applied is not

sufficiently accurate to model the effect of lateral density variations. Implicit within this theory is the assumption that on the addition of laterally varying structure distinct multiplets do not interact with each other. This is not true, with the resulting errors being especially significant for nearly degenerate multiplets. A better approach is to use **normal mode coupling theory**. The idea is simple: the eigenfunctions of the spherical reference model form a complete basis, so any displacement field in the aspherical model, and in particular any of its eigenfunctions, can be expanded in this basis in the form

$$\mathbf{s} = \sum_{k'm'} \langle k'm'|P_0|\mathbf{s}\rangle |k'm'\rangle, \tag{35}$$

where $|km\rangle$ are eigenfunctions in a spherical reference model. Putting such an expansion into the rotating eigenvalue problem we arrive at

$$\sum_{k'm'} [-\omega^2 \langle km|P|k'm'\rangle + \mathrm{i}\omega \langle km|W|k'm'\rangle + \langle km|H|k'm'\rangle] \langle k'm'|P_0|\mathbf{s}\rangle = 0, \tag{36}$$

for all (k, m). This is a non-linear eigenvalue problem for the expansion coefficients $\langle k'm'|P_0|\mathbf{s}\rangle$ relative to the chosen basis. Such an equation cannot be solved directly because the matrices and vectors involved are infinite-dimensional. But it is possible to truncate the basis so as to obtain a finite-dimensional problem that can be solved numerically, and by taking sufficiently many terms in the expansion the results can be made as accurate as desired.

In practice the eigenvalue problem itself is rarely solved. What is observed is a spectrum, and this can be computed directly. Consider the equations of motion in the frequency domain for a given source, whose force term $\tilde{\mathbf{f}}(\omega)$ will in general depend on frequency, with the displacement expanded in the spherical basis as in eq. (35), $\tilde{\mathbf{u}}(\omega) = \sum_{k'm'} \langle k'm'|P_0|\tilde{\mathbf{u}}(\omega)\rangle |k'm'\rangle$. Taking the test function equal to each basis element in turn gives

$$\sum_{k'm'} \left[-\omega^2 \langle km|P|k'm'\rangle + \mathrm{i}\omega \langle km|W|k'm'\rangle + \langle km|H|k'm'\rangle \right] \langle k'm'|P_0|\tilde{\mathbf{u}}(\omega)\rangle = \langle km|\tilde{\mathbf{f}}(\omega)\rangle, \tag{37}$$

which for each frequency is a linear system for the expansion coefficients, the same matrices appearing as in eq. (36). Splitting the kinetic and potential energy forms into their spherically symmetric parts and the perturbations, and recalling that the basis is orthonormal with respect to P_0 and diagonalises H_0, this system takes the form

$$(\omega_k^2 - \omega^2) \langle km|P_0|\tilde{\mathbf{u}}(\omega)\rangle + \sum_{k'm'} A_{km,k'm'}(\omega) \langle k'm'|P_0|\tilde{\mathbf{u}}(\omega)\rangle = \langle km|\tilde{\mathbf{f}}(\omega)\rangle, \tag{38}$$

where the matrix A collects the matrix elements of P_1, W and H_1, all of which are small. For a given frequency ω, the diagonal terms $\omega_k^2 - \omega^2$ are large except for the modes whose eigenfrequencies lie close to ω, and so the system is dominated by its diagonal apart from a small block associated with those modes. This structure means that the system can be solved efficiently by modern iterative methods, at a cost of a few multiplications by A per frequency, without ever forming the inverse or diagonalising the full coupling matrix. This is the **direct solution method** of Al-Attar et al. (2012), and it is now the standard means of computing long-period spectra in laterally heterogeneous earth models. In practice the system is solved not at real frequencies but along a line shifted slightly into the complex plane. This keeps the diagonal terms away from

zero, and corresponds, once the solutions are transformed back to the time domain, to multiplying the seismogram by a decaying exponential, which is harmless over a record of finite length.

The first-order splitting theory is recovered by keeping only the $2l+1$ singlets of a single multiplet in the expansion, and this so-called **self-coupling** approximation is the workhorse of most splitting studies. Within it the effect of lateral heterogeneity on a multiplet is summarised by its **splitting function**, a function on the sphere whose spherical harmonic coefficients are, up to known factors, the matrix elements in eq. (28). It can be thought of as a map of the local perturbation to the degenerate eigenfrequency, obtained by integrating the lateral heterogeneity against the mode's radial sensitivity kernels. Splitting functions are measured from the spectra of many earthquakes, and then form the data for inversions for lateral structure, including density. Their appeal is that they collapse a great many spectra into a few coefficients per multiplet that depend only on Earth structure. Their weakness is that they are defined within the self-coupling approximation, and their sensitivity to density is weak and easily contaminated by errors in that approximation, particularly for multiplets that lie close in frequency to others. Only in the past decade or so has it become possible to compute the spectra of realistic 3D earth models in the full-coupling manner just described, and such calculations have shown that self-coupling is not accurate enough for density (Akbarashrafi et al. 2018), the errors it introduces being comparable to the density signal itself. The problem is more severe still for modes whose energy is concentrated near the core–mantle boundary, exactly those most sensitive to the density of the LLSVPs. Hence the results of Ishii and Tromp cannot be taken at face value, and work continues on making full-coupling calculations efficient enough to use routinely within inversions. This being said, a more recent but related study was led by Harriet Lau using observations of solid Earth tides (Lau et al. 2017). In this case sophisticated and accurate numerical modelling was done, with the result of this analysis still supporting the idea that LLSVPs are dense stable features at the base of the mantle.

Finally, it is worth drawing together the threads of the last few lectures. Once spectra can be computed accurately for a 3D earth model, the natural next step is to fit them directly, in the manner of the waveform inversions of Lecture 20, rather than proceeding through intermediate measurements such as splitting functions. The ingredient needed is the sensitivity of the spectra to the model, and this is provided by the adjoint method: applied to the direct solution method, it yields the kernels of the misfit with respect to Earth structure at the cost of one further solution per frequency (Adourian et al. 2024). Full-waveform inversion of the Earth's free oscillations for its long-wavelength structure, including density, is thus now feasible, and this is where much of the current work in the subject lies.

What you need to know and be able to do

(i) How eigenfunctions of a spherically symmetric model can be classified in terms of mode type, degree, overtone number, and order. You do not need to learn the definition of vector spherical harmonics, and will not be asked to perform detailed calculations with them.

(ii) How the toroidal mode eigenvalue problem reduces to a one-dimensional problem for the radial eigenfunction, and the general properties of its solutions (over-

tone number, nodes, boundary condition at the core–mantle boundary).

(iii) How degenerate perturbation theory can be used to understand mode splitting due to rotation and lateral heterogeneity.

(iv) What the diagonal sum rule says (but not the derivation), and how it was applied in determining spherically averaged density variations.

(v) The idea that convection in the mantle might be driven in part by compositional variations, and how this complicates the dynamic interpretation of tomographic velocity models.

(vi) The idea of full coupling, namely that the eigenfunctions or the forced response of a laterally heterogeneous earth model can be expanded in the modes of a spherical reference model, and why the simpler first-order splitting theory can be inadequate, particularly for density. You do not need the details of how such calculations are done, nor anything about splitting functions or adjoint methods for spectra.

Problem Set 1

Questions, or parts of questions, marked with a dagger (†) are more challenging than the rest. They can be attempted, or their solutions simply read through; everything unmarked should be attempted.

1. Within Lecture 13, we derived the following equation

$$\rho \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0,$$

for an elastic body. Here ρ is the referential density, $v_i = \frac{\partial \varphi_i}{\partial t}$ the velocity with $\boldsymbol{\varphi}$ the motion, and $\mathbf{T}$ the first Piola–Kirchhoff stress tensor.

   (a) For an arbitrary sub-body, U, of the reference body, M, show that the equality

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_U \rho v_i \, \mathrm{d}^3\mathbf{x} = \int_{\partial U} T_{ij} \hat{n}_j \, \mathrm{d}S,$$

   holds, where ∂U is the boundary of U. What is the physical significance of this result? In particular, what does the vector $T_{ij}\hat{n}_j$ represent?

   (b) The angular momentum of the sub-body U about the origin is given by

$$\int_U \rho \epsilon_{ijk} \varphi_j v_k \, \mathrm{d}^3\mathbf{x}.$$

   Show that the angular momentum of every such sub-body changes only through the torque exerted by the tractions on its boundary if and only if $\mathbf{F}\mathbf{T}^T$ is a symmetric tensor. Defining the second Piola–Kirchhoff stress, $\mathbf{S}$, through $\mathbf{T} = \mathbf{F}\mathbf{S}$, conclude that $\mathbf{S}$ is symmetric.

2. Continuing from the previous question, we define the energy density of the body by

$$E = \frac{1}{2}\rho v_i v_i + W(\mathbf{x}, \mathbf{F}),$$

   where W is the strain energy function, so that $T_{ij} = \frac{\partial W}{\partial F_{ij}}$. Show that

$$\frac{\partial E}{\partial t} + \frac{\partial s_i}{\partial x_i} = 0,$$

   where s_i is a vector field to be determined. By integrating this identity over an arbitrary sub-body, U, explain the physical meaning of s_i, and hence show that the total energy of the body is conserved, using the traction-free boundary condition $T_{ij}\hat{n}_j = 0$ on ∂M.

3. The Lagrangian density for the linearised motion of an elastic body subject to a stress glut is given by

$$\mathcal{L} = \frac{1}{2}\rho \frac{\partial u_i}{\partial t}\frac{\partial u_i}{\partial t} - \frac{1}{2}A_{ijkl}\frac{\partial u_i}{\partial x_j}\frac{\partial u_k}{\partial x_l} + \overline{S}_{ij}\frac{\partial u_i}{\partial x_j},$$

where $\mathbf{u}$ is the displacement vector, A_{ijkl} the elastic tensor, and $\overline{\mathbf{S}}$ the stress glut. Using this expression, write down the equations of motion and natural boundary conditions; here you may assume the standard form of the Euler–Lagrange equations.

4. The principle of material frame indifference shows that the strain energy function, W, of an elastic body depends on the deformation gradient, $\mathbf{F}$, only through the right Cauchy–Green deformation tensor, $\mathbf{C} = \mathbf{F}^T\mathbf{F}$, and hence we can write $W(\mathbf{x}, \mathbf{F}) = U(\mathbf{x}, \mathbf{C})$, for some auxiliary function, U. The elastic tensor, A_{ijkl}, is defined by
$$A_{ijkl} = \frac{\partial^2 W}{\partial F_{ij} \partial F_{kl}},$$
where the partial derivative on the right-hand side is evaluated at the equilibrium value $\mathbf{F} = \mathbf{I}$. Assuming that the equilibrium is stress-free, show that the elastic tensor possesses the symmetries $A_{ijkl} = A_{jikl} = A_{ijlk} = A_{klij}$. How many independent components does such an elastic tensor have? Do these symmetries remain if there is a non-zero equilibrium stress?

5. In lectures we found the stress glut for an idealised fault surface Σ to be $\overline{S}_{ij} = A_{ijkl} w_k \hat{n}_l \delta_\Sigma$, where $\mathbf{w}$ is the displacement discontinuity vector, which is tangential to Σ, $\hat{\mathbf{n}}$ is the unit normal to Σ, and δ_Σ is the Dirac surface distribution for Σ.

   (a) Consider a planar fault with unit normal $\hat{\mathbf{n}}$ within an isotropic body. Show that
   $$\overline{S}_{ij} = \mu(w_i \hat{n}_j + w_j \hat{n}_i)\delta_\Sigma,$$
   and comment on the absence of λ. Verify that this stress glut is symmetric and has zero trace.

   (b)† Consider the linearised equations of motion in a body M subject to a body force f_i and an applied traction t_i on ∂M,
   $$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j}\left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = f_i \quad \text{in } M, \qquad A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = t_i \quad \text{on } \partial M,$$
   with the body at rest prior to $t = 0$. The solution of this problem can be expressed in terms of a Green function, $G_{ij}(\mathbf{x}, t; \mathbf{x}', t')$, in the form
   $$u_i(\mathbf{x}, t) = \int_0^t \int_M G_{ij}(\mathbf{x}, t; \mathbf{x}', t') f_j(\mathbf{x}', t') \, \mathrm{d}^3\mathbf{x}' \, \mathrm{d}t' + \int_0^t \int_{\partial M} G_{ij}(\mathbf{x}, t; \mathbf{x}', t') t_j(\mathbf{x}', t') \, \mathrm{d}S' \, \mathrm{d}t',$$
   where the detailed form of G_{ij} will not be needed. Recall that a stress glut, $\overline{S}_{ij}$, enters the linearised equations of motion as an effective body force $f_i = -\partial \overline{S}_{ij}/\partial x_j$ together with an effective traction $t_i = \overline{S}_{ij}\hat{n}_j$, where $\hat{\mathbf{n}}$ is the outward unit normal to ∂M. Note that within the integrals above f_j, t_j and $\hat{\mathbf{n}}$ are all evaluated at the integration point $\mathbf{x}'$. Show that the resulting displacement can be written as a single integral over the stress glut,
   $$u_i(\mathbf{x}, t) = \int_0^t \int_M \frac{\partial G_{ij}}{\partial x'_k}(\mathbf{x}, t; \mathbf{x}', t') \overline{S}_{jk}(\mathbf{x}', t') \, \mathrm{d}^3\mathbf{x}' \, \mathrm{d}t'.$$

(c)† Using the result of part (a), show that the displacement generated by slip on a planar fault within an isotropic body is

$$u_i(\mathbf{x},t) = \int_0^t \int_\Sigma \mu \left[\frac{\partial G_{ij}}{\partial x'_k} + \frac{\partial G_{ik}}{\partial x'_j} \right] (\mathbf{x},t;\mathbf{x}',t')\, w_j(\mathbf{x}',t') \hat{n}_k(\mathbf{x}')\, \mathrm{d}S'\, \mathrm{d}t',$$

and comment on the physical content of this expression. (The normal $\hat{\mathbf{n}}$ is constant for a planar fault, but it has been written as a function of $\mathbf{x}'$ because the result holds unchanged for a general fault surface.)

6. Consider plane wave propagation in a homogeneous and transversely isotropic whole space. For such a material the elastic tensor takes the form

$$\begin{aligned} A_{ijkl} = {} & (A-2N)\delta_{ij}\delta_{kl} + N(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \\ & + (F-A+2N)(\delta_{ij}\hat{\nu}_k\hat{\nu}_l + \hat{\nu}_i\hat{\nu}_j\delta_{kl}) + (A+C-2F-4L)\hat{\nu}_i\hat{\nu}_j\hat{\nu}_k\hat{\nu}_l \\ & + (L-N)(\hat{\nu}_i\hat{\nu}_k\delta_{jl} + \hat{\nu}_j\hat{\nu}_k\delta_{il} + \hat{\nu}_j\hat{\nu}_l\delta_{ik} + \hat{\nu}_i\hat{\nu}_l\delta_{jk}), \end{aligned}$$

where A, C, F, L and N are elastic moduli, and $\hat{\boldsymbol{\nu}}$ is a unit vector along the symmetry axis. Let θ denote the angle between the propagation direction, $\hat{\mathbf{p}}$, and the symmetry axis, and regard the departure of the medium from isotropy as small.

(a) Using first-order perturbation theory, obtain an expression for the phase speed of a quasi P-wave as a function of θ.

(b) Using degenerate perturbation theory, obtain the first-order phase speeds of the two quasi S-waves as functions of θ, and describe their polarisations. Check your results against the exact phase speeds for propagation along and perpendicular to the symmetry axis (these follow from the components $A_{3333} = C$, $A_{1313} = L$, $A_{1111} = A$ and $A_{1212} = N$ quoted in Lecture 14). [Hint: choose one of the unperturbed polarisation vectors to lie in the plane containing $\hat{\mathbf{p}}$ and $\hat{\boldsymbol{\nu}}$, and the other perpendicular to this plane.]

7. Show that the time-dependent Schrödinger equation for a free particle

$$\mathrm{i}\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi,$$

admits plane wave solutions of the form

$$\psi(\mathbf{x},t) = a \exp\left[\frac{-\mathrm{i}(Et - \mathbf{p}\cdot\mathbf{x})}{\hbar} \right]$$

and obtain the relationship between the energy E and momentum $\mathbf{p}$. Consider the corresponding equation for a particle in a non-constant but smoothly varying potential field V:

$$\mathrm{i}\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m}\nabla^2 + V \right) \psi.$$

Starting from the ansatz

$$\psi(\mathbf{x},t) = a(\mathbf{x}) \exp\left\{ \frac{-\mathrm{i}[Et - \varphi(\mathbf{x})]}{\hbar} \right\},$$

where the amplitude a and phase function φ are to be determined, and retaining only the leading-order terms in powers of $1/\hbar$, obtain the eikonal equation

$$H(\mathbf{x}, \nabla\varphi) = E,$$

where H is the classical Hamiltonian. Applying the method of characteristics, show that the particle's phase function can be determined through solution of the classical equations of motion.

8. Consider P-wave rays within an isotropic body. The travel time of a ray between two points $\mathbf{x}_1$ and $\mathbf{x}_2$ can be written

$$T = \int_0^1 \frac{1}{\alpha(\mathbf{x})} \sqrt{\frac{\mathrm{d}x_i}{\mathrm{d}\gamma}\frac{\mathrm{d}x_i}{\mathrm{d}\gamma}}\,\mathrm{d}\gamma,$$

where $\gamma \mapsto \mathbf{x}(\gamma)$ is the ray path between $\mathbf{x}_1$ and $\mathbf{x}_2$ parameterised by a generating parameter $\gamma \in [0,1]$. Regarding this travel time as a functional of the given ray path, Fermat's principle states that the true ray path is a stationary point of this functional. Obtain the corresponding Euler–Lagrange equations, and show that they are equivalent to the Hamiltonian ray tracing equations discussed within lectures. [Hint: it will be useful to express both sets of ordinary differential equations in terms of the arc length along the ray.]

9. Consider a plane wave $u_i = f(t - p_j x_j)a_i$ within a homogeneous anisotropic body, where the slowness vector $\mathbf{p}$ and the unit polarisation vector $\mathbf{a}$ satisfy the Christoffel equation $\mathbf{\Gamma}(\mathbf{p})\mathbf{a} = \mathbf{a}$, with $\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho}A_{ijkl}p_j p_l$. The energy conservation law of Question 2 carries over to the linearised theory if T_{ij} and v_i are replaced by their linearised forms, $A_{ijkl}\frac{\partial u_k}{\partial x_l}$ and $\frac{\partial u_i}{\partial t}$, and only terms of second order in the displacement are retained, giving the energy density and energy flux

$$E = \frac{1}{2}\rho\frac{\partial u_i}{\partial t}\frac{\partial u_i}{\partial t} + \frac{1}{2}A_{ijkl}\frac{\partial u_i}{\partial x_j}\frac{\partial u_k}{\partial x_l}, \quad s_j = -A_{ijkl}\frac{\partial u_k}{\partial x_l}\frac{\partial u_i}{\partial t}.$$

   (a) Show that for the plane wave the kinetic and elastic contributions to E are equal, and that $s_j = Ev_j$, where the *energy velocity* is

$$v_j = \frac{1}{\rho}A_{ijkl}a_i a_k p_l.$$

   Show that $\mathbf{v}\cdot\hat{\mathbf{p}} = c$, the phase speed, and deduce that $\|\mathbf{v}\| \geq c$, with equality if and only if $\mathbf{v}$ is parallel to $\hat{\mathbf{p}}$. Evaluate $\mathbf{v}$ for P- and S-waves in an isotropic body.

   (b)† Let $\lambda(\mathbf{p})$ denote the eigenvalue of $\mathbf{\Gamma}(\mathbf{p})$ associated with $\mathbf{a}$, so that the corresponding sheet of the slowness surface is $\lambda(\mathbf{p}) = 1$ and the ray Hamiltonian for this sheet is $H(\mathbf{p}) = \frac{1}{2}\lambda(\mathbf{p})$. Show that

$$\frac{\partial\lambda}{\partial p_j} = \mathbf{a}\cdot\frac{\partial\mathbf{\Gamma}}{\partial p_j}\mathbf{a},$$

   [Hint: differentiate $\lambda = \mathbf{a}\cdot\mathbf{\Gamma}\mathbf{a}$, remembering that $\mathbf{a}$ depends on $\mathbf{p}$ but has unit length.] Hence show that $\mathbf{v} = \partial H/\partial\mathbf{p}$, and deduce that the energy velocity is normal to the slowness surface and equal to $\mathrm{d}\mathbf{x}/\mathrm{d}\sigma$ along a ray.

(c)† An impulsive point source acts at the origin of the body at time $t = 0$. Using ray theory, show that the wavefront at time t is the surface traced out by $t\,\mathbf{v}(\hat{\mathbf{p}})$ as $\hat{\mathbf{p}}$ ranges over the unit sphere; this is known as the *wave surface*. Describe the geometric relationship between the wave surface and the slowness surface.

Problem Set 2

Questions, or parts of questions, marked with a dagger (†) are more challenging than the rest. They can be attempted, or their solutions simply read through; everything unmarked should be attempted.

1. In Lecture 18 it was shown that, for a P-wave ray in a spherically symmetric earth model of radius b whose turning point lies at radius r_t, the epicentral angle and travel time are

$$\Delta(q) = \int_{r_t}^{b} \frac{2\alpha q}{r\sqrt{r^2 - \alpha^2 q^2}}\,\mathrm{d}r, \quad T(q) = \int_{r_t}^{b} \frac{2r}{\alpha\sqrt{r^2 - \alpha^2 q^2}}\,\mathrm{d}r,$$

where $q = r\sin\theta/\alpha$ is the ray parameter, θ being the angle between the ray and the radial direction, and r_t satisfies $r_t/\alpha(r_t) = q$. This question shows how such a travel-time curve can be inverted for the velocity structure, a method due to Herglotz and Wiechert. Throughout, define $\eta(r) = r/\alpha(r)$ and assume that η increases monotonically with r.

   (a) Show that $\tau(q) = T(q) - q\Delta(q)$ can be written

$$\tau(q) = 2\int_{r_t}^{b} \sqrt{\eta^2 - q^2}\,\frac{\mathrm{d}r}{r},$$

   and hence that $\mathrm{d}\tau/\,\mathrm{d}q = -\Delta$. [Remember that the lower limit r_t depends on q, so that Leibniz's rule for differentiating an integral with respect to a parameter is needed; consider the value of the integrand at the turning point.] Deduce that $\mathrm{d}T/\,\mathrm{d}\Delta = q$ along the travel-time curve, so that the ray parameter of an arrival can be measured from the local slope of the observed curve.

   (b) Using η as the integration variable, show that

$$\Delta(q) = 2\int_{q}^{\eta_b} \frac{q}{\sqrt{\eta^2 - q^2}}\,\frac{\mathrm{d}\ln r}{\mathrm{d}\eta}\,\mathrm{d}\eta,$$

   where $\eta_b = \eta(b)$.

   (c)† Let η_1 be the ray parameter of the ray turning at some radius r_1. Multiply the result of (b) by $(q^2 - \eta_1^2)^{-1/2}$, integrate over q from η_1 to η_b, and exchange the order of integration to show that

$$\ln\frac{b}{r_1} = \frac{1}{\pi}\int_{\eta_1}^{\eta_b} \frac{\Delta(q)}{\sqrt{q^2 - \eta_1^2}}\,\mathrm{d}q.$$

   You may use the identity

$$\int_{\eta_1}^{\eta} \frac{q\,\mathrm{d}q}{\sqrt{(\eta^2 - q^2)(q^2 - \eta_1^2)}} = \frac{\pi}{2}.$$

By integrating by parts, show that this can equally be written in terms of the observed travel-time curve as

$$\ln \frac{b}{r_1} = \frac{1}{\pi} \int_0^{\Delta_1} \cosh^{-1}\left[\frac{q(\Delta)}{\eta_1}\right] \, \mathrm{d}\Delta,$$

where $\Delta_1 = \Delta(\eta_1)$ and $q(\Delta) = \mathrm{d}T/\,\mathrm{d}\Delta$.

(d) Explain how these results allow $\alpha(r)$ to be determined from a travel-time curve, working downwards from the surface. What goes wrong if there is a region in which the velocity decreases with depth so rapidly that η is no longer monotonic? And what is seen in the travel-time curve at a discontinuity across which the velocity increases with depth?

2. Consider the regularised least squares solution to a linear(ised) inverse problem obtained by minimising the function

$$J(\mathbf{m}) = \frac{1}{2}(\mathbf{Am} - \mathbf{d})^T \mathbf{C}^{-1}(\mathbf{Am} - \mathbf{d}) + \frac{\lambda}{2}\mathbf{m}^T \mathbf{Bm}.$$

Suppose that the data take the form $\mathbf{d} = \mathbf{Am}_{\text{in}}$ with $\mathbf{m}_{\text{in}}$ a specified input model. Show that the resulting least squares solution, $\mathbf{m}_{\text{out}}$, takes the form

$$\mathbf{m}_{\text{out}} = \mathbf{R}\mathbf{m}_{\text{in}},$$

where $\mathbf{R}$ is a matrix to be determined. Discuss the significance of this so-called resolution matrix, explaining why, in particular, it would be desirable for its value to be close to the identity matrix.

3. Consider a linear inverse problem of the standard form

$$\mathbf{d} = \mathbf{Am} + \mathbf{e},$$

with $\mathbf{d}$ an n-dimensional data vector, $\mathbf{m}$ an m-dimensional model vector, $\mathbf{A}$ an $n \times m$ matrix, and $\mathbf{e}$ an n-dimensional vector of random errors. We assume $m > n$, which is to say that there are more unknown model parameters than available data. We further suppose that the data have been processed such that the components of the random errors are uncorrelated and have unit standard deviations (this corresponds to the data covariance, $\mathbf{C}$, being the $n \times n$ identity matrix).

(a) A model $\mathbf{m}$ is said to be compatible with the data if

$$\|\mathbf{Am} - \mathbf{d}\|^2 \leq \chi_c^2,$$

where χ_c^2 is a critical value for the so-called chi-squared statistic. What type of geometric region of the model space is defined by this inequality?

(b) Supposing that $\|\mathbf{d}\|^2 > \chi_c^2$, the region of the model space defined by the inequality in (a) does not contain the origin. In such cases, the closest point within this region to the origin must lie on the boundary set defined by

$$\|\mathbf{Am} - \mathbf{d}\|^2 = \chi_c^2.$$

Using the method of Lagrange multipliers, we can determine this *minimum norm solution* by considering the Lagrangian

$$\mathcal{L}(\mathbf{m}, \mu) = \|\mathbf{m}\|^2 + \mu(\|\mathbf{A}\mathbf{m} - \mathbf{d}\|^2 - \chi_c^2),$$

where μ is a Lagrange multiplier. Derive the equations for $\mathbf{m}$ and μ that must hold at the minimum norm solution. Discuss the relationship between this approach and regularised least squares.

4.† Consider the deformation of an isotropic elastic body due to an applied surface load, σ. The relevant linearised equilibrium equations are

$$-\frac{\partial T_{ij}}{\partial x_j} = 0,$$

$$T_{ij} = \lambda \frac{\partial u_k}{\partial x_k}\delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right),$$

within the reference body M, and subject to boundary conditions

$$T_{ij}\hat{n}_j = \sigma g_i$$

on its boundary, ∂M.[1] Here g_i is the acceleration due to gravity prior to the deformation. Note that self-gravitation is being ignored within these equations.

Suppose that observations, $\mathbf{u}_r^{\text{obs}}$, of the displacement vector have been made at surface locations $\mathbf{x}_r$ for $r = 1, \ldots, N$. An inverse problem could then be formulated to estimate the Lamé parameters, λ and μ, as well as the load, σ. To this end, we can seek to minimise the misfit

$$J = \frac{1}{2}\sum_{r=1}^{N} \frac{1}{\varepsilon_r^2}\|\mathbf{u}(\mathbf{x}_r) - \mathbf{u}_r^{\text{obs}}\|^2,$$

where ε_r is the standard deviation for the rth observation. Adapt the approach within Lecture 20 to derive sensitivity kernels for J with respect to λ, μ, and σ, showing how they can be calculated through one solution of the forward problem and one of an appropriate adjoint problem.

5. Recall that the Lagrangian for a self-gravitating elastic body is given by

$$\mathcal{L}(\mathbf{x}, t, \boldsymbol{\varphi}, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x}, t),$$

where ζ is the referential gravitational potential

$$\zeta(\mathbf{x}, t) = -G\int_M \frac{\rho(\mathbf{x}')}{\{[\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)][\varphi_i(\mathbf{x}, t) - \varphi_i(\mathbf{x}', t)]\}^{\frac{1}{2}}}\,\mathrm{d}^3\mathbf{x}'.$$

By calculating the appropriate functional derivative, show that

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = \rho\gamma_i,$$

where γ_i is the referential gravitational acceleration.

[1] For a purely traction-loaded static problem the load must exert no net force or torque and the displacement is determined only up to a rigid-body motion; we ignore this subtlety here.

6. Using results from Lecture 23 along with residue calculus (which you don't need to know for this course), it can be shown that the time-domain solution of the elastodynamic equations in a non-rotating earth model can be expressed as the following eigenfunction expansion

$$\mathbf{u}(\mathbf{x},t) = \sum_{km} \left\{ \int_0^t \frac{\sin[\omega_k(t-t')]}{\omega_k} \int_M \overline{S}_{ij}(\mathbf{x}',t') \frac{\partial |km\rangle_i^*}{\partial x_j'}(\mathbf{x}') \, \mathrm{d}^3\mathbf{x}' \, \mathrm{d}t' \right\} |km\rangle(\mathbf{x}),$$

where $|km\rangle_i$ denotes the ith Cartesian component of the eigenfunction.

   (a) Suppose that the stress glut is non-zero only within a region of dimension L about a point $\mathbf{x}_s$, as it would be for an earthquake on a fault of that size. By expanding $\partial |km\rangle_i^* / \partial x_j'$ in a Taylor series about $\mathbf{x}_s$, show that the contribution of a mode to the sum can be approximated by replacing the stress glut with the *point source*

$$\overline{S}_{ij}(\mathbf{x},t) = M_{ij}(t)\delta(\mathbf{x}-\mathbf{x}_s), \quad M_{ij}(t) = \int_M \overline{S}_{ij}(\mathbf{x},t) \, \mathrm{d}^3\mathbf{x},$$

   provided that L is small compared with the length scale over which the strain of the eigenfunction varies. Estimate this length scale in terms of the eigenfrequency and a wave speed, and hence explain why the point-source approximation is a statement about the frequencies retained in the seismograms. For a fault of length 100 km and a shear wave speed of $5\,\mathrm{km\,s^{-1}}$, below roughly what frequency is the approximation reasonable?

   (b) Simplify the eigenfunction expansion in the case of a point source with a step-function time dependence,

$$\overline{S}_{ij}(\mathbf{x},t) = M_{ij}\delta(\mathbf{x}-\mathbf{x}_s)H(t-t_s),$$

   with M_{ij} known as the moment tensor, $\mathbf{x}_s$ the source location, and t_s the source time. Describe briefly and qualitatively how an inverse problem might be formulated to estimate the ten source parameters $(M_{ij}, \mathbf{x}_s, t_s)$ from recorded seismograms.

7. Consider the eigenvalue problem for a rotating earth model

$$-\omega^2 \langle \mathbf{w}|P|\mathbf{s}\rangle + \mathrm{i}\omega \langle \mathbf{w}|W|\mathbf{s}\rangle + \langle \mathbf{w}|H|\mathbf{s}\rangle = 0,$$

   where $\mathbf{s}$ is the eigenfunction, ω the eigenfrequency, and $\mathbf{w}$ an arbitrary test function. Assuming the eigenfunction normalisation condition

$$\langle \mathbf{s}|P|\mathbf{s}\rangle = 1,$$

   show that the eigenfrequency can be written

$$\omega = \frac{1}{2}\mathrm{i}\langle \mathbf{s}|W|\mathbf{s}\rangle \pm \sqrt{\langle \mathbf{s}|H|\mathbf{s}\rangle - \frac{1}{4}\langle \mathbf{s}|W|\mathbf{s}\rangle^2}.$$

   Use this result to discuss qualitatively the stability of a rotating earth model, including, in particular, the role of the Coriolis and centrifugal forces.

8. Consider a non-rotating planet of radius a with uniform density ρ, and let M_p be its mass.

   (a) Show that the gravitational potential energy (or binding energy) of the planet is
   $$E_g = -\frac{3GM_\mathrm{p}^2}{5a}.$$

   (b) Assuming the planet to be in hydrostatic equilibrium, show that its pressure is
   $$p_0(r) = \frac{2\pi}{3}G\rho^2(a^2 - r^2),$$
   and hence verify that $3\int_M p_0 \,\mathrm{d}^3\mathbf{x} = -E_g$.

   (c) Suppose that the planet is compressed uniformly, so that its radius becomes $a(1-\epsilon)$ with ϵ small and positive, the density remaining uniform. Determine the change in the gravitational potential energy correct to second order in ϵ.

   (d) Let the material respond to compression through $p = p_0 + \kappa s$, where s is the fractional decrease in the volume of a material element and κ is a constant bulk modulus. Recalling that the work done in compressing a material element of initial volume V_0 is $V_0 \int p \,\mathrm{d}s$, show that the change in the internal energy of the planet is
   $$\Delta E_\mathrm{int} = (3\epsilon - 3\epsilon^2)\int_M p_0 \,\mathrm{d}^3\mathbf{x} + \frac{9}{2}\kappa V\epsilon^2 + O(\epsilon^3),$$
   where V is the volume of the planet.

   (e) Show that the total change in energy vanishes at first order in ϵ, and explain why this had to be so. Show that the second-order change is positive, and hence the planet stable with respect to this perturbation, if and only if
   $$\kappa > \frac{4}{3}\bar{p}_0 = \frac{16\pi}{45}G\rho^2a^2,$$
   where $\bar{p}_0$ is the mean pressure. Express this condition in terms of the sound speed $\sqrt{\kappa/\rho}$ and the surface gravity g, and evaluate it for $\rho = 5500\,\mathrm{kg\,m^{-3}}$ and $a = 6371\,\mathrm{km}$. Comment on the result in relation to the bulk moduli within the earth model PREM, which range from roughly 130 GPa in the upper mantle to 650 GPa at the base of the mantle.

9. This question concerns the splitting of a multiplet by the Earth's rotation, using the first-order theory of Lecture 24. Ignore the centrifugal force and the Earth's ellipticity, so that within the multiplet the only perturbation is the Coriolis form W, and take the rotation vector to be $\boldsymbol{\Omega} = \Omega\hat{\mathbf{z}}$.

   (a) It can be shown that the Coriolis matrix elements within a multiplet of degree l are diagonal in the order m, taking the form
   $$\langle km'|W|km\rangle = -2\mathrm{i}m\Omega\beta\,\delta_{mm'},$$

where β is a real number depending on the radial eigenfunctions of the multiplet. Show that the singlets $|km\rangle$ are then eigenfunctions of the first-order splitting problem, with eigenfrequencies

$$\omega_m = \omega_k + m\Omega\beta, \quad -l \le m \le l,$$

so that the multiplet is split into $2l+1$ equally spaced singlets, and verify that the diagonal sum rule holds. Show that this is consistent with the result of question 7.

(b)† Consider a toroidal multiplet with singlets of the form

$$|km\rangle = W(r)\mathbf{C}_{lm}(\theta,\varphi)$$

(the radial eigenfunction $W(r)$, not to be confused with the Coriolis operator W), normalised so that

$$l(l+1)\int_0^b \rho W^2 r^2 \,\mathrm{d}r = 1.$$

Show that $\beta = 1/[l(l+1)]$. You may use

$$\mathbf{C}_{lm} = -\hat{\mathbf{r}} \times \nabla_1 Y_{lm}, \quad \nabla_1 Y_{lm} = \hat{\boldsymbol{\theta}}\,\frac{\partial Y_{lm}}{\partial\theta} + \frac{\hat{\boldsymbol{\varphi}}}{\sin\theta}\frac{\partial Y_{lm}}{\partial\varphi},$$

the dependence $Y_{lm} \propto \mathrm{e}^{\mathrm{i}m\varphi}$, and the fact that Y_{lm} vanishes at the poles for $m \neq 0$. [For a spheroidal multiplet a similar calculation gives $\beta = \int_0^b \rho(2UV+V^2)r^2\,\mathrm{d}r$ with the normalisation $\int_0^b \rho[U^2+l(l+1)V^2]r^2\,\mathrm{d}r = 1$, which for ${}_0S_2$ in PREM gives $\beta \approx 0.40$ and for ${}_0S_3$ gives $\beta \approx 0.19$.]

(c) The Earth rotates once per sidereal day, 86164 s. Estimate the spacing between adjacent singlets of ${}_0T_2$, ${}_0S_2$ and ${}_0S_3$, and compare with the observed spectra in Fig. 6 of Lecture 24. Why might the observed singlets not be exactly equally spaced?

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